

GENERAL STRUCTURAL NOTES

APPLY UNLESS NOTED OTHERWISE

1.0 STRUCTURAL ENGINEERING GENERAL REQUIREMENTS:

BUILDING CODE:

2021 EDITION OF THE INTERNATIONAL BUILDING CODE / ASCE/SEI 7-16 WITH CITY OF THORNTON AMENDMENTS.

DESIGN CRITERIA / LOADS:

ALL COMPONENTS REQUIRING DESIGN BY THE CONTRACTOR AND THEIR SUBCONTRACTORS SHALL USE THE DESIGN CRITERIA DEFINED BELOW. ALL FORCES AND PRESSURES LISTED BELOW SHALL BE THE MINIMUM PERMISSIBLE UNLESS SUBSTANTIATING CALCULATIONS USING THE DESIGN CRITERIA SHOWN ARE PROVIDED TO THE FOR REVIEW AND ACCEPTANCE PRIOR TO FABRICATION.

RISK CATEGORY = IV

FLOOR:
TYPICAL FLOOR LIVE LOAD = 100 PSF (REDUCIBLE), EXCEPT AS FOLLOWS:
LOBBY, CORRIDOR AND STAIR LIVE LOAD = 100 PSF (NON-REDUCIBLE)
STORAGE LIVE LOAD = 125 PSF (NON-REDUCIBLE)
SUPERIMPOSED DEAD LOAD ON FLOOR STRUCTURE = 12 PSF.
APPARATUS BAY = 140 PSF (NON-REDUCIBLE)

ROOF:
ROOF LIVE LOAD = 20 PSF (NON-REDUCIBLE)
SUPERIMPOSED DEAD LOAD ON ROOF STRUCTURE = 19 PSF.
NET UPLIFT ON ROOF JOISTS/TRUSSES:
ZONE 1 = 18.1 PSF (ASD)
ZONE 2 = 27.0 PSF (ASD)
ZONE 3 = 40.3 PSF (ASD)

SNOW:
GROUND SNOW LOAD, Pg = 30 PSF
ROOF SNOW LOAD, Pt = 30 PSF X 1.2 = 36 PSF (NON-REDUCIBLE)
SNOW DRIFT PER ASCE-7 (Ce = 1.0, Ct = 1.1, Cs = 1.0)
IMPORTANCE FACTOR, I = 1.2

WIND DESIGN DATA:
RISK CATEGORY = IV
ULTIMATE DESIGN WIND SPEED, Vult = 120 MPH (3 SEC. GUST)
EQUIVALENT NOMINAL DESIGN WIND SPEED, Vasd = 88 MPH (3 SEC. GUST, USING I = 1.15)
EXPOSURE C
MEAN ROOF HEIGHT USED FOR DESIGN, h = 25 FT
Kd = 0.85
Kzt = 1.0
VELOCITY PRESSURE CONSTANT = 0.83 X 0.00256 = 0.00212 (AMBIENT AIR DENSITY ADJUSTMENT)
INTERNAL PRESSURE COEFFICIENT, Gcpi = ±0.18
COMPONENTS AND CLADDING DESIGN WIND PRESSURES (BASED ON 10 SF):

STRENGTH DESIGN:
-28.8 PSF (WALL, FIELD); -35.4 PSF (WALL, CORNER);
-46.2 PSF (ROOF, ZONE 1); -61.0 PSF (ROOF, ZONE 2); -83.1 PSF (ROOF, ZONE 3)

ALLOWABLE STRESS DESIGN:
-17.3 PSF (WALL, FIELD); -21.2 PSF (WALL, CORNER);
-27.7 PSF (ROOF, ZONE 1); -36.6 PSF (ROOF, ZONE 2); -49.9 PSF (ROOF, ZONE 3)

SEISMIC DESIGN DATA:
RISK CATEGORY = IV
SEISMIC IMPORTANCE FACTOR, Ie = 1.5
Ss = 0.189, S1 = 0.055
SITE CLASS: D
SDS = 0.202, SD1 = 0.088
SEISMIC DESIGN CATEGORY C
BASIC SEISMIC FORCE RESISTING SYSTEM = STEEL SYSTEMS NOT SPECIFICALLY DESIGNED FOR SEISMIC
RESISTANCE/INTERMEDIATE REINFORCED MASONRY SHEAR WALLS.
DESIGN BASE SHEAR, V = 88 K (ULT)
SEISMIC RESPONSE COEFFICIENT, STRENGTH DESIGN, Cs = 0.101
RESPONSE MODIFICATION FACTOR, R = 3
ANALYSIS PROCEDURE = EQUIVALENT LATERAL FORCE PROCEDURE

NON-STRUCTURAL COMPONENT SEISMIC BRACING CRITERIA:

FOR DESIGN CATEGORY C.

NON-STRUCTURAL COMPONENT SEISMIC BRACING CRITERIA: IT IS THE CONTRACTOR'S RESPONSIBILITY TO PROVIDE AND DESIGN ALL NON-STRUCTURAL COMPONENT SEISMIC BRACING FOR THE PROJECT IN ACCORDANCE WITH CHAPTER 13 OF THE ASCE 7-16. EXEMPTIONS PER SECTION 13.1.4 OF ASCE 7-16 APPLY AS FOLLOWS:

- FURNITURE.
- TEMPORARY OR MOVABLE EQUIPMENT.
- MECHANICAL AND ELECTRICAL COMPONENTS WHERE Ip IS EQUAL TO 1.0.

REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND PLUMBING DRAWINGS AND SPECIFICATIONS FOR ADDITIONAL INFORMATION, SUBMITTALS, AND REQUIREMENTS FOR NON-STRUCTURAL COMPONENTS. NON-STRUCTURAL COMPONENT SEISMIC BRACING AND ITS REQUIREMENTS SHALL BE REVIEWED BY DISCIPLINES OTHER THAN THE STRUCTURAL ENGINEER.

REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND PLUMBING DRAWINGS AND SPECIFICATIONS FOR ADDITIONAL INFORMATION, SUBMITTALS, AND REQUIREMENTS FOR NON-STRUCTURAL COMPONENTS. NON-STRUCTURAL COMPONENT SEISMIC BRACING AND ITS REQUIREMENTS SHALL BE REVIEWED BY DISCIPLINES OTHER THAN THE STRUCTURAL ENGINEER.

WALL STUD DEFLECTIONS HAVE BEEN LIMITED TO L/360 FOR STUCCO, EIFS AND DRYMALL FINISH, L/600 FOR MASONRY AND STONE FINISH.

COORDINATION:

THESE DRAWINGS ARE A SCHEMATIC REPRESENTATION OF THE STRUCTURAL SYSTEM AND REQUIREMENTS FOR THE PROJECT, AND ARE ONLY A PORTION OF THE COMPLETE CONTRACT DOCUMENTS. THE STRUCTURAL SYSTEMS REQUIRE CAREFUL COORDINATION BETWEEN ALL STRUCTURAL COMPONENTS AND MATERIALS SHOWN IN THESE STRUCTURAL DRAWINGS, AND CAREFUL COORDINATION OF INFORMATION SHOWN ON OTHER DISCIPLINES' DRAWINGS IN ORDER TO BE CONSTRUCTED. THE CONTRACTOR SHALL COORDINATE CONSTRUCTION OPERATIONS WITH ALL SUBCONTRACTORS AND OTHER RELATED ENTITIES TO ENSURE EFFICIENT AND ORDERLY INSTALLATION OF EACH PART OF THE WORK. EACH CONTRACTOR SHALL COORDINATE ITS OPERATIONS WITH THE OPERATIONS OF OTHER CONTRACTORS FOR PROPER INSTALLATION, CONNECTION AND OPERATION.

SHOP DRAWINGS/SUBMITTALS:

SHOP DRAWINGS SHALL BE SUBMITTED FOR ALL STRUCTURAL ITEMS IN ADDITION TO ITEMS REQUIRED BY ARCHITECTURAL SPECIFICATIONS. THE STRUCTURAL SHOP DRAWING REVIEW IS INTENDED TO HELP THE ENGINEER VERIFY THE DESIGN CONCEPT. THE CONTRACTOR SHALL REVIEW ALL SHOP DRAWINGS PRIOR TO SUBMITTAL. ITEMS NOT IN ACCORDANCE WITH CONTRACT DOCUMENTS SHALL BE IDENTIFIED UPON HIS REVIEW AND REVISED PRIOR TO FORWARDING TO ARCHITECT. THE STRUCTURAL SHOP DRAWINGS WILL BE RETURNED FOR RESUBMITTAL IF A CURSORY REVIEW SHOWS MAJOR ERRORS WHICH SHOULD HAVE BEEN FOUND BY THE CONTRACTOR'S CHECKING.

VERIFY ALL DIMENSIONS WITH ARCHITECT AND ALL FINISHED GRADE WITH CIVIL DRAWINGS.

ANY CHANGES, SUBSTITUTIONS, OR DEVIATIONS FROM CONTRACT DOCUMENTS SHALL BE CLOUDED BY MANUFACTURER OR FABRICATOR, ANY OF THE AFOREMENTIONED WHICH ARE NOT CLOUDED OR FLAGGED BY SUBMITTING PARTIES, SHALL NOT BE CONSIDERED APPROVED AFTER ENGINEER'S REVIEW, UNLESS NOTED ACCORDINGLY.

THE ENGINEER HAS THE RIGHT TO APPROVE OR DISAPPROVE ANY CHANGES TO CONTRACT DOCUMENTS AT ANY TIME BEFORE OR AFTER SHOP DRAWING REVIEW.

THE SHOP DRAWINGS DO NOT REPLACE THE CONTRACT DOCUMENTS. ITEMS OMITTED OR SHOWN INCORRECTLY AND ARE NOT FLAGGED BY THE STRUCTURAL ENGINEER OR ARCHITECT ARE NOT TO BE CONSIDERED CHANGES TO CONTRACT DOCUMENTS. IT IS THE CONTRACTOR'S RESPONSIBILITY TO MAKE SURE ITEMS ARE CONSTRUCTED TO CONTRACT DOCUMENTS.

THE ADEQUACY OF ENGINEERING DESIGNS AND LAYOUT PERFORMED BY OTHERS RESTS WITH THE DESIGNING OR SUBMITTING AUTHORITY.

REVIEWING IS INTENDED ONLY AS AN AID TO THE CONTRACTOR IN OBTAINING CORRECT SHOP DRAWINGS. RESPONSIBILITY FOR CORRECTNESS SHALL REST WITH THE CONTRACTOR.

REPRODUCTION OF ANY PORTION OF THE CONTRACT DOCUMENTS FOR USE IN SUBMITTALS IS NOT PERMITTED AND MAY RESULT IN REJECTION.

DEFERRED SUBMITTALS: (PER GOVERNING BUILDING CODE)

FOR THE PURPOSES OF THIS SECTION, DEFERRED SUBMITTALS ARE DEFINED AS THOSE PORTIONS OF THE DESIGN WHICH ARE NOT SUBMITTED AT THE TIME OF THE BUILDING PERMIT APPLICATION AND WHICH ARE TO BE SUBMITTED TO THE BUILDING OFFICIAL WITHIN A SPECIFIED PERIOD BY THE OWNER OR CONTRACTOR.

DEFERRAL OF ANY SUBMITTAL ITEMS SHALL HAVE PRIOR APPROVAL OF THE BUILDING OFFICIAL, AND THE ARCHITECT OR ENGINEER OF RECORD SHALL LIST THE DEFERRED SUBMITTALS ON THE PLANS.

SUBMITTAL DOCUMENTS FOR DEFERRED ITEMS SHALL BE SUBMITTED TO THE ARCHITECT OR ENGINEER OF RECORD WHO SHALL REVIEW THEM AND INCLUDE NOTATION INDICATING THAT THE DEFERRED SUBMITTAL DOCUMENTS HAVE BEEN REVIEWED AND THAT THEY HAVE BEEN FOUND TO BE IN GENERAL CONFORMANCE WITH THE DESIGN OF THE BUILDING. THE DEFERRED SUBMITTAL DOCUMENTS SHALL THEN BE SUBMITTED TO THE BUILDING OFFICIAL, AND THE DEFERRED SUBMITTAL ITEMS SHALL NOT BE INSTALLED UNTIL THEIR DESIGN AND SUBMITTAL DOCUMENTS HAVE BEEN APPROVED BY THE BUILDING OFFICIAL.

DEFERRED SUBMITTAL ITEMS:

STEEL JOISTS / JOIST GIRDERS
STEEL STAIRS

GENERAL REQUIREMENTS:

ENTIRE CONTRACT DOCUMENTS SHALL BE USED TO BUILD BUILDING. SOME CRITICAL ITEMS REQUIRED BY OTHER DISCIPLINES MAY NOT BE SHOWN ON STRUCTURAL DRAWING (I.E. WALL, FLOOR AND ROOF OPENING, ARCHITECTURAL, MECHANICAL AND PLUMBING LOADS, SUPPORT PLATES ETC.). IT IS THE CONTRACTOR'S SOLE RESPONSIBILITY TO PERFORM CAREFUL COORDINATION BETWEEN TRADES DURING CONSTRUCTION.

WHERE DISCREPANCIES OCCUR BETWEEN PLANS, DETAILS, GENERAL STRUCTURAL NOTES AND SPECIFICATIONS, THE MORE STRINGENT REQUIREMENTS SHALL GOVERN. SPECIFIC DETAILS ON DRAWINGS SHALL TAKE PRECEDENCE OVER TYPICAL DETAILS AND GENERAL STRUCTURAL NOTES. BRING ALL DISCREPANCIES TO THE ATTENTION OF THE ENGINEER IMMEDIATELY UPON DISCOVERY.

DO NOT SCALE DRAWINGS FOR THE PURPOSE OF ESTABLISHING DIMENSIONS OF ANY KIND.

ALL PROPRIETARY ITEMS, MATERIALS AND COMPONENTS SHALL BE INSTALLED IN STRICT ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS, GUIDELINES AND/OR RECOMMENDATIONS.

ITEMS SHOWN BY OTHER DISCIPLINES WITH REFERENCE TO STRUCTURAL DRAWING BUT NOT SHOWN ON THESE STRUCTURAL DOCUMENTS SHALL BE CONSIDERED DESIGN BUILD ITEMS AND COSTS REQUIRED FOR THESE ITEMS SHALL BE INCLUDED IN CONTRACTOR'S PRICING EXERCISES AND BID. CONTRACTOR SHALL SUBMIT DESIGN BY OTHERS FOR REVIEW.

THE STRUCTURAL CONSTRUCTION DOCUMENTS REPRESENT THE FINISHED STRUCTURE. THEY DO NOT INDICATE THE METHOD OF CONSTRUCTION. THE CONTRACTOR SHALL PROVIDE ALL MEASURES NECESSARY TO PROTECT THE STRUCTURE DURING CONSTRUCTION. SUCH MEASURES SHALL INCLUDE, BUT NOT BE LIMITED TO, BRACING, SHORING FOR LOADS DUE TO CONSTRUCTION EQUIPMENT, ETC. THE STRUCTURAL ENGINEER SHALL NOT BE RESPONSIBLE FOR THE CONTRACTOR'S MEANS, METHODS, TECHNIQUES, SEQUENCES FOR PROCEDURE OF CONSTRUCTION, OR THE SAFETY PRECAUTIONS AND THE PROGRAMS INCIDENT THERETO, NOR SHALL OBSERVATION VISITS TO THE SITE INCLUDE INSPECTION OF THESE ITEMS.

CONSTRUCTION MATERIALS SHALL BE SPREAD OUT IF PLACED ON FRAMED CONSTRUCTION. LOAD SHALL NOT EXCEED THE DESIGN LIVE LOAD PER SQUARE FOOT.

WHERE REFERENCE IS MADE TO VARIOUS TEST STANDARDS FOR MATERIALS, SUCH STANDARDS SHALL BE THE LATEST EDITION AND/OR ADDENDA.

ESTABLISH AND VERIFY ALL OPENINGS AND INSERTS FOR ARCHITECTURAL, MECHANICAL, PLUMBING AND ELECTRICAL WITH APPROPRIATE TRADES, DRAWINGS AND SUBCONTRACTORS PRIOR TO CONSTRUCTION. SHOW PENETRATIONS AND BLOCKOUT LOCATIONS ON SHOP DRAWINGS AND OTHER SUBMITTALS PRIOR TO ENGINEER'S REVIEW.

OPTIONS ARE FOR CONTRACTOR'S CONVENIENCE. IF HE CHOOSES AN OPTION, CONTRACTOR SHALL BE RESPONSIBLE FOR ALL NECESSARY CHANGES AND SHALL COORDINATE ALL DETAILS.

NOTES AND DETAILS ON DRAWINGS SHALL TAKE PRECEDENCE OVER GENERAL STRUCTURAL NOTES AND TYPICAL DETAILS. TYPICAL DETAILS MAY NOT NECESSARILY BE CUT ON PLANS, BUT APPLY UNLESS NOTED OTHERWISE. WHERE NO DETAILS ARE SHOWN, CONSTRUCTION SHALL CONFORM TO SIMILAR WORK ON THE PROJECT.

DETAILS INDICATED AS TYPICAL (I.E., TYP) APPLY TO ALL SIMILAR CONDITIONS.

SOME DETAILS ARE DEFINED AS TYPICAL TO THE PROJECT AND NOT NECESSARILY CUT OR CALLED OUT ON PLANS. IT IS THE CONTRACTOR'S RESPONSIBILITY TO APPLY AND COORDINATE THESE DETAILS WITH CONDITIONS DEFINED THROUGHOUT THE CONTRACT DOCUMENTS AS REQUIRED.

ALL DIMENSIONS SHOWN (INCLUDING ELEVATIONS) ON STRUCTURAL DRAWINGS ARE TO ASSIST CONTRACTOR IN VERIFICATION.

CONTRACTOR SHALL BE RESPONSIBLE FOR VERIFICATION OF ALL DIMENSIONS WITH ARCHITECTURAL DRAWINGS PRIOR TO START OF CONSTRUCTION – RESOLVE ANY DISCREPANCY WITH THE ARCHITECT. CONTRACTOR SHALL COORDINATE ALL FINISHED GRADE ELEVATIONS SHOWN IN CIVIL DRAWINGS WITH MINIMUM FOOTING DEPTHS SHOWN IN STRUCTURAL DRAWINGS – NOTIFY ENGINEER OF ANY DISCREPANCY PRIOR TO START OF CONSTRUCTION.

ANY ENGINEERING DESIGN, PROVIDED BY OTHERS AND SUBMITTED FOR REVIEW, SHALL BEAR THE SEAL OF AN ENGINEER REGISTERED IN THE STATE OF THE PROJECT.

SUPPLIER OF ENGINEERED STRUCTURAL COMPONENTS (I.E. STEEL JOISTS, STAIRS, PRECAST ITEMS) SHALL BE RESPONSIBLE FOR COMPLETE DESIGN AND SHALL USE ENTIRE CONTRACT DOCUMENTS TO INCLUDE ALL LOADS AND DETAIL REQUIREMENTS FROM ALL DISCIPLINES. SUPPLIER SHALL PROVIDE ADDITIONAL MATERIAL REQUIRED TO MEET ALL THEIR REQUIREMENTS FOR INSTALLATION (I.E. WIDER BEARING PLATES, SHIMS, ERECTION BOLTS ETC.).

STRUCTURAL STEEL SUPPLIER SHALL FURNISH BOLTS FOR OSHA CONNECTIONS.

SPECIAL INSPECTION AND MATERIAL TESTING:

THE CONTRACTOR SHALL EMPLOY ONE OR MORE SPECIAL INSPECTORS OR AGENCIES WHO SHALL PROVIDE INSPECTIONS DURING CONSTRUCTION FOR THE ITEMS LISTED BELOW. THE CONTRACTOR SHALL COOPERATE WITH THE SPECIAL INSPECTOR AND SHALL FURNISH TOOLS, EQUIPMENT AND ASSISTANCE AS REQUESTED. THE CONTRACTOR SHALL NOTIFY THE INSPECTOR AT LEAST 24 HOURS PRIOR TO EXPECTED TIME FOR OPERATIONS REQUIRING TESTING OR INSPECTION SERVICES. THE INSPECTOR SHALL FAMILIARIZE HIMSELF WITH ALL APPLICABLE PORTIONS OF THE CONTRACT DOCUMENTS PERTAINING TO THE AREA OF INVESTIGATION PRIOR TO PERFORMING SERVICES. COMPENSATION FOR SPECIAL INSPECTION SERVICES SHALL BE PROVIDED BY THE CONTRACTOR.

THE ENGINEER IS NOT THE SPECIAL INSPECTOR, AND SHALL NOT PERFORM THESE SERVICES.

INSPECTION SHALL NOT RELIEVE THE CONTRACTOR OF HIS RESPONSIBILITY TO PERFORM THE WORK IN ACCORDANCE WITH THE CONTRACT DOCUMENTS.

PER IBC SECTION 1704, SPECIAL INSPECTION IS REQUIRED FOR THE ITEMS DEFINED IN THE SPECIAL INSPECTION MATRIX FOR EACH TYPE OF CONSTRUCTION IN THESE STRUCTURAL CONSTRUCTION DOCUMENTS.

DUTIES AND RESPONSIBILITIES OF THE SPECIAL INSPECTOR:
A) THE SPECIAL INSPECTOR SHALL OBSERVE THE WORK ASSIGNED TO BE CERTAIN IT CONFORMS WITH THE APPROVED DESIGN DRAWINGS AND SPECIFICATION.
B) THE SPECIAL INSPECTOR SHALL FURNISH INSPECTION REPORTS TO THE BUILDING OFFICIAL, AND TO THE ENGINEER OR ARCHITECT OF RECORD. ALL DISCREPANCIES SHALL BE BROUGHT TO THE IMMEDIATE ATTENTION OF THE CONTRACTOR FOR CORRECTION, THEN, IF UNCORRECTED, TO THE DESIGN AUTHORITY AND THE BUILDING OFFICIAL.
C) UPON COMPLETION OF THE WORK THE SPECIAL INSPECTOR SHALL COMPLETE AND SIGN THE APPROPRIATE FORMS CERTIFYING THAT TO THE BEST OF THEIR KNOWLEDGE THE WORK IS IN CONFORMANCE WITH THE APPROVED PLANS AND SPECIFICATIONS AND THE APPLICABLE WORKMANSHIP PROVISIONS OF THE CODE.

MATERIAL TESTING REQUIREMENTS ARE DEFINED IN THE CONTRACT DRAWINGS AND PROJECT SPECIFICATIONS. COMPENSATION FOR ALL MATERIAL TESTING SHALL BE PROVIDED BY THE CONTRACTOR.

ALL TEST & INSPECTION REPORTS SHALL BE COPIED TO THE STRUCTURAL ENGINEER WITHIN 3 DAYS OF INSPECTION OR TEST.

2.0 FOUNDATIONS:

GEOTECHNICAL REPORT BY KUMAR & ASSOCIATES, INC ; JOB NO. 24-1-229 DATED 04/23/2024.

THE FOUNDATION AND FLOOR SLAB SYSTEMS SHOWN ON THE STRUCTURAL DRAWINGS HAVE BEEN DESIGNED IN ACCORDANCE WITH THE GEOTECHNICAL REPORT RECOMMENDATIONS. THE FOLLOWING FOUNDATION AND SLAB MOVEMENTS HAVE BEEN ESTIMATED BY THE GEOTECHNICAL ENGINEER USING THE DESIGN CRITERIA PRESENTED IN THE SOILS REPORT:

- TOTAL SETTLEMENT OF INTERIOR SLABS < = 1"
- TOTAL SETTLEMENT OF EXTERIOR SLABS < = 1"
- DIFFERENTIAL SETTLEMENT = 1/2 TO 3/4 OF THE TOTAL SETTLEMENT

ALL RISKS ASSOCIATED WITH THESE DESIGN REQUIREMENTS HAVE BEEN FULLY EVALUATED AND ACCEPTED BY THE OWNER.

THE SLAB-ON-GRADE HAS NOT BEEN DESIGNED TO ACT AS A FLOOR DIAPHRAGM.

PERIMETER FOUNDATION DRAINS AND UNDERSLAB DRAINS ARE NON-STRUCTURAL ITEMS RELATED TO SOIL PERFORMANCE AND WATER TRANSPORT AWAY FROM THE BUILDING. THESE ITEMS ARE NOT THE RESPONSIBILITY OF THE STRUCTURAL ENGINEER AND ARE NOT DEFINED OR SHOWN IN THE STRUCTURAL DRAWINGS. REFER TO OTHER DISCIPLINES' DRAWINGS AND THE GEOTECHNICAL REPORT FOR REQUIREMENTS.

SPREAD FOOTINGS SHALL BEAR ON SOILS WHICH HAVE BEEN IMPROVED WITH RAMMED AGGREGATE PIERS (RAP) AS DEFINED IN THE SOILS REPORT, 36" MINIMUM BELOW FINISHED GRADE AT EXTERIOR FOOTINGS; 24" MINIMUM BELOW GRADE AT INTERIOR FOOTINGS. FINISHED GRADE IS DEFINED AS TOP OF SLAB FOR INTERIOR FOOTINGS AND LOWEST ADJACENT GRADE WITHIN 5 FEET FOR PERIMETER FOOTINGS. DESIGN SOIL BEARING VALUE = 6000 PSF AFTER RAP SYSTEM HAS BEEN INSTALLED. SOIL BEARING VALUES HAVE NOT BEEN INCREASED 33% FOR SEISMIC OR WIND LOADING. FOUNDATION EXCAVATIONS AND RAP INSTALLATIONS SHALL BE INSPECTED BY SOILS ENGINEER PRIOR TO PLACEMENT OF CONCRETE.

EQUIVALENT FLUID PRESSURE USED FOR WALL DESIGN = 55 PCF FOR "ACTIVE" CONDITION, 65 PCF FOR "AT REST" CONDITION, AND 175 PCF FOR "PASSIVE" CONDITION.

FOUNDATION SUBMITTALS:

THE FOLLOWING ITEMS SHALL BE SUBMITTED FOR REVIEW FOR THE LIMITED PURPOSE OF CHECKING FOR CONFORMANCE WITH INFORMATION GIVEN AND THE DESIGN CONCEPT EXPRESSED IN THE CONTRACT DOCUMENTS. REFER TO PROJECT SPECIFICATIONS FOR ADDITIONAL REQUIRED SUBMITTALS.

MIX DESIGNS
STEEL REINFORCEMENT SHOP DRAWINGS
WELDING CERTIFICATES
MATERIAL TEST REPORTS
MATERIAL CERTIFICATES
FIELD QUALITY CONTROL TEST AND INSPECTION REPORTS
COLD-WEATHER AND HOT-WEATHER PROCEDURES
RAP SYSTEM DESIGN AND SPECIFICATIONS

ABBREVIATIONS			
AFF	ABOVE FINISHED FLOOR	IFW	INSIDE FACE OF WALL
ALT	ALTERNATE	IT	PRECAST INVERTED TEE BEAM
AB	ANCHOR BOLT	K (KIP)	1000 POUNDS
ADD'L	ADDITIONAL	LB	PRECAST ELL BEAM
ARCH	ARCHITECT	LBS (#)	POUNDS
ARCH'L	ARCHITECTURAL	LL	LIVE LOAD
@	AT (MEASUREMENT)	LLH	LONG LEG HORIZONTAL
BM	BEAM	LLV	LONG LEG VERTICAL
BFF	BELOW FINISHED FLOOR	LOC	LOCATION
BOB	BOTTOM OF BEAM	LVL	LAMINATED VENEER LUMBER
BOD	BOTTOM OF DECK	MAS	MASONRY
BOF	BOTTOM OF FOOTING	MAS CJ	MASONRY CONTROL JOINT
BOT (B)	BOTTOM	MAX	MAXIMUM
BRS	BEARING	MBM	METAL BUILDING MANUFACTURER
CFS	COLD FORMED STEEL	MECH'L	MECHANICAL
COS	CENTER OF GRAVITY STRAND	MFR('S)	MANUFACTURER('S)
CIP	CAST IN PLACE	MIN	MINIMUM
CL	CENTERLINE	MPIT	MFR PRINTED INSTALLATION INSTRUCTIONS
CJ	CONTROL JOINT	N/A	NOT APPLICABLE
CJP	COMPLETE JOINT PENETRATION	NTS	NOT TO SCALE
		NSNM	NON-SHRINK NON-METALLIC
GTR'D	CENTERED	OC	ON CENTER
CLB	CENTERLINE OF BEAM	OFW	OUTSIDE FACE OF WALL
CLC	CENTERLINE OF COLUMN	OH	OPPOSITE HAND
CLF	CENTERLINE OF FOOTING	OPP	OPPOSITE
CLW	CENTERLINE OF WALL	OPNG	OPENING
CLB	CLEAR	OS	OVERSIZED
COL	COLUMN	OSB	ORIENTED STRAND BOARD
CONC	CONCRETE	PAF	POWDER ACTUATED FASTENER
CMU	CONCRETE MASONRY UNIT	PC	PRECAST CONCRETE
CONN	CONNECTION	PEMB	PRE-ENGINEERED METAL BUILDING
CONT	CONTINUOUS	PFWT	PREFABRICATED WOOD TRUSS
d	PENNY (SIZE)	PJ	PANEL JOINT
DAS	DEFORMED ANCHOR STUD	PL	PLATE
DEG	DEGREE	PLF	POUNDS PER LINEAR FOOT
DIA	DIAMETER	PLYWD	PLYWOOD
DL	DEAD LOAD	PREFAB	PREFABRICATED
DP	DEEP OR DRILLED PIER	PSF	POUNDS PER SQUARE FOOT
DN	DOWN	PSI	POUNDS PER SQUARE INCH
DTL	DETAIL	PT	POST TENSION
DWG(S)	DRAWING(S)	PTL	PRESSURE TREATED LUMBER
EA	EACH	RB	PRECAST RECTANGLE BEAM
EC	EPoxy COATED	REF	REFERENCE
EE	EACH END	REINF	REINFORCING
EL	ELEVATION	REQ'D	REQUIRED
ENGR	ENGINEER	RS	ROUGH SAWN
EDS	EDGE OF SLAB	SCHED	SCHEDULE
EQ	EQUAL	SLH	SHORT LEG HORIZONTAL
EQUIP	EQUIPMENT	SLV	SHORT LEG VERTICAL
EXIST (E)	EXISTING	STM	SIMILAR
EXP ANCH	EXPANSION ANCHOR	SOG	SLAB ON GRADE
EXP JT (EJ)	EXPANSION JOINT	SOV	SLAB ON VOID
EW	EACH WAY	SQ	SQUARE
FDN	FOUNDATION	STD	STANDARD
FF	FINISHED FLOOR	STL	STEEL
FOM	FACE OF MEMBER	T&B	TOP AND BOTTOM
FOW	FACE OF WALL	TL	TOTAL LOAD
FP	FIRE PROOFING	TJL	PLYWOOD WEB JOIST
FTG	FOOTING	TOS	TOP OF BEAM
FV	FIELD VERIFY	TOC	TOP OF CONCRETE
GA	GAGE	TOO	TOP OF DECK
GALV	GALVANIZED	TOF	TOP OF FOOTING
GSN	GENERAL STRUCTURAL NOTES	TOL	TOP OF LEDGER
GLB (GLULAM)	GLUED-LAMINATED BEAM	TOM	TOP OF MASONRY
GT	GIRDER TRUSS	TOP	TOP OF PLATE
HC	PRECAST HOLLOW CORE PLANK	TOS	TOP OF STEEL
HAS	HEADED ANCHOR STUD	TOW	TOP OF WALL
HOG	HOT-DIP GALVANIZED	TYP	TYPICAL
HK	HOOK	UNO	UNLESS NOTED OTHERWISE
HORIZ	HORIZONTAL	VERT	VERTICAL
HT	HEIGHT	WP	WORK POINT
		WWF	WELDED WIRE FABRIC
		W/ (w/)	WITH
		W/O (w/o)	WITHOUT

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MASONRY, STEEL, AND MECHANICAL SUBCONTRACTORS NOTE:
STRUCTURAL DRAWINGS DO NOT INDICATE ALL WALL, FLOOR, OR ROOF PENETRATIONS FOR MECH DUCTS, DRAINS, VENTS, ETC.; DRAWINGS INDICATE TYPICAL AND SPECIAL CONDITIONS FOR FRAMING AT THE PENETRATIONS. GENERAL CONTRACTOR AND SUB CONTRACTORS SHALL BE RESPONSIBLE FOR DETERMINING AND/OR MODIFYING OPENING LOCATIONS, ELEVATIONS, AND DIMENSIONS FOR MECH UNLESS NOTED OTHERWISE. COORDINATION TO BE COMPLETED PRIOR TO FABRICATION OF STRUCTURAL STEEL AND ROOF JOISTS.

CITY OF THORNTON
FIRE STATION #8

15343 MONACO STREET
THORNTON, COLORADO, 80620

REVISION NAME	DATE

ISSUE DATE: 2024 0927

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GENERAL
STRUCTURAL NOTES

S0.1

GENERAL STRUCTURAL NOTES - CONTINUED

APPLY UNLESS NOTED OTHERWISE

3.0 CONCRETE:

ALL CONCRETE WORK SHALL CONFORM TO THE LATEST EDITION OF ACI 301, UNLESS MORE STRINGENT REQUIREMENTS ARE DEFINED ELSEWHERE IN THE CONTRACT DOCUMENTS.

CONCRETE DEFINED ON THE STRUCTURAL DRAWINGS SHALL HAVE THE FOLLOWING PROPERTIES:

USE	MIN f'c, 28 DAY	SLUMP	MAX w/c RATIO	AIR CONTENT*	CEMENT TYPE
FOOTINGS	3,000 PSI	4" MAX----	0.50	NO REQUIREMENT	I/II
GRADE BEAMS	3,000 PSI	4" MAX----	0.50	NO REQUIREMENT	I/II
SLABS ON GROUND (INTERIOR)	3,750 PSI	4" MAX----	0.45	NO REQUIREMENT	I/II
SLABS ON GROUND (EXTERIOR)**	3,750 PSI	4" MAX----	0.45	5.0%	I/II
APPARATUS BAY SLAB	4,000 PSI	4" MAX----	0.45	NO REQUIREMENT	I/II
TOPPING FOR STEEL DECK	3,500 PSI	4" MAX----	0.50	NO REQUIREMENT	I/II

SLUMP SHALL BE MEASURED AT POINT OF PLACEMENT (END OF PUMP HOSE, TREMIE, ETC.) - NO EXCEPTIONS.

SLUMP SPECIFIED AS A RANGE SHALL BE PROVIDED WITH TOLERANCE OF +/-0".
SLUMP SPECIFIED AS MAXIMUM SHALL BE PROVIDED WITH TOLERANCE OF +0", -1 1/2" FOR SLUMP 3" OR LESS, AND + 0", -2 1/2" FOR SLUMP MORE THAN 3".
SLUMP SPECIFIED AS A SINGLE VALUE SHALL BE PROVIDED WITH TOLERANCE OF +/- 1" FOR SLUMP 4" OR LESS; AND +/- 1 1/2" FOR SLUMP MORE THAN 4".
SLUMP MAY BE INCREASED THROUGH THE USE OF ADMIXTURES IN THE CONTRACTOR'S MIX DESIGNS. COSTS FOR ALL ADMIXTURES SHALL BE INCLUDED IN THE BID. REFER TO ACI 301 FOR ADDITIONAL INFORMATION.

* SPECIFIED AIR CONTENT IS TOTAL CONCRETE AIR CONTENT (ENTRAPPED + ENTRAINED).

** PAVING AND EXTERIOR FLATWORK - REFER TO ARCHITECTURAL/CIVIL DRAWINGS AND SPECIFICATIONS.

MECHANICALLY VIBRATE ALL CONCRETE WHEN PLACED, EXCEPT THAT SLABS ON GRADE NEED BE VIBRATED ONLY AROUND UNDER-FLOOR DUCTS, ETC. IF PLASTICIZER IS USED, A HIGHER FINAL SLUMP MAY BE ALLOWED UPON STRUCTURAL ENGINEER'S APPROVAL. CAST CLOSURE POUR AROUND COLUMNS AFTER COLUMN DEAD LOAD IS APPLIED. UNLESS APPROVED OTHERWISE IN WRITING BY THE ARCHITECT, ALL CONCRETE SLABS ON GRADE SHALL BE BOUND BY CONTROL JOINTS (KEYED OR SAW CUT), SUCH THAT THE ENCLOSED AREA DOES NOT EXCEED 144 SQUARE FEET. KEYED CONTROL JOINTS NEED ONLY OCCUR AT EXPOSED EDGES DURING POURING, ALL OTHER JOINTS MAY BE SAW CUT. REFER TO ARCHITECTURAL DRAWINGS FOR SPECIFIC JOINT LOCATIONS.

FLY ASH SHALL BE LIMITED TO 20% OF CEMENTITIOUS MATERIALS AND IS NOT PERMITTED FOR INTERIOR SLAB CONCRETE.

"WET STABBING" OF ANY EMBEDDED ITEM OR BOLT IS STRICTLY PROHIBITED.

ALL BUILDING SLABS TO BE WATER CURED INCLUDING THE APPARATUS BAY SLAB, ADMIN SLAB AND MEZZANINE SLAB.

VOID FORMS:

VOID FORM UNITS SHALL HAVE A BIODEGRADABLE PAPER SURFACE, TREATED FOR MOISTURE RESISTANCE. STRUCTURALLY SUFFICIENT TO SUPPORT THE WEIGHT OF PLASTIC CONCRETE AND OTHER SUPERIMPOSED LOADS. CONTRACTOR MUST WORK CLOSELY WITH THE MANUFACTURER TO ENSURE MATERIAL SELECTED DEGRADES QUICKLY AND COMPLETELY AFTER CONCRETE IS PLACED ABOVE. VOID FORM UNITS SHALL NOT BE WRAPPED IN PLASTIC NOR HAVE PLASTIC PLACED ABOVE OR BELOW.

REINFORCING STEEL:

ASTM A615 (Fy = 60 KSI) DEFORMED BARS FOR ALL BARS. ALL GRADE 60 REINFORCING TO BE WELDED OR FIELD BENT SHALL BE ASTM A706. ALL REINFORCING STEEL WELDS SHALL BE COMPLETE JOINT PENETRATION UNLESS NOTED OTHERWISE. WELDED WIRE FABRIC AND WIRE PER ASTM A1064. NO TACK WELDING OF REINFORCING BARS ALLOWED WITHOUT PRIOR REVIEW OF PROCEDURE WITH THE STRUCTURAL ENGINEER. LATEST ACI CODE AND DETAILING MANUAL APPLY. CLEAR CONCRETE COVERAGES AS FOLLOWS:

CAST AGAINST AND PERMANENTLY EXPOSED TO EARTH	3"
EXPOSED TO EARTH OR WEATHER	
#6 OR LARGER	2"
#5 AND SMALLER	1 1/2"
COLUMNS (TO TIES)	1 1/2"
BEAMS (TO STIRRUPS)	1 1/2"
FLAT SLAB	3/4"

ALL OTHERS PER LATEST EDITION OF ACI 318.

LAP SPLICES IN CONCRETE:

LAP SPLICES, UNLESS NOTED OTHERWISE, SHALL BE CLASS "B" TENSION LAP SPLICES PER LATEST EDITION OF ACI 318. LAP SPLICES IN CONCRETE COLUMNS SHALL BE STANDARD COMPRESSION LAP SPLICES. STAGGER SPLICES A MINIMUM OF ONE LAP LENGTH. LAPS IN WELDED WIRE FABRIC SHALL BE MADE SO THAT THE OVERLAP, MEASURED BETWEEN OUTERMOST CROSS WIRES OF EACH FABRIC SHEET, IS NOT LESS THAN THE SPACING OF CROSS WIRES PLUS 2 INCHES. ALL WELDED WIRE FABRIC SHALL BE CHAIRED TO ENSURE PROPER CLEARANCES.

ALL SPLICE LOCATIONS SUBJECT TO APPROVAL BY THE STRUCTURAL ENGINEER. UNLESS NOTED, LAP TOP BARS AT MID-SPAN AND BOTTOM BARS OVER SUPPORT. PROVIDE BENT CORNER BARS TO MATCH AND LAP WITH HORIZONTAL BARS AT ALL CORNERS AND INTERSECTIONS PER TYPICAL DETAILS. REINFORCING BAR SPACING GIVEN ARE MAXIMUM ON CENTERS. ALL BARS PER CRSI SPECIFICATIONS AND HANDBOOK. DOWEL ALL VERTICAL REINFORCING TO FOUNDATION WITH STANDARD 90-DEGREE HOOKS UNLESS NOTED OTHERWISE. SECURELY TIE ALL BARS IN LOCATION BEFORE PLACING CONCRETE. CONCRETE COLUMN DOWEL EMBEDMENT SHALL BE A STANDARD COMPRESSION DOWEL WITH EMBEDMENT LENGTH ACCORDING TO THE LATEST EDITION OF THE ACI 318.

POST INSTALLED ANCHORS TO CONCRETE:

THE INSTALLATION OF POST INTALLED ANCHORS SHALL BE INSPECTED IN ACCORDANCE WITH SECTION 1.3 OF ACI 318, AND THE GOVERNING BUILDING CODE. ANCHORS SHALL BE INSTALLED BY QUALIFIED PERSONNEL IN ACCORDANCE WITH THE CONTRACT DOCUMENTS AND THE MANUFACTURER'S PRINTED INSTALLATION INSTRUCTIONS (MPII). THE FOLLOWING INSTALLATION CONDITIONS ARE REQUIRED:

NORMAL WEIGHT CONCRETE WITH A MIN f'c = 2500 PSI AND MAX f'c = 8000 PSI.
CONCRETE AT TIME OF ANCHOR INSTALLATION SHALL HAVE A MINIMUM COMPRESSION STRENGTH OF 2500 PSI.

ADHESIVE ANCHORS:

THE INSTALLATION OF ADHESIVE ANCHORS HORIZONTALLY OR UPWARDLY INCLINED TO SUPPORT SUSTAINED TENSION LOADS SHALL ONLY BE PERFORMED BY PERSONNEL CERTIFIED BY THE ACI/CRSI ADHESIVE ANCHOR INSTALLER CERTIFICATION PROGRAM. REFER TO THE SPECIAL INSPECTIONS MATRIX FOR ADDITIONAL TESTING AND INSPECTION REQUIREMENTS. THE FOLLOWING INSTALLATION CONDITIONS ARE REQUIRED:

ANCHORS SHALL MEET THE REQUIREMENTS OF ACI 355.4
ANCHORS SHALL BE INSTALLED IN HOLES DRILLED WITH A ROTARY IMPACT DRILL OR ROCK DRILL. CORE DRILLING IS NOT PERMITTED.
CONCRETE AT TIME OF ANCHOR INSTALLATION SHALL HAVE A MINIMUM AGE OF 21 DAYS.
CONCRETE TEMPERATURE AT TIME OF ANCHOR INSTALLATION SHALL BE AT LEAST 50 DEGREES F.
DRY USE CONDITION – UNSATURATED CONCRETE, INSTALLATION HOLE CONTAINS NO WATER.

ACCEPTABLE ADHESIVES:
HILTI HIT-HY 200 ADHESIVE
HILTI RE 500 V3 ADHESIVE
SIMPSON SET-36 ADHESIVE
DEWALT AC208+ ADHESIVE
DEWALT PURE110+ ADHESIVE

TORQUE-CONTROLLED EXPANSION ANCHORS:

REFER TO SPECIAL INSPECTION MATRIX FOR TESTING AND INSPECTION REQUIREMENTS.

ACCEPTABLE TORQUE-CONTROLLED EXPANSION ANCHORS:
HILTI KWIK BOLT-TZ
SIMPSON STRONG-BOLT 2
DEWALT POWER-STUD+SD2

CONCRETE SUBMITTALS:

THE FOLLOWING ITEMS SHALL BE SUBMITTED FOR REVIEW FOR THE LIMITED PURPOSE OF CHECKING FOR CONFORMANCE WITH INFORMATION GIVEN AND THE DESIGN CONCEPT EXPRESSED IN THE CONTRACT DOCUMENTS. REFER TO PROJECT SPECIFICATIONS FOR ADDITIONAL REQUIRED SUBMITTALS.

SHOP DRAWINGS: SUBMIT SHOP DRAWINGS FOR FABRICATION, BENDING AND PLACEMENT OF REINFORCEMENT STEEL. COMPLY WITH ACI DETAILING MANUAL, SP-66, AND PROVIDE 1/4" = 1'-0" SCALE ELEVATIONS OF ALL WALLS, BEAMS, GRADE BEAMS, STEM WALLS, TIE BEAMS, CORE WALLS AND OTHER VERTICAL WORK. SHOW ALL OPENINGS IN VERTICAL WORK, SLABS AND BEAMS; ALL POCKETS IN BEAMS AND WALLS; TOP AND BOTTOM ELEVATIONS OF MEMBERS, SECTIONS THROUGH COLUMNS, PILASTERS AND BEAMS; PLAN VIEW OF ALL WALLS AND BEAM/JOIST INTERSECTIONS; PLACING SEQUENCE FOR MULTIPLE LAYERS OF REINFORCING STEEL; LOCATIONS OF ALL CONSTRUCTION JOINTS. ALL OPENINGS SHALL BE REVIEWED BY THE CONTRACTOR PRIOR TO SUBMITTING DRAWINGS FOR REVIEW. SHOW SPECIAL REINFORCEMENT AT OPENINGS.

CERTIFICATION FOR ANCHORAGE INSTALLER
ANCHORAGE ICC APPROVAL
MIX DESIGNS
STEEL REINFORCEMENT SHOP DRAWINGS
FORMWORK SHOP DRAWINGS
WELDING CERTIFICATES
MATERIAL TEST REPORTS
MATERIAL CERTIFICATES
FIELD QUALITY CONTROL TEST AND INSPECTION REPORTS
COLD-WEATHER AND HOT-WEATHER PROCEDURES

4.0 MASONRY:

ALL MASONRY WORK SHALL CONFORM TO THE LATEST EDITION OF ACI 530.1, UNLESS MORE STRINGENT REQUIREMENTS ARE DEFINED ELSEWHERE IN THE CONTRACT DOCUMENTS.

HOLLOW CONCRETE MASONRY UNITS SHALL BE MEDIUM WEIGHT AND SHALL CONFORM TO ASTM C90, GRADE N, TYPE 1, F'm = 3,250 PSI, F'm = 2,500 PSI, RUNNING BOND, MORTAR TYPE S, GROUT 2,500 PSI. GROUT MIX SHALL CONFORM TO ASTM C476. MORTAR MIX SHALL CONFORM TO ASTM C270. MECHANICALLY VIBRATE GROUT IN VERTICAL SPACES IMMEDIATELY AFTER POURING AND AGAIN ABOUT 5 MINUTES LATER. PROVIDE CLEANOUTS IF GROUT LIFT EXCEEDS 4'-0" IN BLOCK WALLS. MAXIMUM GROUT LIFT SHALL BE 8'-0". UNLESS NOTED OTHERWISE ON THE PLANS, PLACE CONTROL JOINTS IN MASONRY WALLS SUCH THAT NO STRAIGHT RUN OF WALL EXCEEDS 24'-0". CONTROL JOINTS SHALL NOT OCCUR AT WALL CORNERS, INTERSECTIONS, ENDS, WITHIN 24" OF CONCENTRATED POINTS OF BEARING OR JAMBS, OR OVER OPENINGS UNLESS SPECIFICALLY SHOWN ON THE STRUCTURAL DRAWINGS. SOLID GROUT 24" BELOW AND TO EACH SIDE OF POINT LOAD CONDITIONS ON MASONRY MINIMUM. ALL MASONRY BELOW FINISHED FLOOR OR GRADE SHALL BE GROUTED SOLID. "WET STABBING" OF ANY EMBEDDED ITEM OR BOLT IS STRICTLY PROHIBITED.

VERTICAL REINFORCING LAP SPLICES SHALL BE 48 BAR DIAMETERS. LAP SPLICES SHALL BE 1.3 X LAP LENGTH WHEN ADJACENT SPLICES ARE SEPARATED BY 3" OR LESS.

HORIZONTAL REINFORCING LAP SPLICES SHALL BE 48 BAR DIAMETERS. STAGGER SPLICES A MINIMUM OF 40 BAR DIAMETERS. DO NOT SPLICE WITHIN 8'-0" OF CONTROL JOINTS. PROVIDE BENT BARS PER TYPICAL DETAILS, TO MATCH HORIZONTAL BOND BEAM REINFORCING, AT CORNERS AND WALL INTERSECTION TO MAINTAIN BOND BEAM CONTINUITY.

MASONRY LINTELS:

UNLESS NOTED OTHERWISE OR SHOWN, PROVIDE LINTELS IN MASONRY WALLS, PER STEEL OR MASONRY LINTEL SCHEDULES. USE THESE LINTELS FOR OPENINGS REQUIRED BY OTHER DISCIPLINES (MECHANICAL, ELECTRICAL, PLUMBING, ETC.). PROVIDE MINIMUM 5" BEARING OF ANGLES ON JAMBS. UNLESS SPECIFICALLY INDICATED, CONTRACTOR MAY USE EITHER STEEL OR MASONRY LINTELS.

THESE LINTELS, OR THE OPENING THEY SPAN, SHALL NOT BE PLACED SO AS TO INTERFERE WITH THE REQUIREMENTS OF OTHER STRUCTURAL ELEMENTS (I.E. BOND BEAMS, LINTELS, CONTROL JOINTS, CONCENTRATED POINTS OF BEARING, ETC.) WITHOUT THE PRIOR APPROVAL OF THE STRUCTURAL ENGINEER.

SOLID GROUT SHALL BE PROVIDED BETWEEN WEBS AND MASONRY FACE SHELLS FOR FULL LENGTH OF ALL STEEL LINTELS. MORTAR MAY BE USED FOR GROUT FOR THIS PURPOSE ONLY. FACE UNITS, SOAPS, ROMANS, ETC., SHALL BE LAID WITH FULL HEAD AND BED JOINTS. FOR ADDITIONAL INFORMATION AT OPENINGS IN MASONRY WALLS, SEE TYPICAL DETAILS.

MASONRY SUBMITTALS:

THE FOLLOWING ITEMS SHALL BE SUBMITTED FOR REVIEW FOR THE LIMITED PURPOSE OF CHECKING FOR CONFORMANCE WITH INFORMATION GIVEN AND THE DESIGN CONCEPT EXPRESSED IN THE CONTRACT DOCUMENTS. REFER TO PROJECT SPECIFICATIONS FOR ADDITIONAL REQUIRED SUBMITTALS.

SHOP DRAWINGS: SUBMIT SHOP DRAWINGS FOR FABRICATION, BENDING AND PLACEMENT OF REINFORCEMENT STEEL. COMPLY WITH ACI 530, AND PROVIDE 1/4" = 1'-0" SCALE ELEVATIONS OF ALL WALLS, COLUMNS AND STEM WALLS AND OTHER VERTICAL WORK. SHOW ALL OPENINGS IN VERTICAL WORK, AND BEAMS; ALL POCKETS IN BEAMS AND WALLS; TOP AND BOTTOM ELEVATIONS OF MEMBERS, SECTIONS THROUGH COLUMNS, PILASTERS AND BEAMS; PLAN VIEW OF ALL WALLS; PLACING SEQUENCE FOR MULTIPLE LAYERS OF REINFORCING STEEL; LOCATIONS OF ALL CONTROL JOINTS. ALL OPENINGS SHALL BE REVIEWED BY THE CONTRACTOR PRIOR TO SUBMITTING DRAWINGS FOR REVIEW. SHOW SPECIAL REINFORCEMENT AT OPENINGS.

PRODUCT DATA
STEEL REINFORCEMENT SHOP DRAWINGS
QUALIFICATION DATA FOR TESTING AGENCY
MATERIAL CERTIFICATES
MIX DESIGNS (MORTAR AND GROUT)
STATEMENT OF COMPRESSIVE STRENGTH OF MASONRY
COLD-WEATHER AND HOT-WEATHER PROCEDURES
FIELD QUALITY CONTROL TEST AND INSPECTION REPORTS

5.0 METALS:

STRUCTURAL STEEL:

ALL WORK SHALL CONFORM TO THE LATEST EDITION OF AISC MANUAL OF STEEL CONSTRUCTION, AND LATEST EDITION OF AWS D1.1, UNLESS MORE STRINGENT REQUIREMENTS ARE DEFINED ELSEWHERE IN THE CONTRACT DOCUMENTS. UNLESS AN ACTIVITY OR DUTY IS SPECIFICALLY IDENTIFIED AS BEING PERFORMED BY THE PROJECT STRUCTURAL ENGINEER OF RECORD IN OSHA SUBPART R, IT WILL NOT BE PERFORMED BY THE ENGINEER. IT IS TO BE PERFORMED BY OTHERS.

ALL STRUCTURAL STEEL SHALL BE ASTM A992 (Fy = 50 KSI). ALL CHANNELS, ANGLES, AND PLATES SHALL BE ASTM A36 (Fy = 36 KSI). ALL PIPE STEEL SHALL BE ASTM A501 (Fy = 36 KSI) OR ASTM A53, TYPE E OR S, GRADE B (Fy = 35 KSI). ALL TUBE STEEL SHALL BE ASTM A500 (Fy = 46 KSI). ALL ANCHOR RODS SHALL BE ASTM F1554, GRADE 36 UNLESS NOTED OTHERWISE.

ALL REFERENCE TO HEADED STUDS SHALL BE HIGH STRENGTH HEADED STUDS. ATTACHMENT OF HEADED STUDS SHALL CONFORM TO ALL REQUIREMENTS OF THE LATEST EDITION OF THE "RECOMMENDED PRACTICES FOR STUD WELDING" AND THE "STRUCTURAL WELDING CODE" PUBLISHED BY AWS.

ALL BOLTS, ANCHOR BOLTS, EXPANSION BOLTS, ETC. SHALL BE INSTALLED WITH STEEL WASHERS AT FACE OF WOOD OR AT SLOTTED HOLES IN STEEL SECTIONS.

ALL WELDING SHALL BE PERFORMED BY WELDERS HOLDING VALID CERTIFICATES AND HAVING CURRENT EXPERIENCE IN THE TYPE OF WELD SHOWN ON THE DRAWINGS OR NOTES. CERTIFICATES SHALL BE THOSE ISSUED BY AN ACCEPTED TESTING AGENCY. ALL WELDING DONE BY E70 SERIES LOW HYDROGEN RODS UNLESS NOTED OTHERWISE. FOR GRADE 60 REINFORCING BARS, USE E90 SERIES. ALL WELDING PER LATEST AMERICAN WELDING SOCIETY STANDARDS, (EXCEPT STEEL JOISTS AND JOIST GIRDERS SHALL COMPLY WITH SJI STANDARDS). SHOP WELDS AND FIELD WELDS SHALL BE SHOWN ON THE SHOP DRAWINGS SUBMITTED FOR REVIEW. ALL FULL (COMPLETE) PENETRATION WELDS SHALL BE TESTED AND CERTIFIED BY AN INDEPENDENT TESTING LABORATORY. ALL REINFORCING STEEL WELDS SHALL BE COMPLETE JOINT PENETRATION UNLESS NOTED OTHERWISE.

FILLET WELD SIZES SHOWN IN CONTRACT DOCUMENTS ARE THE FILLET WELD LEG SIZE. GROOVE WELD SIZES SHOWN IN CONTRACT DOCUMENTS ARE THE REQUIRED WELD SIZE "(E)".

THE ENGINEER SHALL BE COMPENSATED FOR ANY ANALYSIS, RE-DESIGN AND/OR REVIEW OF CONNECTIONS NOT CONFORMING TO THE CONTRACT DOCUMENTS, WHETHER MADE BY THE CONTRACTOR, THEIR SUBCONTRACTOR(S), FABRICATOR, DETAILER OR ERECTOR.

CONTRACTOR SHALL NOT RECEIVE ADDITIONAL COMPENSATION FOR DETAILING, FABRICATION, ERECTION OR SCHEDULE IMPACTS AS A RESULT OF INCORRECT DETAILING IN THE SHOP DRAWINGS.

HIGH STRENGTH BOLTS:

ALL HIGH STRENGTH BOLTS SHALL BE ASTM 7/8" DIAMETER A325 AND SHALL BE INSTALLED AS BEARING-TYPE CONNECTIONS WITH THREADS INCLUDED IN SHEAR PLANE (I.E. A TYPE "N" CONNECTION) UNLESS NOTED OTHERWISE. UNLESS SPECIFICALLY NOTED OTHERWISE IN THESE STRUCTURAL DRAWINGS, ALL A325 HIGH STRENGTH BOLTS SHALL BE TENSIONED TO THE VALUES SHOWN IN TABLE J3.1 OF THE AISC SPECIFICATION FOR STEEL BUILDINGS, USING ANY AISC APPROVED METHOD. ALL HIGH STRENGTH BOLTING SHALL BE INSPECTED BY AN INDEPENDENT TESTING LABORATORY.

STEEL DECKING:

ALL STEEL DECK SHALL BE DESIGNED, FABRICATED, WELDED AND ERECTED IN ACCORDANCE WITH THE LATEST EDITION OF THE "STANDARD SPECIFICATIONS" OF THE STEEL DECK INSTITUTE.

ALL WELDING SHALL BE PERFORMED BY WELDERS EXPERIENCED IN LIGHT GAGE STEEL DECK WORK. DECK WELDING MAY BE ACHIEVED WITH E60 SERIES LOW HYDROGEN RODS.

STEEL GRATING:

UNLESS NOTED OTHERWISE IN ARCHITECTURAL SPECIFICATIONS, USE GALVANIZED STEEL GRATING WITH 2-1/2" X 3/16" BEARING BARS AT 1 3/16" O.C. AND CROSS BARS AT 4" O.C. GRATING SHALL BE ERECTED AND ATTACHED PER MANUFACTURER'S RECOMMENDATIONS.

DRYPACK:

DRYPACK SHALL BE 5,000 PSI NON-SHRINK GROUT, FIVE STAR OR EQUIVALENT. INSTALL DRYPACK UNDER BEARING PLATES BEFORE FRAMING MEMBER IS INSTALLED. AT COLUMNS, INSTALL DRYPACK UNDER BASE PLATES AFTER COLUMN HAS BEEN PLUMBED BUT PRIOR TO FLOOR OR ROOF INSTALLATION.

COLD FORMED STRUCTURAL STEEL FRAMING:

ALL COLD-FORMED STEEL FRAMING SHALL BE FABRICATED AND ERECTED IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS AND IN ACCORDANCE WITH THE LATEST EDITION OF "SPECIFICATIONS FOR THE DESIGN OF COLD-FORMED STEEL STRUCTURAL MEMBERS" BY THE AMERICAN IRON AND STEEL INSTITUTE.

ALL PROPRIETARY ITEMS, MATERIALS AND COMPONENTS SHALL BE INSTALLED IN STRICT ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS, GUIDELINES AND/OR RECOMMENDATIONS.

STEEL FOR 12, 14 AND 16 GAGE STUDS, JOISTS, TRACKS AND FOR ALL DIAGONAL TENSION STRAPS SHALL HAVE A MINIMUM YIELD STRENGTH OF 50 KSI. STEEL FOR ALL 18 AND 20 GAGE STUDS, JOISTS, AND TRACKS, AND FOR ALL GAGES OF ACCESSORIES AND BRIDGING SHALL HAVE A MINIMUM YIELD STRENGTH OF 33 KSI. STEEL SHALL BE G60 GALVANIZED. STEEL SHEET FOR ALL STRUCTURAL FRAMING SHOWN IN THESE DRAWINGS SHALL CONFORM TO ASTM A 1003/A 1003M, STRUCTURAL GRADE, TYPE H, METALLIC COATED.

UNLESS SPECIFICALLY NOTED ELSEWHERE WITHIN THESE ENGINEERED COLD FORMED METAL FRAMING SHOP DRAWINGS, THE FOLLOWING SHALL APPLY AS A MINIMUM. ALL STUDS SHALL BE SECURELY SEATED FOR FULL END BEARING ON TOP AND BOTTOM TRACK. UNLESS NOTED OTHERWISE, PROVIDE DOUBLE STUDS AT ALL JAMBS, CORNERS, INTERSECTIONS, BEAM BEARINGS AND JOIST BEARINGS. BRIDGING SHALL BE INSTALLED PER MANUFACTURER'S RECOMMENDATION WITH THE FOLLOWING MINIMUM REQUIREMENTS.

BRIDGING SHALL BE STEEL CHANNEL MADE FROM ASTM A1003/A1003M, STRUCTURAL GRADE, TYPE H, METALLIC COATED STEEL SHEET, OF SAME GRADE AND COATING DESIGNATION USED FOR FRAMING MEMBERS.

PROVIDE WALL BRIDGING AS NOTED ON THE DRAWINGS. SOLID BLOCKING SHALL BE INSTALLED IN LIEU OF BRIDGING WHERE NOTED ON THE DRAWINGS.

PROVIDE SOFFIT AND JOIST BRIDGING AS NOTED ON THE DRAWINGS. SOLID BLOCKING SHALL BE INSTALLED IN LIEU OF BRIDGING WHERE NOTED ON THE DRAWINGS.

BRIDGING CLIPS TO PROVIDE ATTACHMENT TO STUD WEB AND WRAP AROUND THE BRIDGING CHANNEL. BRIDGING ACCESSORIES SHALL BE FORMED FROM STRUCTURAL QUALITY STEEL WITH MINIMUM YIELD STRENGTH OF 50 KSI AND HAVE MINIMUM PROTECTIVE COATING COMPLYING WITH ASTM A1003/A1003M (OR ASTM A653/A653M). ALL STRUCTURAL FRAMING ACCESSORIES SHALL BE FORMED FROM STRUCTURAL QUALITY STEEL WITH MINIMUM YIELD STRENGTH OF 50 KSI AND HAVE MINIMUM PROTECTIVE COATING EQUAL TO ASTM A1003 6-G0 GALVANIZED COATING.

WHEN REQUIRED, DEFLECTION CONNECTIONS SHALL ALLOW FOR POSITIVE ATTACHMENT TO STRUCTURE AND STUD WEB AND SHALL PROVIDE FRICTIONLESS, VERTICAL MOVEMENT. CONNECTION PRODUCTS ARE REQUIRED TO HAVE A VALID ICC ES REPORT OR EQUIVALENT COMPLYING WITH ICC ACCEPTANCE CRITERIA AC261. ALL STRUCTURAL FRAMING ACCESSORIES SHALL BE FORMED FROM STRUCTURAL QUALITY STEEL WITH MINIMUM YIELD STRENGTH OF 50 KSI AND HAVE MINIMUM PROTECTIVE COATING COMPLYING WITH ASTM 1003/A1003M (OR ASTM A653/A653M).

RIGID CONNECTIONS FOR ATTACHMENT OF METAL FRAMING TO METAL FRAMING AND TO THE PRIMARY STRUCTURE SHALL BE FORMED FROM STRUCTURAL QUALITY STEEL WITH MINIMUM YIELD STRENGTH OF 50 KSI AND HAVE MINIMUM PROTECTIVE COATING EQUAL TO COMPLYING WITH ASTM 1003/A1003M (OR ASTM A653/A653M).

ALL MEMBERS TO BE MANUFACTURED BY A MEMBER OF "STEEL STUD MANUFACTURERS ASSOCIATION" (SSMA), OR APPROVED EQUAL.

POUNDER ACTUATED FASTENERS (PAF'S) SHALL BE HILTI X-U 0.157" DIA. OR APPROVED EQUAL. RE: DETAILS FOR EMBED INTO CONCRETE; RE: MFR FOR REQ'D EMBED. IN STEEL.

ALL SCREWS TO BE #10 U.N.O., SCREWS ATTACHED PER MFR. INSTRUCTIONS AND SHALL PENETRATE 3 THREAD MINIMUM BEYOND THE ATTACHED MATERIAL.

ALL WELDING SHALL BE PERFORMED BY WELDERS EXPERIENCED IN LIGHT GAGE STRUCTURAL STEEL FRAMING WORK.

DO NOT NOTCH FLANGES OF JOISTS OR STUDS.

HEADERS, JAMBS, STUDS, JOISTS, RAFTERS, KICKERS AND GIRTS SHALL BE INSTALLED IN ONE-PIECE LENGTHS WITH NO SPLICES PERMITTED UNLESS SPECIFICALLY NOTED OTHERWISE IN THESE DRAWINGS.

PERFORMANCE SPECIFICATION FOR STAIRS DESIGN:

STAIRS SHALL BE DESIGNED FOR SELF WEIGHT PLUS A LIVE LOAD EQUAL TO 100 PSF. ALL STAIR TREADS ARE TO BE WELDED. CONTRACTOR SHALL SUBMIT SHOP DRAWINGS WITH DESIGN CALCULATIONS SEALED BY A PROFESSIONAL ENGINEER REGISTERED IN THE GOVERNING JURISDICTION OF THE PROJECT, SUBMITTED FOR REVIEW PRIOR TO MANUFACTURE.

METALS SUBMITTALS:

THE FOLLOWING ITEMS SHALL BE SUBMITTED FOR REVIEW FOR THE LIMITED PURPOSE OF CHECKING FOR CONFORMANCE WITH INFORMATION GIVEN AND THE DESIGN CONCEPT EXPRESSED IN THE CONTRACT DOCUMENTS. REFER TO PROJECT SPECIFICATIONS FOR ADDITIONAL REQUIRED SUBMITTALS.

COLD-FORMED METAL FRAMING (CFS) SHOP DRAWINGS: SUBMIT SHOP DRAWINGS FOR INSTALLATION OF COLD FORMED STEEL SYSTEMS. PROVIDE 1/4" = 1'-0" SCALE ELEVATIONS OF ALL ROOFS, FLOORS, SOFFITS, WALLS AND OTHER WORK. SHOW ALL OPENINGS IN VERTICAL WORK; ALL BEAMS AND POSTS; SECTIONS THROUGH COLD FORMED CONSTRUCTION; AND FRAMING COORDINATED WITH THE BUILDING STRUCTURE AND OTHER TRADES. ALL SHOP DRAWINGS SHALL BE REVIEWED BY THE CONTRACTOR PRIOR TO SUBMITTING DRAWINGS FOR REVIEW. **NO ENGINEERED DRAWINGS OR CALCULATIONS PERMITTED.**

PRODUCT DATA
SHOP DRAWINGS
STRUCTURAL STEEL
STEEL JOIST FRAMING
STEEL DECKING
COLD-FORMED METAL FRAMING (PLANS, DETAILS, ELEVATIONS) – **NO ENGINEERING PERMITTED**
METAL STAIRS
WELDING PROCEDURE SPECIFICATIONS AND PROCEDURE QUALIFICATION RECORDS
QUALIFICATION DATA FOR QUALIFIED INSTALLER, FABRICATOR, PROFESSIONAL ENGINEER, TESTING AGENCY
WELDING CERTIFICATES
MILL TEST REPORTS
PRODUCT TEST REPORTS
SOURCE QUALITY CONTROL REPORTS

INFORMATION ON DRAWINGS

PLAN LEGEND		
SYMBOL	DESCRIPTION	REMARKS
	DETAIL CUTS SHOWN ON PLANS	
	CONCRETE WALL UNO	SEE PLANS AND SCHEDULES FOR REINFORCING
	8" MASONRY WALL UNO	OTHER SIZES ARE DIMENSIONED ON PLANS.
	12" MASONRY WALL UNO	OTHER SIZES ARE DIMENSIONED ON PLANS.
	STUD WALL UNO	SEE GSN, PLANS, AND SCHEDULES FOR SIZE AND SPACING OF STUDS.
	MECHANICAL EQUIPMENT	SEE PLANS FOR UNIT WEIGHTS.
	OPENING IN FRAMING	SEE NOTE #4.
	INDICATES EXTENTS OF MEMBER	SEE PLANS.

NOTES

- FOR MATERIAL STRENGTHS, SEE GENERAL STRUCTURAL NOTES.
- VERIFY ALL DIMENSIONS WITH ARCHITECTURAL DRAWINGS PRIOR TO START OF CONSTRUCTION - RESOLVE ANY DISCREPANCIES WITH ARCHITECT.
- FOR CLARITY, ALL EXTERIOR SLABS AND SIDEWALKS MAY NOT BE SHOWN. FOR EXACT DIMENSIONS, LOCATIONS, JOINT AND SCORE LINES, SEE ARCHITECTURAL DRAWINGS.
- FOR CLARITY, ALL OPENINGS MAY NOT BE SHOWN ON FRAMING PLANS. FOR EXACT SIZE, NUMBER, AND LOCATION OF OPENINGS, SEE ARCHITECTURAL, MECHANICAL, ELECTRICAL AND PLUMBING DRAWINGS. FOR FRAMING AT OPENINGS, SEE TYPICAL STRUCTURAL DETAILS. VERIFY ALL SIZES, WEIGHTS AND LOCATIONS OF MECHANICAL EQUIPMENT WITH MECHANICAL ENGINEER AND MECHANICAL CONTRACTOR THROUGH ARCHITECT.
- DETAILS MARKED "TYPICAL" MAY NOT BE CUT ON PLANS.
- CONC C.J. - SHALL BE EITHER KEYED OR SAW CUT CONTROL JOINT IN SLAB ON GRADE AT CONTRACTOR'S OPTION, SEE GENERAL STRUCTURAL NOTES AND PLANS.
- MAS C.J. - PROVIDE MASONRY CONTROL JOINT IN MASONRY WALL PER G.S.N. AND TYPICAL DETAIL.
- FOR CLARITY, DETAILS MAY SHOW ONLY ONE SIDE OF FRAMING CONDITION.
- CONTRACTOR TO VERIFY, AND BE RESPONSIBLE FOR VARIATIONS IN CONCRETE QUANTITY DUE TO CAMBER, CONSTRUCTION DEAD LOAD DEFLECTIONS AND/OR TOLERANCES OF STRUCTURAL STEEL ELEMENTS (I.E. BEAMS, STEEL DECK, ETC.) AND PRECAST CONCRETE ELEMENTS.
- ALL SCHEDULE MARK DESIGNATIONS MAY NOT NECESSARILY BE FOUND ON THE PLANS WHERE THE SCHEDULES OCCUR. SCHEDULES ARE TYPICAL TO THE PROJECT.

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GENERAL
STRUCTURAL NOTES

S0.2

SPECIAL STRUCTURAL INSPECTIONS						
MATERIAL	VERIFICATION AND INSPECTION TASK		FREQUENCY		REFERENCED STANDARD	IBC REFERENCE
			CONTINUOUS	PERIODIC		
COLD-FORMED STEEL FRAMING	1.	MATERIAL VERIFICATION:				1705.10.2
		A) MEMBER SIZE AND THICKNESS TO MATCH CONTRACT DOCUMENTS INCLUDING TRACKS, STUDS, ASSEMBLIES, CONNECTORS.		X		
		B) FASTENER MATERIAL AND COMPONENTS.		X		
	2.	INSPECTION OF INSTALLATION:				
		A) INSPECT MEMBER LAYOUT, CONNECTION, ORIENTATION.		X		
		B) SPECIAL INSPECTION REQUIRED FOR FASTENERS PER MANUFACTURER.		X		
		C) INSPECTION PRIOR TO SHEATHING:VERIFY FLANGES ARE INTACT, STUDS ARE NOT CUT OR SPLICED.		X		
	3.	INSPECTION OF WELDING.		X		
POST INSTALLED ANCHORS	1.	EPOXY ANCHORS				
		A) ALL EPOXY BOLTS TO HAVE I.C.C. RATING FOR MATERIAL INTO WHICH INSTALLATION TAKES PLACE. ALL EPOXY BOLTS ARE SUBJECT TO THE FOLLOWING SPECIAL INSPECTION AND TESTING.				
		B) VERIFY ANCHOR TYPE, DIMENSIONS, BASE MATERIAL TYPE & STRENGTH, HOLE DIMENSIONS, HOLE CLEANING PROCEDURES, ANCHOR SPACING, EDGE DISTANCES, BASE MATERIAL THICKNESS, ANCHOR EMBEDMENT, AND TIGHTENING TORQUE.		X		
		C) LOAD TEST THE FIRST 10% OF EACH TYPE OF BOLT TO 150% RECOMMENDED ALLOWABLE WORKING LOAD IN TENSION. IF AT ANYTIME THE NUMBER OF REJECTED BOLTS EXCEEDS 10%, TEST 100% OF REMAINING BOLTS UNTIL NOT MORE THAN 10% FAIL LOAD TEST. COSTS FOR ADDITIONAL TESTING BEYOND THE FIRST 10% OF ANY BOLT TYPE SHALL BE BORNE BY THE CONTRACTOR.		X		
	2.	EXPANSION ANCHORS				
		A) ALL EXPANSION BOLTS TO HAVE I.C.C. RATING FOR MATERIAL INTO WHICH INSTALLATION TAKES PLACE. ALL EXPANSION BOLTS ARE SUBJECT TO THE FOLLOWING SPECIAL INSPECTION AND TESTING.				
		B) VERIFY ANCHOR TYPE, ANCHOR DIMENSIONS, HOLE CLEANLINESS, EMBEDMENT DEPTH, BASE MATERIAL TYPE & STRENGTH, DRILL BIT DIAMETER, HOLE DEPTH, EDGE DISTANCES, ANCHOR SPACING, BASE MATERIAL THICKNESS, AND TIGHTENING TORQUE.	X			
NOTE: THIS TABLE IS NOT COMPREHENSIVE. ALL TESTING AND INSPECTION REQUIREMENTS DEFINED ELSEWHERE IN THE DRAWINGS AND SPECIFICATIONS ARE REQUIRED. CONTRACTOR SHALL COORDINATE ALL REQUIREMENTS AND BRING ANY DISCREPANCIES TO THE ENGINEER'S ATTENTION PRIOR TO BEGINNING THE WORK.						

SPECIAL STRUCTURAL INSPECTIONS					
MATERIAL	VERIFICATION AND INSPECTION TASK	FREQUENCY		REFERENCED STANDARD	IBC REFERENCE
		CONTINUOUS	PERIODIC		
SOILS (RE: IBC TABLE 1705.6)	1. VERIFY MATERIALS BELOW FOOTINGS ARE ADEQUATE TO ACHIEVE THE DESIGN BEARING CAPACITY.	---	X		
	2. VERIFY EXCAVATIONS ARE EXTENDED TO PROPER DEPTH AND HAVE REACHED PROPER MATERIAL.	---	X		
	3. PERFORM CLASSIFICATION AND TESTING OF CONTROLLED FILL MATERIALS.	---	X		
	4. VERIFY USE OF PROPER MATERIALS, DENSITIES AND LIFT THICKNESSES DURING PLACEMENT AND COMPACTION OF COMPACTED FILL.	X	---		
	5. PRIOR TO PLACEMENT OF COMPACTED FILL, INSPECT SUBGRADE AND VERIFY THAT SITE HAS BEEN PREPARED PROPERLY.	---	X		
CONCRETE (RE: IBC TABLE 1705.3)	1. INSPECTION REINFORCEMENT, INCLUDING PRESTRESSING TENDONS, AND VERIFY PLACEMENT.	---	X	ACI318 Ch: 20, 25.2, 25.3, 26.5.1-26.5.3	1908.4
	2. REINFORCING BAR WELDING:				
	a. VERIFY WELDABILITY OF REINFORCING BARS OTHER THAN ASTM A706;	---	X	AWS D1.4 ACI318: 26.5.4	
	b. INSPECT SINGLE-PASS FILLET WELDS, MAXIMUM 5/16"; AND	---	X		
	c. INSPECT ALL OTHER WELDS.	X	---		
	3. INSPECT ANCHORS CAST IN CONCRETE.	X	---	ACI 318: 17.8.2	
	4. INSPECT ANCHORS POST-INSTALLED IN HARDENED CONCRETE MEMBERS.				
	a. ADHESIVE ANCHORS INSTALLED IN HORIZONTALLLY OR UPWARDLY INCLINED ORIENTATIONS TO RESIST SUSTAINED TENSION LOADS.	X	---	ACI318: 17.8.2.4	
	b. MECHANICAL ANCHORS AND ADHESIVE ANCHORS NOT DEFINED IN 4.a.	---	X	ACI318: 17.8.2	
	5. VERIFY USE OF REQUIRED DESIGN MIX.	---	X	ACI318: Ch.19, 26.4.3, 26.4.4	1904.1-2, 1908.2, 1908.3
	6. PRIOR TO CONCRETE PLACEMENT, FABRICATE SPECIMENS FOR STRENGTH TEST,PERFORM SLUMP AND AIR CONTENT TESTS, AND DETERMINE THE TEMPERATURE OF THE CONCRETE.	X	---	ASTM C 172 ASTM C 31 ACI318: 26.4.5, 26.12	1908.10
	7. INSPECT CONCRETE AND SHOTCRETE PLACEMENT FOR PROPER APPLICATION TECHNIQUES.	X	---	ACI318: 26.4.5	1908.6, 1908.7, 1908.8
	8. VERIFY MAINTENANCE OF SPECIFIED CURING TEMPERATURE AND TECHNIQUES.	---	X	ACI318: 26.4.7, -26.4.9	1908.9

SPECIAL STRUCTURAL INSPECTIONS					
MATERIAL	VERIFICATION AND INSPECTION TASK	FREQUENCY		REFERENCED STANDARD	
		CONTINUOUS	PERIODIC	TMS 402/ ACI 530/ ASCE 5	TMS 602/ ACI 530.1/ ASCE 6
MASONRY LEVEL B (TMS)	1. VERIFY COMPLIANCE WITH THE APPROVAED SUBMITTALS.	---	X	---	ART. 1.5
	2. AS MASONRY CONSTRUCTION BEGINS, THE FOLLOWING SHALL BE VERIFIED TO ENSURE COMPLIANCE:				
	A. PROPORTIONS OF SITE- PREPARED MORTAR.	---	X	---	ART. 2.1, 2.6A
	B. CONSTRUCTION OF MORTAR JOINTS.	---	X	---	ART 3.3B
	C. GRADE AND SIZE OR PRESTRESSING TENDONS AND ANCHORAGES.	---	X		ART. 2.4B, 2.4H
	D. LOCATION OF REINFORCEMENT, CONNECTORS,PRESTRESSING TENDONS AND ANCHORAGES.	---	X	---	ART. 3.4, 3.6A
	3. PRIOR TO GROUTING, VERIFY THAT THE FOLLOWING ARE IN COMPLIANCE:				
	A. GOUT SPACE.	---	X	---	ART. 3.2D, 3.2F
	B. GRADE, TYPE, AND SIZE OF REINFORCEMENT AND ANCHOR BOLTS, AND PRESTRESSING TENDONS, AND ANCHORAGES.	---	X	SEC. 6.1	ART. 2.4, 3.4
	C. PLACEMENT OF REINFORCEMENT, CONNECTORS, AND PRESTRESSING TENDONS AND ANCHORAGES.	---	X	SEC. 6.1, 6.2.1, 6.2.6, 6.2.7	ART. 3.2E, 3.4, 3.6A
	D. PROPORTIONS OF SITE-PREPARED GROUT AND PRESTRESSING GROUT FOR BONDED TENDONS.	X	---	---	ART. 2.6B, 2.4G.1.b
	E. CONSTRUCTION OF MORTAR JOINTS.	---	X	---	ART. 3.3B
	4. VERIFY DURING CONSTRUCTION:				
	A. SIZE AND LOCATION OF STRUCTURAL ELEMENTS.	---	X	---	ART. 3.3F
	B. TYPE, SIZE, AND LOCATION OF ANCHORS, INCLUDING OTHER DETAILS OF ANCHORAGE OF MASONRY TO STRUCTURAL MEMBERS, FRAMES, OR OTHER CONSTRUCTION.	---	X	SEC. 1.2.1(e), 6.1.4.3, 6.2.1	---
	D. PREPARATION, CONSTRUCTION, AND PROTECTION OF MASONRY DURING COLD WEATHER (TEMPERATURE BELOW 40 DEG. F OR HOT WEATHER (TEMPERATURE ABOVE 90 DEG. F)).	---	X	---	ART. 1.8C, 1.8D
	F. PLACEMENT OF GROUT AND PRESTRESSING GROUT FOR BONDED TENDONS IS IN COMPLIANCE.	X	---	---	ART. 3.5, 3.6C
	5. OBSERVE PREPARATION OF GROUT SPECIMENS, MORTAR SPECIMENS, AND/OR PRISMS.	---	X	---	ART. 1.4B.2.a.3, 1.4B.2.b.3, 1.4B.2.c.3, 1.4B.3, 1.4B.4
* REQUIRED FOR THE FIRST 5000 SQUARE FEET OF AAC MASONRY.					

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REVISION NAME	DATE

STRUCTURAL STEEL CONSTRUCTION OTHER THAN STRUCTURAL STEEL REFERENCE: IBC TABLE 1705.2.2	EXTENT OF SERVICE (CONTINUOUS OR PERIODIC)	
1. MATERIAL VERIFICATION OF COLD-FORMED STEEL DECK: A. IDENTIFICATION MARKINGS TO CONFORM TO ASTM STANDARDS SPECIFIED IN THE APPROVED CONSTRUCTION DOCUMENTS. B. MANUFACTURER'S CERTIFIED TEST REPORTS.	P	
2. INSPECTION OF WELDING AND FASTENERS: A. COLD-FORMED STEEL DECK: 1) FLOOR AND ROOF DECK WELDS AND FASTNERS. B. REINFORCING STEEL: 1) VERIFICATION OF WELDABILITY OF REINFORCING STEEL OTHER THAN A706. 2) REINFORCING STEEL RESISTING FLEXURAL AND AXIAL FORCES IN INTERMEDIATE AND SPECIAL MOMENT FRAMES, AND BOUNDARY ELEMENTS OF SPECIAL STRUCTURAL WALLS OF CONCRETE AND SHEAR REINFORCEMENT. 3) SHEAR REINFORCEMENT. 4) OTHER REINFORCING STEEL.	P	
INSPECTION OF FABRICATORS AND FABRICATION PROCEDURES REFERENCE: IBC SECTION 1704.2.5.		

STRUCTURAL STEEL REFERENCE: IBC SECTION 1705.2.1. INSPECTIONS FOR STRUCTURAL STEEL SHALL BE IN ACCORDANCE WITH QUALITY ASSURANCE REQUIREMENTS OF AISC 360, CHAPTER N.
FABRICATOR AND ERECTOR QUALITY CONTROL PROGRAM REFERENCE AISC 360, CHAPTER N, SECTION N2
THE FABRICATOR'S QUALITY CONTROL INSPECTOR SHALL INSPECT THE FOLLOWING AS A MINIMUM, AS APPLICABLE: 1. SHOP WELDING, HIGH-STRENGTH BOLTING, AND DETAILS IN ACCORDANCE WITH AISC 360 SECTION N5. 2. SHOP CUT AND FINISHED SURFACES IN ACCORDANCE WITH AISC 360, SECTION M2. 3. SHOP HEATING FOR STRAIGHTENING, CAMBERING AND CURVING IN ACCORDANCE WITH AISC 360, SECTION M2.1. 4. TOLERANCES FOR SHOP FABRICATION IN ACCORDANCE WITH SECTION 6 OF THE CODE OF STANDARD PRACTICE. THE ERECTOR'S QUALITY CONTROL INSPECTOR SHALL INSPECT THE AS A MINIMUM, AS APPLICABLE: 1. FIELD WELDING, HIGH-STRENGTH BOLTING, AND DETAILS IN ACCORDANCE WITH AISC 360, SECTION N5. 2. STEEL DECK AND HEADED STEEL STUD ANCHOR PLACEMENT AND ATTACHMENT IN ACCORDANCE WITH AISC 360, SECTION N6. 3. FIELD CUT SURFACES IN ACCORDANCE WITH AISC 360, SECTION M2.2. 4. FIELD HEATING FOR STRAIGHTENING IN ACCORDANCE WITH AISC 360, SECTION M2.1. 5. TOLERANCES FOR FIELD ERECTION IN ACCORDANCE WITH SECTION 7.13 OF THE CODE OF STANDARD PRACTICE.
FABRICATOR AND ERECTOR DOCUMENTS REFERENCE AISC 360, CHAPTER N, SECTION N3
SUBMITTALS FOR STEEL CONSTRUCTION AND AVAILABLE DOCUMENTS FOR STEEL CONSTRUCTION SHALL CONFORM TO AISC 360, SECTION N3.
INSPECTION AND NONDESTRUCTIVE TESTING PERSONNEL REFERENCE AISC 360, CHAPTER N, SECTION N4
QUALITY CONTROL INSPECTOR (FABRICATOR OR ERECTOR) QUALIFICATIONS, QUALITY ASSURANCE INSPECTOR (SPECIAL INSPECTOR) QUALIFICATIONS AND NONDESTRUCTIVE TESTING PERSONNEL (INSPECTION AGENCY PERSONNEL) QUALIFICATIONS SHALL CONFORM TO AISC 360, SECTION N4.
MINIMUM REQUIREMENTS FOR INSPECTION OF STRUCTURAL STEEL BUILDINGS REFERENCE AISC 360, CHAPTER N, SECTION N5
QUALITY CONTROL INSPECTIONS BY THE FABRICATOR'S OR ERECTOR'S QUALITY CONTROL INSPECTOR (QC1) AND QUALITY ASSURANCE INSPECTIONS OF FABRICATED ITEMS AND THE ERECTED STEEL SYSTEM BY THE SPECIAL INSPECTOR (SI), SHALL CONFORM TO AISC 360, SECTION N5 AND TABLES N5.4-1, N5.4-2, N5.4-3, N5.6-1, N5.6-2, N5.6-3. IN THESE TABLES INSPECTION TASKS ARE AS FOLLOWS: O-OBSERVE THESE ITEMS ON A RANDOM BASIS. OPERATIONS NEED NOT BE DELAYED PENDING THESE INSPECTIONS. P-PERFORM THESE TASKS FOR EACH WELDED JOINT OR MEMBER.
NONDESTRUCTIVE TESTING OF WELDED JOINTS
NONDESTRUCTIVE TESTING OF WELDED JOINTS SHALL CONFORM TO AISC 360, SECTION N5 AND SHALL BE PERFORMED BY THE SPECIAL INSPECTOR (QUALITY ASSURANCE INSPECTOR) IN ACCORDANCE WITH AWS D1.1.
MINIMUM REQUIREMENTS FOR INSPECTION OF COMPOSITE CONSTRUCTION REFERENCE AISC 360, CHAPTER N, SECTION N6
INSPECTION OF COMPOSITE CONSTRUCTION SHALL CONFORM TO AISC 360, SECTION N6 AND TABLE N6.1
INSPECTION OF FABRICATORS AND FABRICATION PROCEDURES REFERENCE IBC SECTION 1704.2.5
INSPECTION OF FABRICATORS AND FABRICATION PROCEDURES SHALL BE PERFORMED BY THE QUALITY ASSURANCE INSPECTOR (SPECIAL INSPECTOR) AND SHALL CONFORM TO IBC SECTIONS 1704.2.5 AND 1704.2.5.1.
NONCONFORMING MATERIALS AND WORKMANSHIP REFERENCE AISC 360, CHAPTER N, SECTION N8
IDENTIFICATION AND REJECTION OF MATERIALS OR WORKMANSHIP THAT IS NOT IN CONFORMANCE WITH THE CONSTRUCTION DOCUMENTS SHALL BE PERMITTED AT ANY TIME DURING THE PROGRESS OF THE WORK. NONCONFORMING MATERIAL AND WORKMANSHIP SHALL BE BROUGHT TO THE IMMEDIATE ATTENTION OF THE GENERAL CONTRACTOR AND THE FABRICATOR OR ERECTOR, AS APPLICABLE. NONCONFORMING MATERIAL OR WORKMANSHIP SHALL BE BROUGHT INTO CONFORMANCE, OR MADE SUITABLE FOR ITS INTENDED PURPOSE AS DETERMINED BY THE STRUCTURAL ENGINEER OF RECORD.

TABLE N5.4-1 INSPECTION TASKS PRIOR TO WELDING REFERENCE AISC 360, CHAPTER N				
INSPECTION TASKS PRIOR TO WELDING	QC		SI	
WELDING PROCEDURE SPECIFICATIONS (WPS'S) AVAILABLE	P		P	
MANUFACTURER CERTIFICATIONS FOR WELDING CONSUMABLES AVAILABLE	P		P	
MATERIAL IDENTIFICATION (TYPE/GRADE)	O		O	
WELDER IDENTIFICATION SYSTEM*	O		O	
FIT-UP OF GROOVE WELDS (INCLUDING JOINT GEOMETRY - JOINT PREPARATION - DIMENSIONS (ALIGNMENT, ROOT OPENING, ROOT FACE, BEVEL) - CLEANLINESS (CONDITION OF STEEL SURFACES) - TACKING (TACK WELD QUALITY AND LOCATION) - BACKING TYPE AND FIT (IF APPLICABLE)	O		O	
CONFIGURATION AND FINISH OF ACCESS HOLES	O		O	
FIT-UP OF FILLET WELDS - DIMENSIONS (ALIGNMENT, GAPS AND ROOT) - CLEANLINESS (CONDITION OF STEEL SURFACES) - TACKING (TACK WELDS QUALITY AND LOCATION)	O		O	
CHECK WELDING EQUIPMENT	O		-	
* THE FABRICATOR OR ERECTOR, AS APPLICABLE, SHALL MAINTAIN A SYSTEM BY WHICH A WELDER WHO HAS WELDED A JOINT OR MEMBER CAN BE IDENTIFIED. STAMPS, IF USED, SHALL BE THE LOW-STRESS TYPE.				
WHERE: O - OBSERVE THESE ITEMS ON A RANDOM BASIS. OPERATIONS NEED NOT BE DELAYED PENDING THESE INSPECTIONS. P - PERFORM THESE TASKS FOR EACH WELDED JOINT OR MEMBER. QC - QUALITY CONTROL INSPECTOR (FABRICATOR OR ERECTOR) SI - SPECIAL INSPECTOR (QUALITY ASSURANCE INSPECTOR)				

TABLE N5.4-2 INSPECTION TASKS DURING WELDING REFERENCE AISC 360, CHAPTER N				
INSPECTION TASKS DURING WELDING	QC		SI	
USE OF QUALIFIED WELDERS	O		O	
CONTROL AND HANDLING OR WELDING CONSUMABLES - PACKAGING - EXPOSURE CONTROL	O		O	
NO WELDING OVER CRACKED TACK WELDS	O		O	
ENVIRONMENTAL CONDITIONS - WIND SPEED WITHIN LIMITS - PRECIPITATION AND TEMPERATURE	O		O	
WPS FOLLOWED - SETTINGS ON WELDING EQUIPMENT - TRAVEL SPEED - SELECTED WELDING MATERIALS - SHIELDING GAS TYPE/FLOW RATE - PREHEAT APPLIED - INTERPASS TEMPERATURE MAINTAINED (MIN./MAX.) - PROPER POSITION (F, V, H, OH)	O		O	
WELDING TECHNIQUES - INTERPASS AND FINAL CLEANING - EACH PASS WITHIN PROFILE LIMITATIONS - EACH PASS MEETS QUALITY REQUIREMENTS	O		O	
WHERE: O - OBSERVE THESE ITEMS ON A RANDOM BASIS. OPERATIONS NEED NOT BE DELAYED PENDING THESE INSPECTIONS. P - PERFORM THESE TASKS FOR EACH WELDED JOINT OR MEMBER. QC - QUALITY CONTROL INSPECTOR (FABRICATOR OR ERECTOR) SI - SPECIAL INSPECTOR (QUALITY ASSURANCE INSPECTOR)				

TABLE N5.4-3 INSPECTION TASKS AFTER WELDING REFERENCE AISC 360, CHAPTER N				
INSPECTION TASKS AFTER WELDING	QC		SI	
WELDS CLEANED	O		O	
SIZE, LENGTH AND LOCATION OF WELDS	P		P	
WELDS MEET VISUAL ACCEPTANCE CRITERIA - CRACK PROHIBITION - WELD/BASE-METAL FUSION - CRATER CROSS SECTION - WELD PROFILES - WELD SIZE - UNDERCUT - POROSITY	P		P	
ARC STRIKES	P		P	
K-AREA*	P		P	
BACKING REMOVED AND WELD TABS REMOVED (IF REQUIRED)	P		P	
REPAIR ACTIVITIES	P		P	
DOCUMENT ACCEPTANCE OR REJECTION OF WELDED JOINT OF MEMBER	P		P	
* WHEN WELDING OF DOUBLER PLATES, CONTINUITY PLATES OR STIFFENERS HAS BEEN PERFORMED IN THE K-AREA, VISUALLY INSPECT THE WEB K-AREA FOR CRACKS WITHIN 3 IN. OF THE WELD.				
WHERE: O - OBSERVE THESE ITEMS ON A RANDOM BASIS. OPERATIONS NEED NOT BE DELAYED PENDING THESE INSPECTIONS. P - PERFORM THESE TASKS FOR EACH WELDED JOINT OR MEMBER. QC - QUALITY CONTROL INSPECTOR (FABRICATOR OR ERECTOR) SI - SPECIAL INSPECTOR (QUALITY ASSURANCE INSPECTOR)				

TABLE N5.6-1 INSPECTION TASKS PRIOR TO BOLTING REFERENCE AISC 360, CHAPTER N				
INSPECTION TASKS PRIOR TO BOLTING	QC		SI	
USE OF QUALIFIED WELDERS	O		P	
FASTENERS MARKED IN ACCORDANCE WITH ASTM REQUIREMENTS	O		O	
PROPER FASTENERS SELECTED FOR THE JOINT DETAIL (GRADE, TYPE, BOLT LENGTH IF THREADS ARE TO BE EXCLUDED FROM SHEAR PLANE)	O		O	
PROPER BOLTING PROCEDURE SELECTED FOR JOINT DETAIL	O		O	
CONNECTING ELEMENTS, INCLUDING THE APPROPRIATE FAYING SURFACE CONDITION AND HOLE PREPARATION, IF SPECIFIED, MEET APPLICABLE REQUIREMENTS	O		O	
PRE-INSTALLED VERIFICATION TESTING BY INSTALLATION PERSONNEL OBSERVED AND DOCUMENTED FOR FASTENER ASSEMBLIES AND METHOD USED	P		O	
PROPER STORAGE PROVIDED FOR BOLTS, NUTS, WASHERS AND OTHER FASTENER COMPONENTS	O		O	
WHERE: O - OBSERVE THESE ITEMS ON A RANDOM BASIS. OPERATIONS NEED NOT BE DELAYED PENDING THESE INSPECTIONS. P - PERFORM THESE TASKS FOR EACH WELDED JOINT OR MEMBER. QC - QUALITY CONTROL INSPECTOR (FABRICATOR OR ERECTOR) SI - SPECIAL INSPECTOR (QUALITY ASSURANCE INSPECTOR)				

TABLE N5.6-2 INSPECTION TASKS DURING BOLTING REFERENCE AISC 360, CHAPTER N				
INSPECTION TASKS DURING BOLTING	QC		SI	
FASTENER ASSEMBLIES, OF SUITABLE CONDITION, PLACED IN ALL HOLES AND WASHERS (IF REQUIRED) ARE POSITIONED AS REQUIRED	O		O	
JOINT BROUGHT TO THE SNUG-TIGHT CONDITION PRIOR TO THE PRETENSIONING OPERATION	O		O	
FASTENER COMPONENT NOT TURNED BY THE WRENCH PREVENTED FROM ROTATING	O		O	
FASTENERS ARE PRETENSIONED IN ACCORDANCE WITH RCSC SPECIFICATION, PROGRESSING SYSTEMATICALLY FROM THE MOST RIGID POINT TOWARD THE FREE EDGES	O		O	
WHERE: O - OBSERVE THESE ITEMS ON A RANDOM BASIS. OPERATIONS NEED NOT BE DELAYED PENDING THESE INSPECTIONS. P - PERFORM THESE TASKS FOR EACH WELDED JOINT OR MEMBER. QC - QUALITY CONTROL INSPECTOR (FABRICATOR OR ERECTOR) SI - SPECIAL INSPECTOR (QUALITY ASSURANCE INSPECTOR)				

TABLE N5.6-3 INSPECTION TASKS AFTER BOLTING REFERENCE AISC 360, CHAPTER N				
INSPECTION TASKS AFTER BOLTING	QC		SI	
DOCUMENT ACCEPTANCE OR REJECTION OF BOLTED CONNECTIONS	P		P	
WHERE: O - OBSERVE THESE ITEMS ON A RANDOM BASIS. OPERATIONS NEED NOT BE DELAYED PENDING THESE INSPECTIONS. P - PERFORM THESE TASKS FOR EACH WELDED JOINT OR MEMBER. QC - QUALITY CONTROL INSPECTOR (FABRICATOR OR ERECTOR) SI - SPECIAL INSPECTOR (QUALITY ASSURANCE INSPECTOR)				

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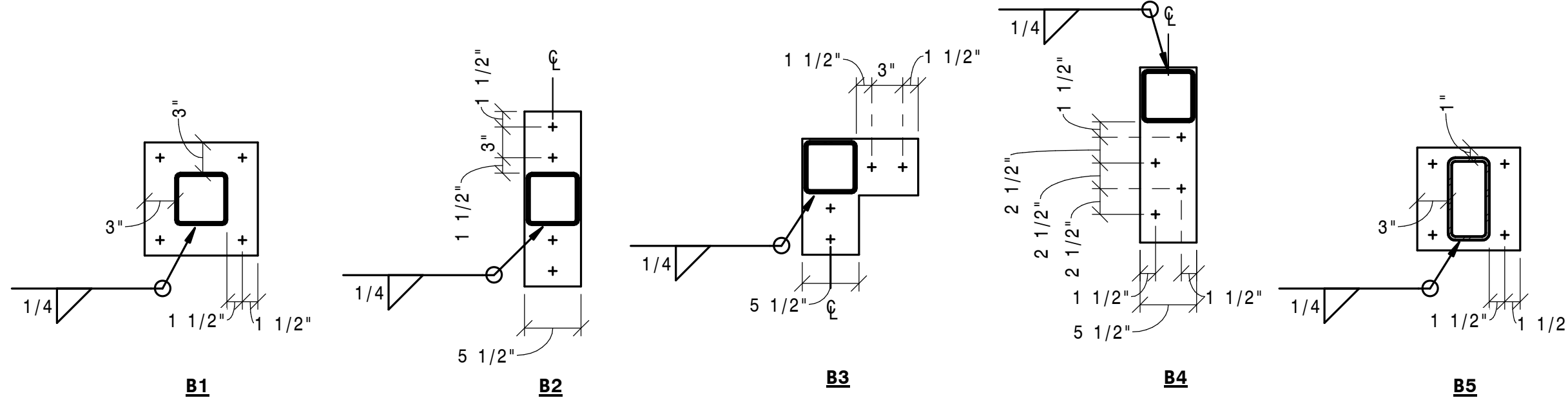
ACI TENSION LAP SPLICE LENGTHS													
BAR SIZE	LAP CLASS	f'c = 3,000 PSI OR LESS				f'c = 4,000 PSI				f'c = 5,000 PSI OR LESS			
		TOP BARS		OTHER BARS		TOP BARS		OTHER BARS		TOP BARS		OTHER BARS	
		CASE 1	CASE 2	CASE 1	CASE 2	CASE 1	CASE 2	CASE 1	CASE 2	CASE 1	CASE 2	CASE 1	CASE 2
#3	A	22	32	17	25	19	28	15	22	17	25	13	19
	B	28	42	22	32	24	36	19	28	22	33	17	25
#4	A	29	43	22	33	25	37	19	29	22	33	17	26
	B	37	56	29	43	32	48	25	37	29	43	22	33
#5	A	36	54	28	41	31	47	24	36	28	42	22	32
	B	47	70	36	54	40	60	31	47	36	54	28	42
#6	A	43	64	33	50	37	56	29	43	33	50	26	38
	B	56	84	43	64	48	72	37	56	43	65	33	50
#7	A	63	94	48	72	54	81	42	63	49	73	37	56
	B	81	122	63	94	70	106	54	81	63	94	49	73
#8	A	72	107	55	82	62	93	48	71	55	83	43	64
	B	93	139	72	107	80	121	62	93	72	108	55	83
#9	A	81	121	62	93	70	105	54	81	63	94	48	72
	B	105	157	81	121	91	136	70	105	81	122	63	94
#10	A	91	136	70	105	79	118	61	91	70	105	54	81
	B	118	177	91	136	102	153	79	118	91	137	70	105
#11	A	101	151	78	116	87	131	67	101	78	117	60	90
	B	131	196	101	151	113	170	87	131	101	152	78	117
#14	N/A	121	181	93	139	105	157	81	121	94	140	72	108
#18	N/A	161	241	124	186	139	209	107	161	125	187	96	144

- NOTE:
- TABULATED VALUES ARE BASED ON GRADE 60 REINFORCING BARS AND NORMAL WEIGHT CONCRETE.
 - TENSION DEVELOPMENT LENGTHS AND TENSION LAP SPLICE LENGTHS ARE CALCULATED PER ACI 318-14, SECTIONS 12.2 AND 12.15, RESPECTIVELY. TABULATED VALUES FOR BEAMS OR COLUMNS ARE BASED ON TRANSVERSE REINFORCEMENT AND CONCRETE COVER MEETING MINIMUM CODE REQUIREMENTS. LENGTHS ARE IN INCHES.
 - CASES 1 AND 2, WHICH DEPEND ON THE TYPE OF STRUCTURAL ELEMENT, CONCRETE COVER, AND THE CENTER-TO-CENTER SPACING OF THE BARS, ARE DEFINED AS:
BEAMS OR COLUMNS: CASE 1: COVER AT LEAST 1.0 d_b AND c.-c. SPACING AT LEAST 2.0 d_b
CASE 2: COVER LESS THAN 1.0 d_b OR c.-c. SPACING LESS THAN 2.0 d_b
ALL OTHERS: CASE 1: COVER AT LEAST 1.0 d_b AND c.-c. SPACING AT LEAST 3.0 d_b
CASE 2: COVER LESS THAN 1.0 d_b OR c.-c. SPACING LESS THAN 3.0 d_b
 - LAP SPLICE LENGTHS ARE MULTIPLES OF TENSION DEVELOPMENT LENGTHS: CLASS A = 1.0d AND CLASS B = 1.3d (ACI 318-14, SECTION 12.15.1). ACI 318-14 DOES NOT ALLOW LAP SPLICES OF #14 OR #18 BARS. THE TABULATED VALUES FOR THOSE BAR SIZES ARE THE TENSION DEVELOPMENT LENGTHS.
 - TOP BARS ARE HORIZONTAL BARS WITH MORE THAN 12 INCHES OF CONCRETE CAST BELOW THE BARS.
 - FOR LIGHTWEIGHT AGGREGATE CONCRETE, MULTIPLY THE TABULATED VALUES BY 1.3.
 - FOR EPOXY-COATED BARS, MULTIPLY THE TABULATED VALUES BY ONE OF THE FOLLOWING FACTORS:
CONCRETE COVER AND SPACING: TOP BARS OTHER BARS
COVER < 3.0 d_b OR c.-c. SPACING < 7.0 d_b 7/16-1.31 1.50
COVER > 3.0 d_b AND c.-c. SPACING > 7.0 d_b 1.20 1.20

STEEL COLUMN SCHEDULE (C)			
MARK	SIZE	BASE PLATE THICKNESS	COMMENTS
C1	HSS5X5X3/8	3/4" U.N.O.	
C2	HSS8X4X3/8	3/4" U.N.O.	
C3	HSS4X4X1/4	3/4" U.N.O.	

- NOTES:
- ALL ANCHOR RODS TO BE 3/4" DIA. UNO.
 - RE: DETAILS FOR EMBED PLATES, ANCHOR RODS AND OTHER REQUIREMENTS AT BRACED FRAMES.
 - STEEL COLUMNS AT EXTERIOR WALLS ARE SUPPORTED ON STEM WALLS WITHIN WALL STUD CAVITY - TYP. CONTR TO COORD BASE PLATE TYPES B2, B3, & B4 AS REQUIRED TO MATCH GEOMETRY OF STEM WALL AND WALL OR STOREFRONT CONDITIONS PER ARCH DRAWINGS - TYP.

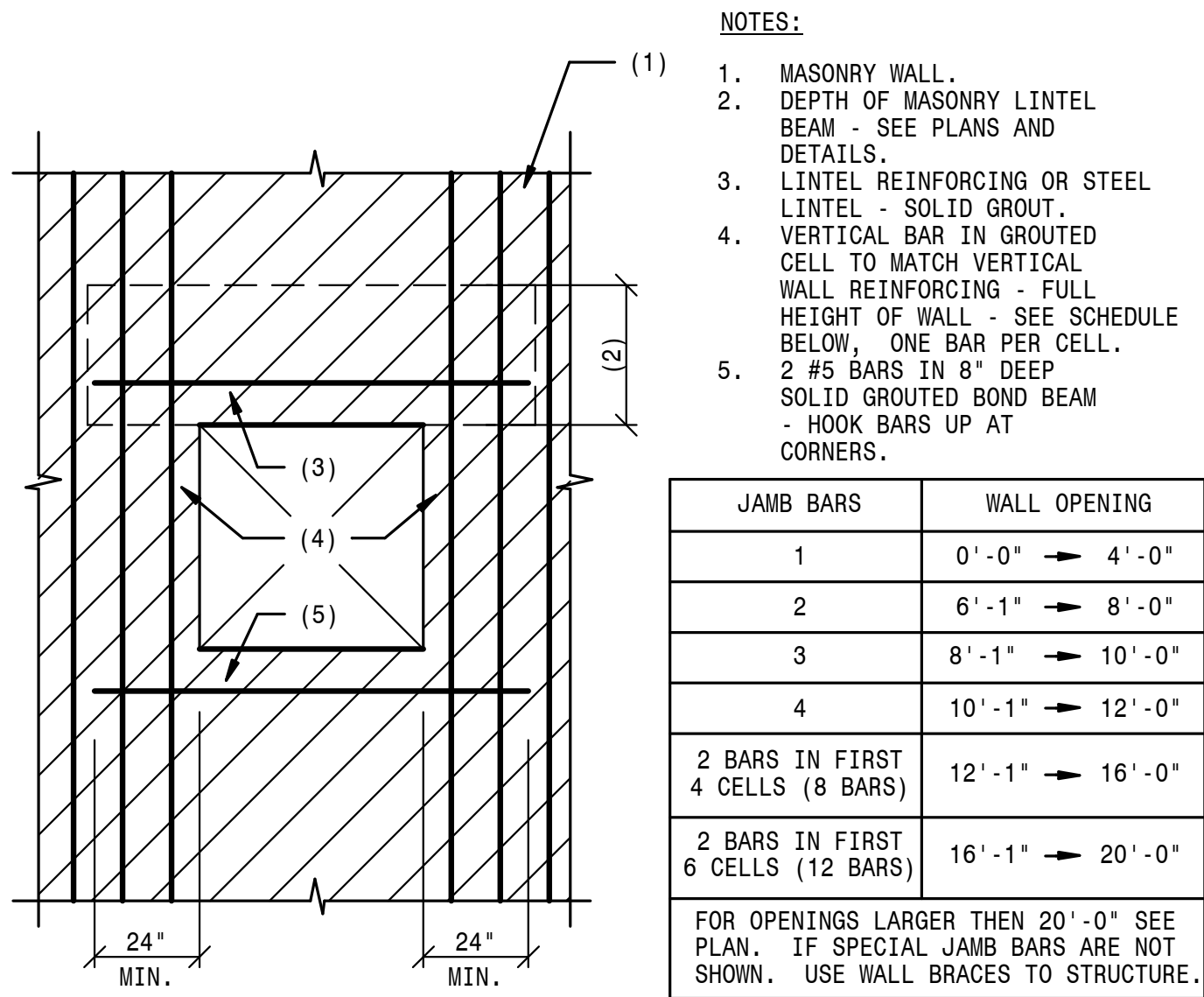
BASE PLATE TYPES



MASONRY WALL SCHEDULE (MW)

MARK	SIZE	VERT REINFORCING
MW1	10"	#6 AT 8" OC
MW2	10"	#5 AT 16" OC
MW3	8"	#5 AT 32" OC
MW4	6"	#4 AT 8" OC

- NOTES:
- RE: PLANS AND DETAILS FOR ADDL REINF.



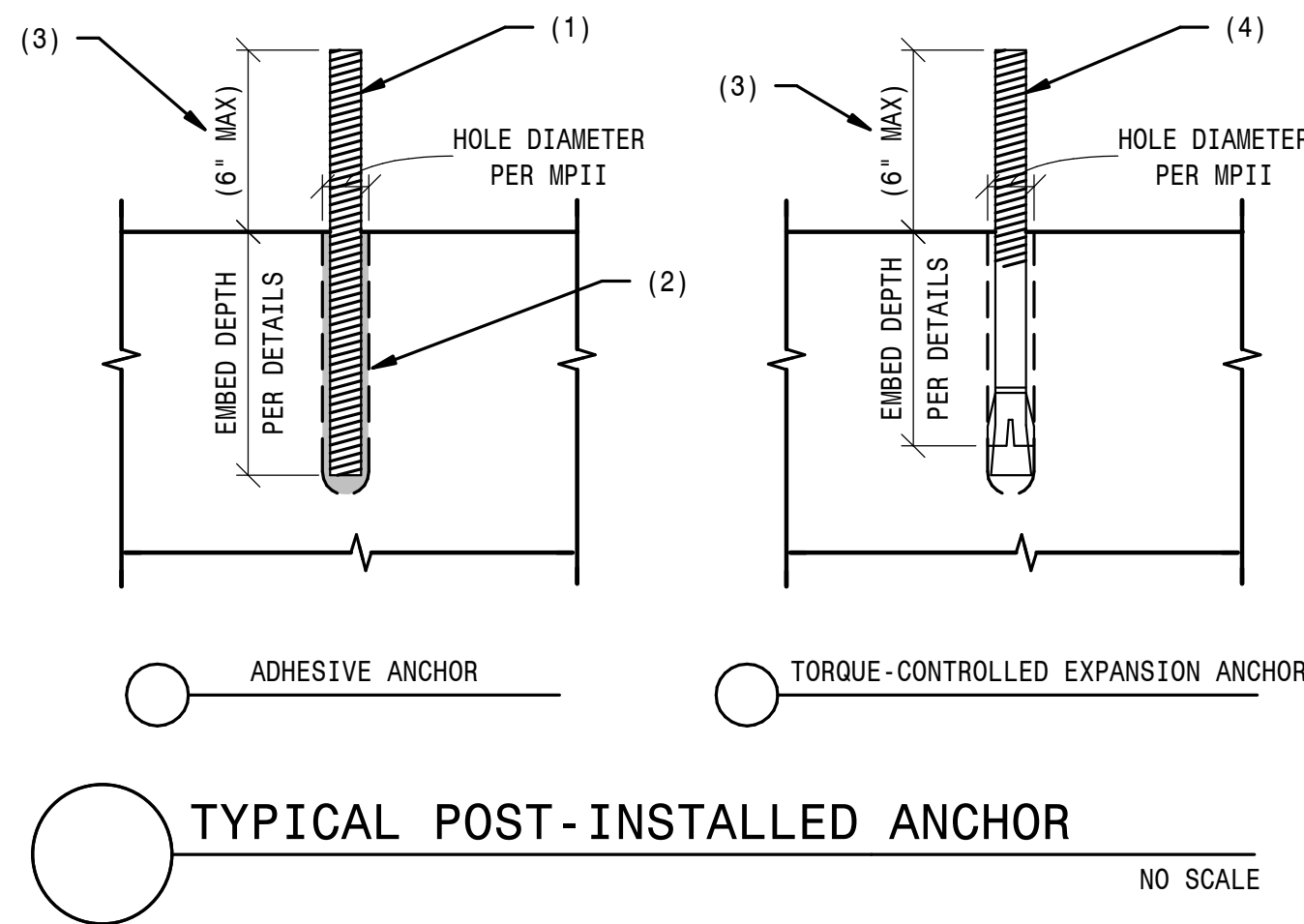
REINFORCED MASONRY LINTEL SCHEDULE

CLEAR SPAN	NOMINAL DEPTH	REINFORCING	TYPICAL DETAIL
0'-0" TO 4'-0"	8"	2 #4 BOTTOM	
OVER 4'-0" TO 6'-8"	16"	2 #4 TOP & BOTTOM	
OVER 6'-8" TO 8'-8"	16"	2 #5 TOP & BOTTOM	
OVER 8'-8" TO 10'-8"	24"	2 #5 TOP & BOTTOM	
OVER 10'-8" TO 12'-8"	32"	2 #6 TOP & BOTTOM	
OVER 12'-8" TO 14'-8"	40"	2 #6 TOP & BOTTOM	

- NOTES:
- SEE ARCH'L DWG'S FOR LOCATION AND CLEAR SPAN.
 - LINELS SHALL SPAN CONTINUOUS BETWEEN BEARINGS EACH SIDE.
 - PROVIDE 8" MIN. BEARINGS FOR CLEAR SPAN 8'-0" OR LESS. 16" MIN. BEARING FOR CLEAR SPAN GREATER THAN 8'-0".
 - EXTEND BOTTOM REINFORCING TO END OF BEARING EACH SIDE - EXTEND TOP REINFORCING WHERE POSSIBLE, 40 BAR DIA. INTO WALL EACH SIDE TERMINATE TOP REINFORCING W/ STD. HOOK AT CONTROL JT. OR FREE EDGES.
 - PROVIDE SOLID GROUTED OR SOLID MAS JAMB UNDER LINTEL EACH SIDE OF OPG.

- NOTES:
- THREADED ROD RE: GSN U.N.O.
 - STRUCTURAL ADHESIVE RE: DETAIL INSTALL PER MPII.
 - PROJECTION TO BE DETERMINED BY CONTRACTOR.
 - TORQUE-CONTROLLED EXPANSION ANCHOR RE: DETAIL.

DRILLED HOLES FOR POST-INSTALLED ANCHORS SHALL NOT PENETRATE THROUGH EXISTING STEEL REINFORCEMENT. CONTRACTOR SHALL COORDINATE SCAN, OR ANTICIPATE DRILLING ADDITIONAL HOLES IF REBAR IS HIT WITH THE DRILL.

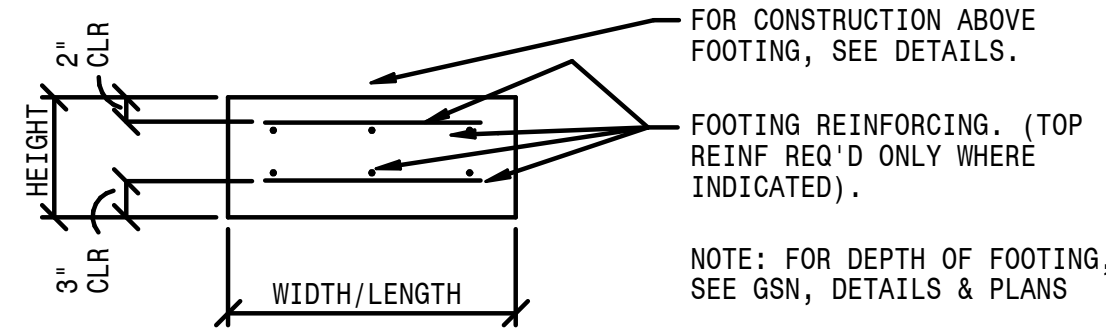


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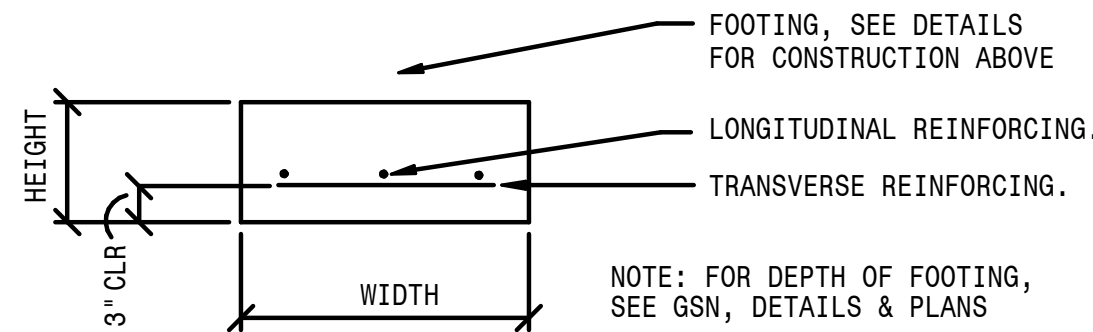
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ISOLATED FOOTING SCHEDULE (F)					
MARK	THICKNESS	DIMENSIONS		REINFORCING	COMMENTS
		WIDTH	LENGTH		
F1	1'-0"	3'-0"	3'-0"	(3) #5 EW BOT	
F2	1'-0"	5'-0"	5'-0"	(5) #5 EW BOT	
F3	2'-0"	6'-0"	12'-0"	#7 AT 12" OC EW T&B	
F4	2'-0"	6'-0"	18'-0"	#7 AT 12" OC EW T&B	



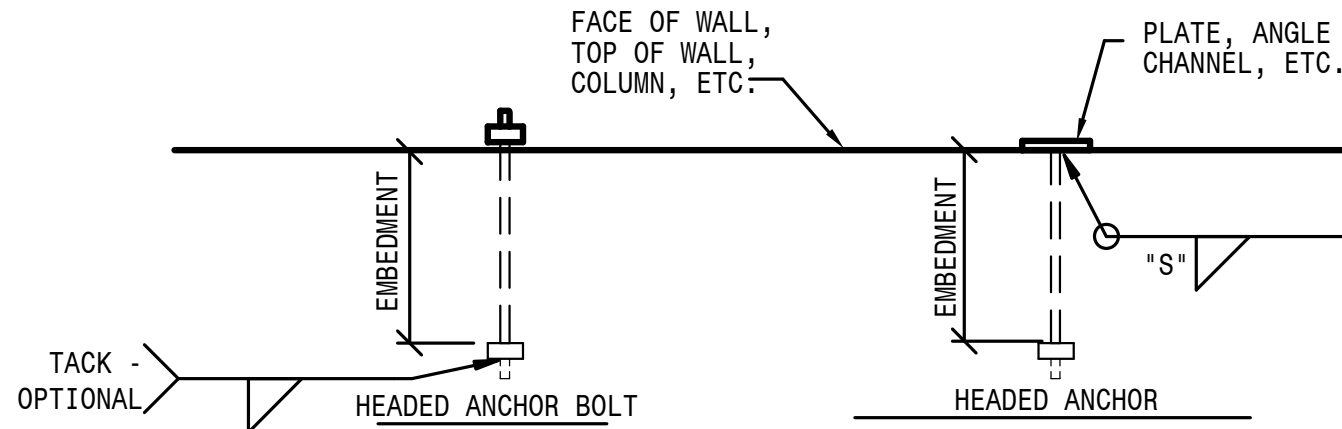
WALL FOOTING SCHEDULE(WF)					
MARK	DIMENSIONS		REINFORCING		REMARKS
	THICKNESS	WIDTH	LONGITUDINAL	TRANSVERSE	
WF1	1'-0"	1'-6"	(2) #5 X CONT	#5 AT 16" oc	
WF2	1'-0"	2'-0"	(2) #5 X CONT	#5 AT 16" oc	



WALL STUD SCHEDULE			
WALL HEIGHT	FRAMING	DEFLECTION TRACK	ALTERNATE SLOTTED DEFLECTION TRACK
<=14'-0"	600S162-54 AT 16" O.C.	600T250-68	600CST300-54
<=20'-0"	600S162-68 AT 16" O.C.	600T250-68	600CST300-54
JAMB SCHEDULE			
OPENING WIDTH	JAMB		
<=4'-0"	600S162-54		
<=7'-0"	600S162-68		
HORIZONTAL FRAMING SCHEDULE			
OPENING WIDTH	HEADER	SILL	
<=4'-0"	(1) 600S162-54 H + (1) 600T125-54 H	600T125-54	
<=7'-0"	(1) 600S162-54 H + (1) 600T125-54 H	600T125-54	
NOTES:			
1. BOT. WALL TRACK TO BE T125 w/ THICKNESS TO MATCH WALL STUD U.N.O. - TYP.			

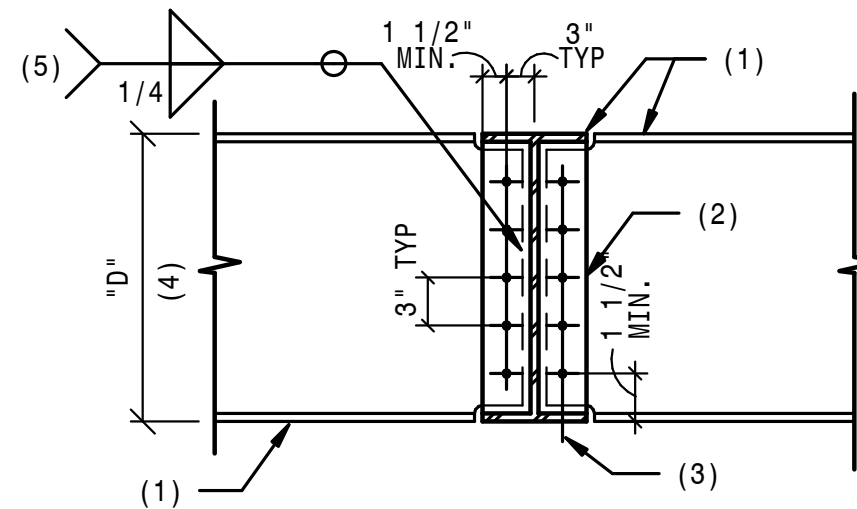
TYPICAL ANCHOR BOLT EMBEDMENT SCHEDULE			
BOLT DIAMETER (d _b)	VERT BOLT EMBEDMENT LENGTH	HORIZ BOLT EMBEDMENT LENGTH	HEADED STUD FILLET WELD SIZE, "S".
1/2"	7"	4"	1/4"
5/8"	7"	4"	5/16"
3/4"	7"	5"	5/16"
7/8"	8"	6"	5/16"
1"	9"	7"	3/8"
1 1/8"	10"	8"	-----
1 1/4"	11"	9"	-----

- NOTES:
- PROVIDE C.I.P. ANCHORS AND C.I.P. ANCHOR BOLTS PER THIS SCHEDULE UNLESS NOTED ON PLANS OR DETAILS.
 - AT "ANCHORS" USE 3/16" FILLET WELD("S").
 - THICKNESS OF DRYPACK DOES NOT APPLY TOWARDS EMBEDMENT.
 - UNLESS NOTED OTHERWISE, HEADED BOLTS SHALL BE USED AT ALL ANCHOR LOCATIONS EXCEPT THE FOLLOWING WHERE HOOKED ANCHORS MAY BE USED:
 - SHEAR WALLS (EXCLUDING HOLD-DOWNS)
 - BEAM/JOIST BEARING PLATES ON CONCRETE OR MASONRY
 - EXTERIOR WOOD AND METAL STUD WALLS BOTTOM ATTACHMENT TO FOUNDATION
 - STEEL AND WOOD LEDGERS NOT SUBJECT TO TENSION
 - HEADED ANCHOR BOLTS MAY BE CONSTRUCTED OF THREADED ROD MATERIAL WITH A NUT TACK WELDED TO THE THREADS ON THE EMBEDDED END OF THE ANCHOR.



BOLTED CONNECTION SCHEDULE	
NOMINAL BEAM DEPTH "D"	NUMBER OF 7/8" DIA. ASTM, A325N BOLTS
UP TO 7"	2
8" - 11"	2
12" - 14"	3
15" - 17"	4
18" - 20"	5
21" - 23"	6
24" - 29"	7
30" - 32"	8
33" - 35"	9
36"	10

- FRAMING NOTES:
- THE TYPICAL STEEL BEAM TO STEEL COLUMN OR STEEL BEAM TO STEEL BEAM CONNECTION CONSISTS OF 3/8" SINGLE SHEAR PLATES WITH 7/8" DIA. ASTM A325N BOLTS. USE 5/8" SHEAR PLATES WHERE "D" = 27" OR GREATER.
 - ALL BOLTS SHALL BE INSTALLED USING SHORT SLOTTED HOLES IN EITHER THE BEAM WEB OR THE SHEAR PLATE PER LATEST AISC SPECIFICATIONS.
 - CONNECTIONS REQUIRING DOUBLE ANGLES OR BENT PLATES WILL BE MARKED ON THE PLANS WITH AN ASTERISK (*).
- ALL DOUBLE ANGLES SHALL BE L4x4x1/4 AND ALL BENT PLATES SHALL BE 1/4" THICK - DETAIL PER SECTION 4 OF THE AISC STEEL CONSTRUCTION MANUAL.

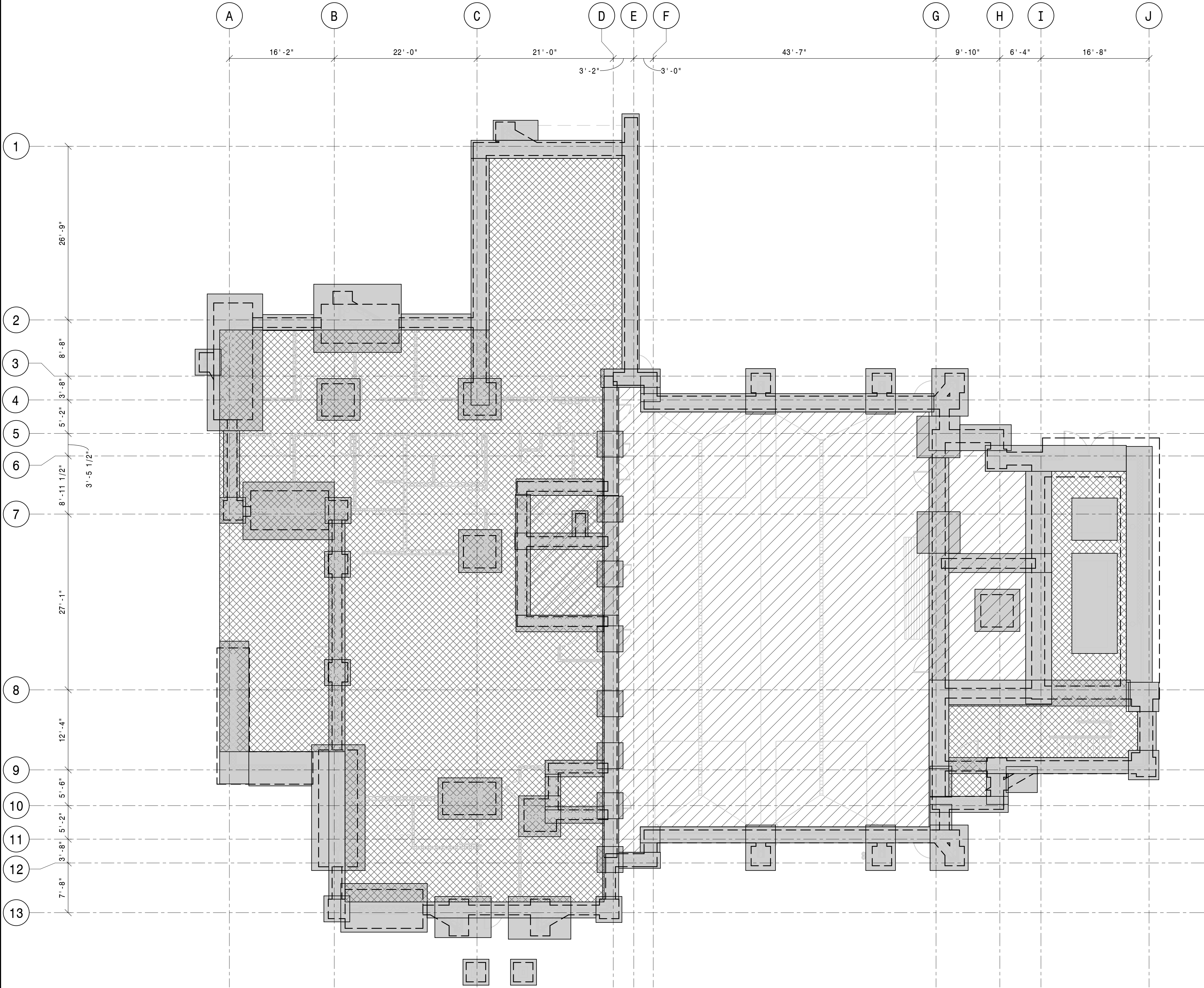


- NOTES:
- STEEL BEAM.
 - 3/8" STEEL SHEAR PLATE -USE 5/8" STEEL SHEAR PLATE WHERE "D" = 27" OR GREATER.
 - FOR SIZE, TYPE & NUMBER OF BOLTS, SEE TYPICAL BOLT SCHEDULE.
 - "D" = LESSER OF BEAM DEPTHS AS OCCURS.
 - WELD 3 SIDES TYPICAL - USE SAME CONNECTION AT STIFFENER PLATES WHEN REQUIRED.

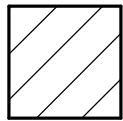
NOTE:
AT ONE SIDED CONNECTIONS, INSTALL 3/8" STIFFENER PLATE OPPOSITE SHEAR PLATE.

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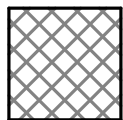
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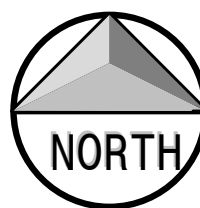
- RAMMED AGGREGATE PIER NOTES:
1. VERIFY ALL DIMENSIONS WITH ARCHITECTURAL DRAWINGS. ALL DIMENSIONS ARE TO CENTER OF PIER U.N.O.
 2. RAMMED AGGREGATE SUBCONTRACTOR TO PROVIDE RAMMED AGGREGATE PIERS BENEATH ALL APREAD FOOTINGS TO ALLOW FOR A BEARING PRESSURE VALUE OF 6,000 PSF TO BE USED.
 3. THE PROPOSED DESIGN IS GRAPHIC IN NATURE AND SHALL BE COMPLETED BY A QUALIFIED LICENSED ENGINEER FOR A COMPLETE SYSTEM MEETING THE GEOTECHNICAL REPORT REQUIREMENTS.



RAP SYSTEMS AT FLOOR SLABS: PIERS SHALL BE INSTALLED IN A GRID LAYOUT WITH A MAXIMUM SPACING BETWEEN PIERS OF 5 FT, AND SHALL ACHIEVE THE FOLLOWING PERFORMANCE CRITERIA:



INDICATES AREAS TO ACHIEVE 1,500 PSF BEARING PRESSURE



RAMMED AGGREGATE PIER PLAN

SCALE: 1/8" = 1'-0"

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RAMMED AGGREGATE
PIER PLAN

S1.0

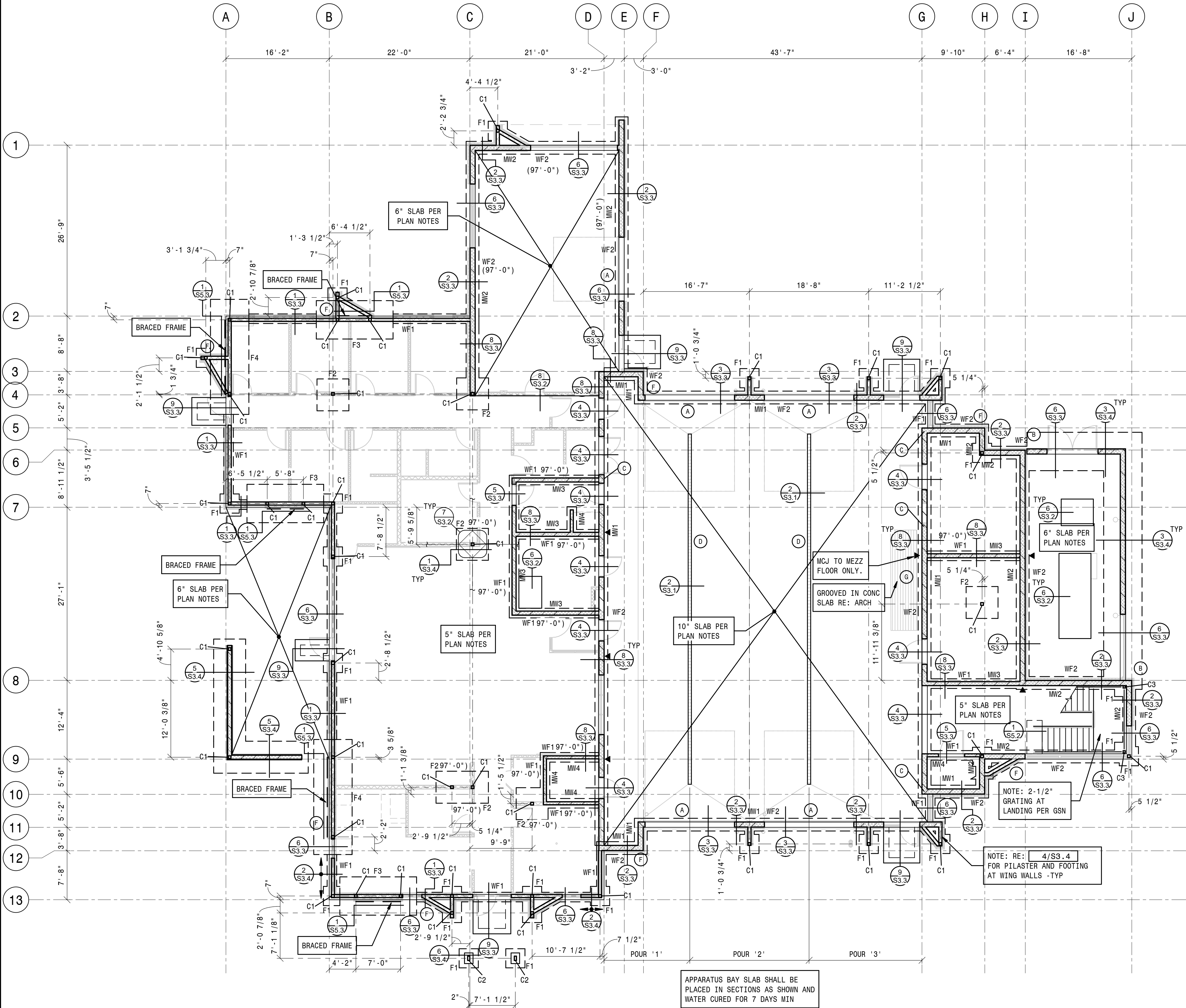
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- FOUNDATION PLAN NOTES:
- ELEVATIONS SHOWN ON PLAN ARE BASED ON DATUM ELEVATION SPECIFIC TO THE PROJECT. RE: ARCH/CIVIL DRAWINGS FOR ACTUAL U.S.G.S. ELEVATIONS AND BENCHMARK LOCATION.
 - CONCRETE SLAB ON GRADE IS 5" THICK REINFORCED WITH #3 AT 16" OC EW CENTERED IN SLAB UNO. TOP OF SLAB ELEVATION IS 100'-0" UNO. APPARATUS BAY CONCRETE SLAB ON GRADE IS 10" THICK REINFORCED WITH #4 AT 12" OC EW CENTERED IN SLAB UNO. TOP OF SLAB ELEVATION IS 100'-0" UNO. EXTERIOR SLAB INDICATED AS 6" SLAB ON STRUCTURAL PLANS IS 6" THICK REINFORCED WITH #3 @ 12" OC EW CENTERED. RE: ARCH FOR ELEVATION. ALL BUILDING SLABS TO BE WATER CURED INCLUDING THE APPARATUS BAY SLAB, ADMIN SLAB AND MEZZANINE SLAB.
 - VERIFY ALL DIMENSIONS WITH ARCHITECTURAL DRAWINGS.
 - SCHEDULED MARK DESIGNATIONS ARE TYPICAL TO THE PROJECT AND MAY NOT NECESSARILY BE FOUND ON THIS PLAN.
 - TOP OF FOOTING SHOWN AS (XX'-XX"). FOOTING ELEVATIONS SHOWN ARE MAXIMUMS AND MAY NEED TO BE LOWERED DUE TO SOIL CONDITIONS. VERIFY CHANGES WITH ENGINEER OF RECORD. TOP OF EXTERIOR FOOTING ELEVATION SHALL BE 97'-6" UNO. TOP OF INTERIOR FOOTING ELEVATION SHALL BE 99'-0" UNO.
 - WF1, WF2, ETC - AS SHOWN ON PLAN INDICATES CONTINUOUS WALL FOOTING, SEE SCHEDULE FOR SIZE AND REINFORCING. DIMENSIONS ARE TO FACE OF WALL UNO.
 - PROVIDE KEYS CONSTRUCTION JOINTS IN WALLS, GRADE BEAMS AND SLABS IN STRICT ACCORDANCE WITH THE REQUIREMENTS SHOWN IN THESE DOCUMENTS. HORIZONTAL REINFORCING SHALL BE CONTINUOUS THROUGH VERTICAL KEYS CONTROL JOINTS.
 - F1, F2, ETC - AS SHOWN ON PLAN INDICATES ISOLATED FOOTING, SEE SCHEDULE FOR SIZE AND REINFORCING. ISOLATED FOOTINGS ARE CENTERED ON GRID OR STEM WALL UNO.
 - C1, C2, ETC - AS SHOWN ON PLAN INDICATES STEEL COLUMN, SEE SCHED FOR MORE INFORMATION. COLUMNS ARE TYPE C2 UNO. COLUMNS ARE IDENTIFIED AT EACH COLUMN'S LOWEST LEVEL ONLY. COLUMNS ARE CENTERED ON GRID UNO.
 - MW1, MW2, ETC - AS SHOWN ON PLAN INDICATES MASONRY WALL, SEE SCHEDULE FOR SIZE AND REINFORCING.
 - VERIFY EXACT LOCATION OF DEPRESSED SLABS WITH ARCHITECTURAL DRAWINGS.
 - FOR SIDEWALK LOCATION AND DETAILS, SEE ARCHITECTURAL DRAWINGS.
 - ▲ INDICATES CMU CONTROL JOINT. PROVIDE MASONRY CONTROL JOINTS IN MASONRY WALLS PER G.S.N. AND TYPICAL DETAILS.
 - VERIFY EXACT SIZE AND LOCATION OF OPENINGS WITH ARCHITECTURAL DRAWINGS.

- KEY NOTES:
- NOTE: ALL KEY NOTES DO NOT NECESSARILY APPLY TO ALL PLANS.
- EMBEDDED PL1/2 x CONT. JAMB AND HEAD PLATE WITH 3/4" DIA x 0'-6" ANCHORS AT 2'-0" OC MAX - TYP AT OH DOOR. NO VISIBLE ANCHORS ARE PERMITTED, ARCH FOR ADD'L REQUIREMENTS.
 - PROVIDE 3/4" ISOLATION JOINT BETWEEN SITE WALL AND BUILDING FINISH. FOOTINGS SHALL BE MONOLITHIC. RE: ARCH FOR SEALANT INFO - TYP
 - PROVIDE (2) #6 FULL HT UNDER BEAM BEARING.
 - RE: ARCH & PLUMBING DRAWINGS FOR TRENCH DRAIN - TYP
 - 10,000LB IMPACT LOADS APPLIED ALONG THIS BEAM HAVE BEEN INCLUDED IN THE DESIGN OF THIS FRAMING. RE: 10/S5.2
 - ROOF OVERFLOW DRAIN PENETRATION THROUGH EXTERIOR WALL & ROOF DRAIN THROUGH FOUNDATION. COORDINATE DRAIN LINES AND SHOW ON FOUNDATION REINFORCING SHOP DRAWINGS FOR REVIEW.
 - TRAINING WINDOW ANCHORS PER DTL: 5/S4.1 RE: ARCH FOR LOCATIONS.
 - ANTENNA SUPPORT PER DETAIL: 11/S5.2 RE: ARCH AND ELECTRICAL FOR LOCATIONS.

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FOUNDATION PLAN

S1.1

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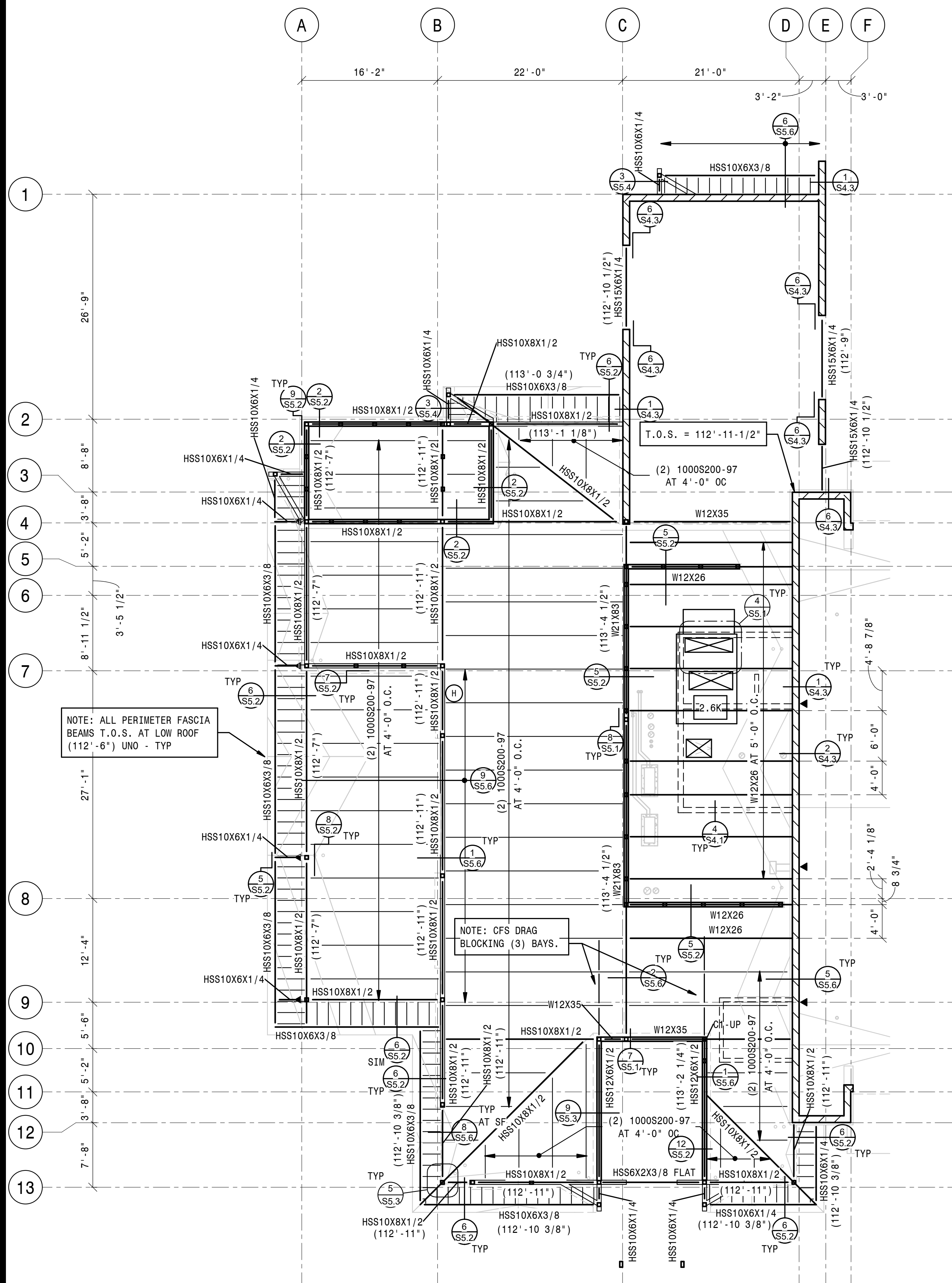
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FOUNDATION PLAN

SCALE: 1/8" = 1'-0"

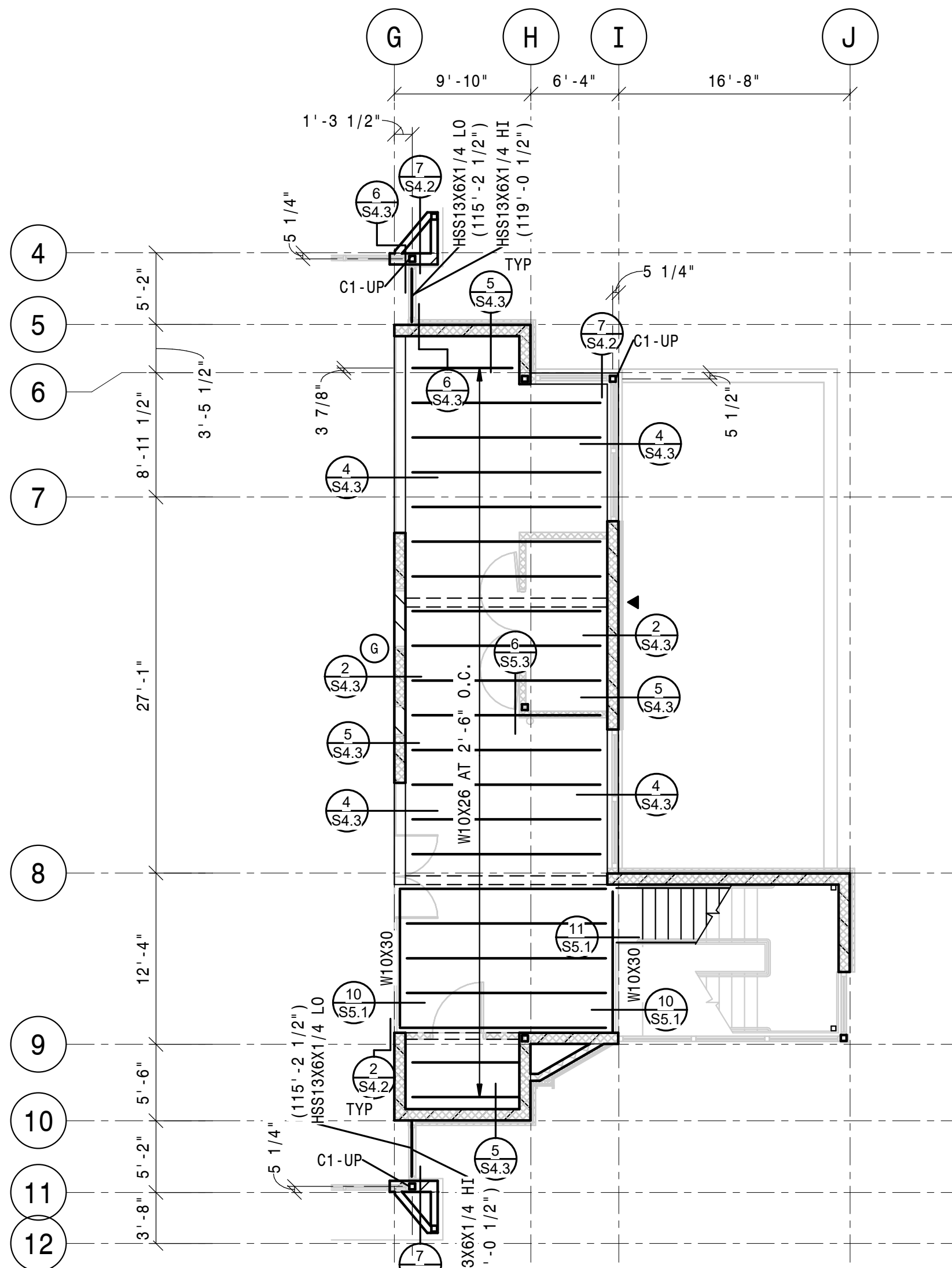
APPARATUS BAY SLAB SHALL BE PLACED IN SECTIONS AS SHOWN AND WATER CURED FOR 7 DAYS MIN



LOW ROOF FRAMING PLAN

SCALE: 1/8" = 1'-0"

- KEY NOTES:**
NOTE: ALL KEY NOTES DO NOT NECESSARILY APPLY TO ALL PLANS.
- (A) EMBEDDED PL1/2 x CONT. JAMB AND HEAD PLATE WITH 3/4" DIA x 0'-6" ANCHORS AT 2'-0" OC MAX - TYP AT OH DOOR. NO VISIBLE ANCHORS ARE PERMITTED, ARCH FOR ADD'L REQUIREMENTS.
 - (B) PROVIDE 3/4" ISOLATION JOINT BETWEEN SITE WALL AND BUILDING FINISH. FOOTINGS SHALL BE MONOLITHIC. RE: ARCH FOR SEALANT INFO - TYP
 - (C) PROVIDE (2) #6 FULL HT UNDER BEAM BEARING.
 - (D) RE: ARCH & PLUMBING DRAWINGS FOR TRENCH DRAIN - TYP
 - (E) 10,000LB IMPACT LOADS APPLIED ALONG THIS BEAM HAVE BEEN INCLUDED IN THE DESIGN OF THIS FRAMING. RE: 10/S5.2
 - (F) ROOF OVERFLOW DRAIN PENETRATION THROUGH EXTERIOR WALL & ROOF DRAIN THROUGH FOUNDATION. COORDINATE DRAIN LINES AND SHOW ON FOUNDATION REINFORCING SHOP DRAWINGS FOR REVIEW.
 - (G) TRAINING WINDOW ANCHORS PER DTL: 5/S4.1 RE: ARCH FOR LOCATIONS.
 - (H) ANTENNA SUPPORT PER DETAIL: 11/S5.2 RE: ARCH AND ELECTRICAL FOR LOCATIONS.



MEZZANINE FRAMING PLAN

SCALE: 1/8" = 1'-0"

FLOOR FRAMING NOTES

- ELEVATIONS SHOWN ON PLAN ARE BASED ON DATUM ELEVATION VERIFY ALL DIMENSIONS WITH ARCHITECTURAL DRAWINGS. TOP OF MEZZANINE CONCRETE = 110'-4" UNO. TOP OF STEEL = 110'-0" UNO.
- SCHEDULED MARK DESIGNATIONS ARE TYPICAL TO THE PROJECT AND MAY NOT NECESSARILY BE FOUND ON THIS PLAN.
- C1, C2, ETC - AS SHOWN ON PLAN INDICATES STEEL COLUMN, SEE SCHEDULE FOR MORE INFORMATION.
- FLOOR DECK SHALL BE 1" DEEP TYPE C, 32"-36" WIDE, 22 GAGE GALVANIZED STEEL, WITH MINIMUM YIELD STRESS OF 60 KSI, WITH MINIMUM S = .131 IN3 AND I = .070 IN4 PER FOOT OF WIDTH. DECK SHALL BE DEFORMED/INDENTED TO PROVIDE A MECHANICAL BOND WITH THE CONCRETE. DECK SHALL BE ERECTED IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS AS (3) SPAN MINIMUM.
- WELD FLOOR DECK TO SUPPORTING MEMBERS WITH (4) 5/8" DIAMETER PUDDLE WELDS PER SHEET AT ENDS, END LAPS AND AT INTERMEDIATE SUPPORTS, AND AT 12" O.C. AT PERIMETER BEAMS AND OPENING EDGES RUNNING PARALLEL TO THE DECK. SIDE SEAM ATTACHMENT SHALL BE #10 AT 12" O.C.
- FLOOR SLAB SHALL BE 3" NORMAL WEIGHT CONCRETE ON 1" DECK (4" TOTAL THICKNESS) REINFORCED WITH 6X6 W2.1XW2.1 WWF U.N.O. ALL BUILDING SLABS TO BE WATER CURED INCLUDING THE APPARATUS BAY SLAB, ADMIN SLAB AND MEZZANINE SLAB.
- FOR MASONRY LINTELS SEE SCHEDULE ON TYPICAL DETAIL SHEET. CONTRACTOR SHALL COORDINATE WITH OTHER TRADES FOR EXACT SIZE AND LOCATION REQUIREMENTS.
- FOR CLARITY, DETAILS MAY SHOW ONLY ONE SIDE OF FRAMING CONDITION.
- FOR CLARITY, ALL FLOOR OPENINGS MAY NOT BE SHOWN ON FLOOR FRAMING PLAN. FOR EXACT SIZE, NUMBER AND LOCATION OF OPENINGS, SEE ARCHITECTURAL, MECHANICAL, PLUMBING, AND ELECTRICAL DRAWINGS. FOR FRAMING AT OPENINGS, SEE TYPICAL DETAILS.
- CONTRACTOR TO VERIFY, AND BE RESPONSIBLE FOR VARIATIONS IN CONCRETE QUANTITY DUE TO CAMBER, CONSTRUCTION DEAD LOAD DEFLECTIONS, AND/OR STRUCTURAL STEEL TOLERANCES OF STEEL BEAMS AND STEEL DECK. FOR CONCRETE CONTROL JOINTS, IN CONCRETE SLAB, SEE G.S.N. AND TYPICAL DETAILS.
- ALL STAIR STRINGERS ARE MC12X10.6; ALL STAIR HEADERS ARE CBX11.5.
- AS SHOWN ON PLAN INDICATES MOMENT CONNECTION SEE DETAIL 4/S5.2

ROOF FRAMING NOTES:

- VERIFY ALL DIMENSIONS WITH ARCHITECTURAL DRAWINGS. ROOF ELEVATIONS, WHERE SHOWN, ARE TO BE PROVIDED AND VERIFIED BY THE ARCHITECT.
- SCHEDULED MARK DESIGNATIONS ARE TYPICAL TO THE PROJECT AND MAY NOT NECESSARILY BE FOUND ON THIS PLAN.
- C1, C2, ETC - AS SHOWN ON PLAN INDICATES STEEL COLUMN, SEE SCHEDULE FOR MORE INFORMATION.
- STEEL ROOF DECK SHALL BE 1.5" DEEP, 36" WIDE, 20 GAGE GALVANIZED STEEL, WITH MINIMUM YIELD STRESS OF 50 KSI, WITH MINIMUM S = 0.229 IN3 AND I = 0.217 IN4 PER FOOT OF WIDTH. DECK SHALL BE ERECTED IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS AS (3) SPAN MINIMUM.
- WELD ROOF DECK TO STEEL SUPPORTING MEMBERS WITH (7) - 5/8" DIAMETER PUDDLE WELDS PER SHEET AT ENDS, END LAPS AND AT INTERMEDIATE SUPPORTS, AND AT 12" O.C. AT PERIMETER BEAMS AND OPENING EDGES RUNNING PARALLEL TO THE DECK. SCREW ROOF DECK TO CFS SUPPORTING MEMBERS WITH (7) #12 SCREWS AT ENDS, END LAPS AND AT INTERMEDIATE SUPPORTS, AND AT 12" O.C. AT PERIMETER BEAMS AND OPENING EDGES RUNNING PARALLEL TO THE DECK. SIDE SEAM ATTACHMENT SHALL BE #10 AT 12" O.C. MINIMUM DIAPHRAGM SHEAR CAPACITY = 554 PLF.
- FOR STEEL LINTELS, SEE SCHEDULE ON TYPICAL DETAIL SHEET. CONTRACTOR SHALL COORDINATE WITH OTHER TRADES FOR EXACT SIZE AND LOCATION REQUIREMENTS.
- FOR MASONRY LINTELS SEE SCHEDULE ON TYPICAL DETAIL SHEET. CONTRACTOR SHALL COORDINATE WITH OTHER TRADES FOR EXACT SIZE AND LOCATION REQUIREMENTS.
- FOR CLARITY, DETAILS MAY SHOW ONLY ONE SIDE OF FRAMING CONDITION.
- FOR CLARITY, ALL ROOF OPENINGS MAY NOT BE SHOWN ON ROOF FRAMING PLAN. FOR EXACT SIZE, NUMBER AND LOCATION OF OPENINGS, SEE ARCHITECTURAL, MECHANICAL, PLUMBING, AND ELECTRICAL DRAWINGS. FOR FRAMING AT OPENINGS, SEE TYPICAL DETAILS.
- AS SHOWN ON PLAN INDICATES MOMENT CONNECTION SEE DETAIL 4/S5.2
- EQUIPMENT WEIGHTS SHOWN AS [XXK] WHERE XX IS IN KIPS. WEIGHT SHOWN INCLUDES ALL INERTIA BASES AND/OR HOUSEKEEPING PADS. IF WEIGHT OF ACTUAL EQUIPMENT IS GREATER THAN SHOWN, OR ADDITIONAL EQUIPMENT NOT SHOWN EXISTS, NOTIFY STRUCTURAL ENGINEER IMMEDIATELY.

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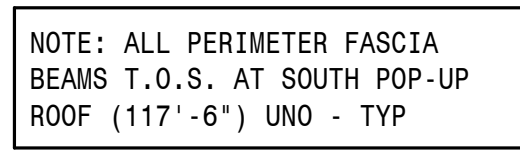
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LOW ROOF AND
MEZZANINE FRAMING
PLAN

S2.1



NOTE: ALL PERIMETER FASCIA BEAMS
T.O.S. AT NORTH POP-UP ROOF
(117'-5") UNO - TYP

NOTE: ALL PERIMETER FASCIA
BEAMS T.O.S. AT APPARATUS BAY
(125'-0") UNO - TYP

NOTE: ALL PERIMETER FASCIA
BEAMS T.O.S. AT EAST ROOF
(122'-7") UNO - TYP

NOTE: ALL PERIMETER FASCIA
BEAMS T.O.S. AT APPARATUS BAY
(125'-0") UNO - TYP

- KEY NOTES:
NOTE: ALL KEY NOTES DO NOT NECESSARILY APPLY TO ALL PLANS.

- (A) EMBEDDED PL1/2 x CONT. JAMB AND HEAD PLATE WITH 3/4" DIA x 0'-6" ANCHORS AT 2'-0" OC MAX - TYP AT OH DOOR. NO VISIBLE ANCHORS ARE PERMITTED, ARCH FOR ADL REQUIREMENTS.
- (B) PROVIDE 3/4" ISOLATION JOINT BETWEEN SITE WALL AND BUILDING FINISH. FOOTINGS SHALL BE MONOLITHIC. RE: ARCH FOR SEALANT INFO - TYP
- (C) PROVIDE (2) #6 FULL HT UNDER BEAM BEARING.
- (D) RE: ARCH & PLUMBING DRAWINGS FOR TRENCH DRAIN - TYP
- (E) 10,000LB IMPACT LOADS APPLIED ALONG THIS BEAM HAVE BEEN INCLUDED IN THE DESIGN OF THIS FRAMING. RE: 10/SS.2
- (F) ROOF OVERFLOW DRAIN PENETRATION THROUGH EXTERIOR WALL & ROOF DRAIN THROUGH FOUNDATION. COORDINATE DRAIN LINES AND SHOW ON FOUNDATION REINFORCING SHOP DRAWINGS FOR REVIEW.
- (G) TRAINING WINDOW ANCHORS PER DTL: 5/S4.1 RE: ARCH FOR LOCATIONS.
- (H) ANTENNA SUPPORT PER DETAIL: 11/SS.2 RE: ARCH AND ELECTRICAL FOR LOCATIONS.

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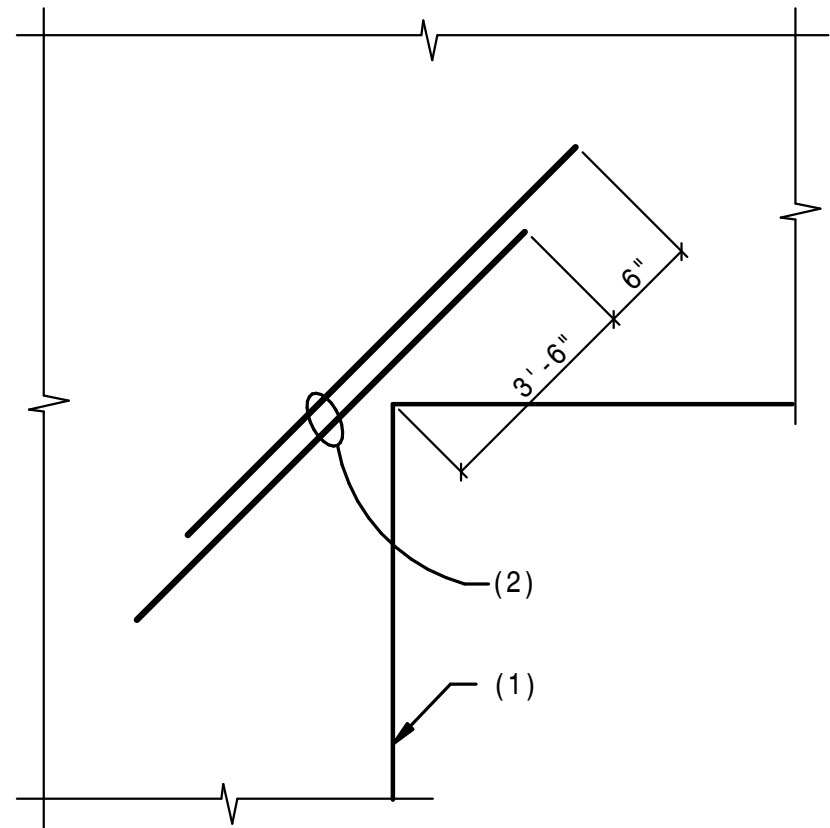
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HIGH ROOF FRAMING PLAN

225

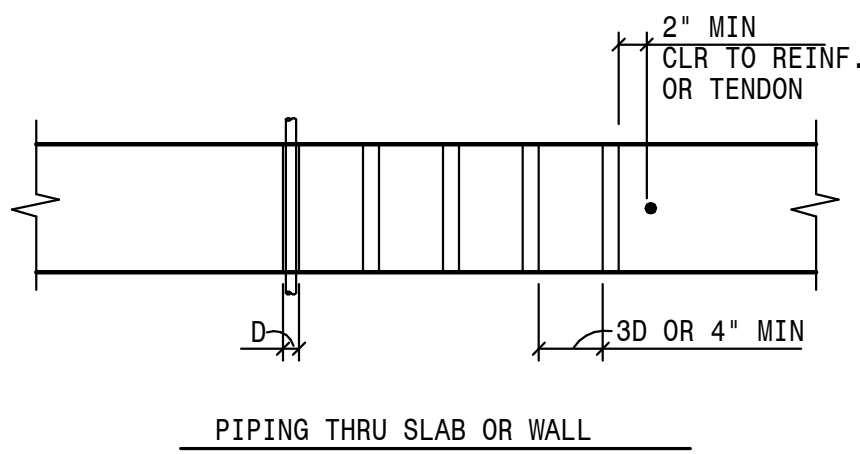


- NOTES:
1. SLAB EDGE.
 2. 2 #4 X 4'-0" LONG AT MID DEPTH.

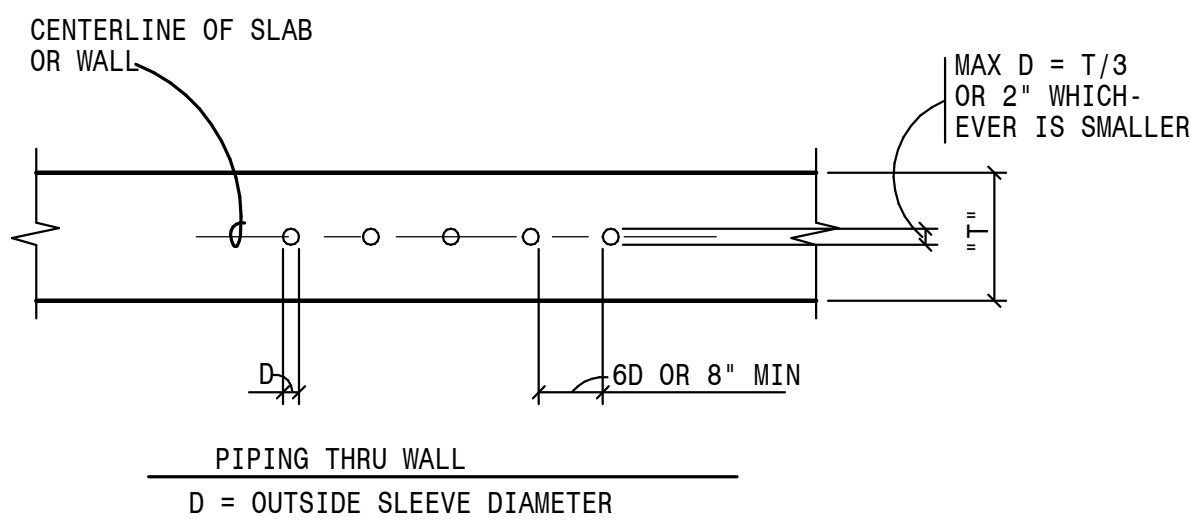
7 PLAN - REENTRANT CORNER REINFORCEMENT

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE



NOTE: WHERE CLEAR DISTANCE BETWEEN SLEEVES IS IMPOSSIBLE THIS AREA SHALL BE TREATED AS A SLAB OR WALL OPENING REINFORCED AS PER TYPICAL OPENING DETAILS. PRESET ALL SLEEVES PRIOR TO POURING CONCRETE. NO CORE DRILLING ALLOWED WITHOUT STRUCTURAL ENGINEERS APPROVAL. NO PIPING OR CONDUIT PERMITTED IN OR THROUGH CONCRETE BEAMS, JOISTS OR COLUMNS UNLESS SPECIFICALLY DETAILED ON STRUCTURAL DRAWINGS OR APPROVED BY THE STRUCTURAL ENGINEER.

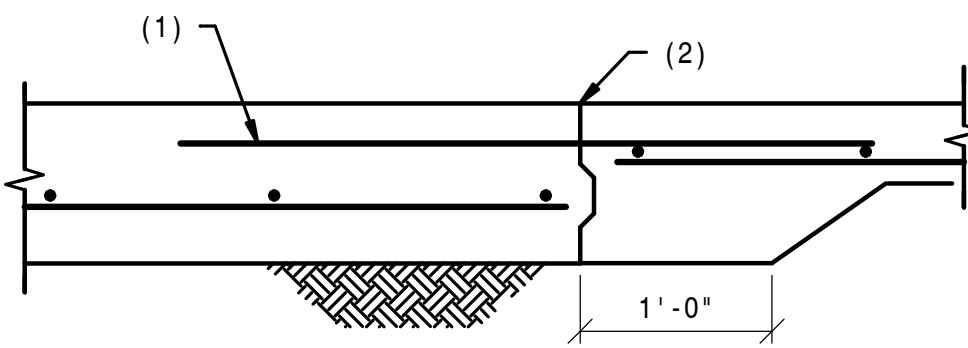


PIPING THRU WALL
D = OUTSIDE SLEEVE DIAMETER

8 PIPING AND CONDUIT IN OR THRU SLAB OR WALL

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE

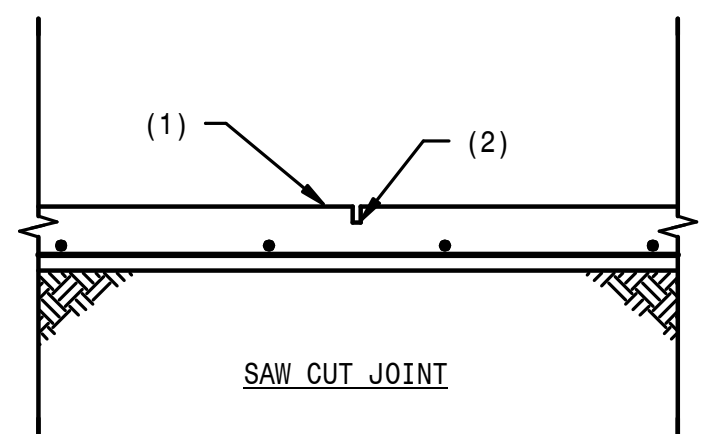


- NOTES:
1. ADD 'L #6 X 6'-0" LONG AT 12" OC AT THRESHOLDS.
 2. CONSTRUCTION JOINT.

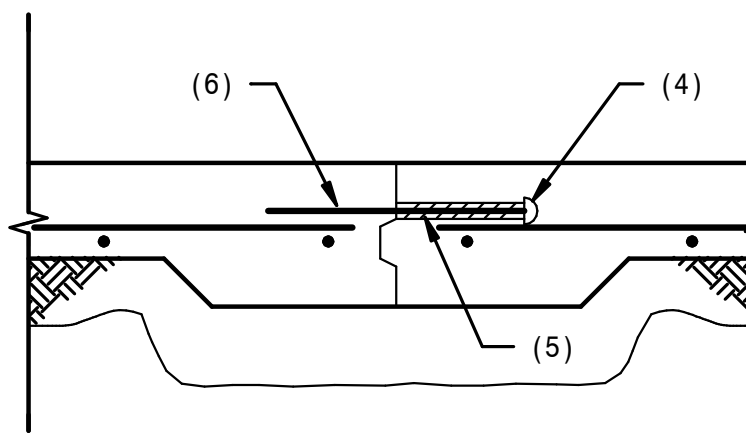
4 SLAB CONSTRUCTION JOINT

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

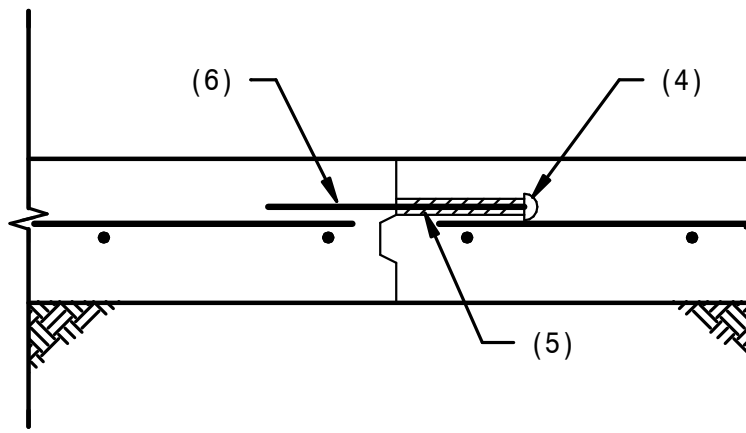
NO SCALE



SAW CUT JOINT



CONSTRUCTION JOINT



CONSTRUCTION JOINT AT APPARATUS BAY

- NOTES:
1. CONCRETE SLAB ON GRADE.
 2. SAWCUT- 1/8" WIDE X 1/4 SLAB THICKNESS IN DEPTH- CUT SHALL BE MADE SOON ENOUGH TO PREVENT SHRINKAGE CRACKING, BUT NOT SO SOON AS TO CAUSE SPALLING OF THE CONCRETE WHILE SAWING. WORK MUST BE ACCOMPLISHED WITHIN 12 HOURS OF CONCRETE PLACEMENT.
 3. NOT USED.
 4. 1 1/2" LONG MASTIC CAP INSTALL WITH 1/2" GAP AT END.
 5. POLYTHENE WRAP TO ASSURE SLIPPAGE.
 6. 3/4" Ø X 16" LONG SMOOTH DOWEL AT 12" OC- DOWELS MUST BE CAREFULLY ALIGNED AND SUPPORTED DURING CONCRETING OPERATIONS. MISALIGNED DOWELS CAUSE CRACKING.

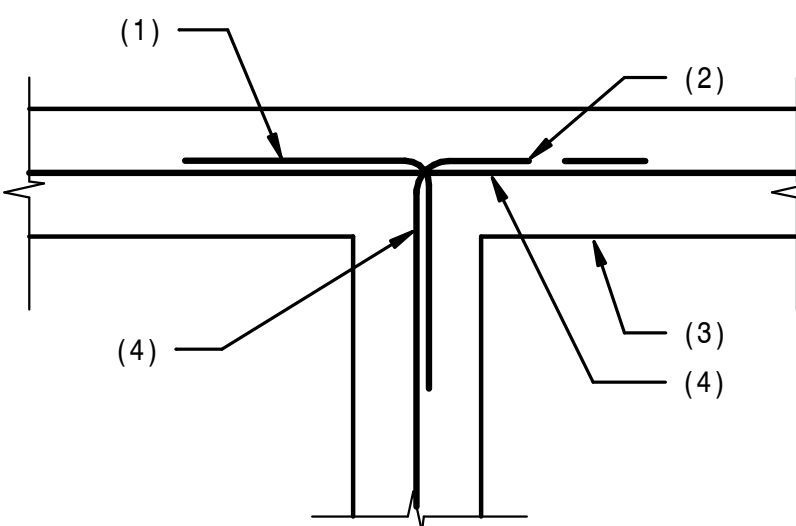
NOTE:

CONSTRUCTION JOINTS NEED ONLY OCCUR AT EXPOSED EDGES DURING PLACEMENT.

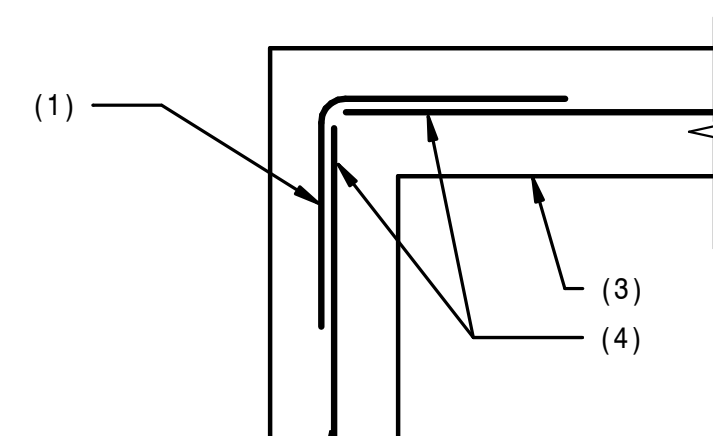
5 CONTROL JOINTS IN CONCRETE SLAB ON GRADE

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE



INTERSECTION



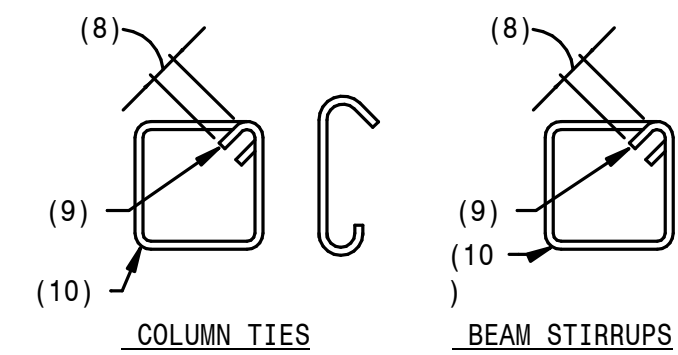
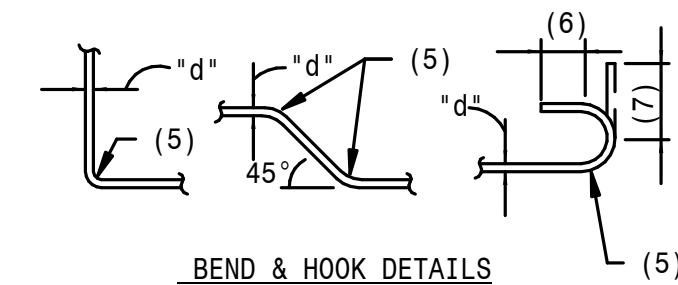
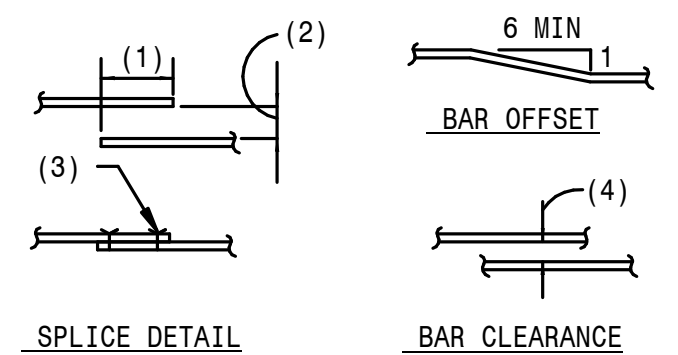
CORNER

- NOTES:
1. CORNER BARS SAME SIZE, QUANTITY AND SPACING AS HORIZONTAL REINFORCING. LAP PER G.S.N. (24" MINIMUM).
 2. ALTERNATE BEND.
 3. CONCRETE STEM WALL OR FOOTING.
 4. REINFORCING PER PLANS AND /OR G.S.N.

6 PLAN - CORNER REINFORCING IN CONCRETE FOOTING AND / OR STEM WALL

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

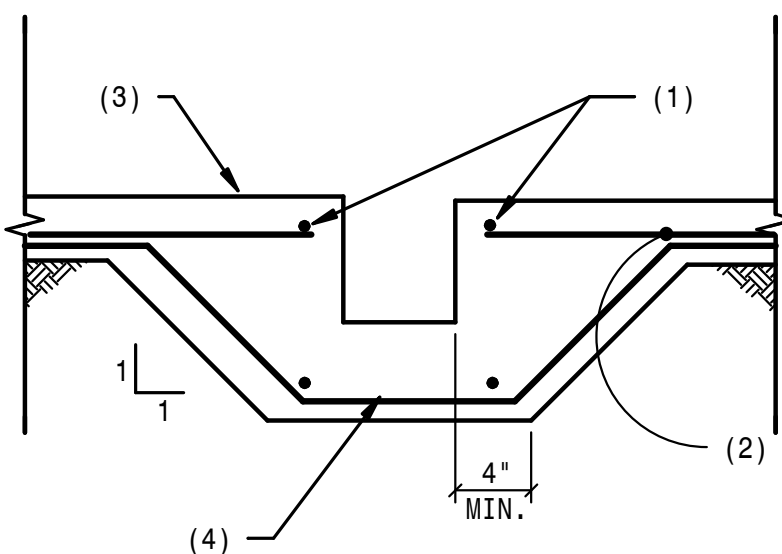
NO SCALE



1 TYPICAL CONCRETE REINFORCING BAR DETAILS

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE



NOTES:

1. (1) #3 CONT. (4 LOCS SHOWN)
2. LAP MIN 18".
3. CONCRETE SLAB.
4. CONT. REINF. - TYP. 24" 24"

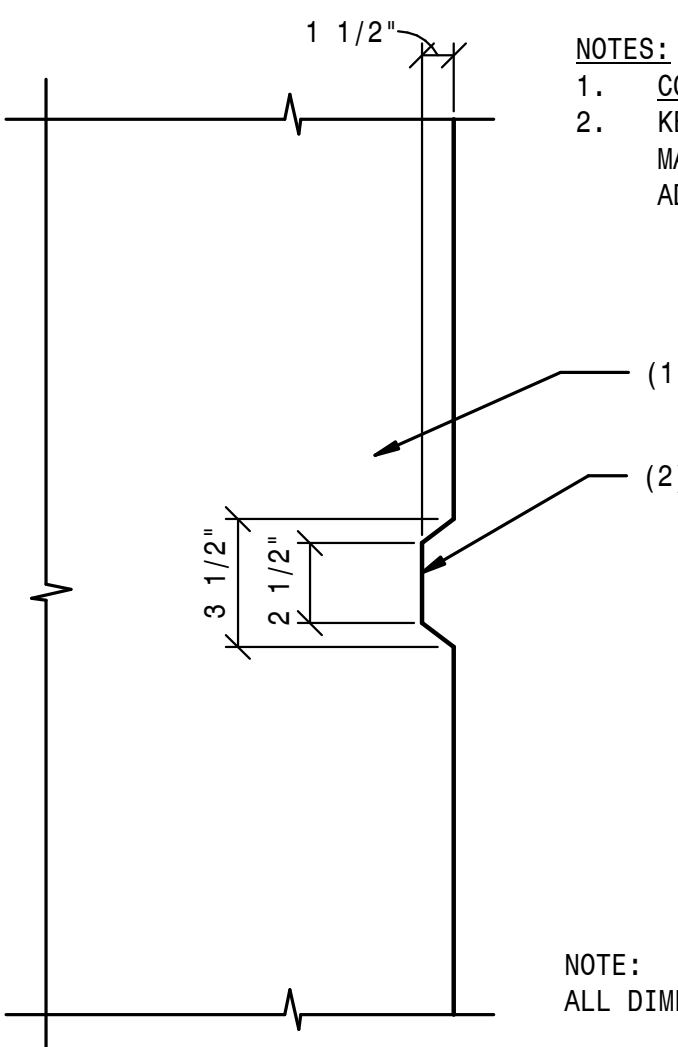
GRATING NOTES:

5. GRATING BY OHIO GRATINGS, INC. OR EQUAL. SUPPLIER TO DESIGN GRATING TO SUPPORT 2000# LIVE LOAD OVER 4 POINT LOADS. SUBMIT PRODUCT AND DESIGN DATA PRIOR TO CONSTRUCTION.
6. L2x2x3/16 MIN x CONT. W/ 3/8" DIA x 4" LONG HDAS AT 12" OC. (HDG) INCREASE ANGLE VERT LEG AS REQ'D TO MATCH GRATE THICKNESS.

2 CONCRETE SLAB AT TRENCH

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE



NOTE:
ALL DIMENSIONS ARE ± 1/2".

3 TYPICAL KEY IN CONCRETE

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE

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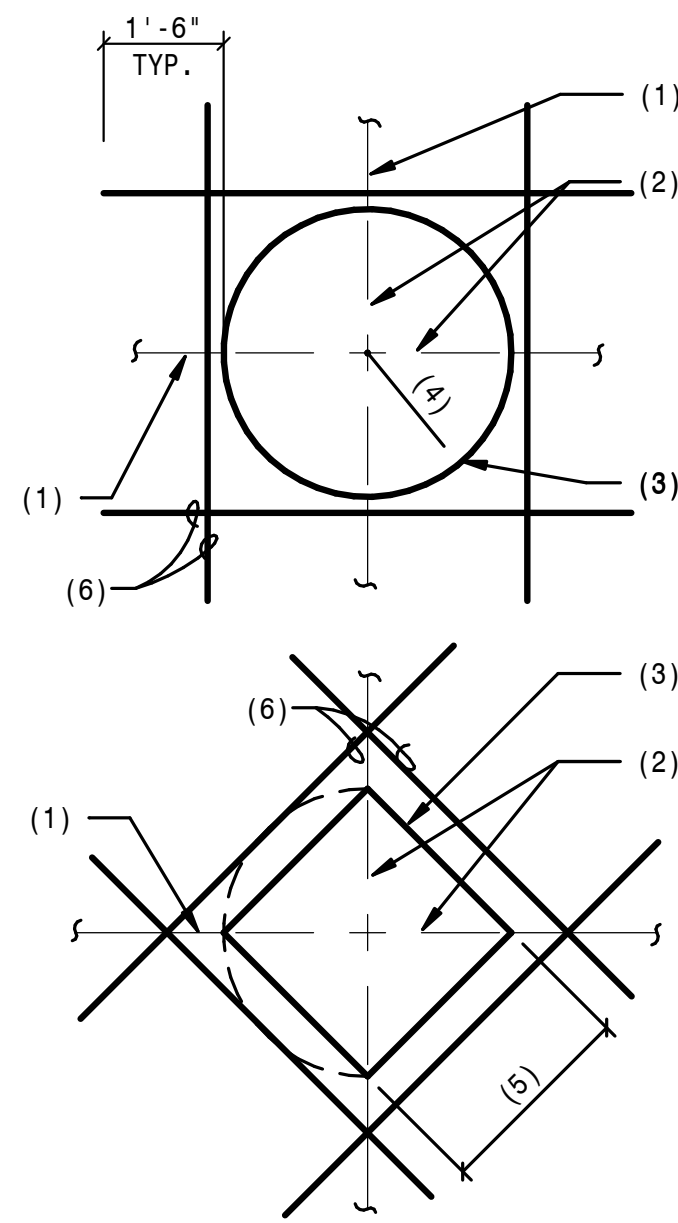
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CONCRETE DETAILS

S3.1



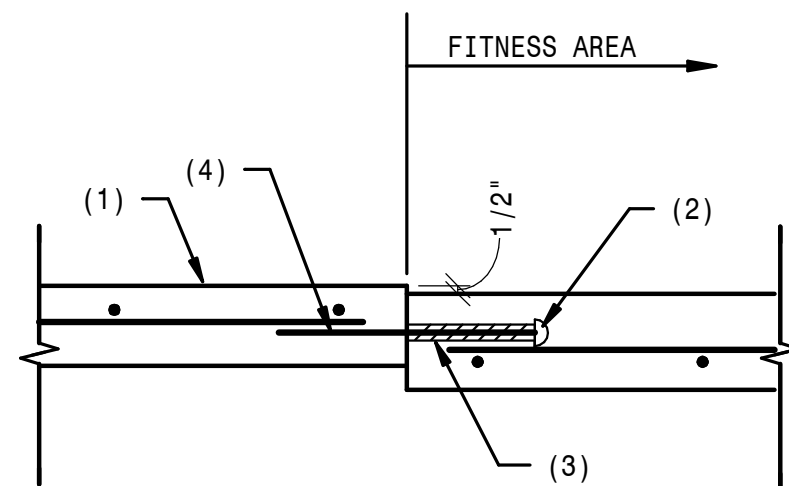
- NOTES:
1. *CONC. C.J.* WHERE SHOWN ON PLAN.
 2. CENTERLINE OF COLUMN.
 3. EXPANSION JOINT - SEE TYPICAL THICKENED SLAB AT CONCRETE CLOSURE POUR DETAIL.
 4. RADIUS 1'-6" MINIMUM / 3'-0" MAXIMUM.
 5. 2'-0" MINIMUM / 4'-0" MAXIMUM.
 6. (1) #5 TRIM BAR TYP.

NOTE:
FOR CLARITY, COLUMNS OMITTED. FOR CONFIGURATION OF SPECIFIC CLOSURE POURS, SEE PLAN.

7 TYPICAL COLUMN CLOSURE POUR AT CONCRETE SLAB ON GRADE

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE

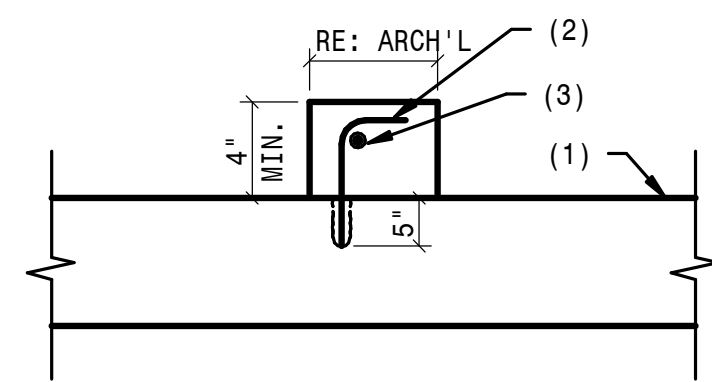


- NOTES:
1. CONCRETE SLAB ON GRADE PER PLAN.
 2. 1 1/2" LONG MASTIC CAP INSTALL WITH 1/2" GAP AT END.
 3. POLYTHENE WRAP TO ASSURE SLIPPAGE.
 4. 3/4" DIAx32" LONG SMOOTH DOWEL AT 8" OC - DOWELS MUST BE CAREFULLY ALIGNED AND SUPPORTED DURING CONCRETING OPERATIONS. MISALIGNED DOWELS CAUSE CRACKING.

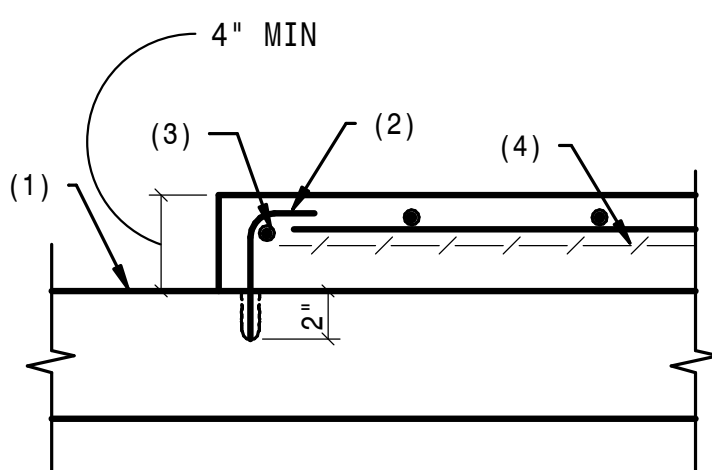
ISOLATION JOINT

8 ISOLATION JOINT AT FITNESS ROOM

NO SCALE



CURB



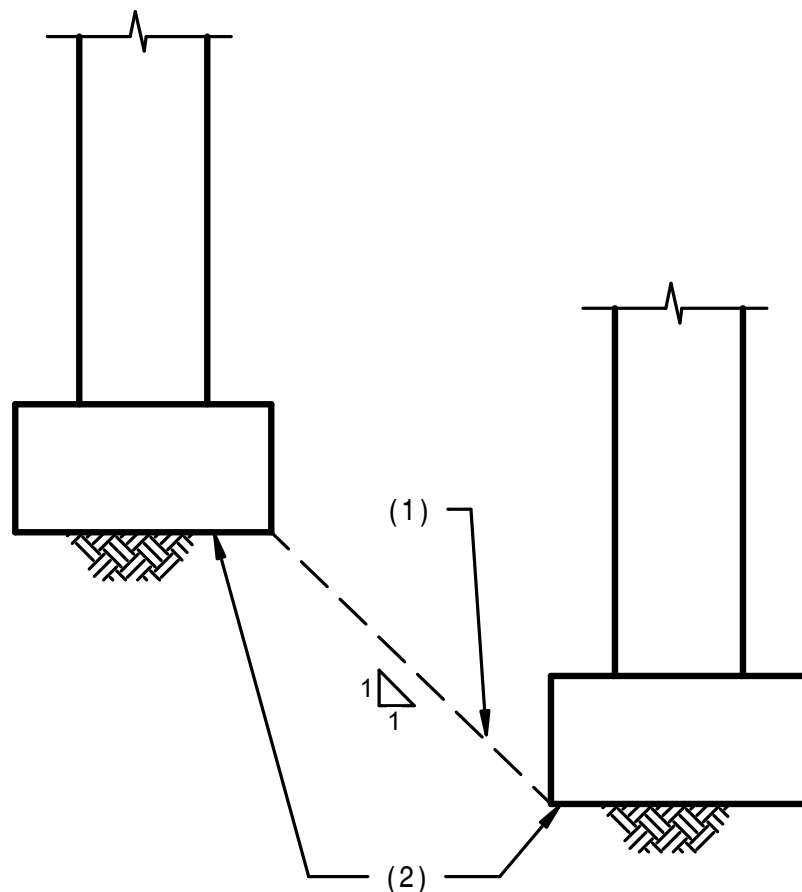
EQUIPMENT PAD

9 CURB AND EQUIPMENT PAD DETAIL

NO SCALE

- NOTES:
1. CONC. SLAB ON GRADE.
 2. #4x 8" AT 12" OC - INSTALL W/ STRUCTURAL ADHESIVE PER GSN OR PRIOR TO PLACING CONCRETE SLAB.
 3. #4XCONT.
 4. #4 AT 12 OC EA WAY TOP.

RE: PLANS FOR ISOLATED PAD PER: 6/S3.2



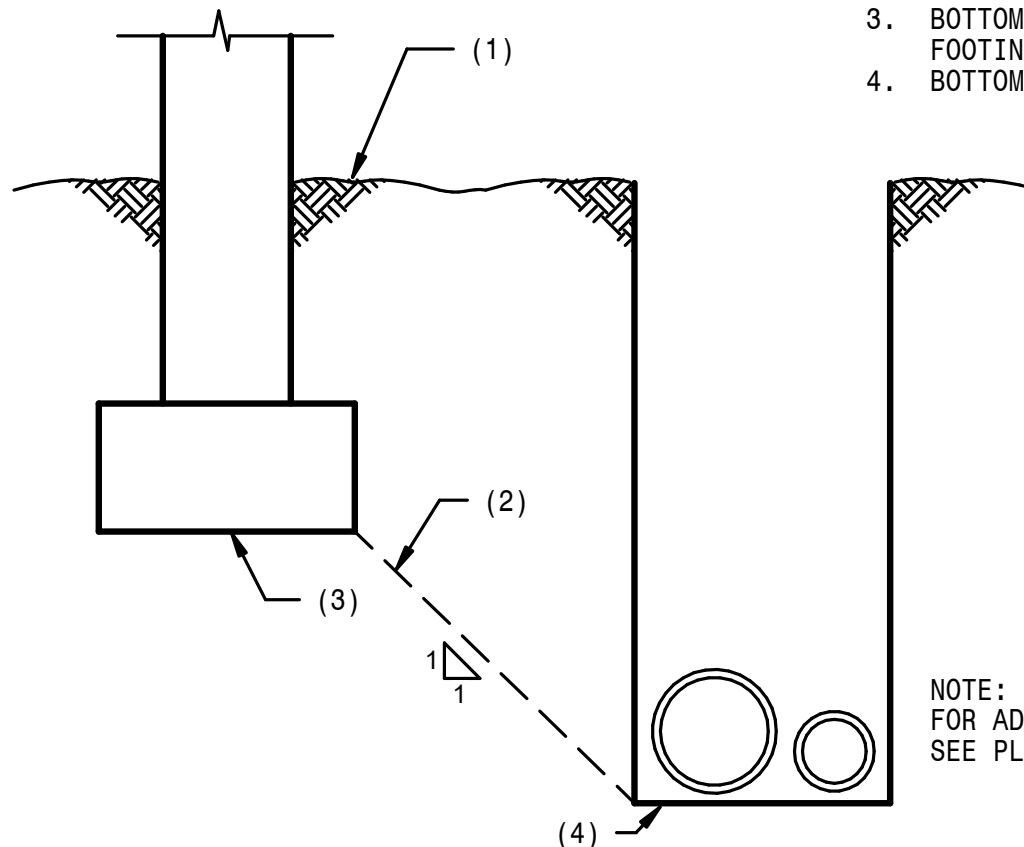
- NOTES:
1. MAXIMUM SLOPE BETWEEN BOTTOM OF FOOTINGS SHALL BE 45 DEGREES. STEP FOOTINGS AS REQUIRED - SEE TYPICAL DETAIL.
 2. BOTTOM OF CONCRETE FOOTING.

NOTE:
FOR ADDITIONAL INFORMATION, SEE PLANS AND DETAILS.

4 MAXIMUM SLOPE BETWEEN ADJACENT FOOTING

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE



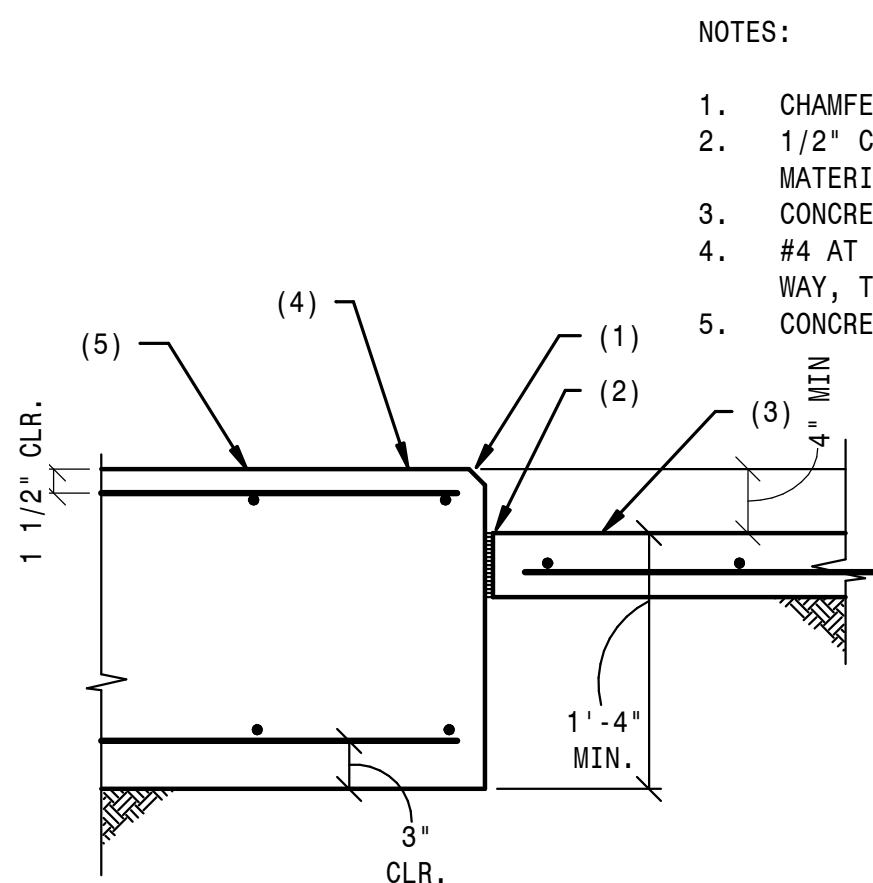
- NOTES:
1. FINISHED GRADE WHERE OCCURS.
 2. DO NOT EXCAVATE A TRENCH CLOSER THAN A 45 DEGREE ANGLE BELOW BOTTOM OF FOOTING OR FOUNDATION.
 3. BOTTOM OF CONCRETE FOOTING.
 4. BOTTOM OF TRENCH.

NOTE:
FOR ADDITIONAL INFORMATION, SEE PLANS AND DETAILS.

5 TRENCH PARALLEL TO FOUNDATION

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE



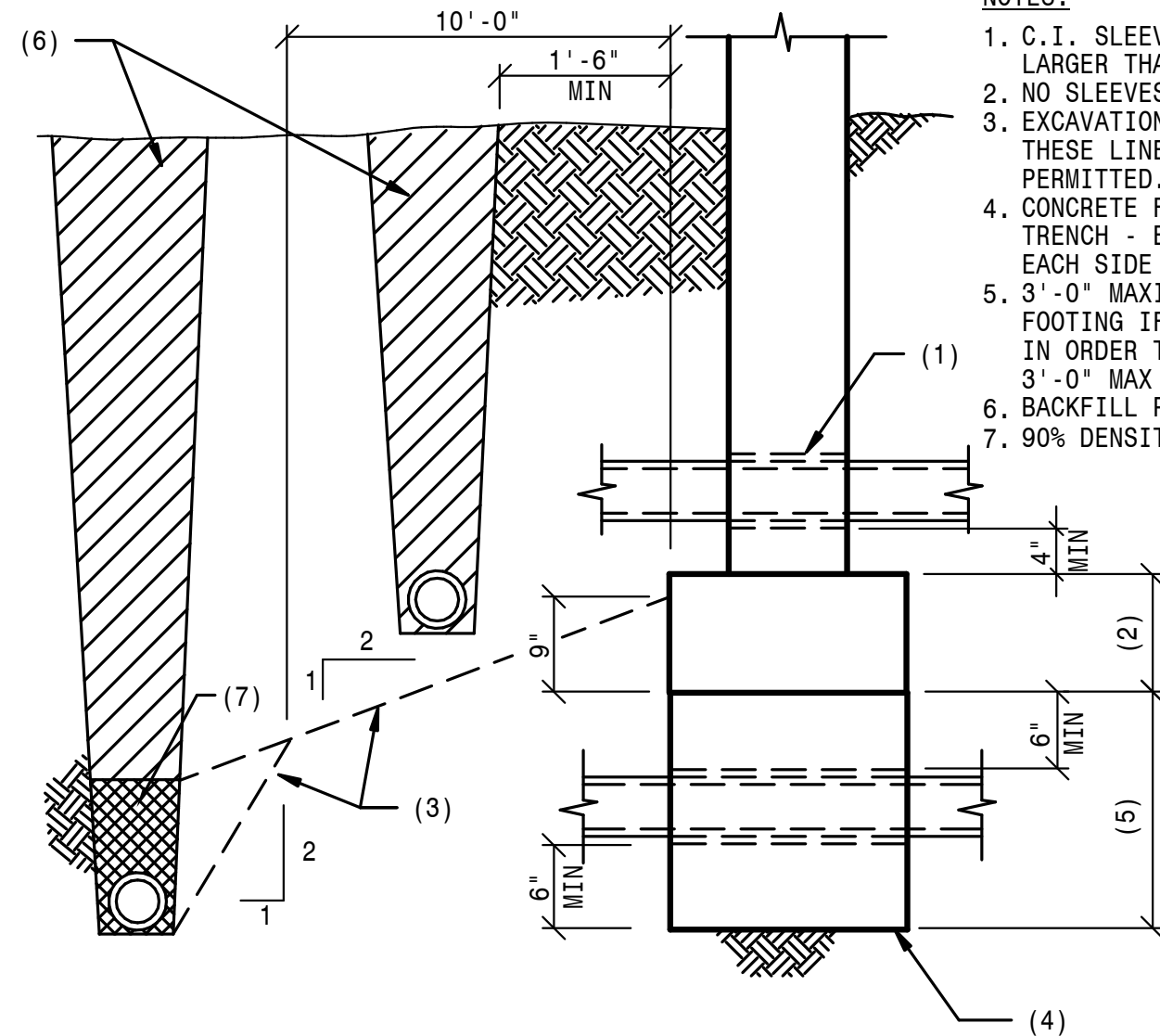
- NOTES:
1. CHAMFER EDGES.
 2. 1/2" COMPRESSIBLE MATERIAL JOINT FILLER.
 3. CONCRETE SLAB ON GRADE.
 4. #4 AT 12" O.C. EACH WAY, TOP AND BOTTOM.
 5. CONCRETE EQUIPMENT PAD.

12" AT EXTRACTOR COORDINATE REQUIREMENTS WITH EQUIPMENT. - TYP.

6 CONCRETE EQUIPMENT PAD ON GRADE

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE



- NOTES:
1. C.I. SLEEVE 2" LARGER THAN PIPE.
 2. NO SLEEVES PERMITTED.
 3. EXCAVATION BELOW THESE LINES NOT PERMITTED.
 4. CONCRETE FILL PIPE TRENCH - EXTEND 2'-0" EACH SIDE OF SLEEVE.
 5. 3'-0" MAXIMUM - STEP FOOTING IF NECESSARY IN ORDER TO MAINTAIN 3'-0" MAX HEIGHT.
 6. BACKFILL PER SPECS.
 7. 90% DENSITY (ASTM D-698).

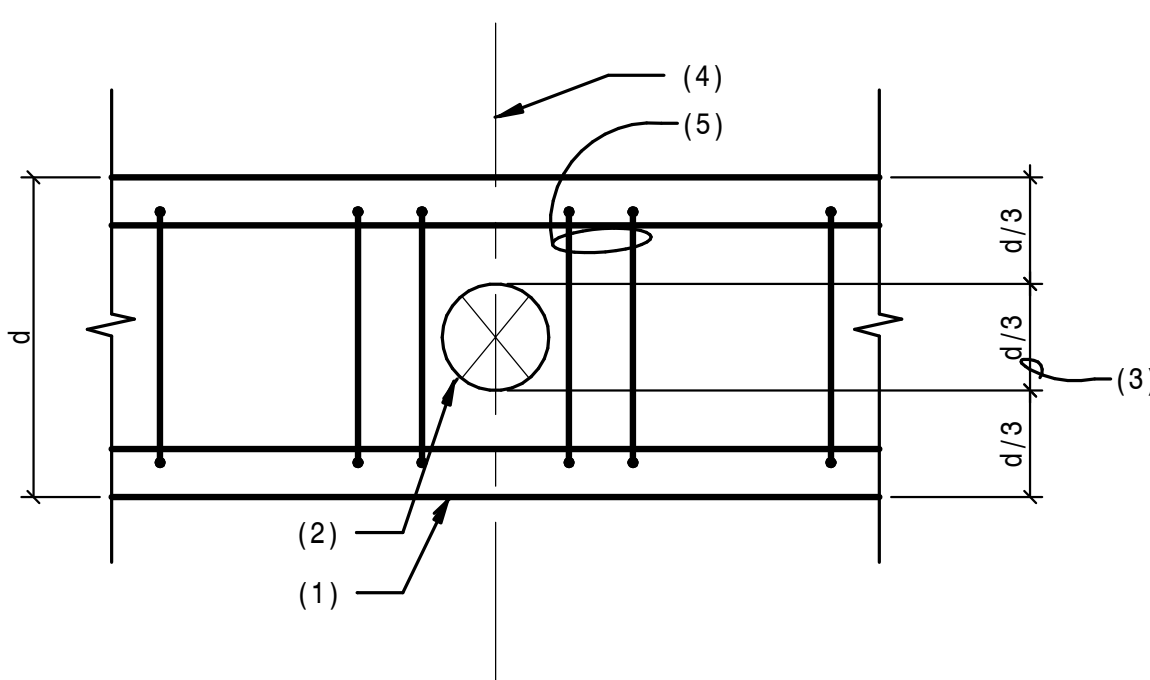
NOTE:
PIPES LOCATED GREATER THAN 3'-0" BELOW FOOTING BASE MAY BE BACKFILLED WITH COMPACTED FILL PER SPECIFICATIONS

NOTE:
NO PIPE SHALL PASS THROUGH FOOTINGS OR UNDER COLUMN FOOTINGS.

1 PIPE THROUGH FOOTING AND TRENCH

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE

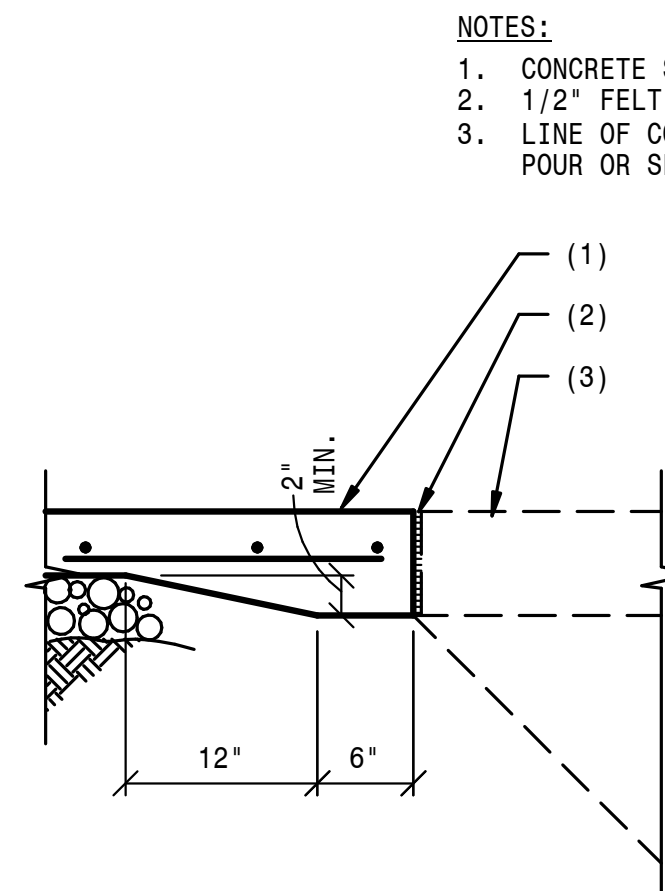


- NOTES:
1. CONC. GRADE BEAM.
 2. MAXIMUM 12" SQ. OR DIAMETER OPENING.
 3. OPENING TO BE LOCATED IN MIDDLE OF 1/3 DEPTH OF BEAM.
 4. LOCATE OPENING IN MIDDLE 1/3 OF BEAM SPAN.
 5. PROVIDE 2 STIRRUPS OR VERT BARS AT 3" EACH SIDE OF OPENING.

2 OPENING IN CONCRETE STEM WALL

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE



- NOTES:
1. CONCRETE SLAB ON GRADE.
 2. 1/2" FELT EXPANSION JOINT.
 3. LINE OF CONCRETE CLOSURE POUR OR SLAB AS OCCURS.

3 TYPICAL THICKENED SLAB AT CONC. CLOSURE POUR

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE

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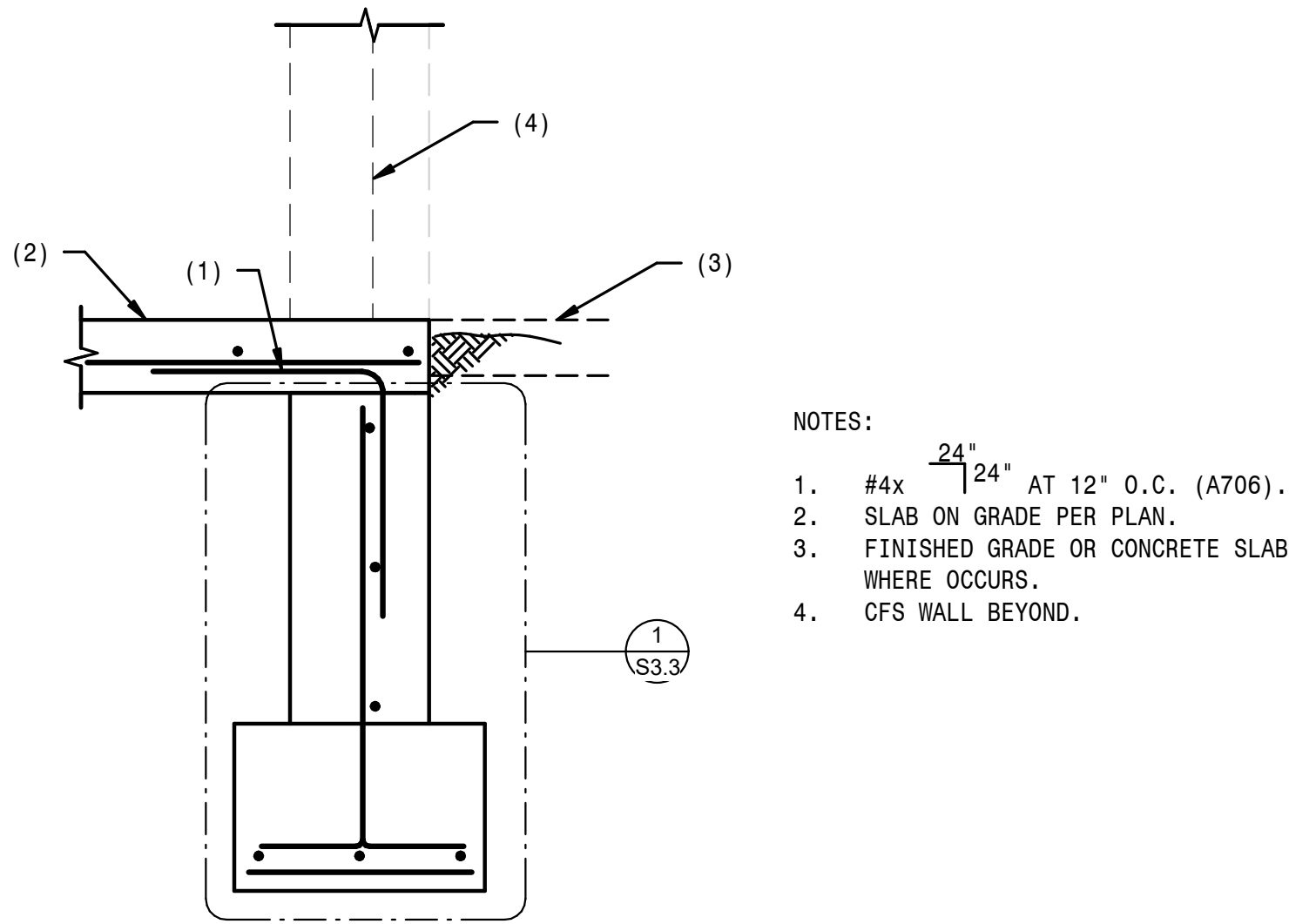
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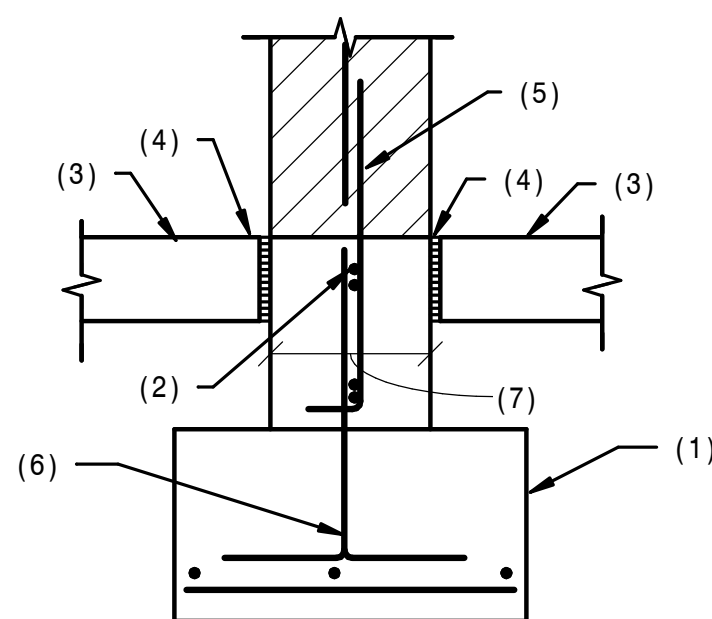
FOUNDATION DETAILS

S3.2



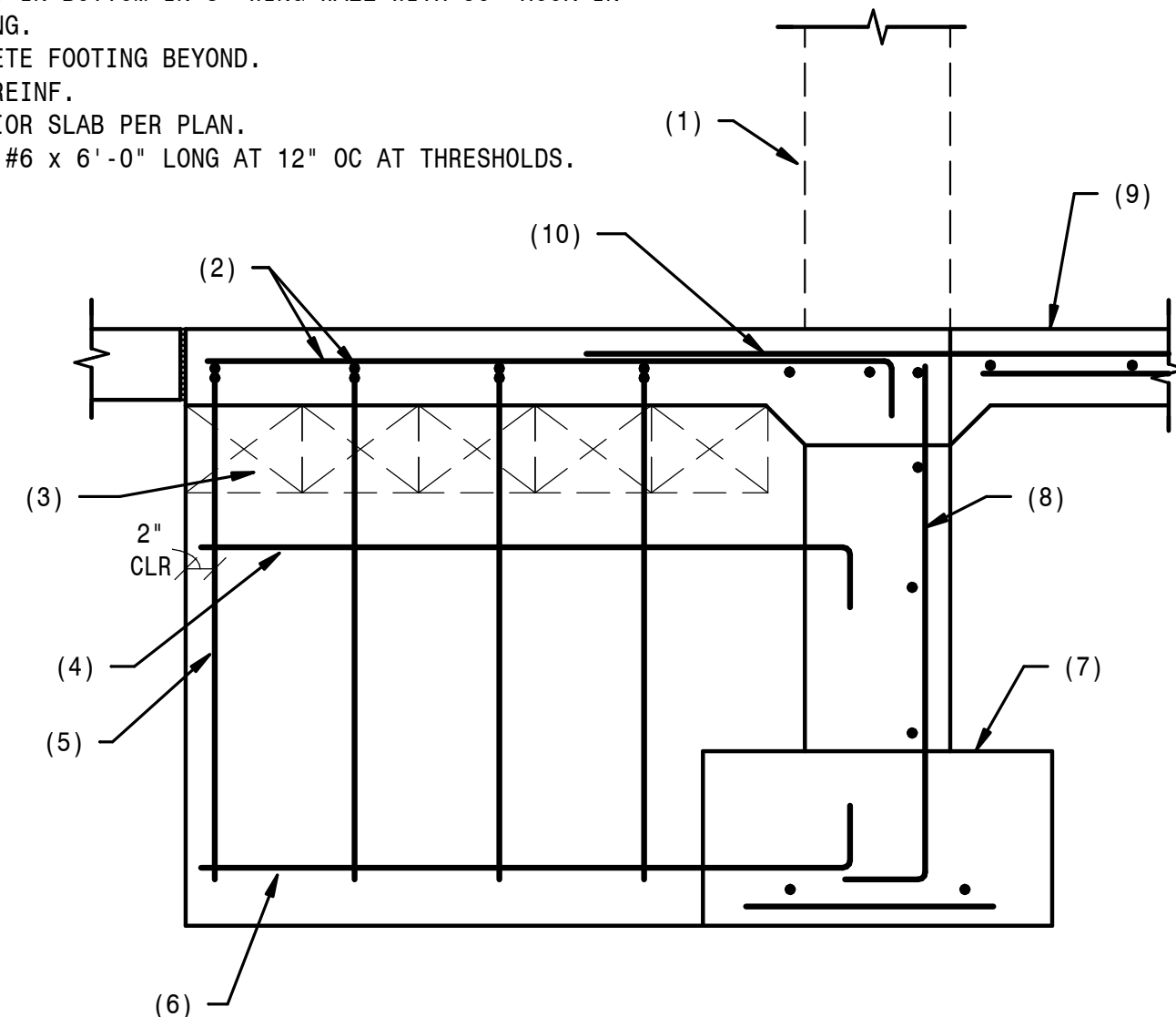
7 THRESHOLD AT METAL STUD WALL NO SCALE

- NOTES:
- FOOTING PER SCHEDULE.
 - (2) #5 x CONT TOP AND BOT.
 - SLAB PER PLANS.
 - 1/2" EXPANSION JOINT MAT'L.
 - #5 DOWELS TO MATCH WALL REINF.
 - #4 HOOKED DOWELS AT 16" OC.
 - 10" WIDE STEM WALL WHERE CMU WALL ABOVE IS 10", 8" WIDE STEM WALL WHERE CMU WALL ABOVE IS 8", 6" WIDE STEM WALL WHERE CMU WALL ABOVE IS 6"



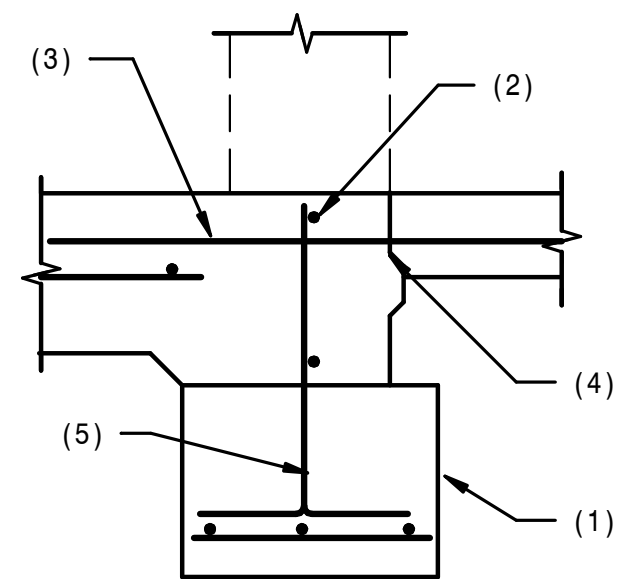
8 INTERIOR STEM WALL AT MASONRY NO SCALE

- NOTES:
- WALL BEYOND.
 - #5 AT 8" OC EACH WAY AT MID DEPTH.
 - 4" VOID FORM UNDER POUR OVER BETWEEN WING WALLS.
 - #5 AT MID HEIGHT OF 8" WING WALL WITH 90° HOOK IN FOOTING.
 - #5 AT 8" VERT IN WING WALL.
 - (2) #5 IN BOTTOM IN 8" WING WALL WITH 90° HOOK IN FOOTING.
 - CONCRETE FOOTING BEYOND.
 - WALL REINF.
 - INTERIOR SLAB PER PLAN.
 - ADD'L #6 x 6'-0" LONG AT 12" OC AT THRESHOLDS.



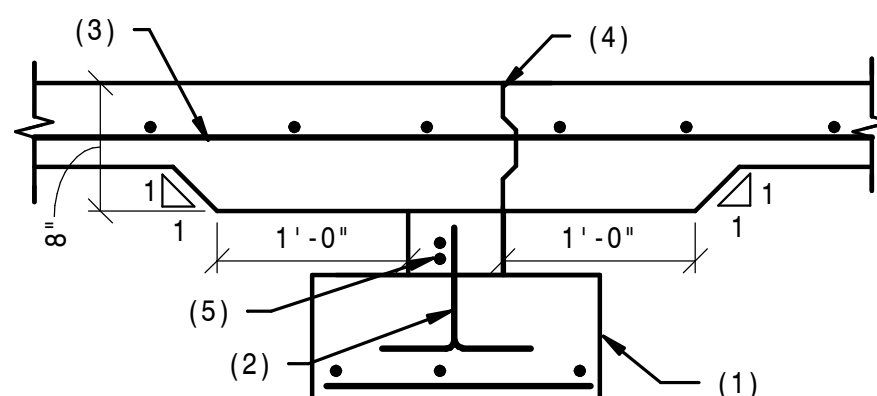
9 STRUCTURAL STOOP NO SCALE

- NOTES:
- FOOTING PER SCHEDULE.
 - (2) #5 x CONT.
 - ADD'L #6 x 6'-0" LONG AT 12" OC AT THRESHOLDS.
 - OPTIONAL CONSTR JOINT.
 - #4 HOOKED DOWELS AT 16" OC.



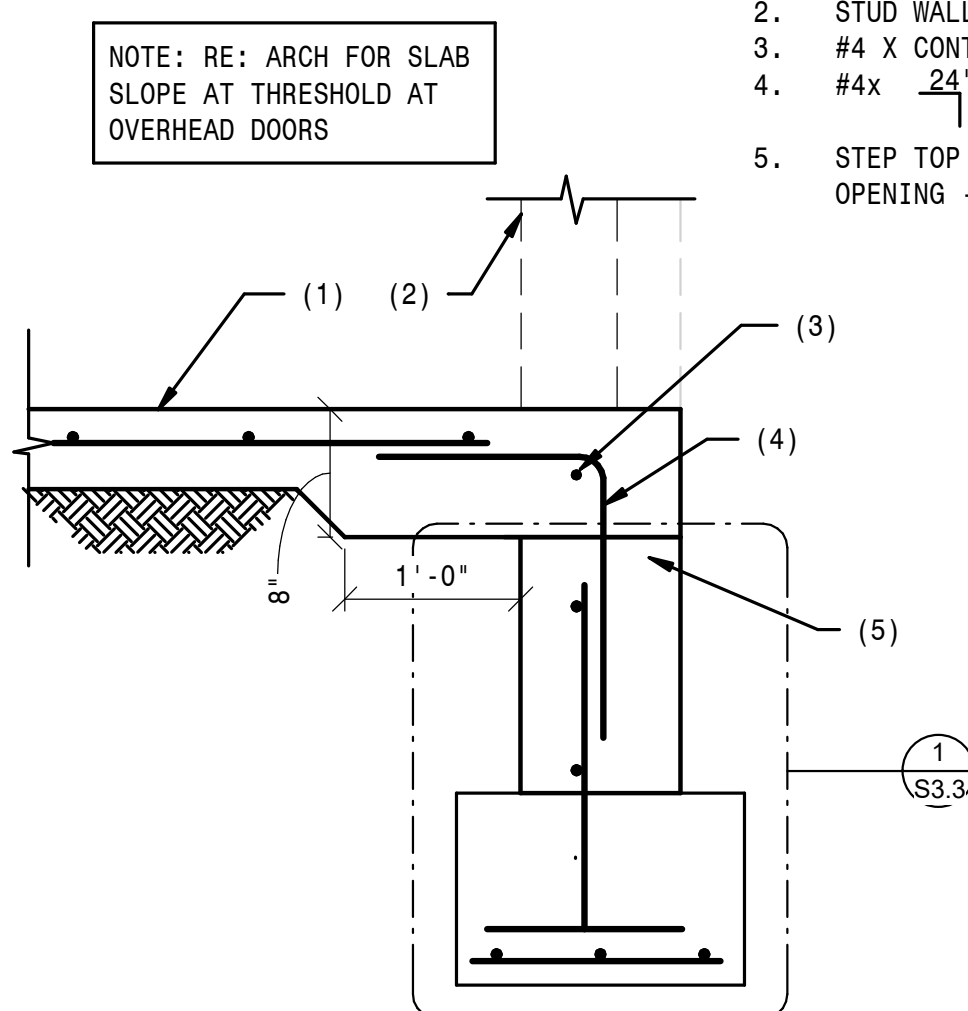
4 APPARATUS BAY SLAB POUR OVER AT INTERIOR OPNG NO SCALE

- NOTES:
- FOOTING PER SCHEDULE.
 - #4 HOOKED DOWELS AT 16" OC.
 - ADD'L #5 x 6'-0" LONG AT 12" OC.
 - OPTIONAL CONSTR JOINT.
 - (2) #5 x CONT.

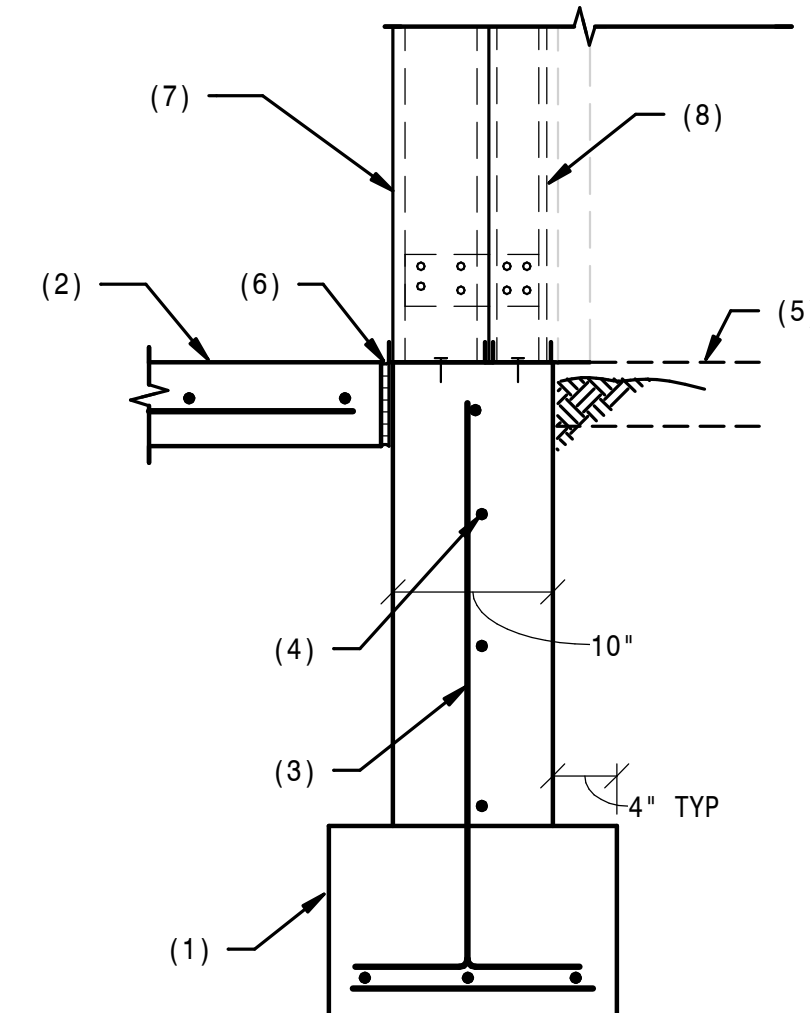


5 STEM WALL AT OPNG POUR OVER NO SCALE

- NOTES:
- CONCRETE SLAB ON GRADE PER PLAN.
 - STUD WALL OR MAS WALL BEYOND.
 - #4 x CONT ACROSS OPNG.
 - #4x $\frac{24}{16}$ AT 12" O.C. (A706).
 - STEP TOP OF FTG DOWN AT OPENING - TYP.

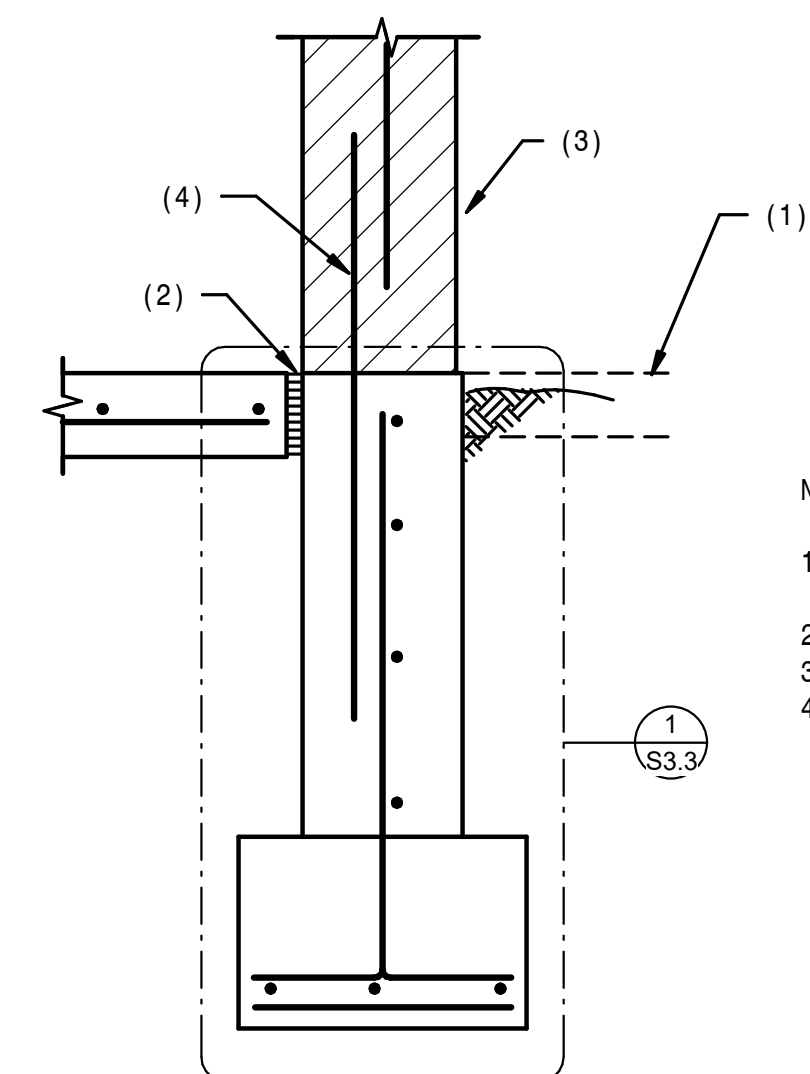


6 POUR OVER AT OPENING OR STOREFRONT NO SCALE



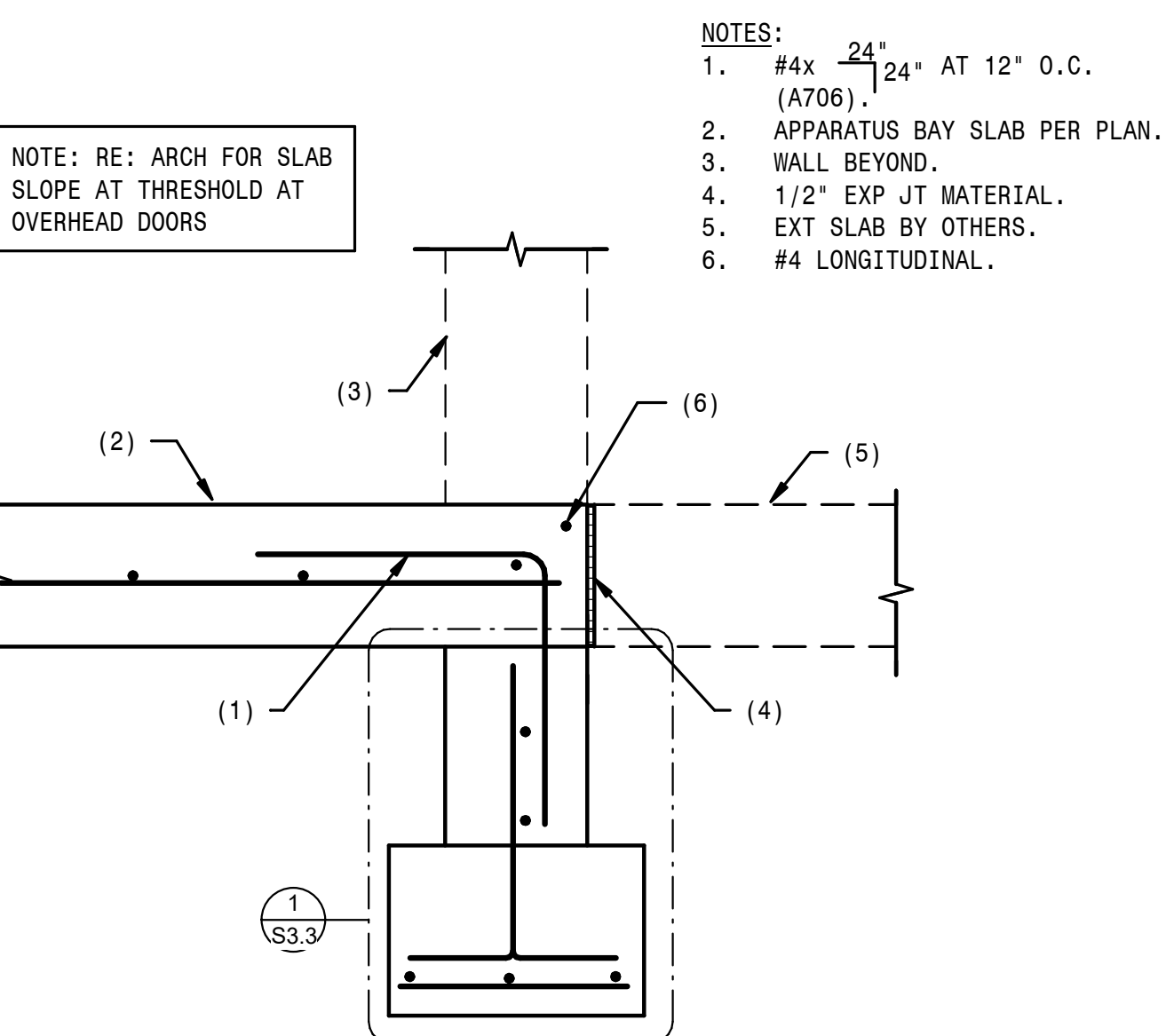
1 STEM WALL AT CFS WALL NO SCALE

- NOTES:
- FTG PER PLAN.
 - SLAB ON GRADE PER PLAN.
 - #4 HOOKED DOWELS AT 16" OC.
 - #4 x CONT T&B AND AT 12" OC BTWN.
 - FINISHED GRADE OR CONCRETE SLAB WHERE OCCURS.
 - 1/2 EXP JOINT MAT'L.
 - CFS WALL - RE: PLAN.
 - 362S162-54 AT 16" OC W/ 362T125-54 TOP & BOT AS OCCURS PER PLAN. TIEBACK WALL TOP AND BOT AND 4'-0" OC BTWN W/ 362S162-54 LAPPED W/ (4) SCREWS TO EA STUD. (TOP OF WALL STUDS TO BE 1" BELOW BOT OF BEAM).



2 STEM WALL AT CMU WALL NO SCALE

- NOTES:
- FINISHED GRADE OR CONCRETE SLAB WHERE OCCURS.
 - 1/2 EXP. JOINT MAT'L.
 - MASONRY WALL - RE: PLAN.
 - DOWELS TO MATCH SIZE AND SPACING OF MW REINFORCEMENT.



3 APPARATUS BAY THRESHOLD NO SCALE

- NOTES:
- #4x $\frac{24}{16}$ AT 12" O.C. (A706).
 - APPARATUS BAY SLAB PER PLAN.
 - WALL BEYOND.
 - 1/2" EXP JT MATERIAL.
 - EXT SLAB BY OTHERS.
 - #4 LONGITUDINAL.

NOTE: RE: ARCH FOR SLAB SLOPE AT THRESHOLD AT OVERHEAD DOORS

NOTE: RE: ARCH FOR SLAB SLOPE AT THRESHOLD AT OVERHEAD DOORS

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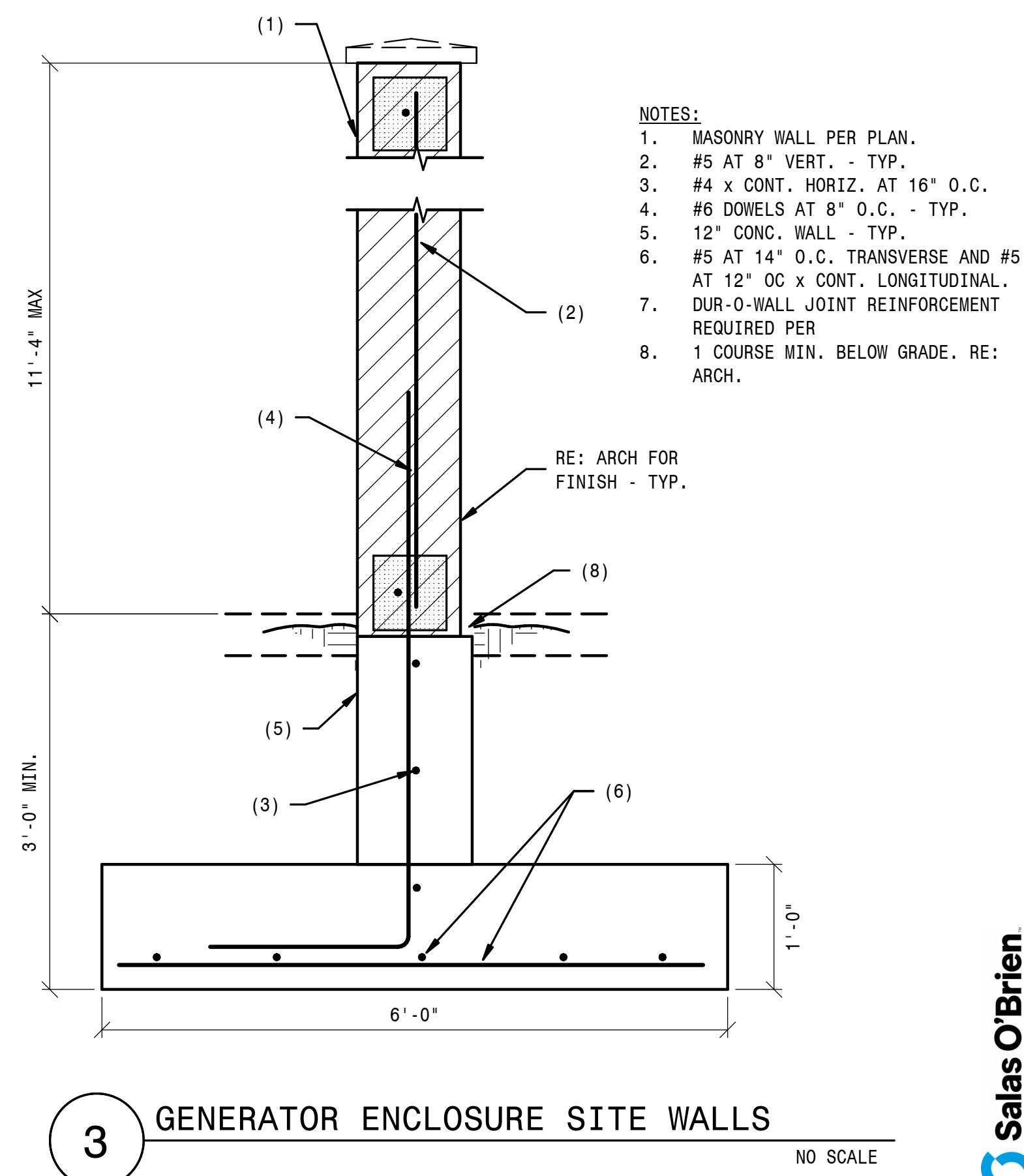
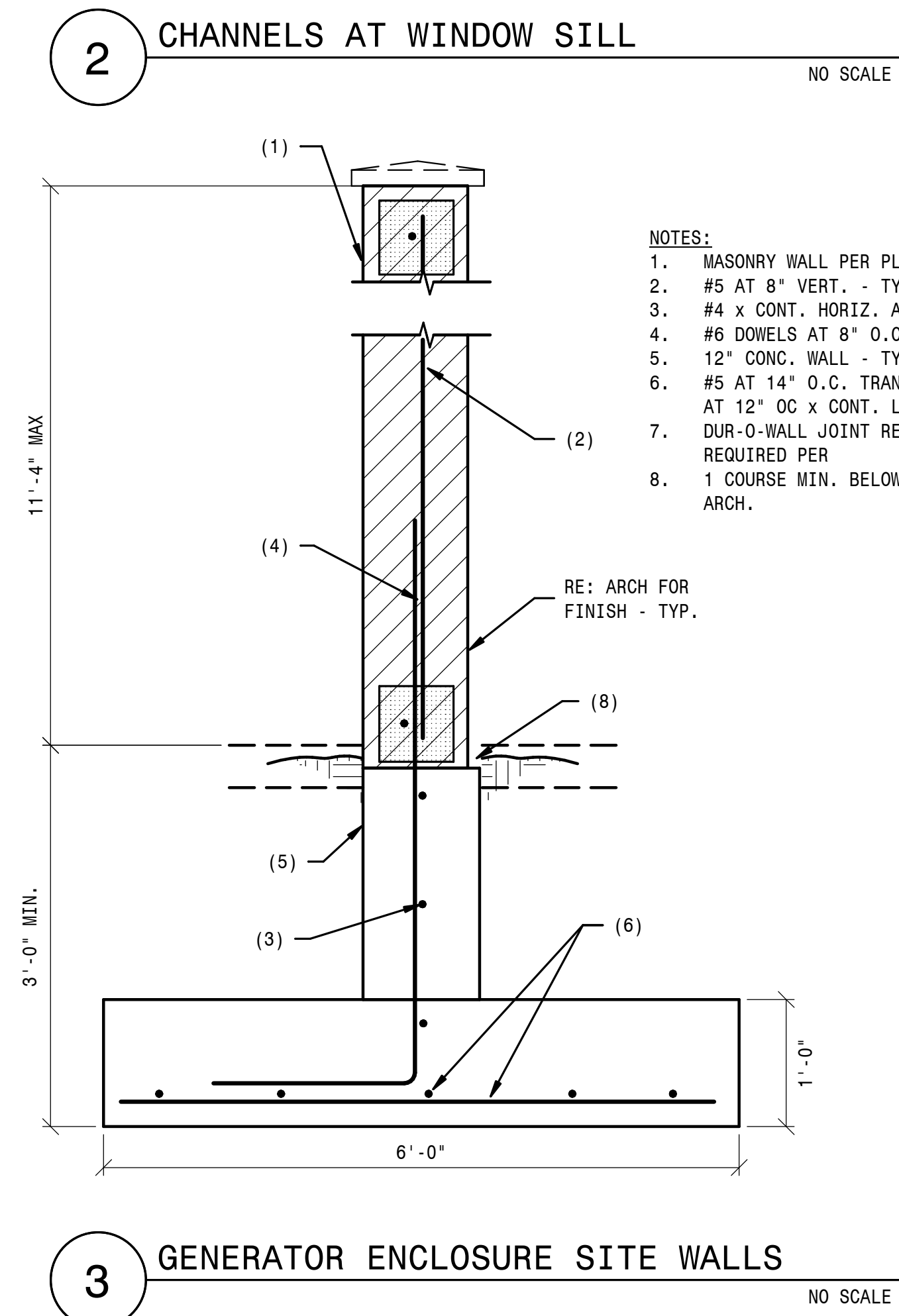
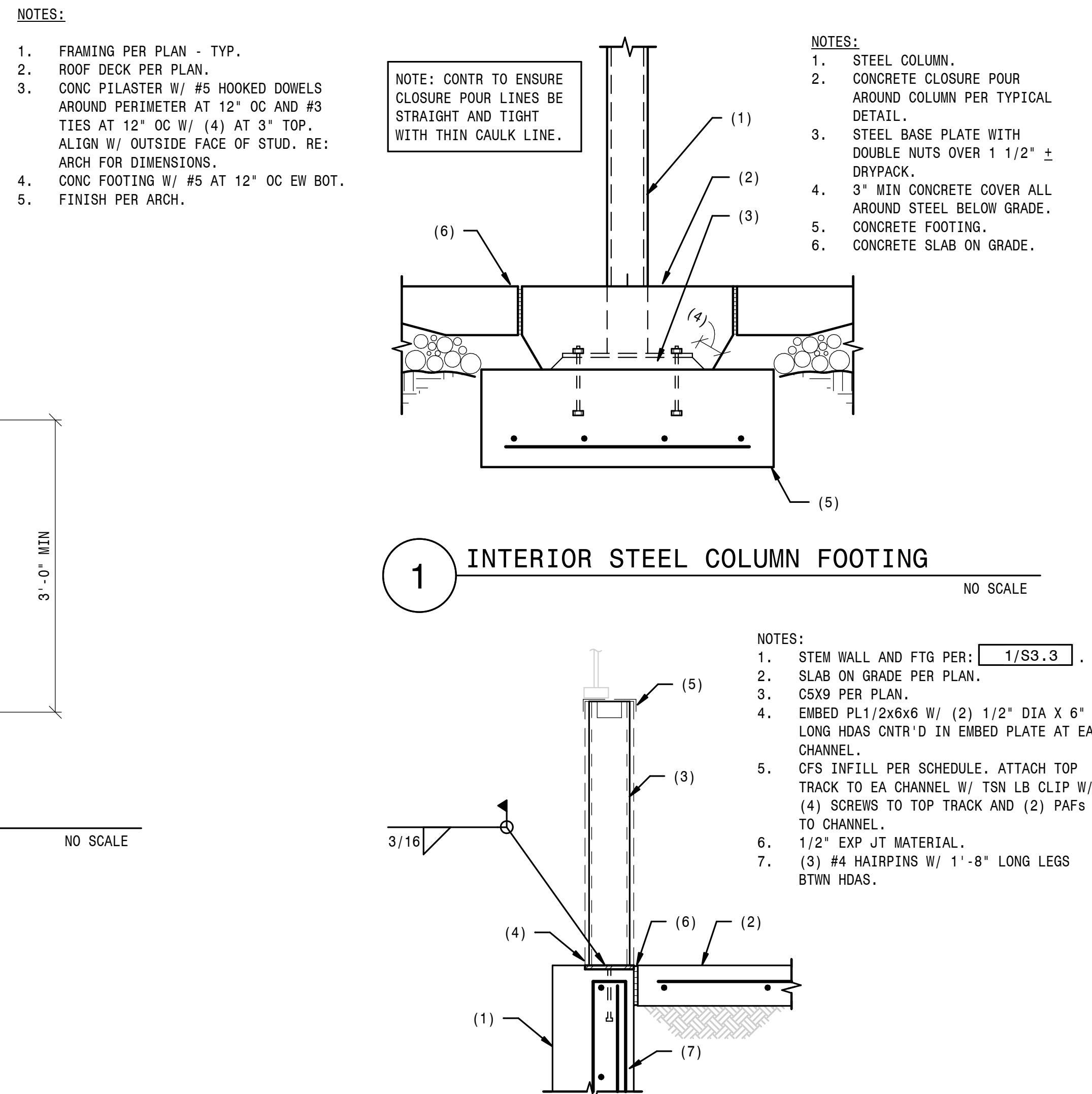
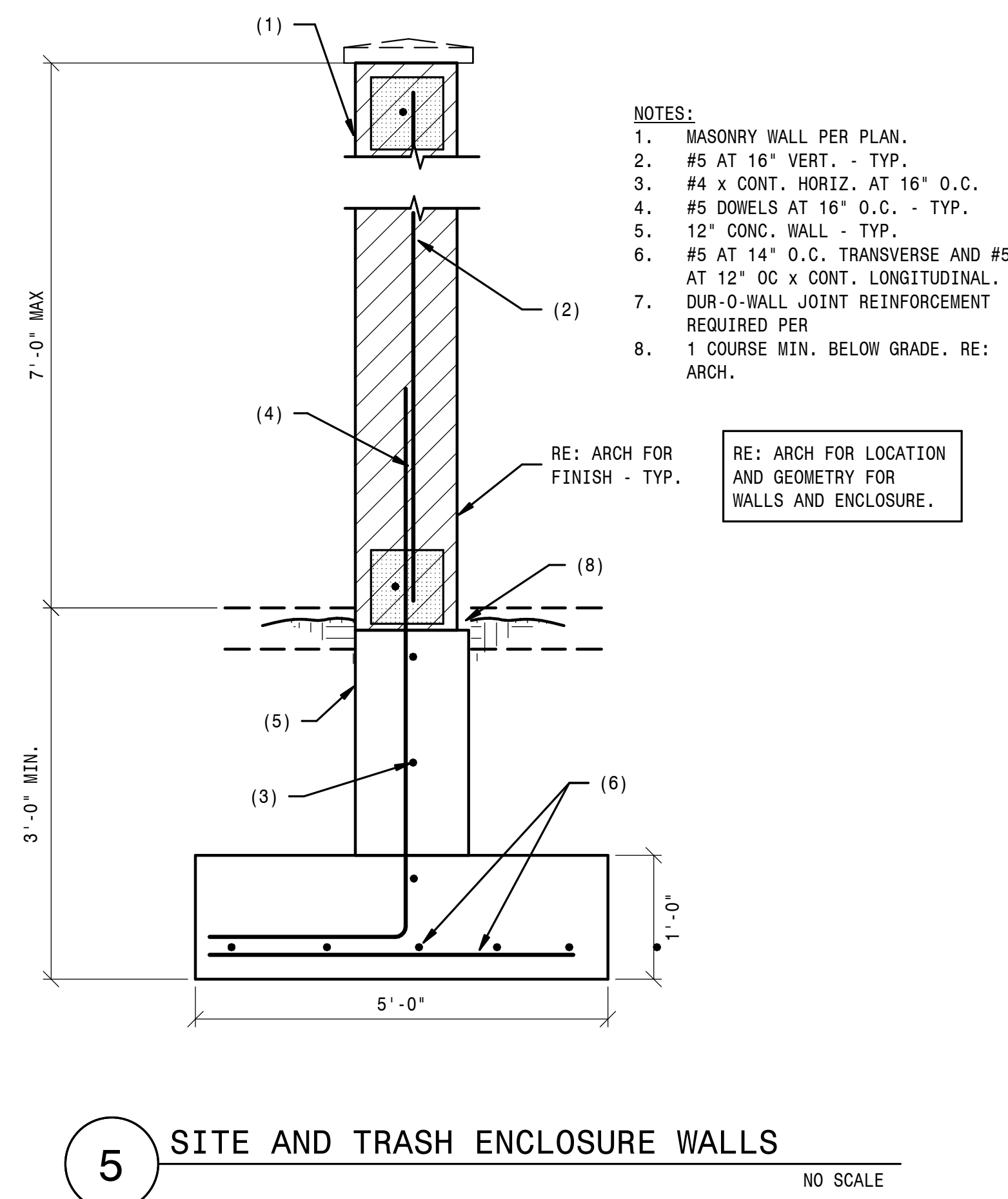
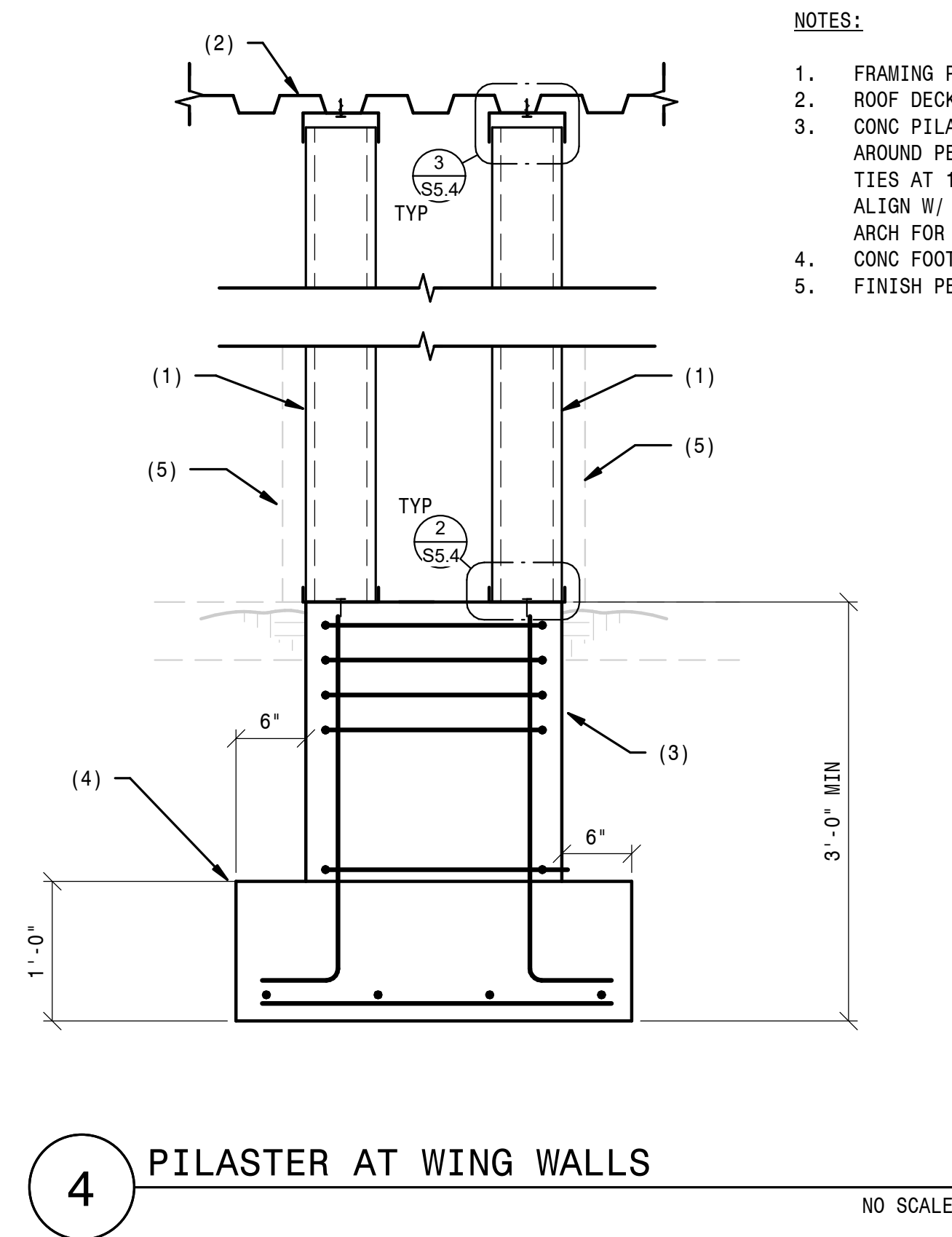
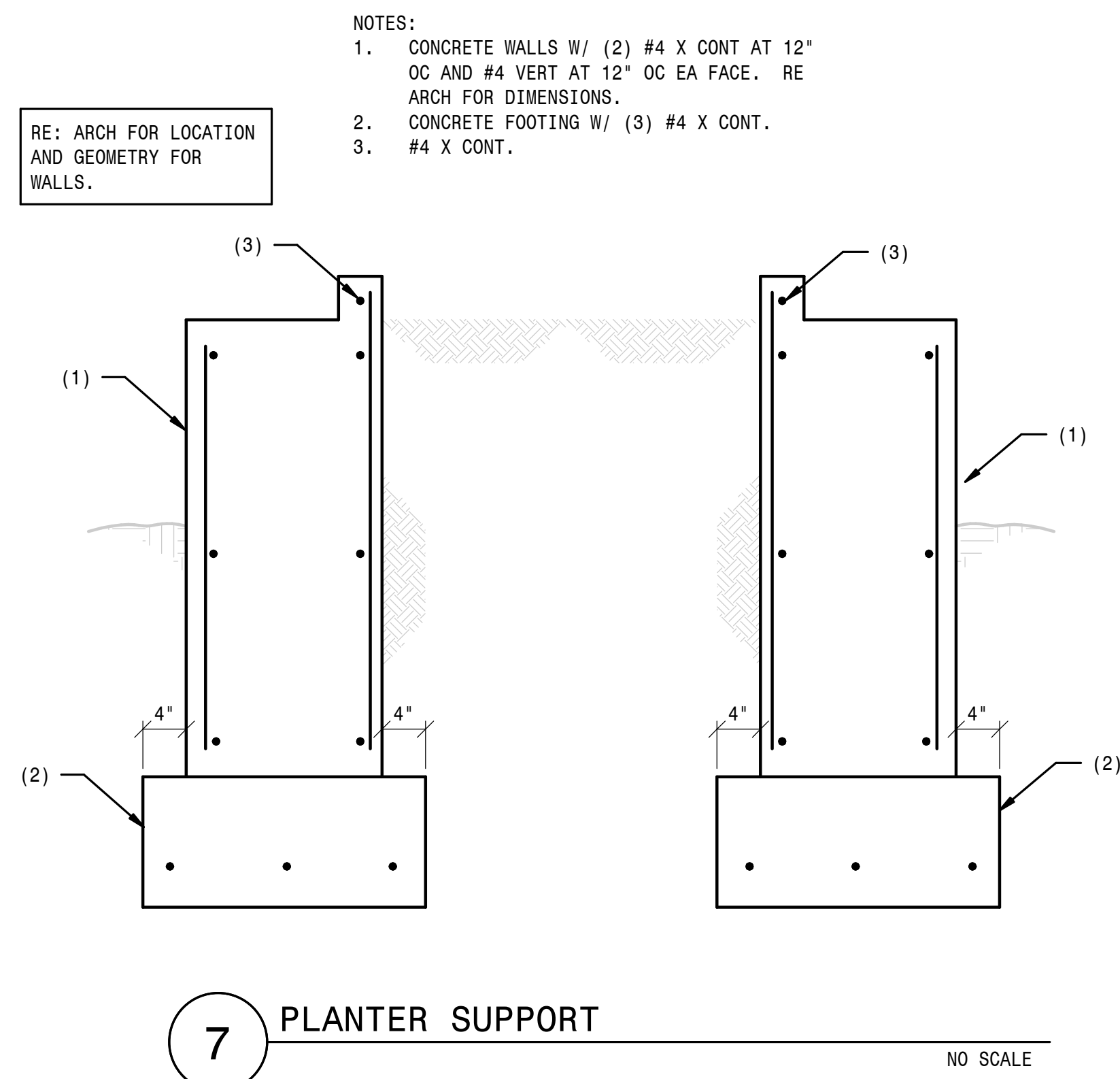
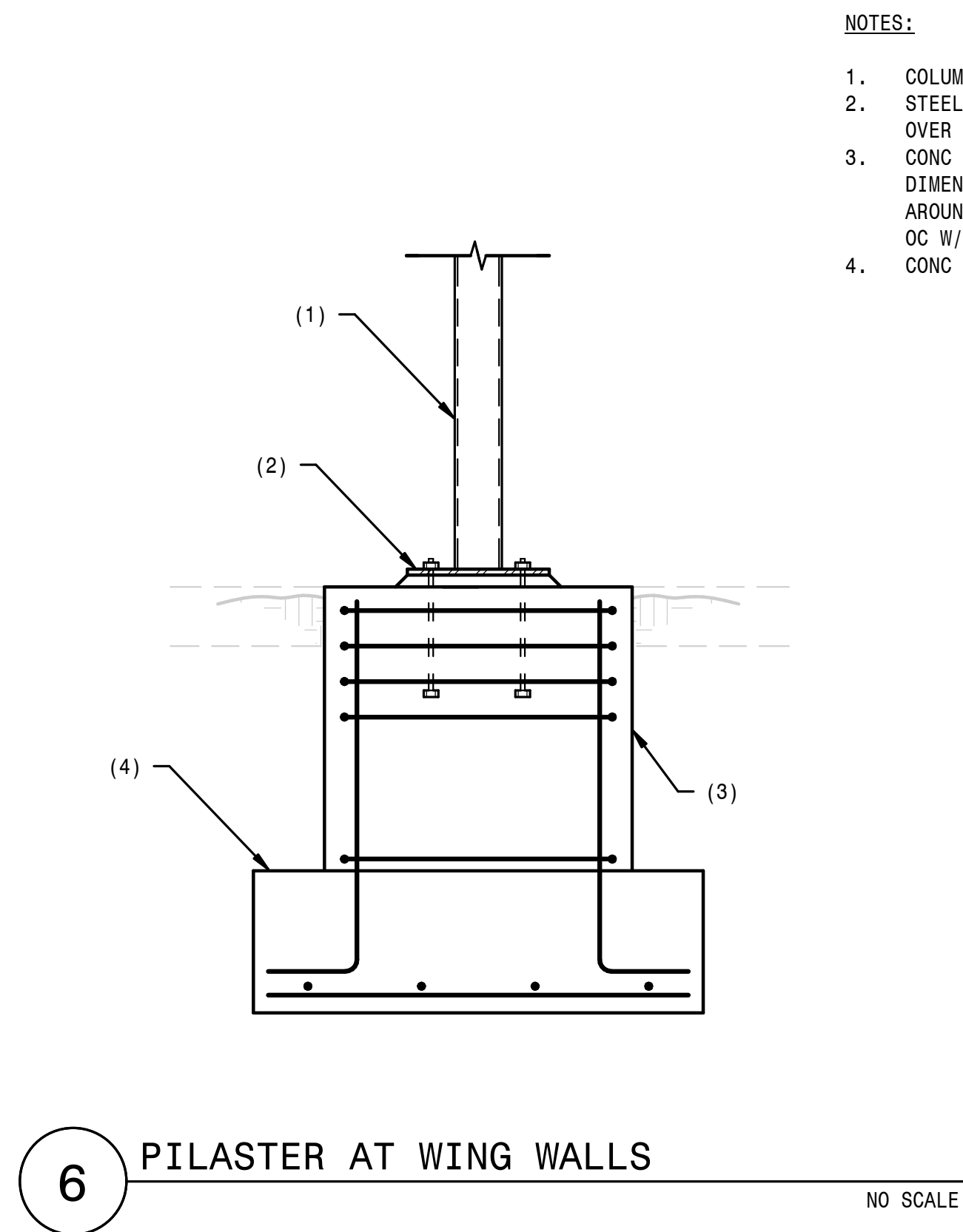
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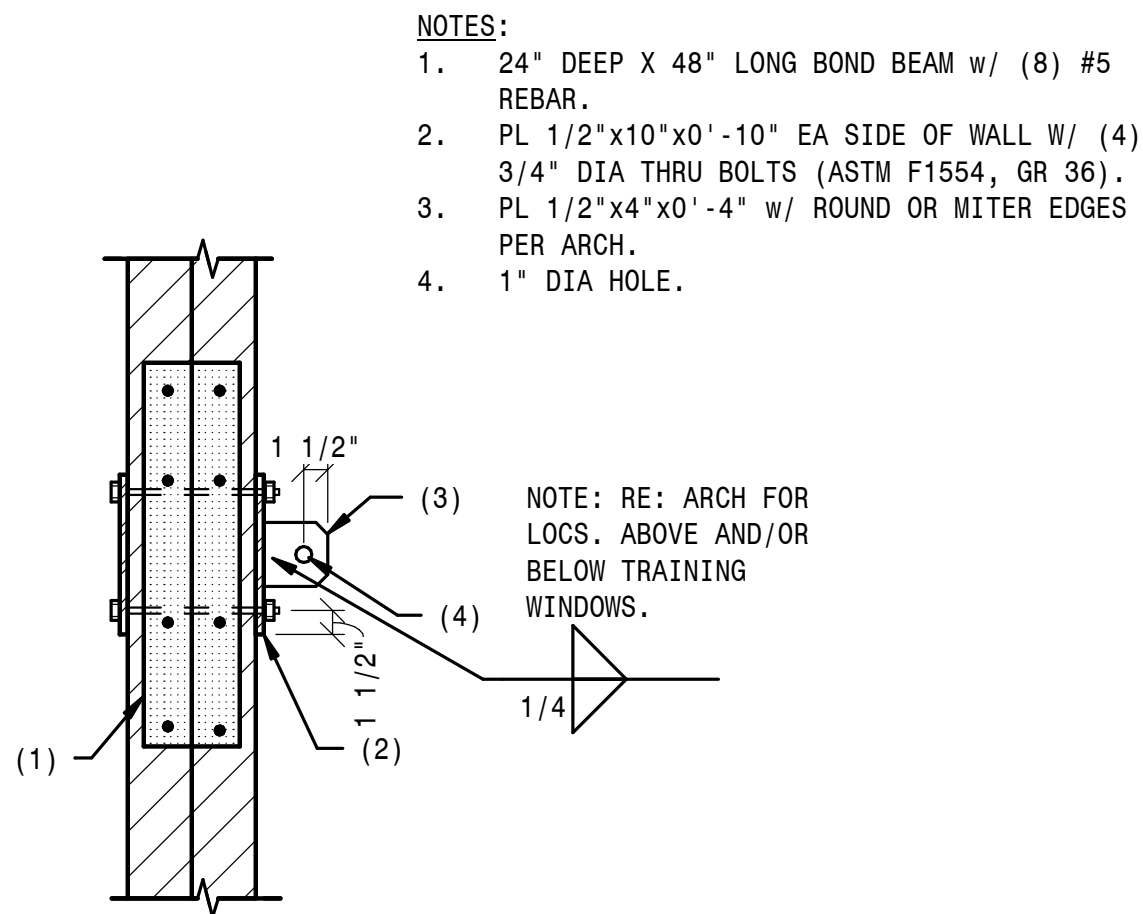
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FOUNDATION DETAILS

S3.3

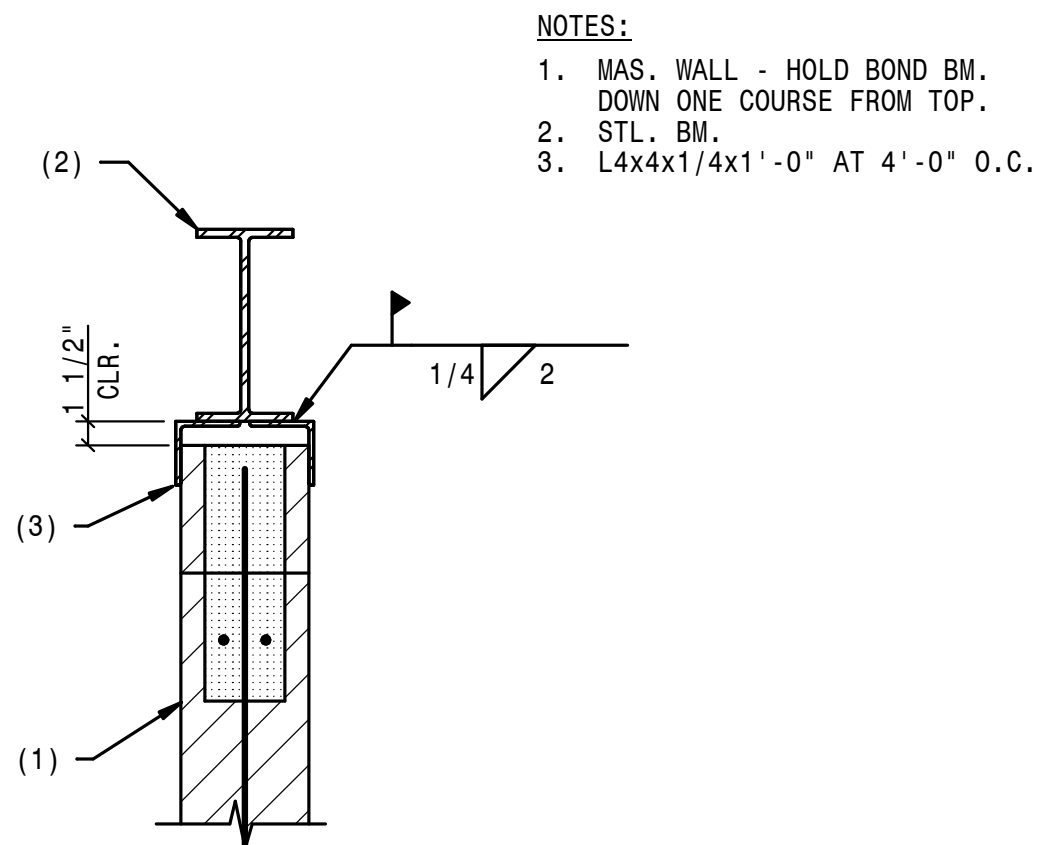




5 TRAINING WINDOW WALL ANCHORS

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

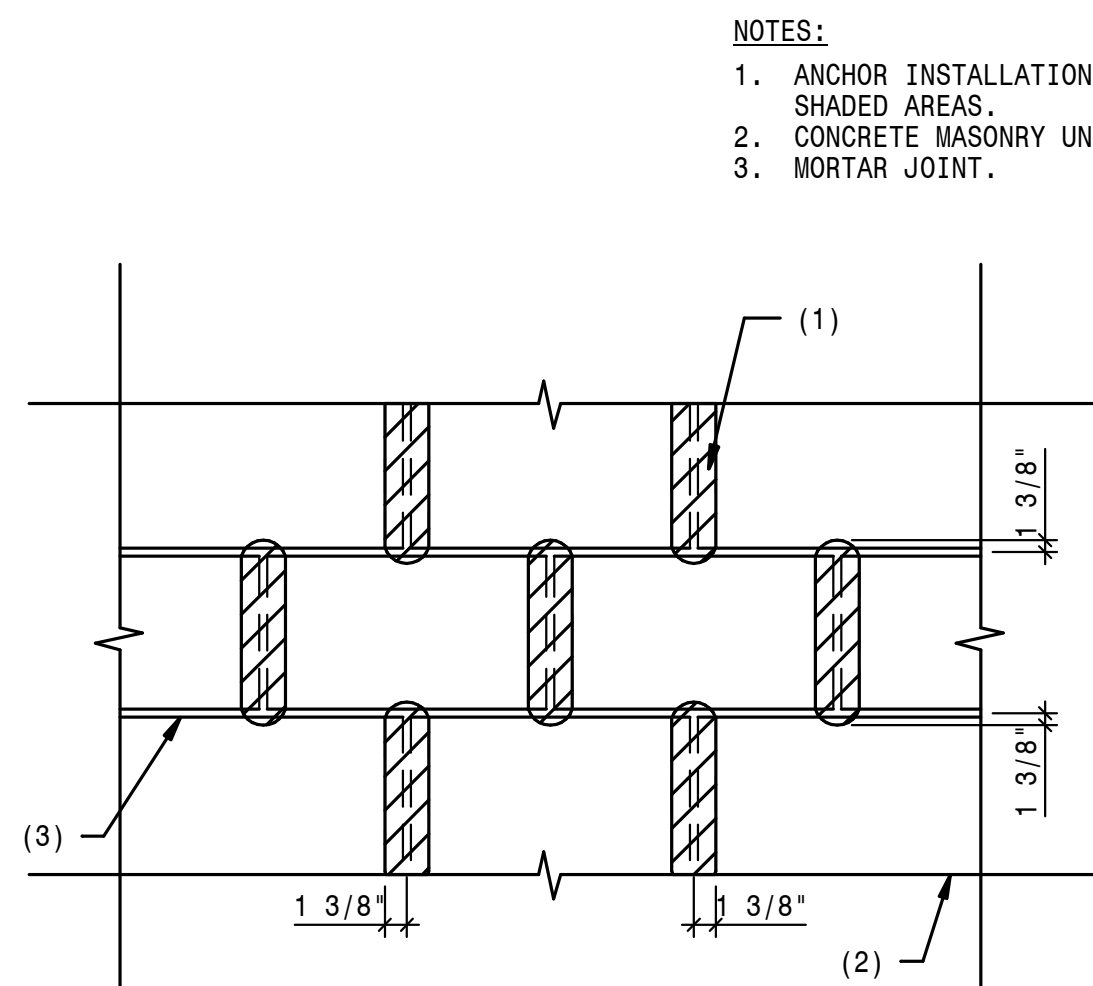
NO SCALE



6 MASONRY WALL BRACING AT STEEL BEAM

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

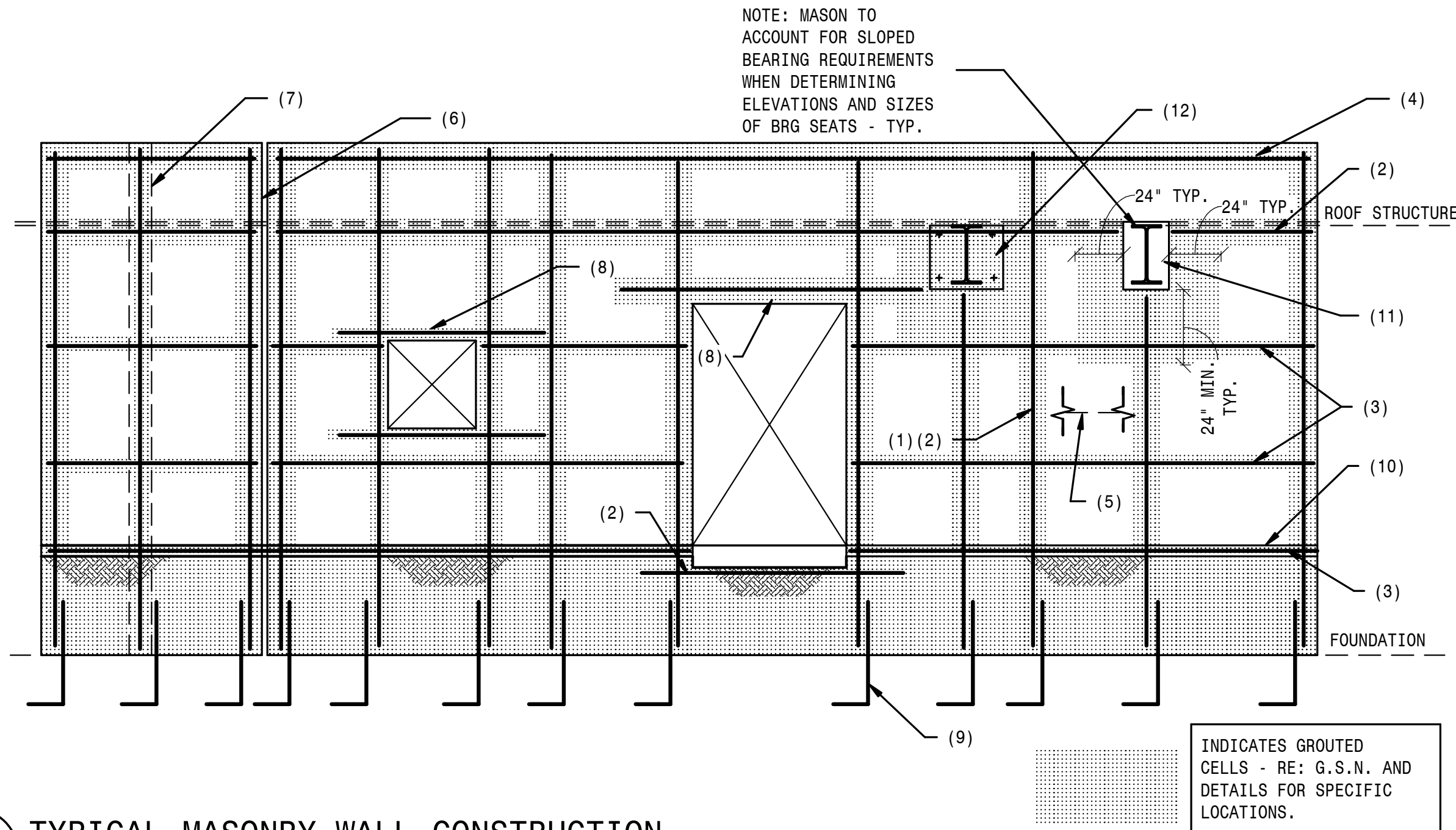
NO SCALE



7 INSTALLATION OF EXPANSION ANCHORS IN MASONRY WALLS

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

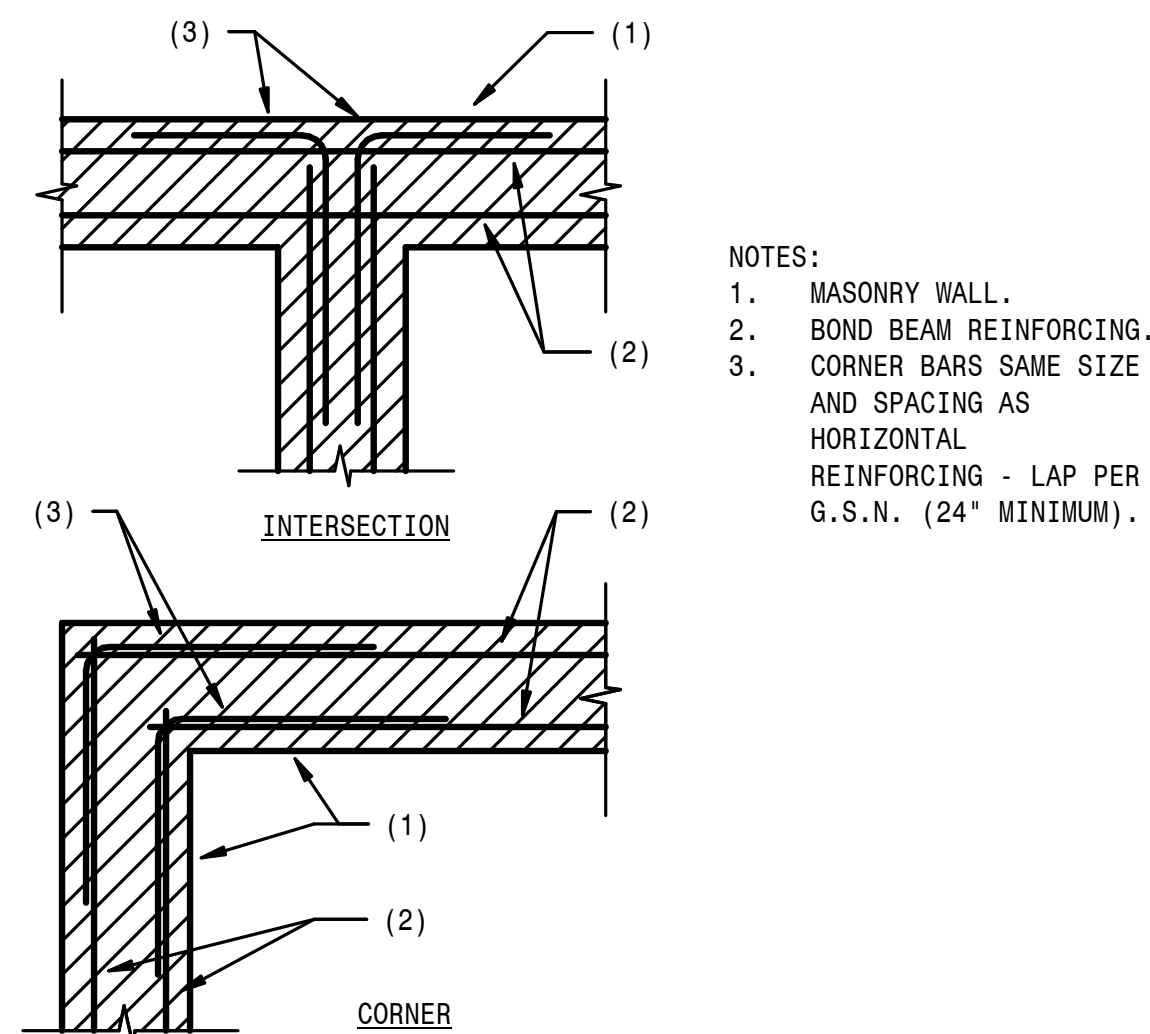
NO SCALE



1 TYPICAL MASONRY WALL CONSTRUCTION

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

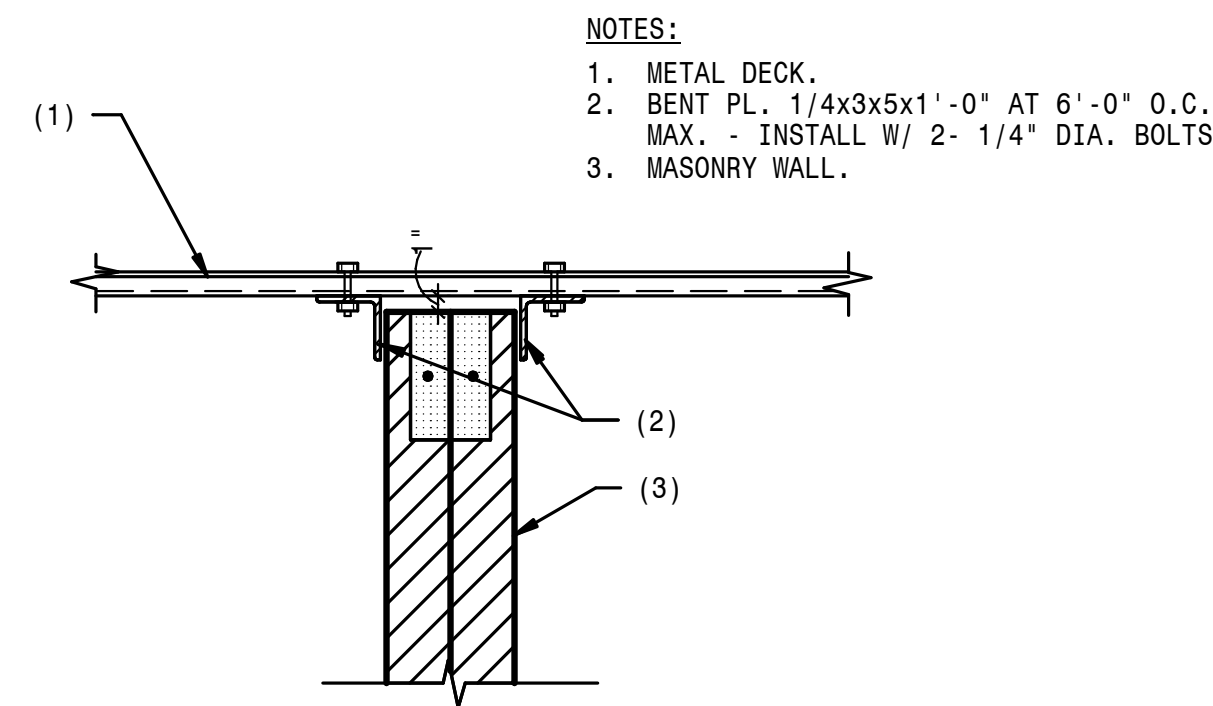
NO SCALE



3 MASONRY BOND BEAM AT INTERSECTING WALLS

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

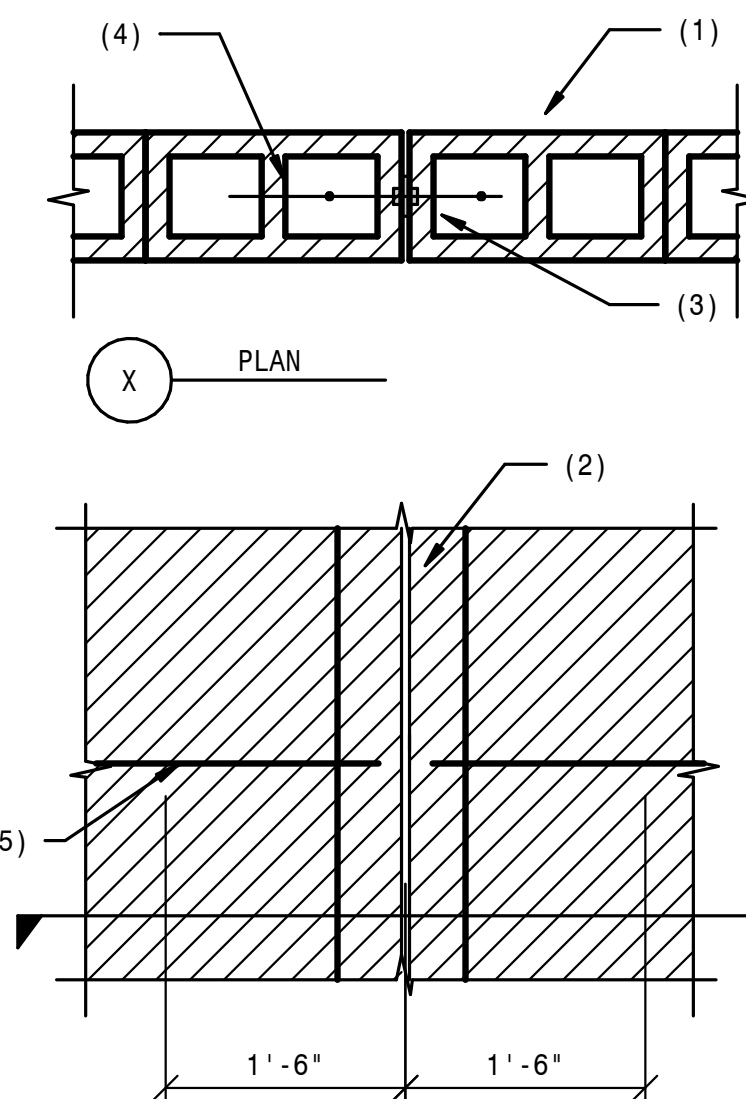
NO SCALE



4 BRACING AT NON STRUCTURAL MASONRY WALL

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE



2 CONTROL JOINT IN MASONRY WALL

THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.

NO SCALE

- NOTES:
- VERTICAL WALL REINF. - UNLESS NOTED OTHERWISE IN THE STRUCTURAL DRAWINGS, PROVIDE (1) #5 IN CENTER OF GROUT AT CENTER OF WALL, CONTINUOUS FULL HEIGHT OF WALL AT ALL CORNERS, INTERSECTIONS, WALL ENDS, BEAM BEARINGS, JAMBS, EACH SIDE OF CONTROL JOINTS AND AT INTERVALS NOT TO EXCEED 32" O.C. NON-STRUCTURAL PARTITION WALLS SHALL BE REINFORCED WITH #4 VERT AT 48" O.C.
 - TIE VERTICAL REINFORCING AT 8'-0" O.C. VERTICALLY, WITH SINGLE WIRE LOOP TIE BY A.A. WIRE PRODUCTS COMPANY.
 - HORIZONTAL WALL REINF. - UNLESS NOTED OTHERWISE IN THE STRUCTURAL DRAWINGS, PROVIDE (2) #5 IN MINIMUM 8" DEEP GROUTED CONTINUOUS BOND BEAM AT ELEVATED FLOORS AND ROOF LINE AND AT 4'-0" O.C. PROVIDE (1) #5 IN 8" DEEP BOND BEAM AT TOP OF PARAPET OR FREESTANDING WALL.
 - STANDARD WEIGHT (NO. 9 GAGE WIRE) DUR-O-WAL OR DUR-O-WIRE (OR EQUIVALENT) LADDER TYPE JOINT REINFORCEMENT AT 16" O.C. IN RUNNING BOND MASONRY WALLS ABOVE GRADE; 8" O.C. IN STACKED BOND MASONRY WALLS ABOVE GRADE; 8" O.C. BELOW GRADE.
 - CONTROL JOINT PER PLAN.
 - INTERSECTING MASONRY WALL BEYOND.
 - LINTEL - RE: LINTEL SCHEDULE.
 - DOWEL ALL VERTICAL REINFORCING TO FOUNDATION WITH DOWELS TO MATCH VERTICAL REINFORCING SIZE AND LOCATION.
 - SLAB ON GRADE.
 - BEAM POCKET - RE: PLAN FOR LOCATIONS.
 - EMBED PLATE - RE: PLAN FOR LOCATIONS.

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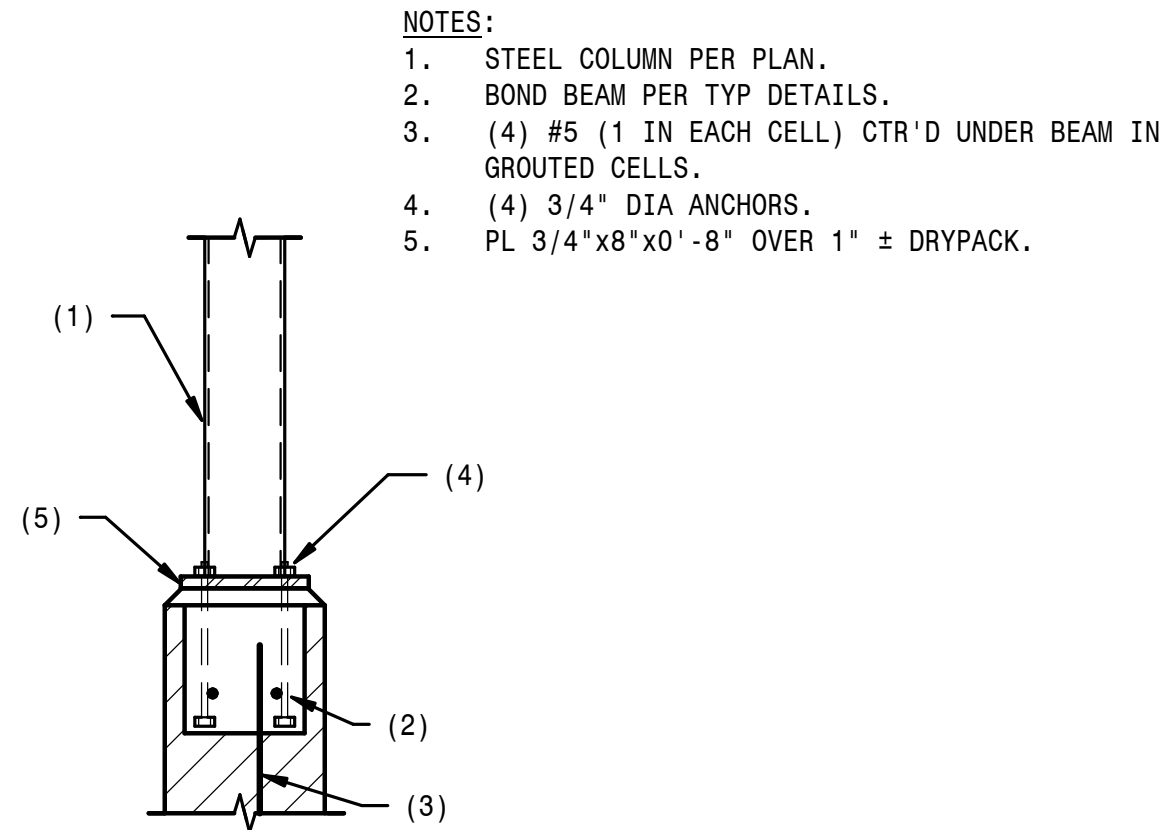
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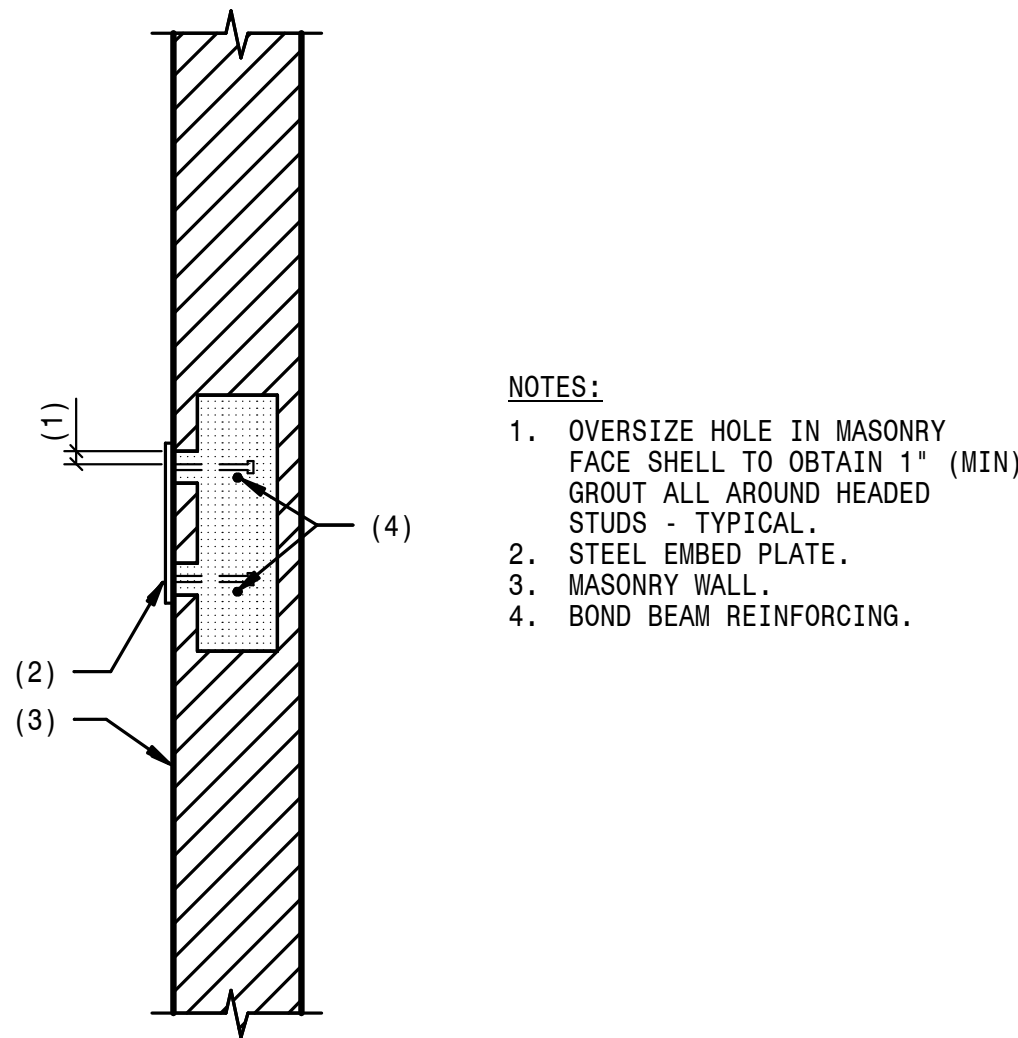
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MASONRY DETAILS

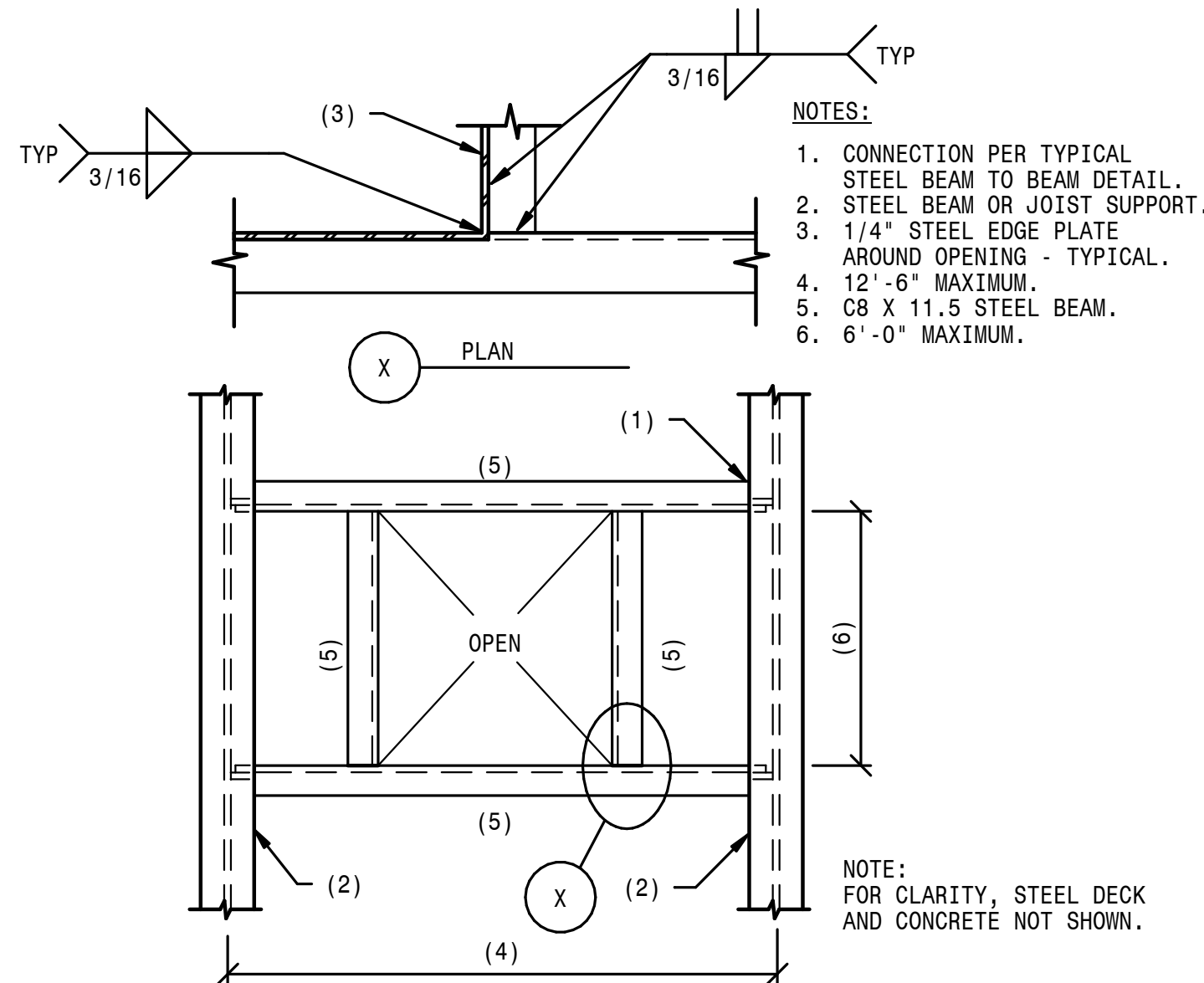
S4.1



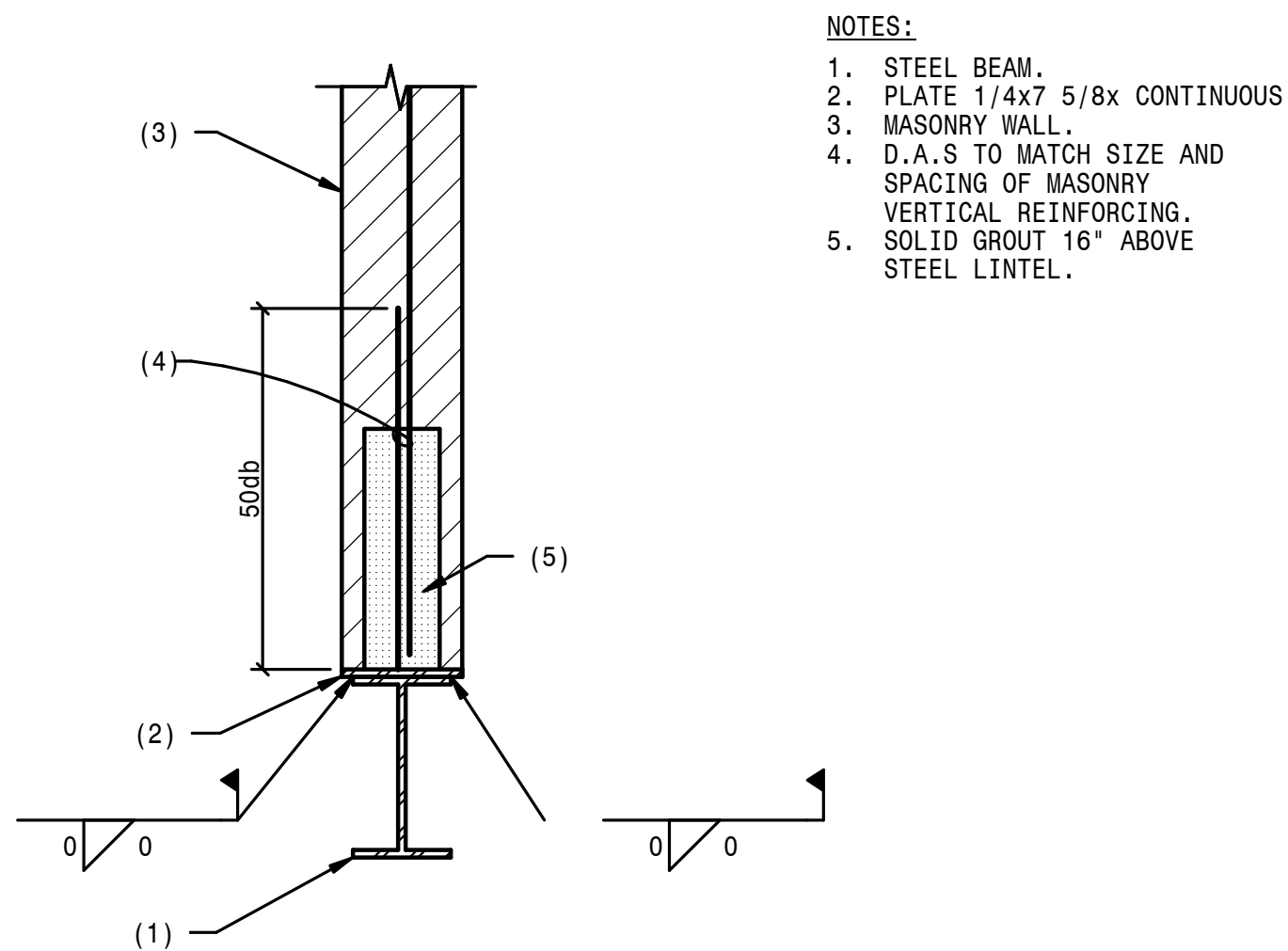
7 STEEL COLUMN AT MASONRY WALL
NO SCALE



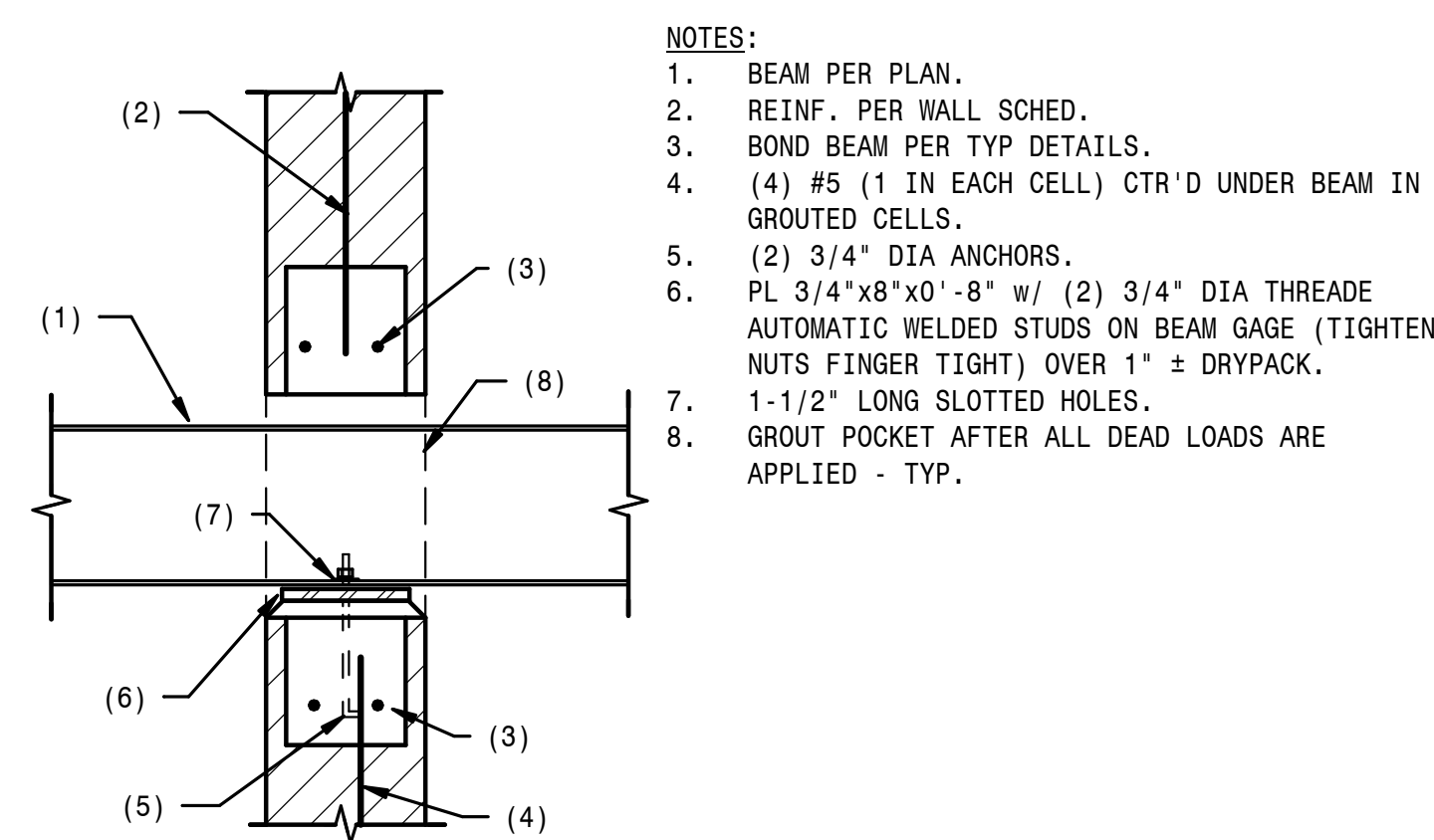
4 TYPICAL EMBED PLATE AT MASONRY WALL
NO SCALE



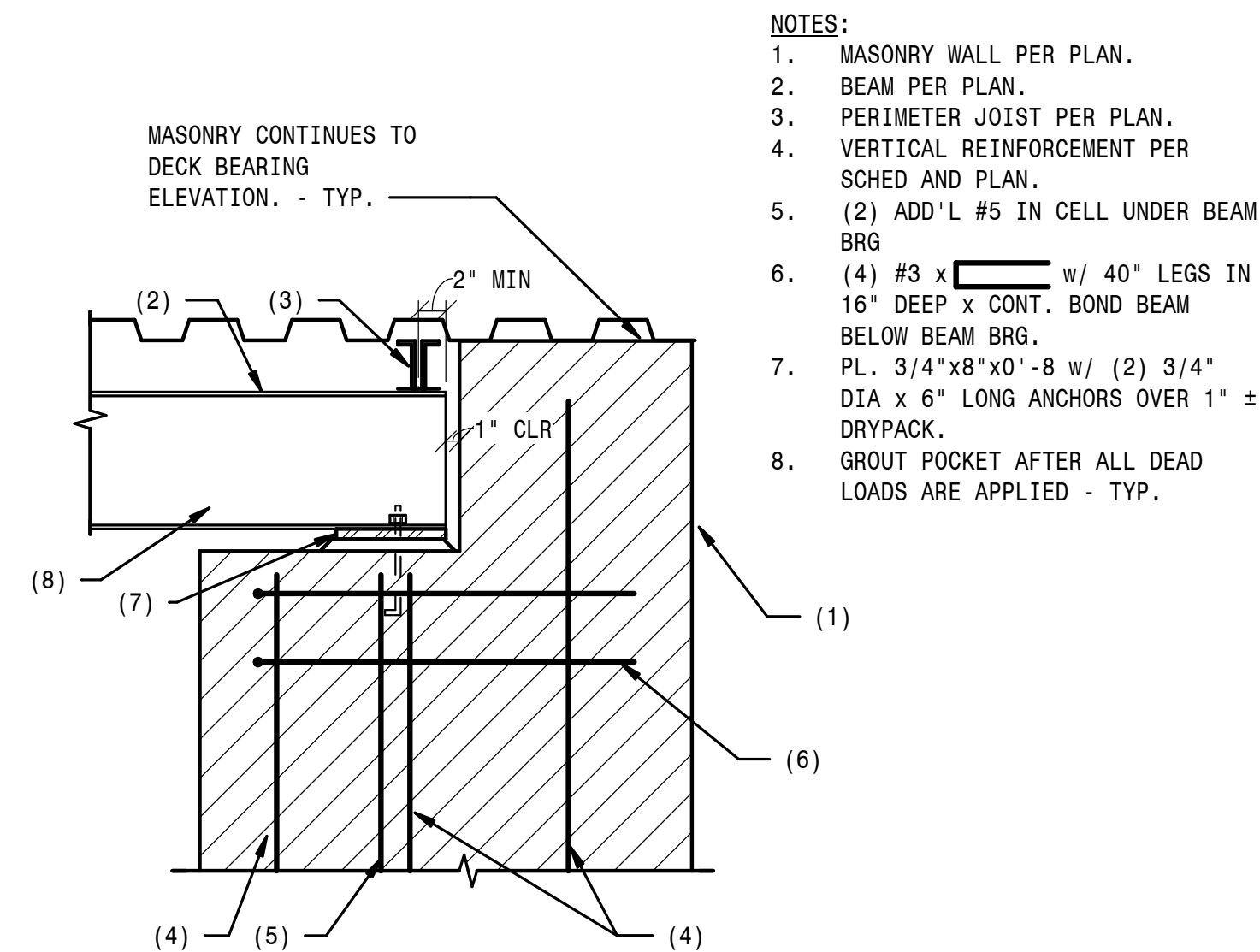
5 PLAN VIEW - TYPICAL FRAME OPENING AT LARGE OPENING
NO SCALE



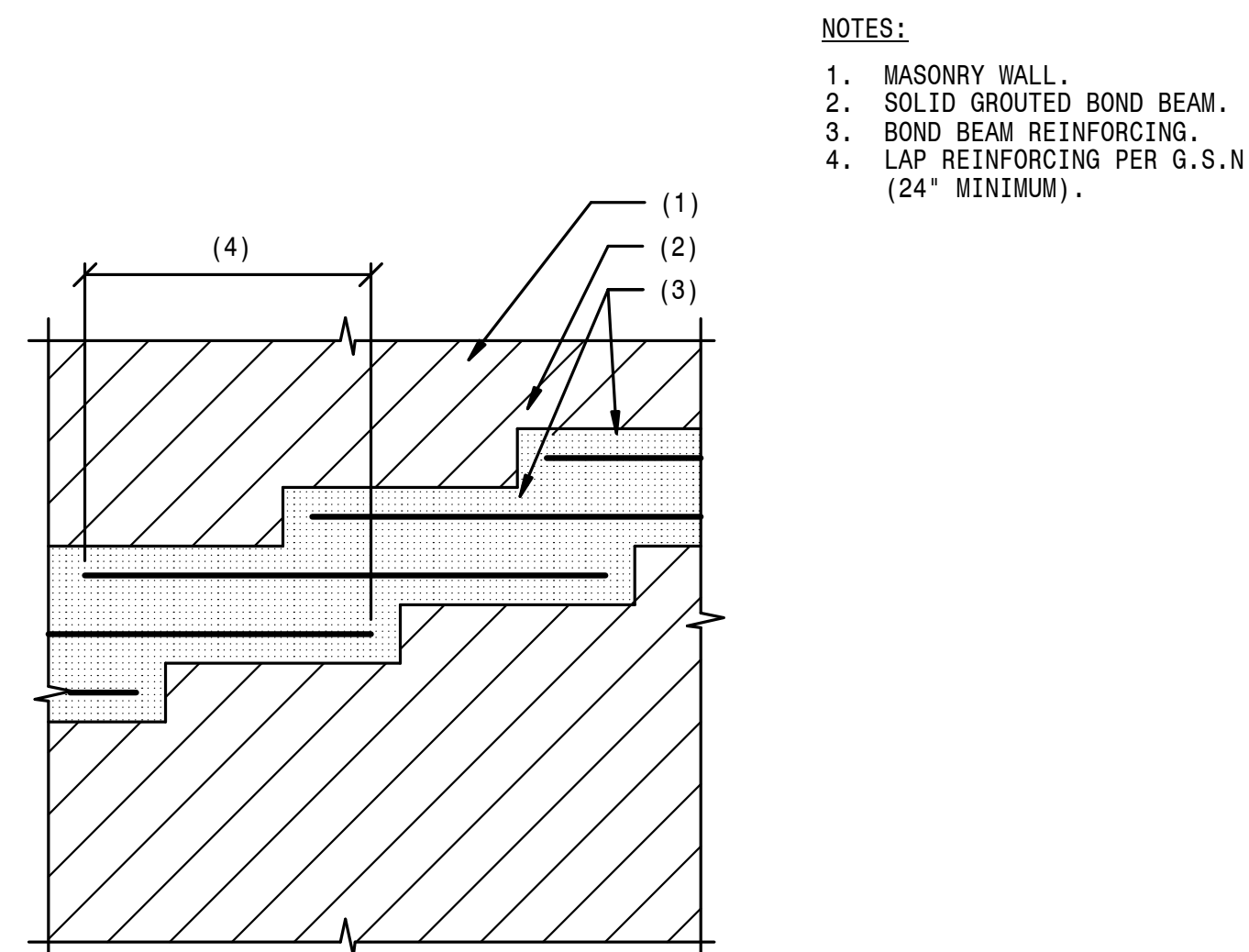
6 TYPICAL MASONRY WALL AT STEEL BEAM
NO SCALE



1 STEEL BEAM THRU MASONRY WALL
NO SCALE



2 BEAM BEARING
NO SCALE



3 STEPPED MASONRY WALL BOND BEAM
NO SCALE

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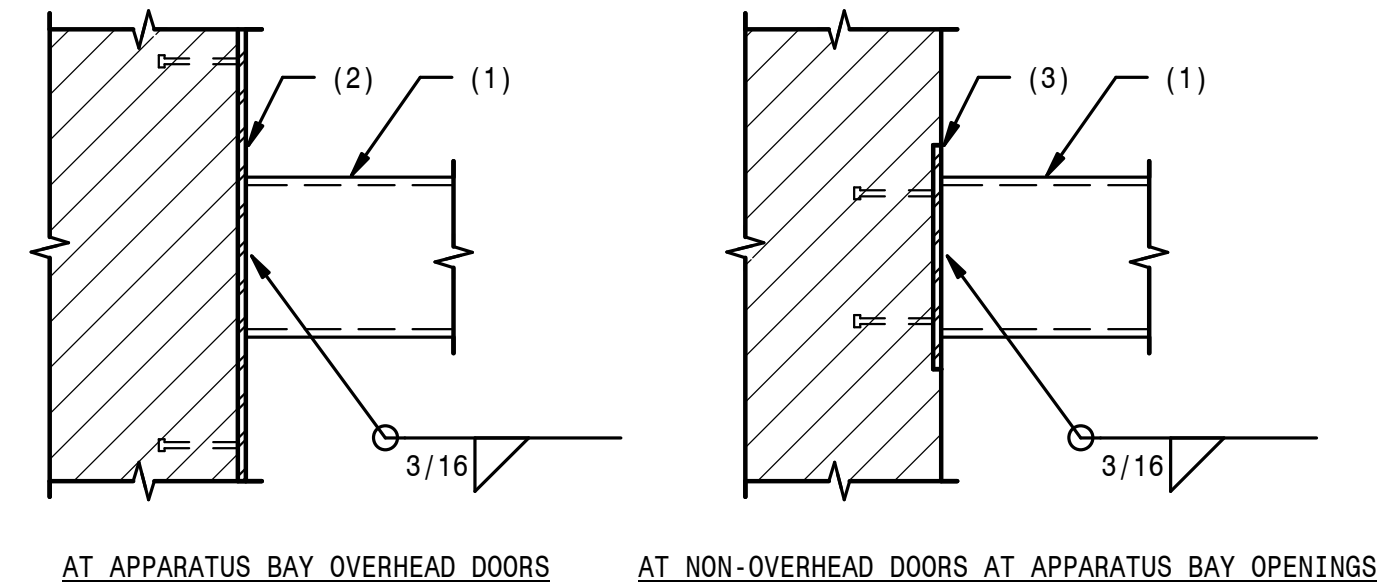
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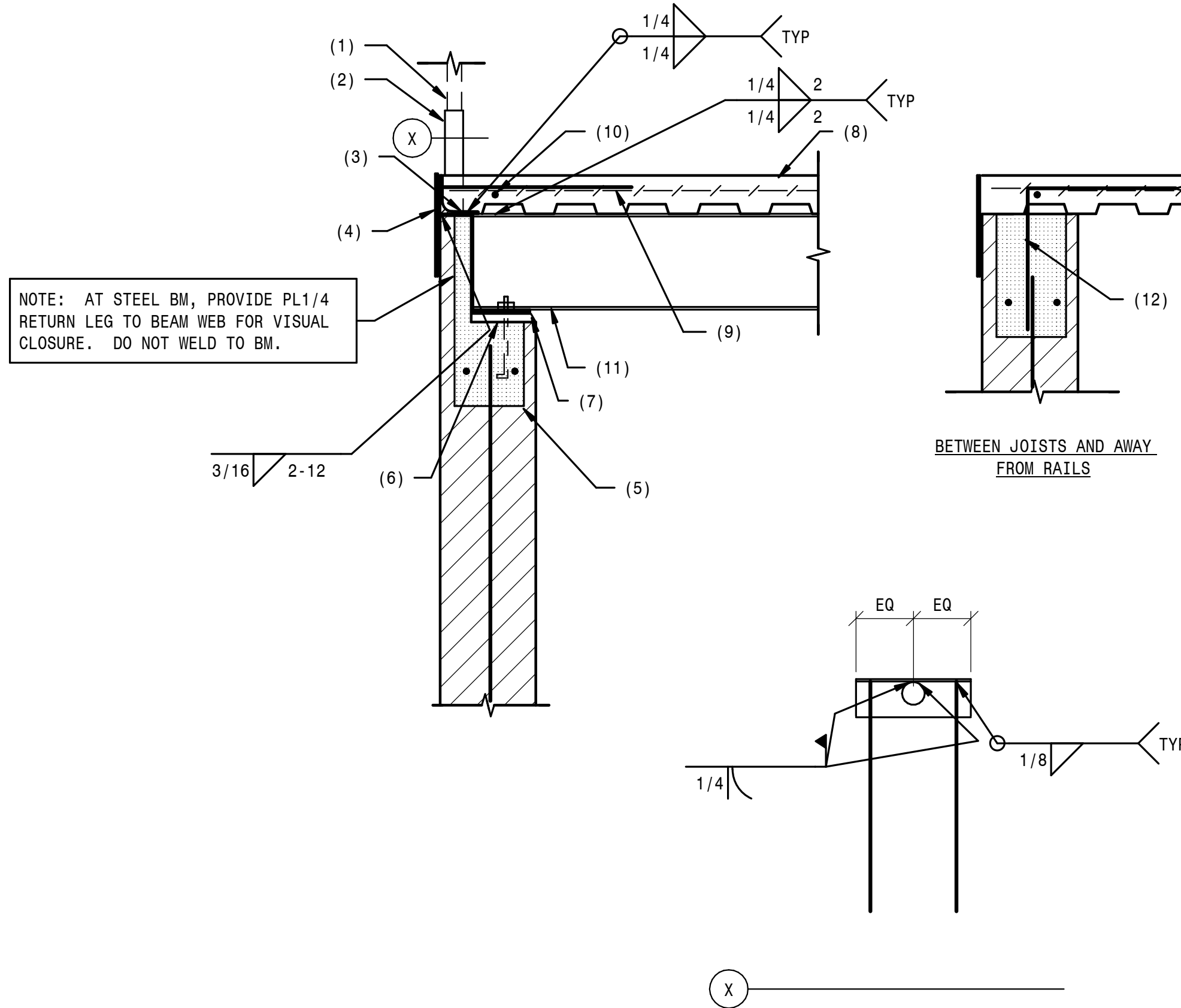
S4.2

- NOTES:
1. HSS GIRT PER PLAN.
 2. JAMB PLATE AND EMBEDS PER PLAN KEYED NOTES.
 3. EMBED PL PER 4/S4.2.
- NOTE: EXTERIOR FACE OF STEEL TO ALIGN WITH EXTERIOR FACE OF MASONRY WALL.



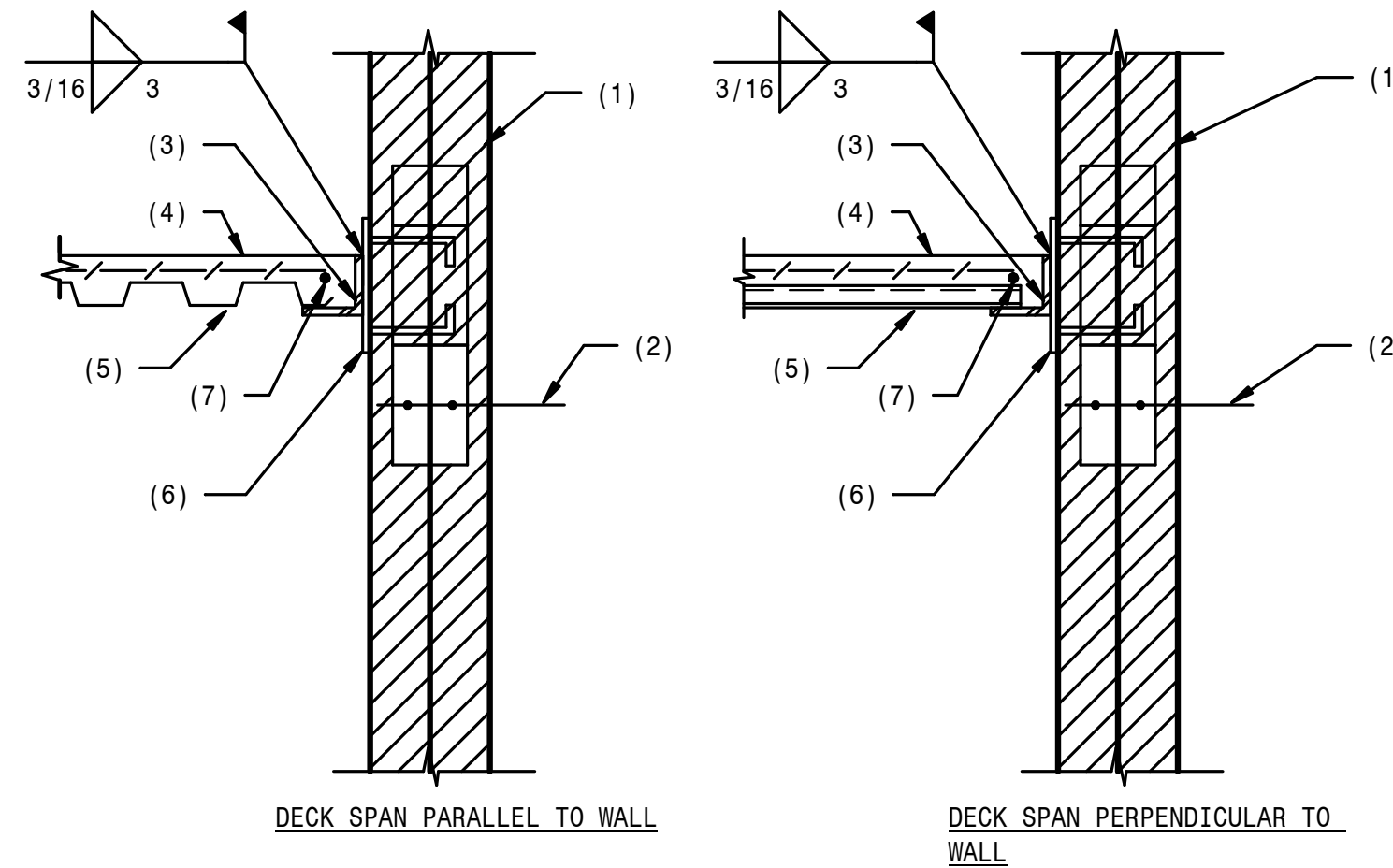
6 HSS GIRT TO MASONRY WALL
NO SCALE

- NOTES:
1. GUARDRAIL VERT SPACING PER ARCH.
 2. PIPE 1 1/2" STD (INSIDE DIA = 1.61") TO FIT REMOVABLE RAILING. PROVIDE CAP PL1/4 W/ WEEP HOLE - TYP.
 3. L4X4X1/4X1'-0" LONG.
 4. PL1/4X6XCONT.
 5. CONT BOND BEAM.
 6. 3" MIN BEARING.
 7. PL1/2X5-1/2X1'-2 W/ (2) 3/4" DIA ANCHORS OVER 1" ± DRYPACK.
 8. CONC ON MTL DECK PER PLAN.
 9. (1) #5X2'-0" WELDED TO ANGLE EACH SIDE OF SLEEVE (2 TOTAL REBAR AT EACH SLEEVE).
 10. (1) #4XCONT.
 11. BEAM PER PLAN.
 12. #4 DOWEL X 2' AT 6'-4" OC MAX. LOCATE DOWELS AT VERT WALL REINF LOGS.



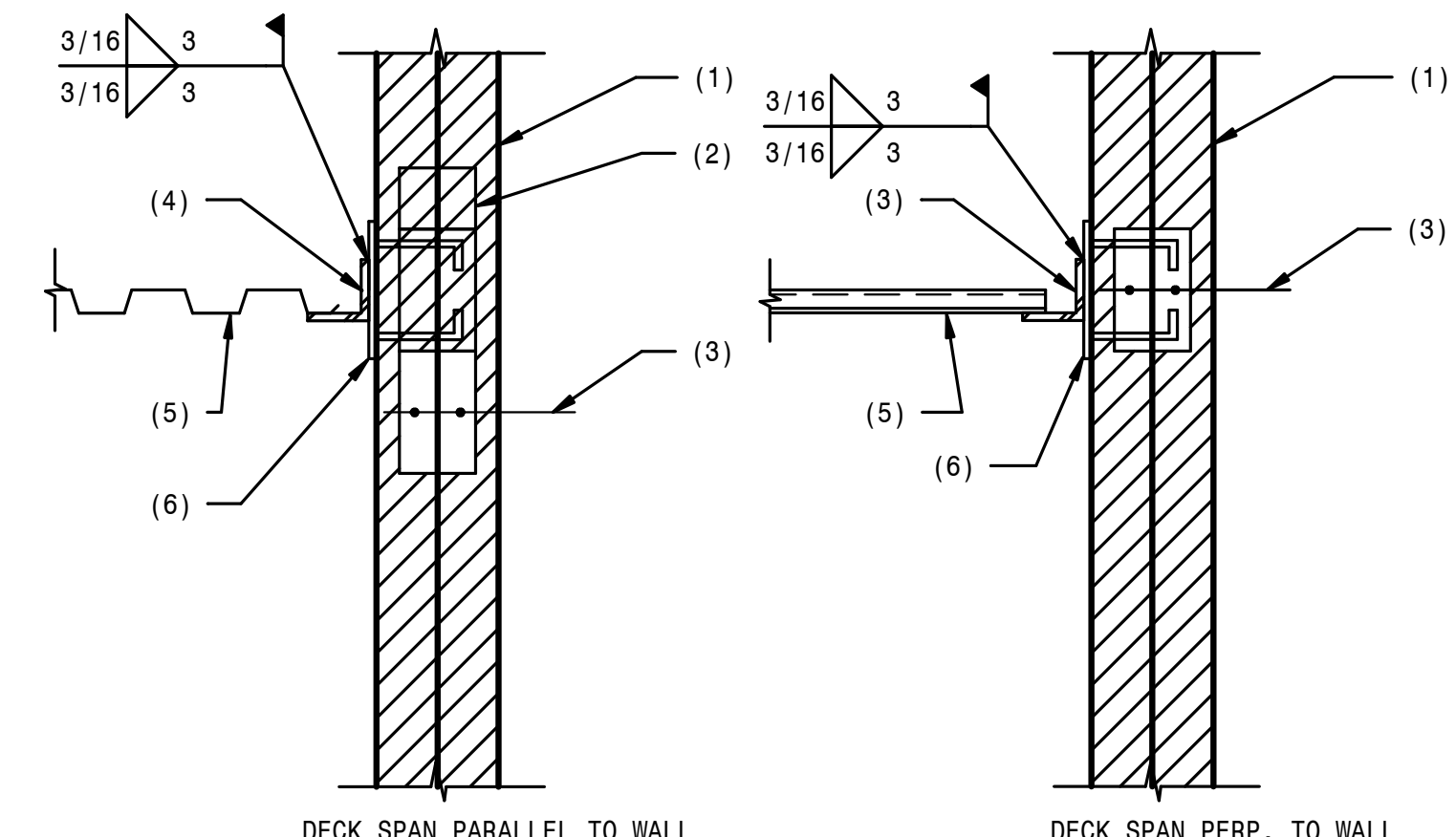
4 MEZZANINE EDGE AT GUARDRAIL
NO SCALE

- NOTES:
1. MAS WALL.
 2. CONT GROUTED BOND BM PER GSN.
 3. CONT LEDGER ANGLE.
 4. CONC OVER STEEL DECK.
 5. STEEL DECK - FOR DIRECTION SEE PLANS.
 6. EMBED PL3/8X4X0'-9" W/ (2) 1/2" DIA ANCHORS AT 6" OC - GROUT 4" MIN ABOVE AND BELOW ANCHORS.
 7. (1) #4XCONT.



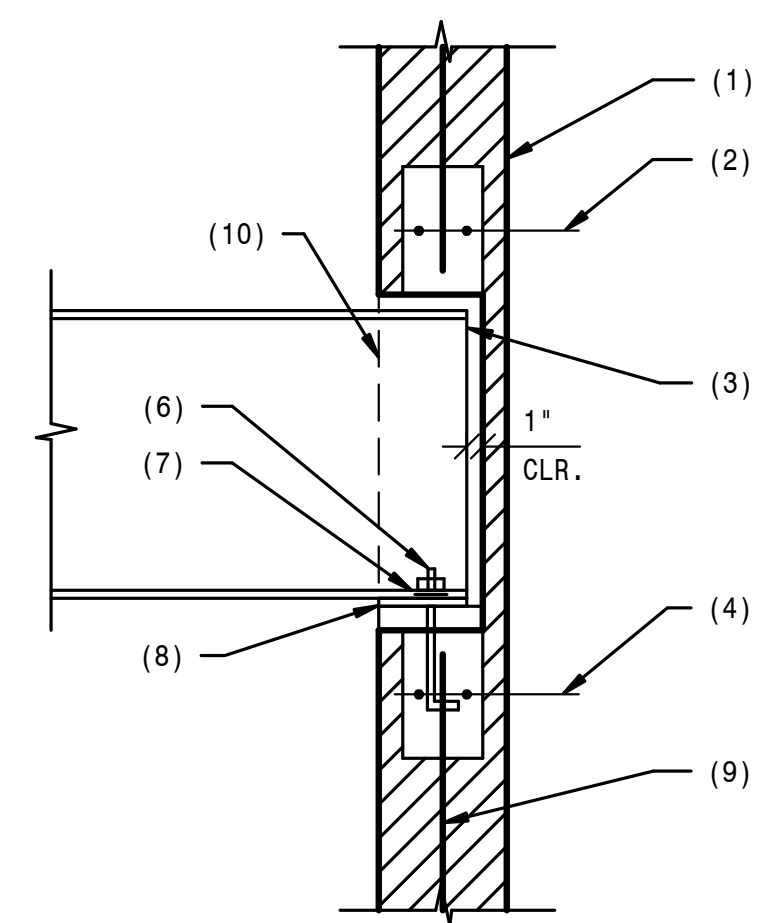
5 LEDGER ANGLE AND STEEL DECK AT MASONRY WALL
NO SCALE

- NOTES:
1. MASONRY WALL.
 2. GROUT CONTINUOUS FROM BOND BEAM TO 4" ABOVE ROOF LINE.
 3. CONTINUOUS GROUTED BOND BEAM PER GSN.
 4. L3X3X1/4XCONT.
 5. STEEL DECK OR BAR GRATING - FOR DIRECTION, SEE PLAN.
 6. (2) 3/4" DIAx5" LONG ANCHORS AT 6" OC - GROUT MINIMUM 4" ABOVE AND BELOW ANCHORS. SPACE PLATES AT 4'-0" OC.



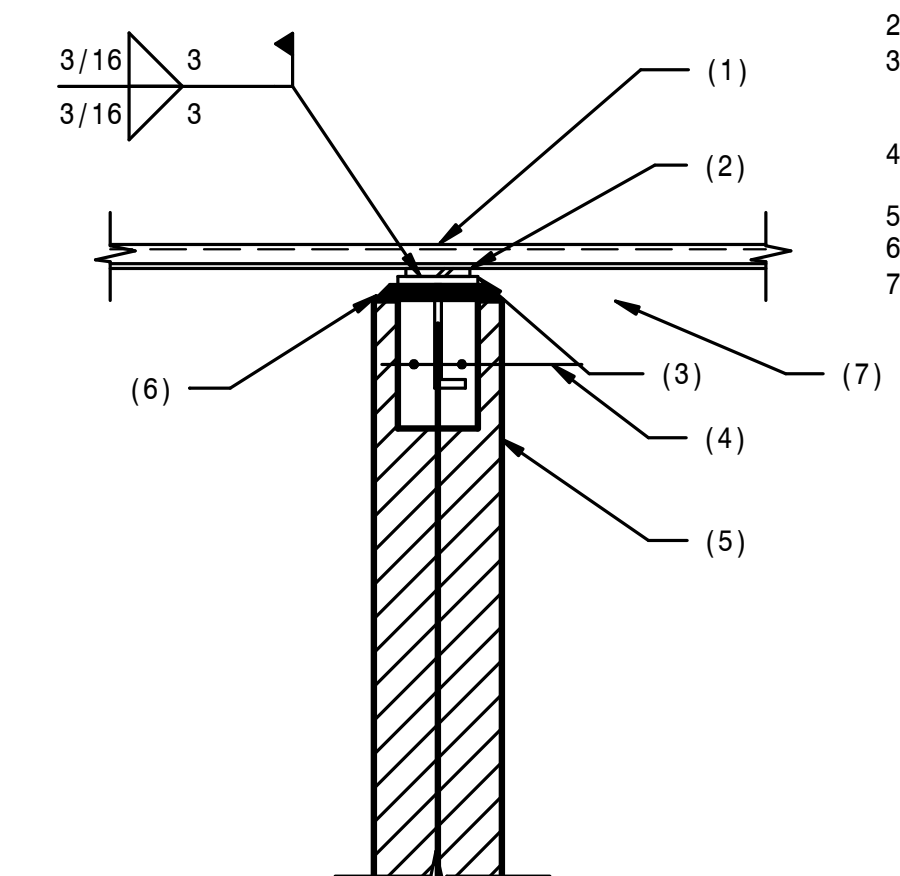
1 CHORD ANGLE AND STEEL DECK AT MASONRY WALL
NO SCALE

- NOTES:
1. MASONRY WALL.
 2. BOND BEAM PER GSN.
 3. STEEL BEAM.
 4. (2) #5X3'-6" LONG IN 8" DEEPX4'-0" LONG GROUTED BOND BEAM.
 5. NOT USED.
 6. (2) 3/4" DIA. AUTOMATIC WELDED THREADED STUDS ON BEAM GAGE (TIGHTEN NUTS FINGERTIGHT).
 7. 1 1/2" LONG HORIZONTALLY SLOTTED HOLE ON BEAM GAGE - EACH SIDE OF WEB.
 8. PL3/4X5 1/2X1'-2" WITH (2) 3/4" DIA ANCHORS OVER 1" ± DRYPACK.
 9. ADD (4) #5 (1 IN EA. CELL) CTR'D UNDER BM. IN GROUTED CELLS.
 10. GROUT POCKET AFTER ALL DEAD LOADS ARE APPLIED - TYP.



2 STEEL BEAM AT MASONRY WALL
NO SCALE

- NOTES:
1. STEEL DECK - FOR DIRECTION OF DECK, SEE PLAN.
 2. PL3/8X4XCONT.
 3. EMBED PL1/2X6X0'-6" AT 48" OC WITH (2) 3/4" DIA ANCHORS.
 4. CONTINUOUS GROUTED BOND BEAM PER GSN.
 5. MASONRY WALL.
 6. 1" ± DRYPACK.
 7. RE: ARCH FOR SHORT CANTILEVERS AT ROOF EDGE - TYP



3 STEEL DECK AT MASONRY WALL
NO SCALE

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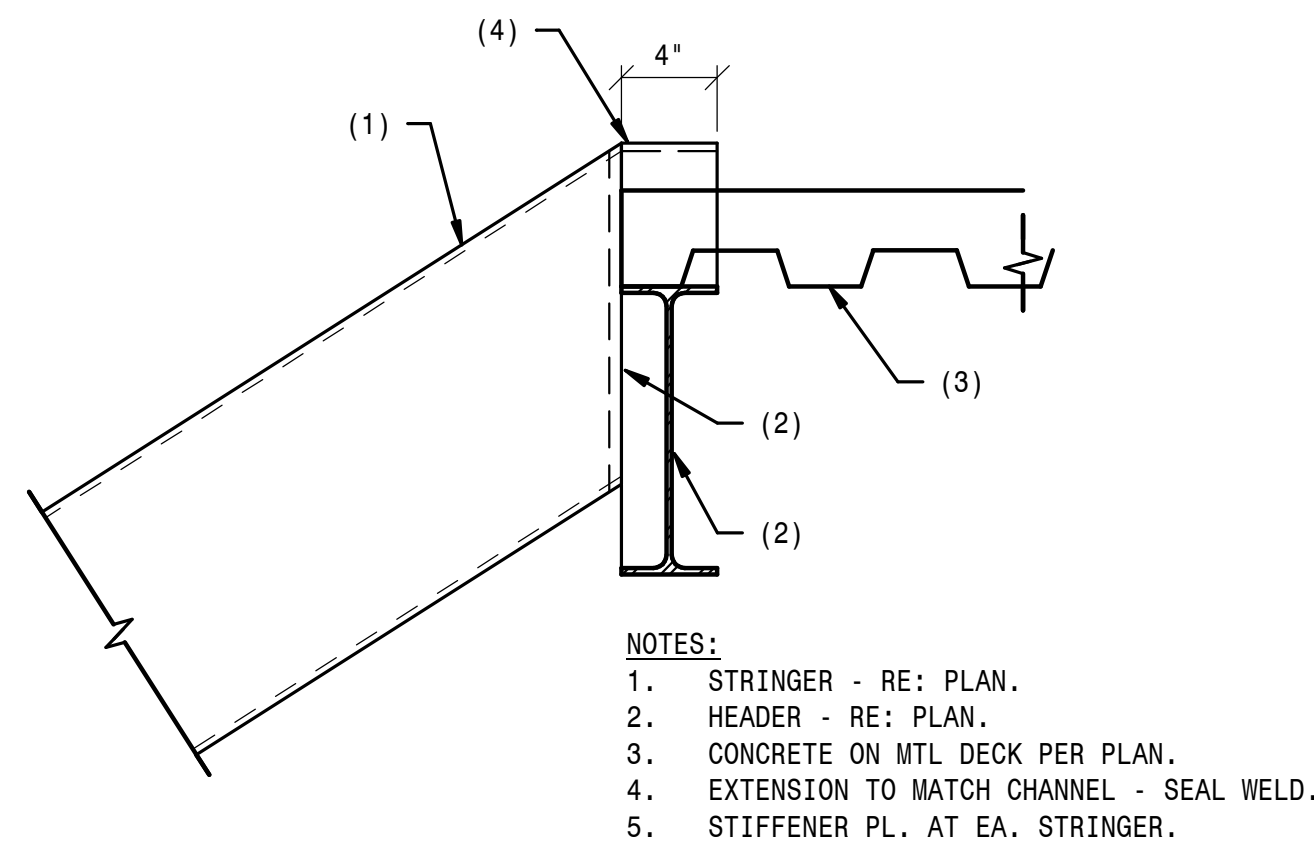
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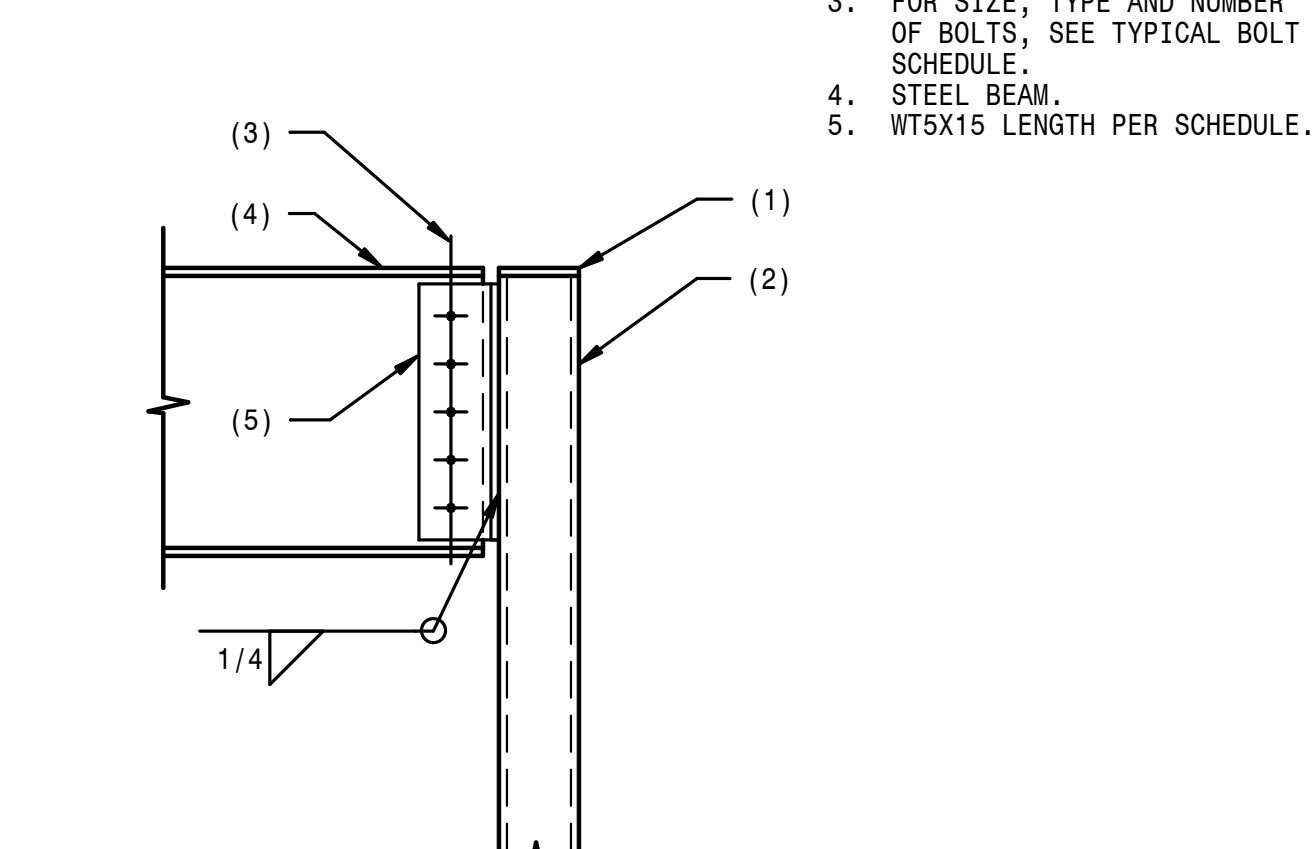
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MASONRY DETAILS

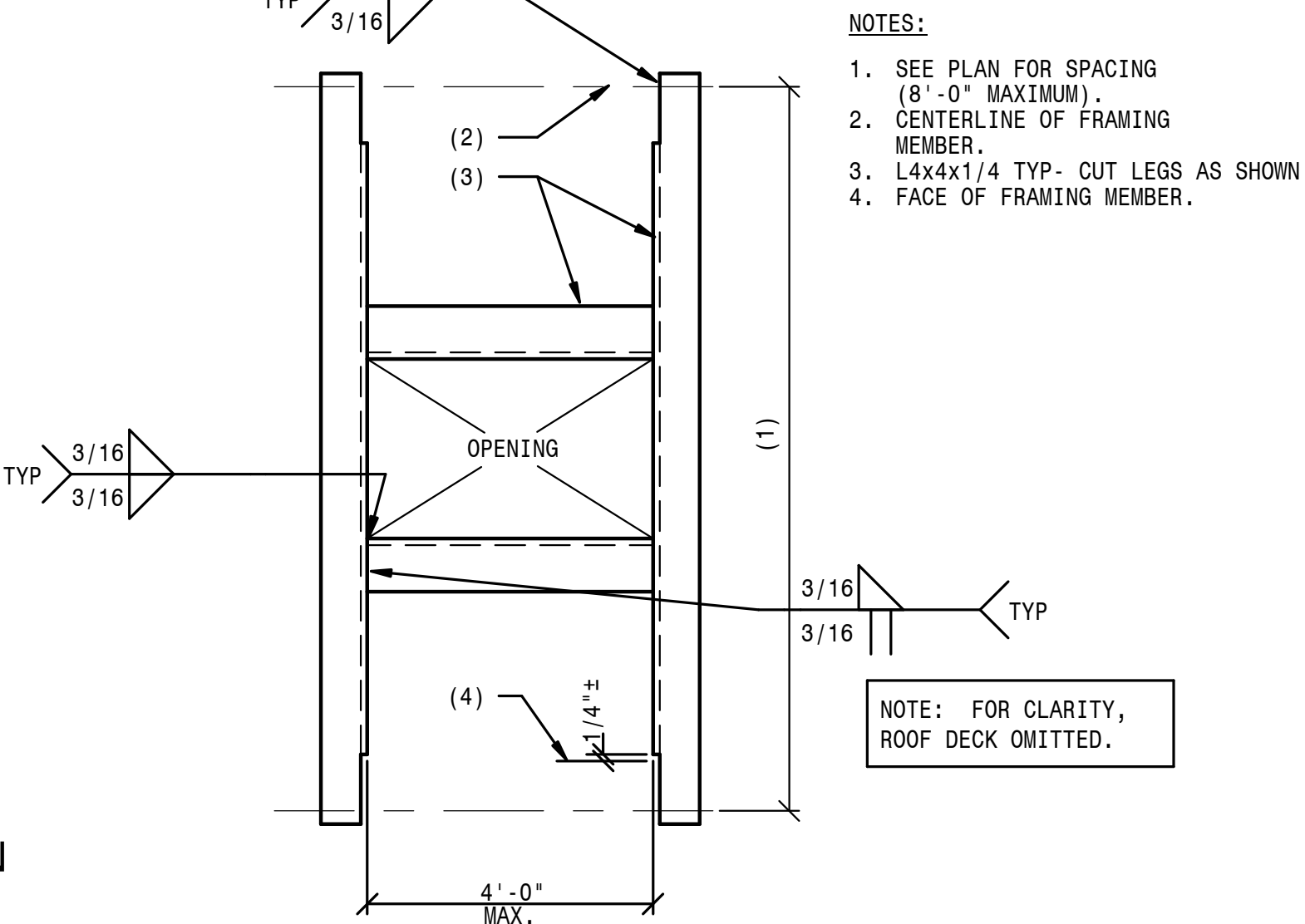
S4.3



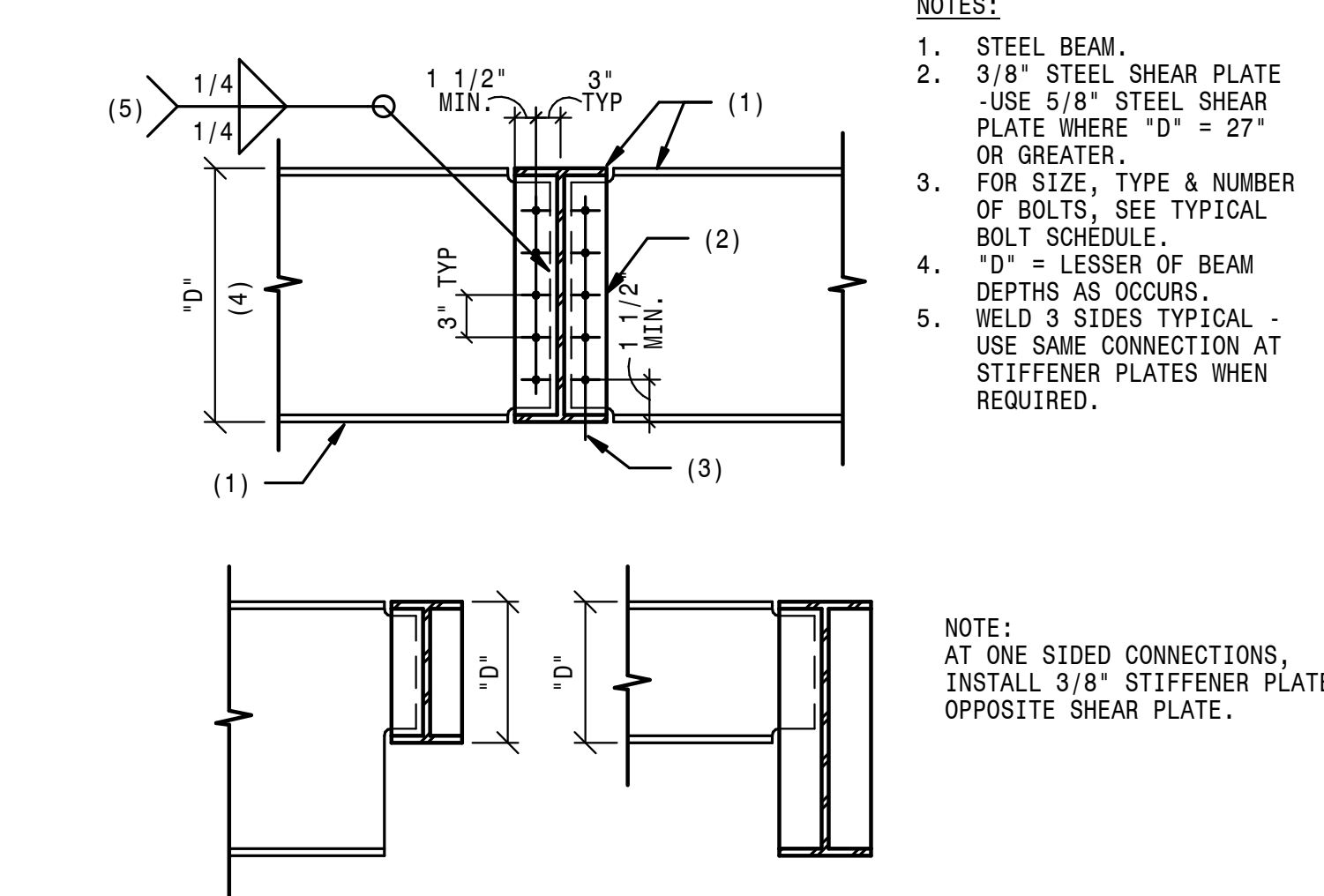
11 TYPICAL STRINGER AT HEADER
NO SCALE



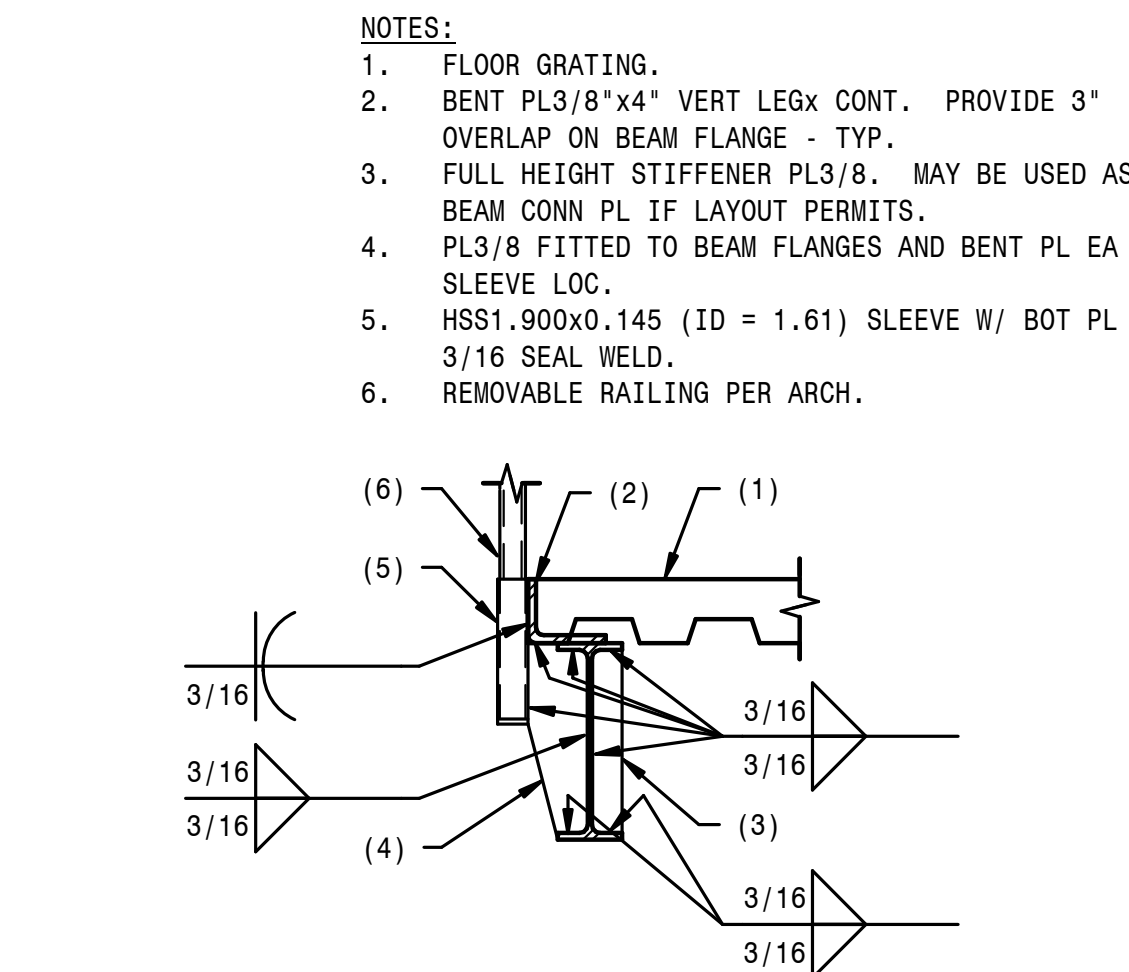
7 STEEL BEAM TO STEEL COLUMN CONNECTION
THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.
NO SCALE



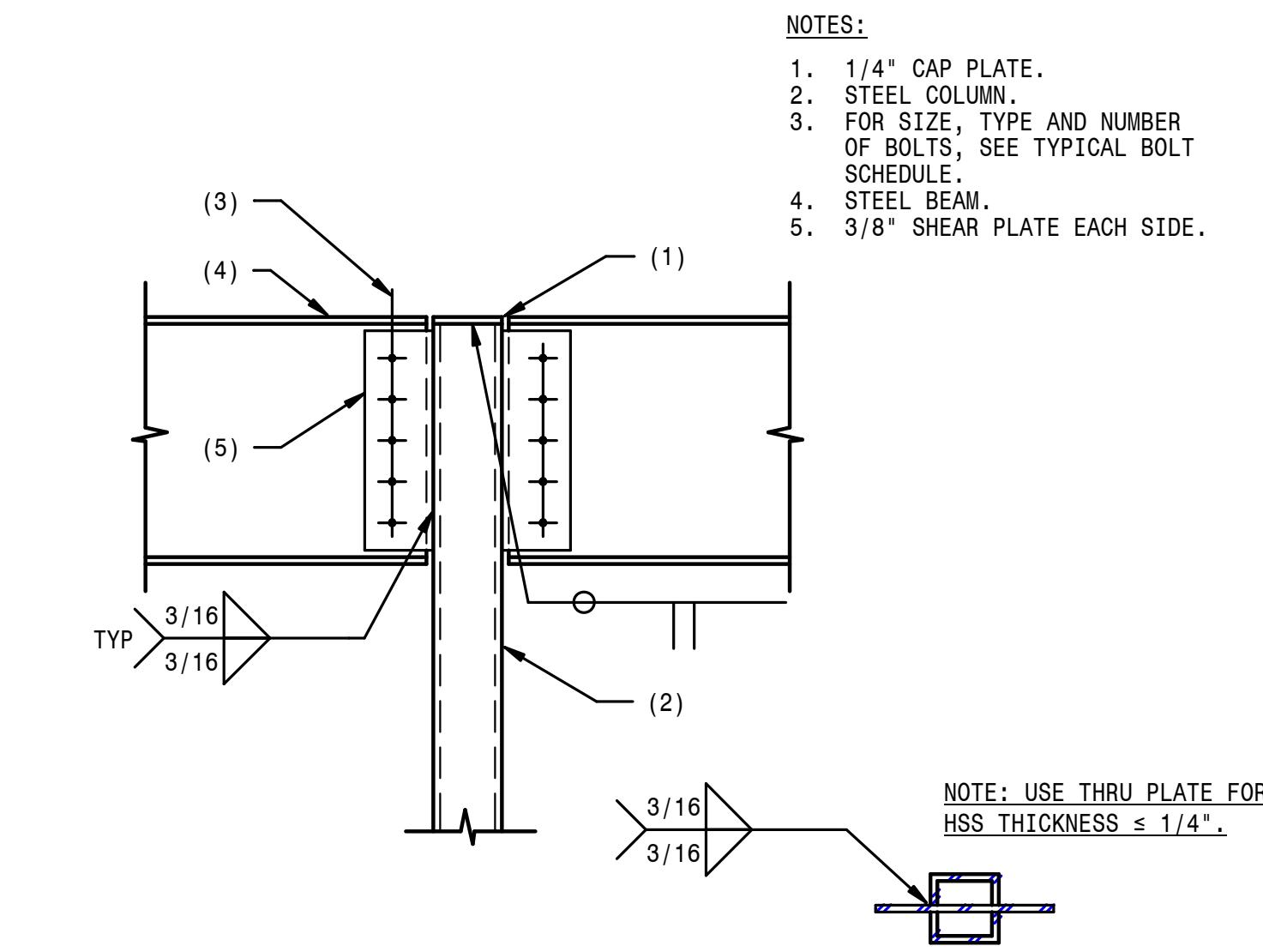
4 PLAN - TYPICAL OPENING IN ROOF FRAMING
THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.
NO SCALE



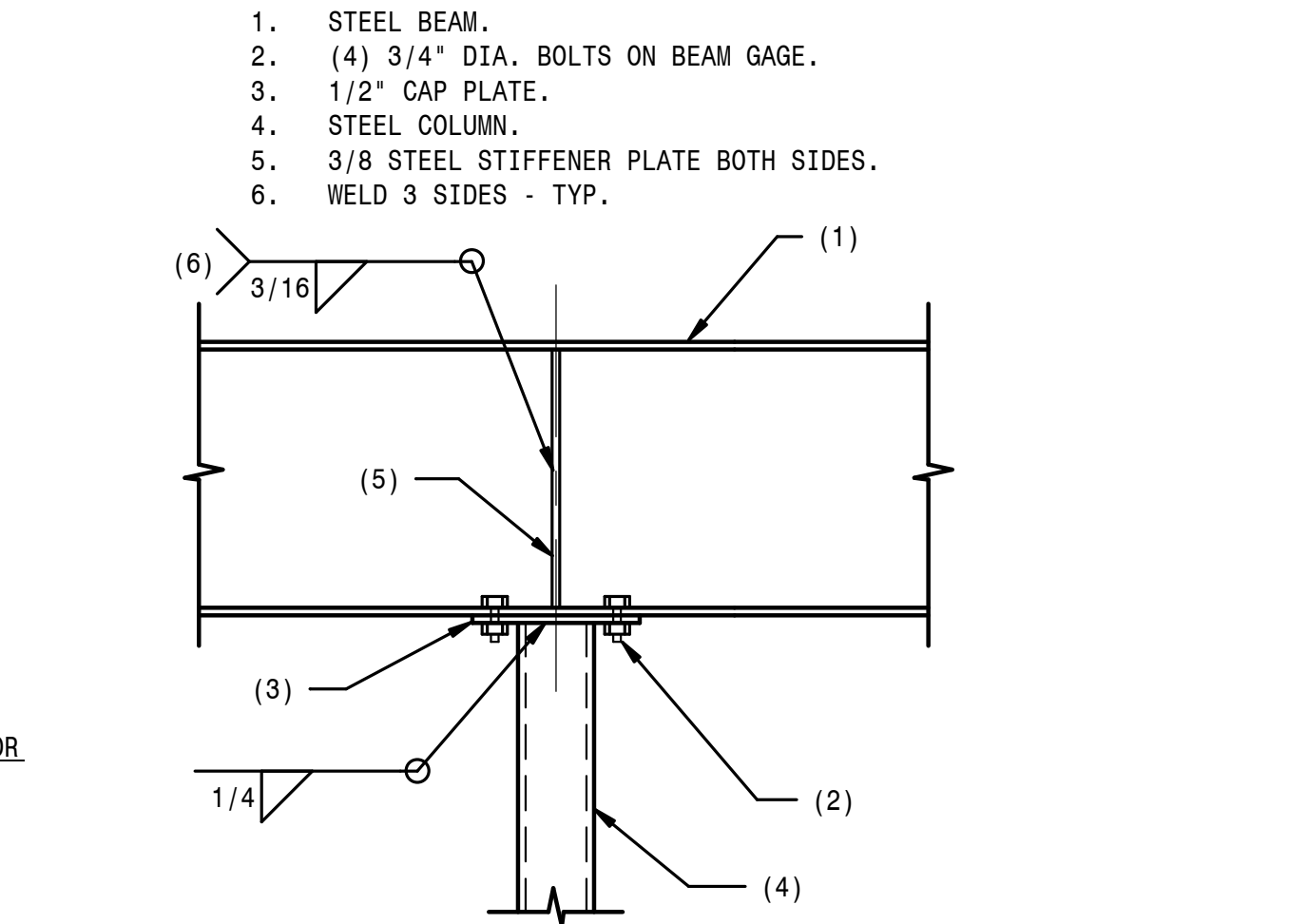
1 TYPICAL CONNECTION WIDE FLANGE BEAM TO BEAM
THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.
NO SCALE



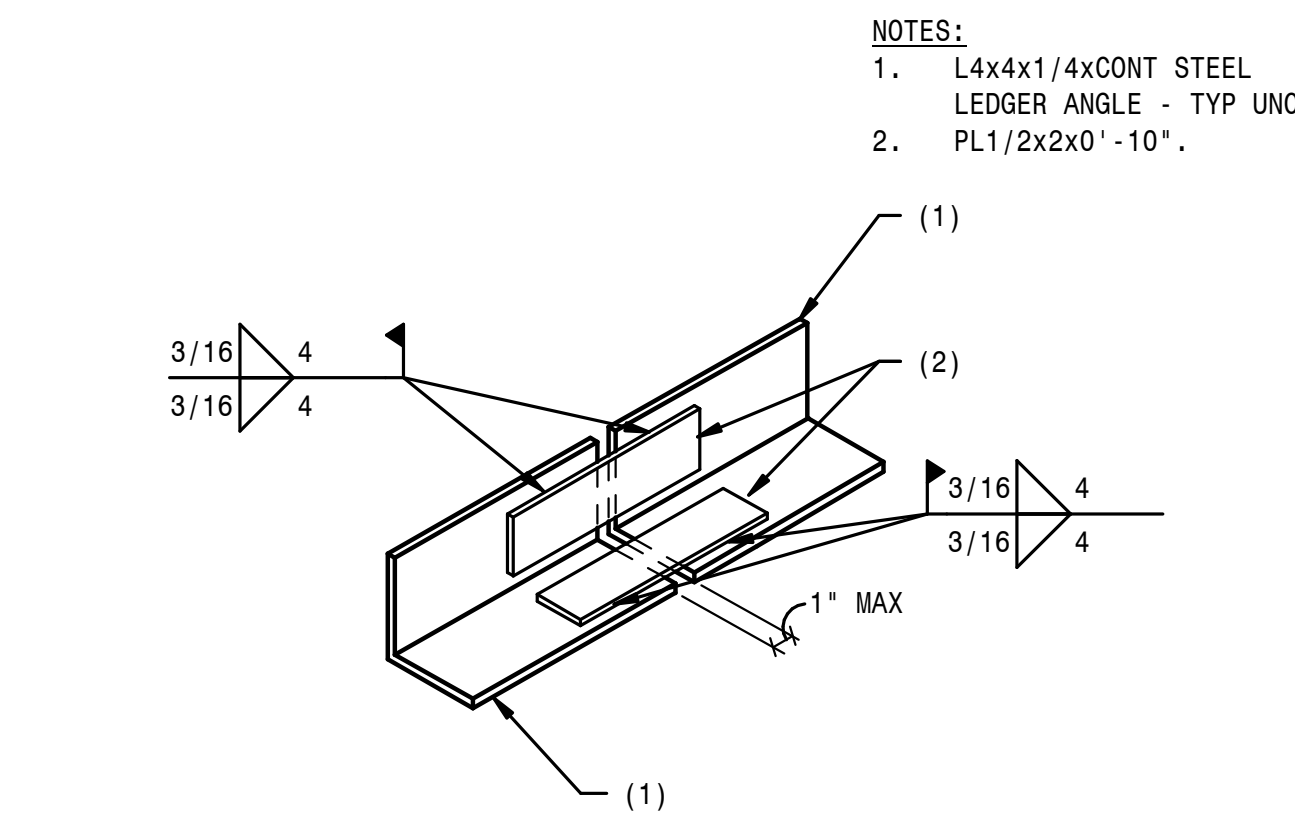
10 MEZZANINE FLOOR EDGE
NO SCALE



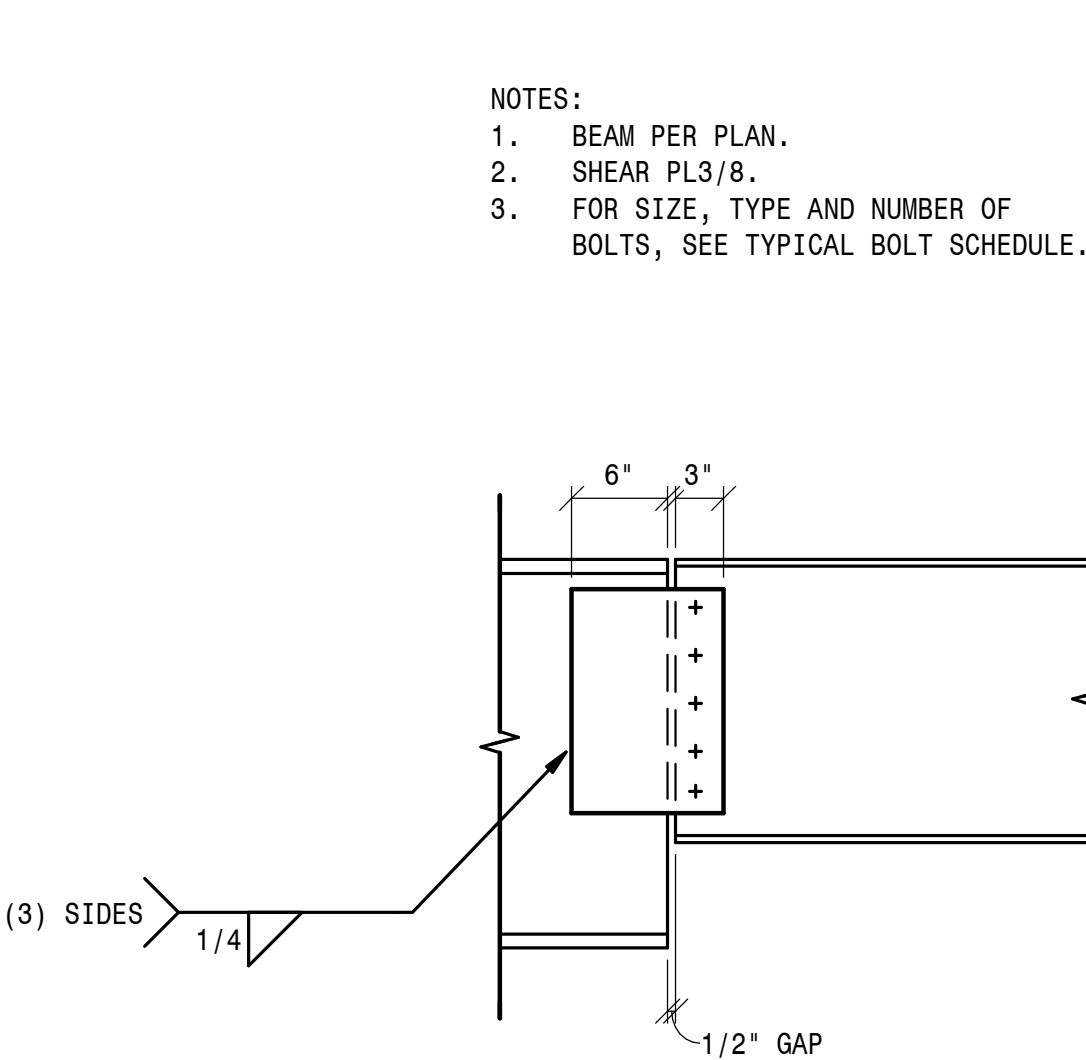
8 STEEL BEAMS TO STEEL COLUMN CONNECTION
THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.
NO SCALE



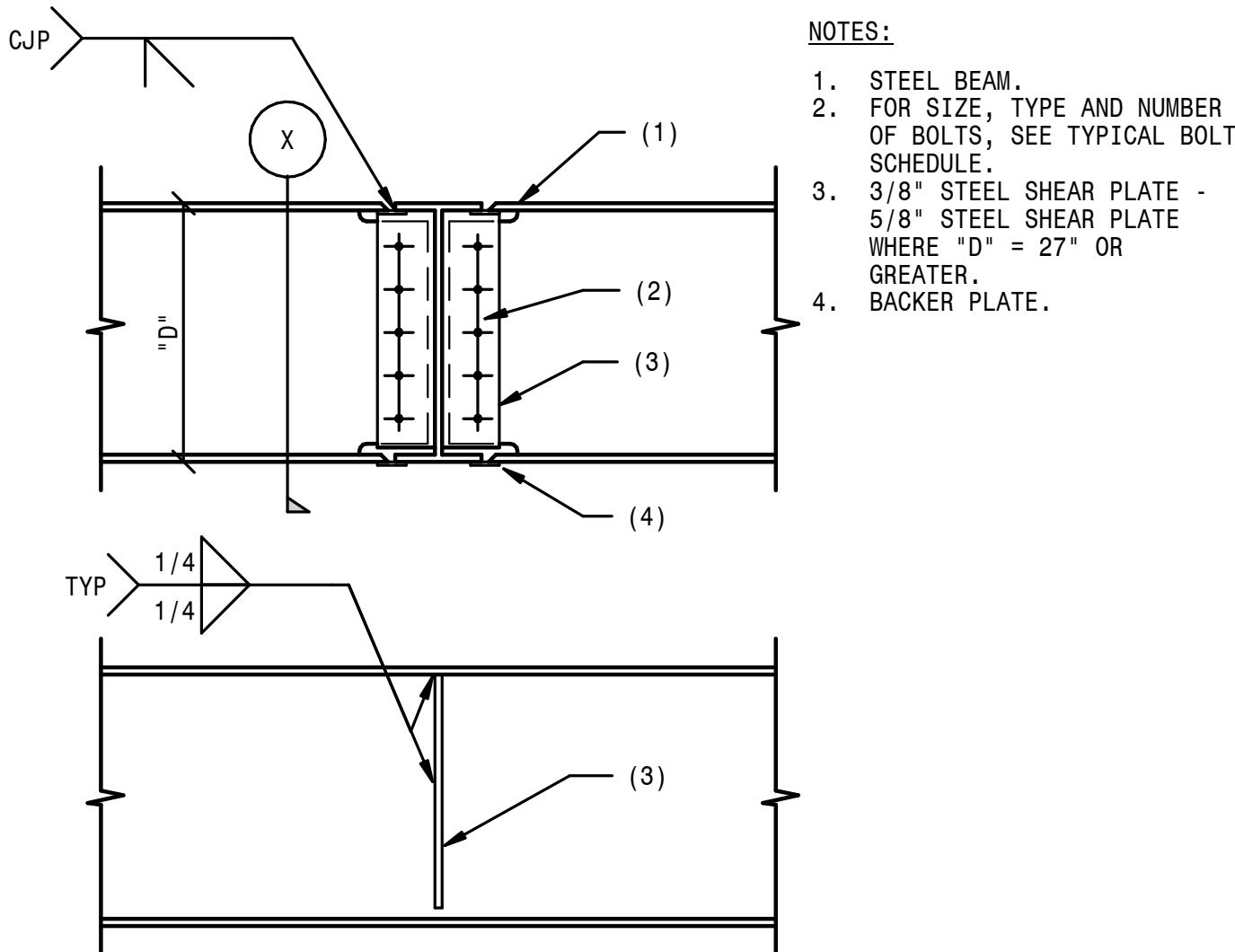
5 STEEL BEAMS TO STEEL COLUMN CONNECTION
NO SCALE



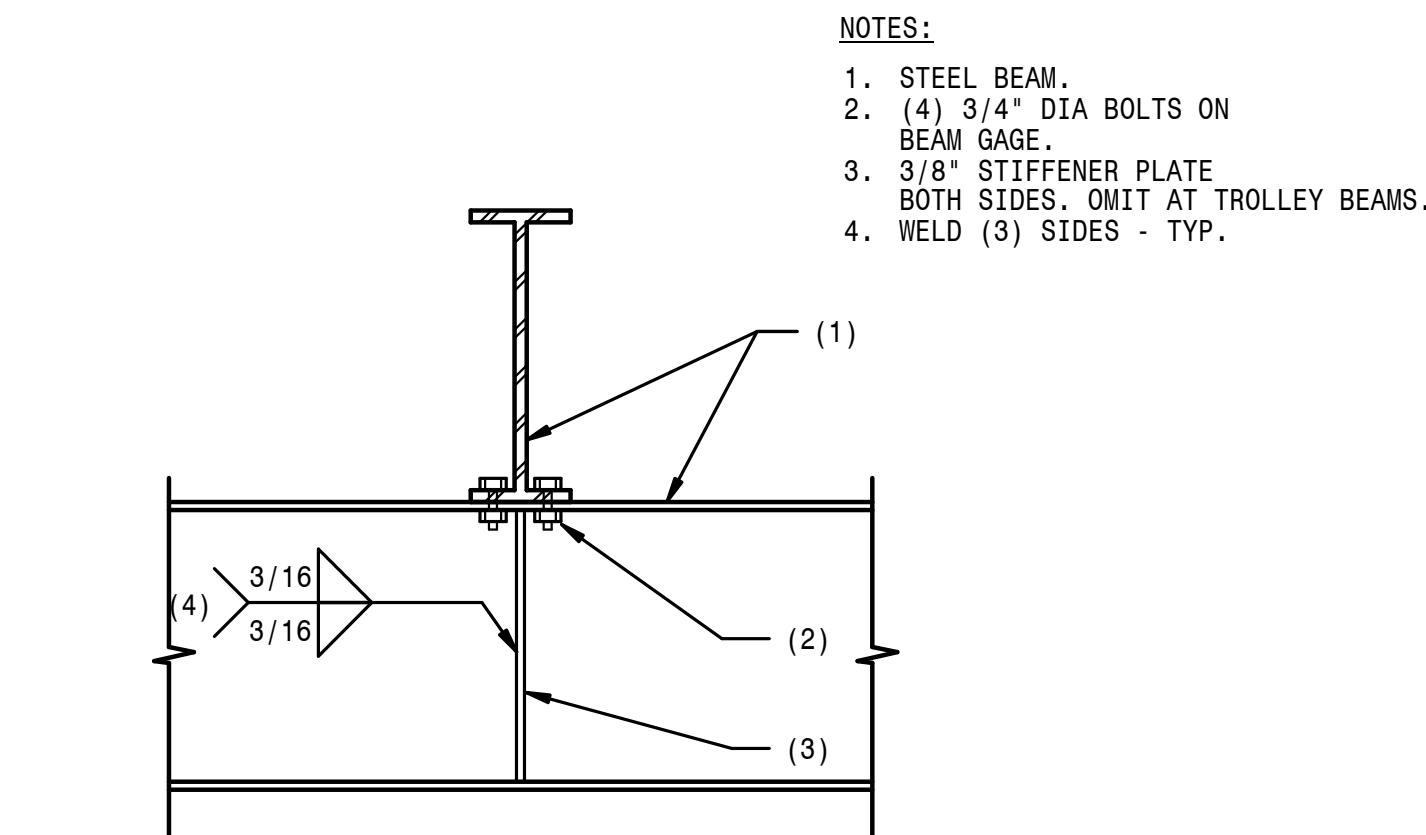
2 CONTINUOUS CHORD ANGLE/LEDGER ANGLE SPLICE
THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.
NO SCALE



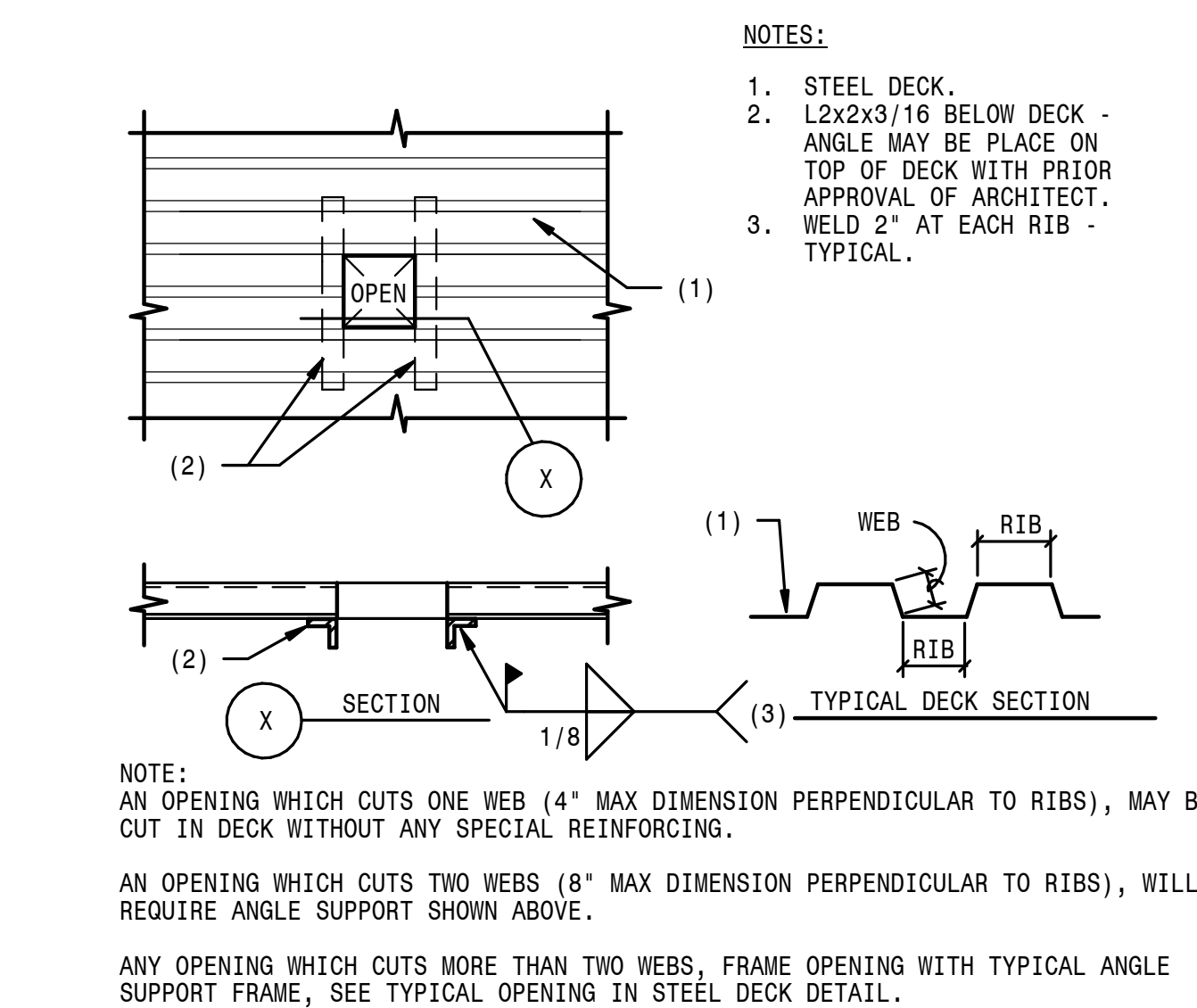
12 BEAM TO BEAM CONNECTION AT APP BAY
NO SCALE



9 TYPICAL MOMENT CONNECTION - WIDE FLANGE BEAM TO BEAM
NO SCALE



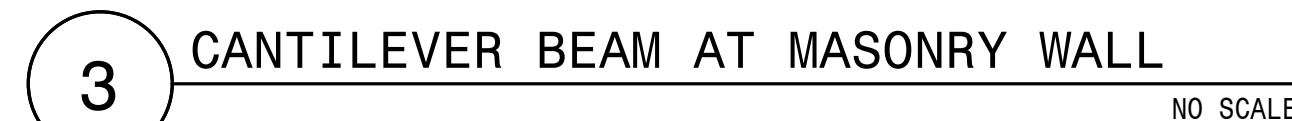
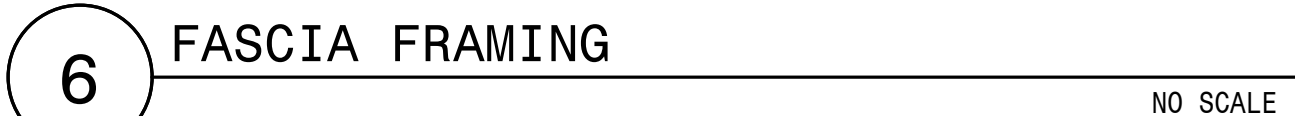
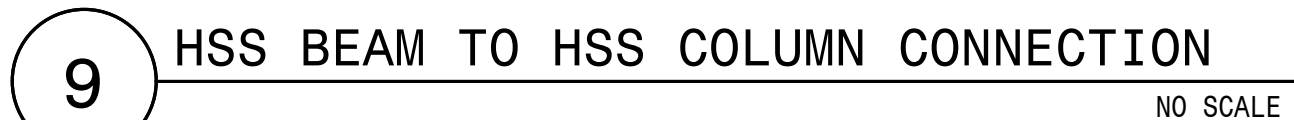
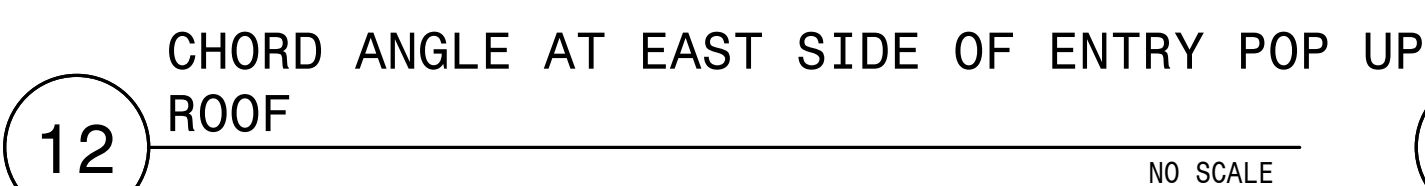
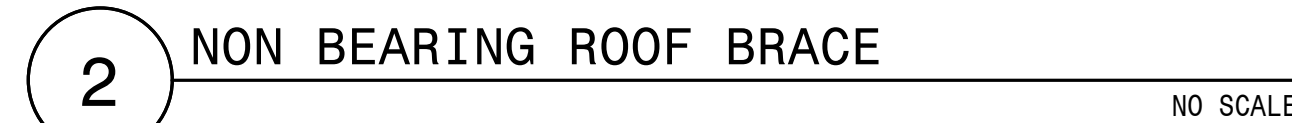
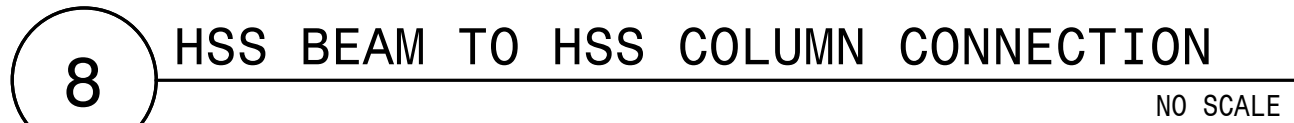
6 STEEL BEAM TO STEEL BEAM CONNECTION
THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.
NO SCALE



3 TYPICAL SMALL OPENING IN STEEL DECK
THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS.
NO SCALE

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1. TROLLEY BEAM PER PLAN. DO NOT INSTALL UNTIL ALL ROOFING DL HAS BEEN APPLIED.
2. BEAM PER PLAN.
3. 3/8" STIFFENER EACH SIDE ABOVE TROLLEY LOCATION.
4. AFTER ALL ROOF DEAD LOAD IS APPLIED, USE LEVELING PL1/2+X5X0'-8 AS REQ'D - VIF.
5. JOISTS PER PLAN.
6. H58x6x3/8 x VIF. DO NOT INSTALL UNTIL ROOF BEAM HAS FULL ROOFING DEAD LOAD.
7. LEVELING PL1/2+X5X1'-0 AS REQ'D - VIF.
8. L4X4X1/4 WHEEL STOP EA SIDE.



1. TROLLEY BEAM PER PLAN.
2. L4X4X1/2.
3. L8X4X1/2XCONT ATTACHED TO EA BEAM.
4. 2" LONG SLOT WITH 3/4" DIA BOLT
FINGER TIGHT AND DING.
5. PL1/2X4X0'-6
6. TOP FLANGE OF BM - BEYOND.



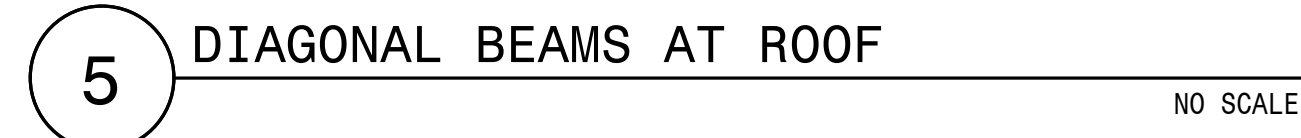
1. BEAM PER PLAN.
2. JOIST PER PLAN,
3. WALL FRAMING PER SCHEDULE.
4. L4X4X1/4 CHORD ANGLE.
5. ROOF DECK PER PLAN.
6. 1000S200-97 W/ (2) SCREWS AT
EA STUD.



1. STEEL BEAM.
2. 3/4" GUSSET PLATE.
3. BEAM CONTINUES - RE: PLAN.
4. 2 ROWS OF (4) - 3/4" DIA. THRD STUDS ON BEAM
GAGE. STUDS SHALL BE CJP WELDED TO HSS.
5. 3/4" CAP PLATE.
6. STEEL COLUMN.
7. WELD 3 SIDES - TYP.
8. HSS SLOTTED BRACE.



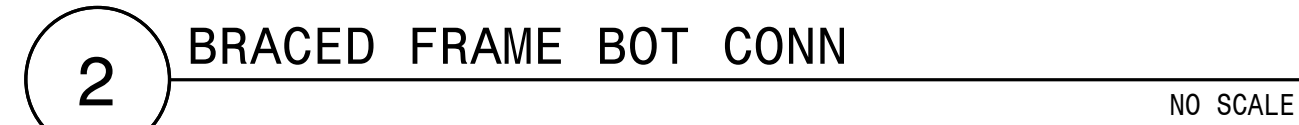
1. BEAM PER PLAN - TYP.
2. COLUMN PER PLAN.
3. PL1/2X8X0'-10". SLOPE TOP TO MATCH BEAM.



1. CONC ON MTL DECK PER PLAN.
2. BEAM PER PLAN.
3. COLUMN PER PLAN.
4. L4X4X1/4 EA SIDE OF COLUMN W/
LONG SLOTTED HOLE.
5. 3/4" DIA THRU BOLT, DING THREADS.



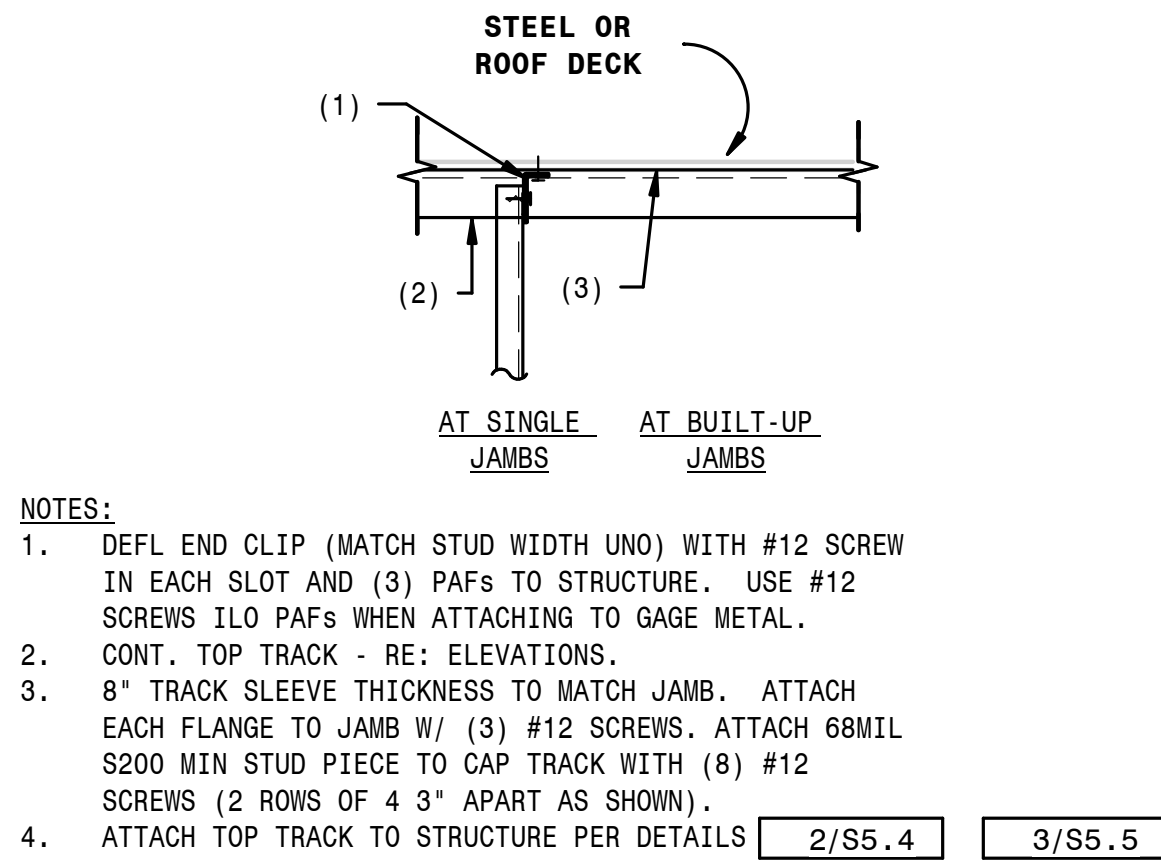
1. HSS FRAME.
2. HSS DIAGONAL.
3. CONCRETE STEM WALL OR FOOTING.
4. EMBED PLATE 1X6.
5. (4) #5 (A706) X 1'-6" VERTICAL WELDED TO EMBED PLATE.
6. 3/4" GUSSET PLATE.
7. (2) #6 X 6' 0" TOP CENTERED ON COLUMN.



1. HSS SLOTTED BRACED FRAME PER:
2. GUSSET PL 3/4".



3.5.3



DEFLECTION END CLIP OPTIONS	
MFR	CLIP DESIGNATION
THE STEEL NETWORK (TSN)	VERTICLIP SL
CLIP NOTES:	
1. MATCH CLIP DEPTH WITH STUD DEPTH.	

5 TYPICAL JAMB CONNECTION TO HEAD OF WALL
THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS. NO SCALE

- NOTES:
- WALL STUDS PER ELEV.
 - JAMB STUDS PER ELEV.
 - TRACK TO STRUCT PER

3/S5.5	(AT CONC) OR
2/S5.4	(AT STL)
 - REQUIRED GAP = 1"
 - BRIDGING PER

4/S5.4

 - RE:

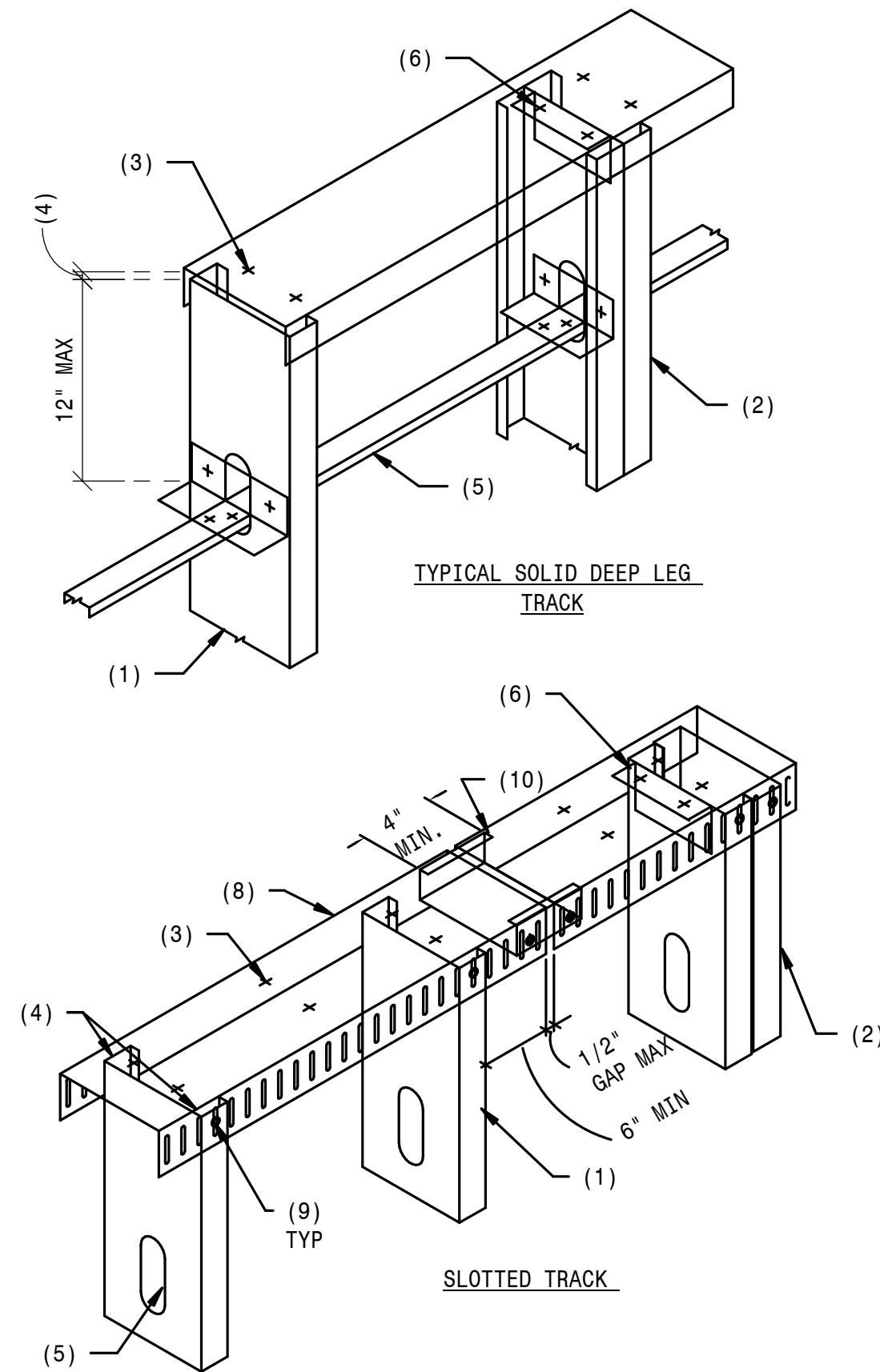
5/S5.4

 FOR TOP OF JAMB CONNECTION.

- SLOTTED TRACK NOTES:
- SLOTTED TRACK PER SCHED.
 - (1) #10 SCREW CENTERED IN SLOT EA FLANGE OF EA STUD.
 - SPLICE TRACK USING 4" LONG MIN. PIECE OF WALL STUD INSIDE TRACK WEB. ATTACH THROUGH SOLID FLANGE AREA (BTWN SLOTS) WITH (1) #10 SCREW EA SIDE OF SPLICE EA FLANGE (4 TOTAL SCREWS).

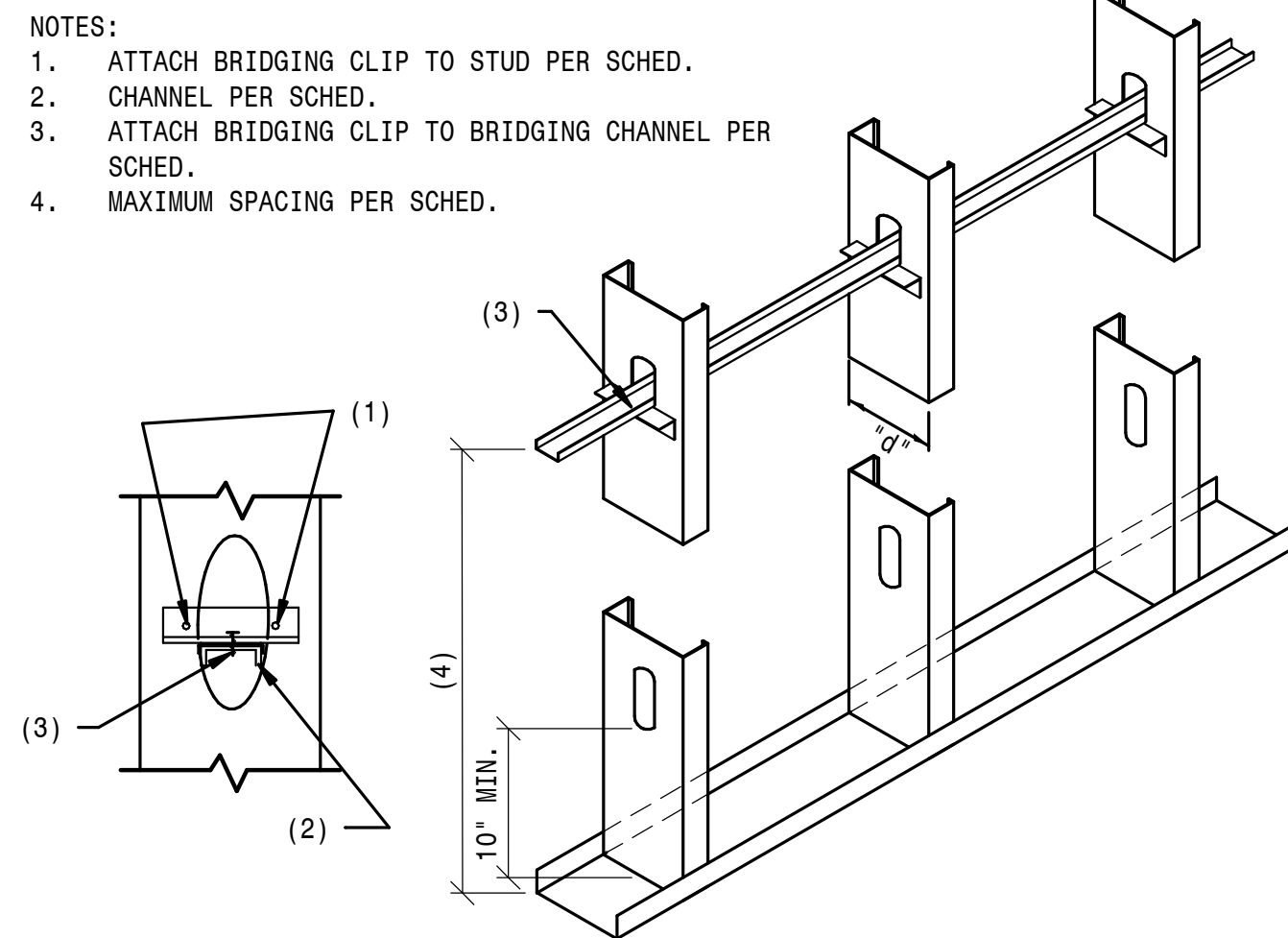
NOTE: SLOTTED TRACKS NOT TO BE USED FOR SLOPES GREATER THAN 1/2:12.

NOTE: SLOTTED TRACK MAY ONLY BE USED AS TOP TRACK.

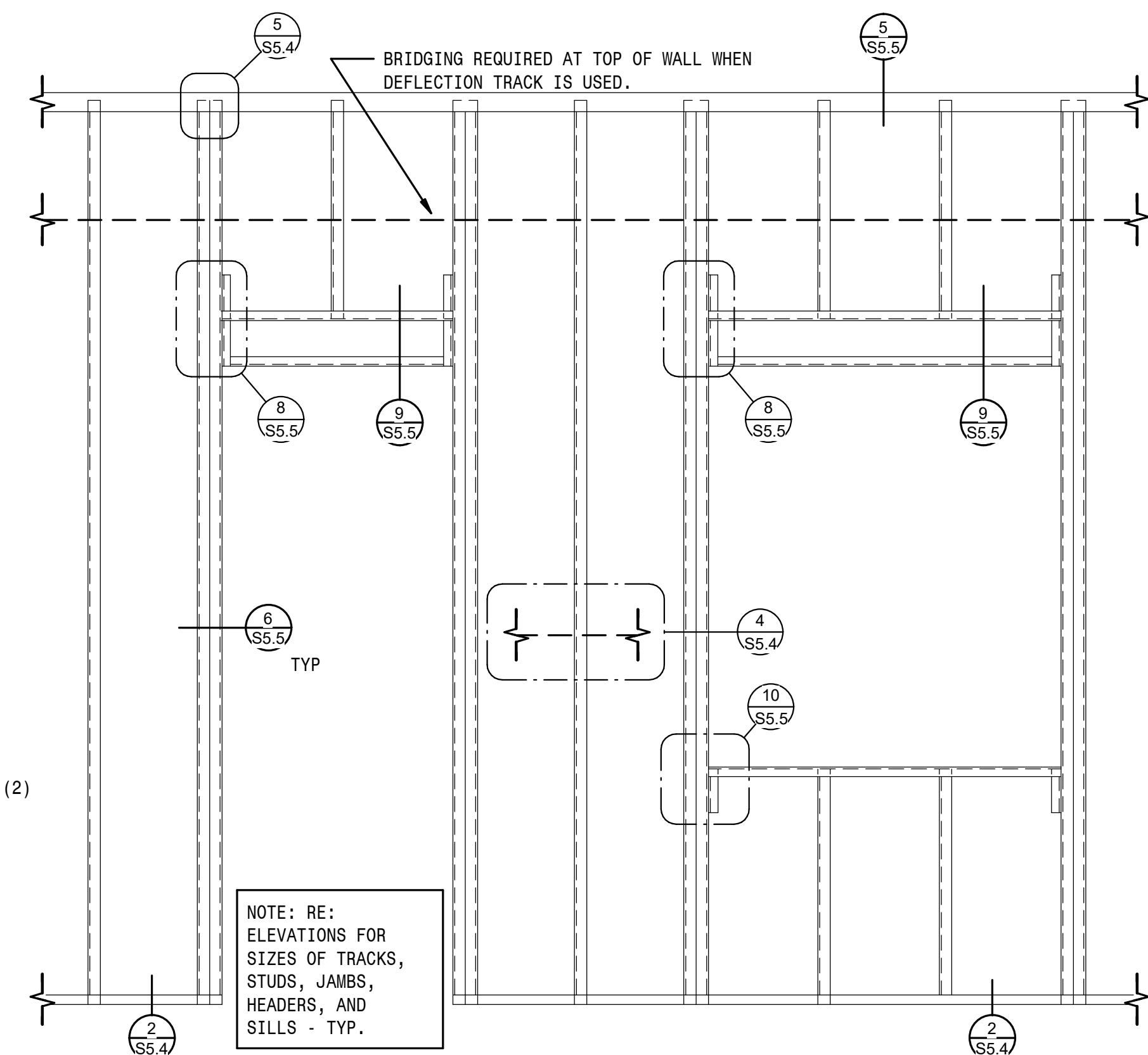


3 TOP OF WALL DEFLECTION TRACK
THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS. NO SCALE

BRIDGING SCHEDULE			
CLIP	CHANNEL	SCREWS FROM CLIP TO CRC	SCREWS FROM CLIP TO STUD
BRIDGECLIP BC43,3	1 1/2" U-CHANNEL	(1)	(2)
BRIDGING SPACING REQUIREMENTS			
WALL TYPE OR CONDITION	SPACING		
WALL OR SOFFIT SPANS < 8' - 0"	BRIDGING NOT REQUIRED		
ALL OTHER WALLS UNO BELOW	4' - 0" OC MAX		
NOTES:			
1. BRIDGING PRODUCTS OTHER THAN THOSE LISTED ABOVE MUST BE REVIEWED BY THE ENGINEER PRIOR TO USE.			

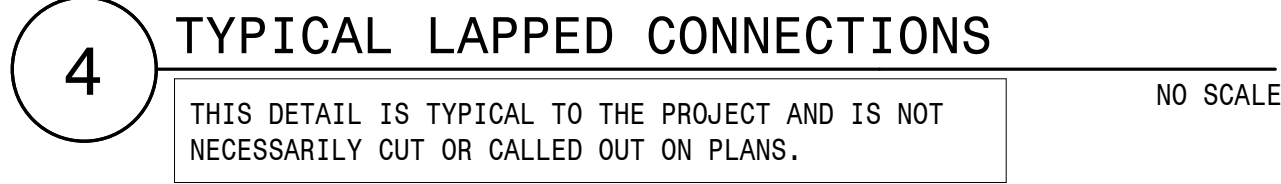
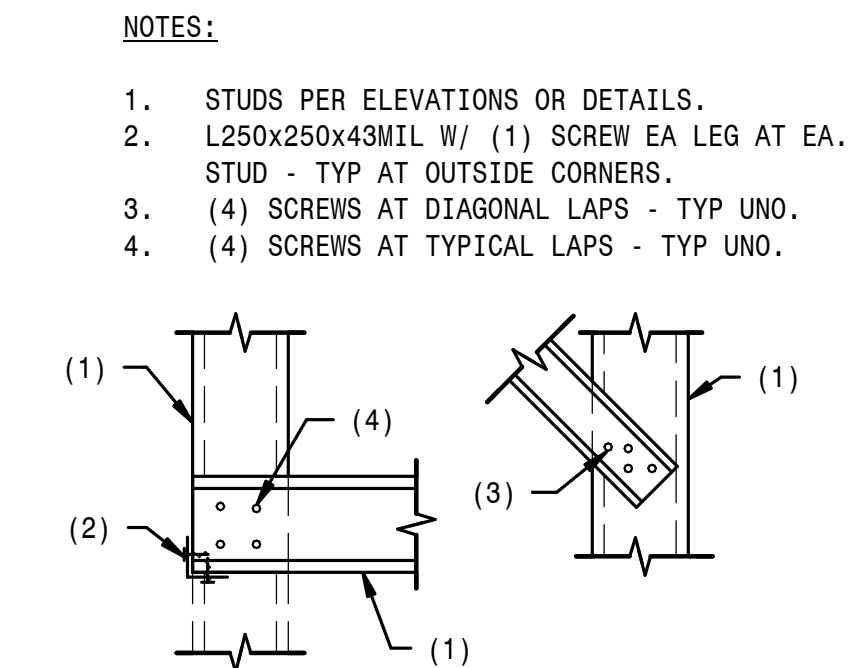
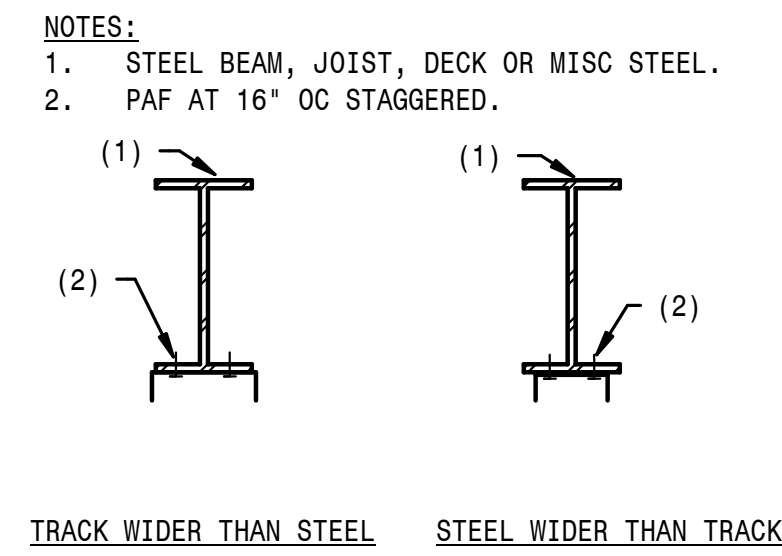
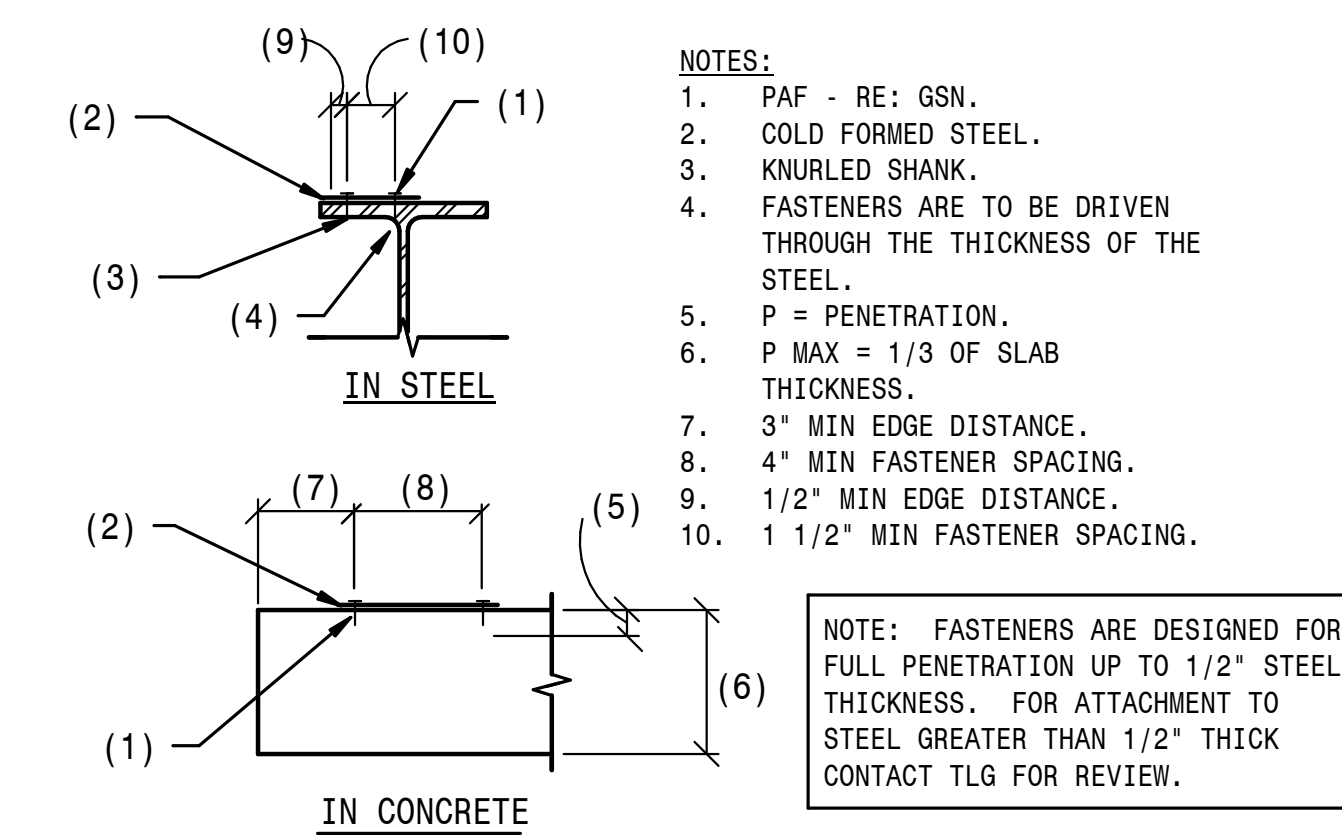
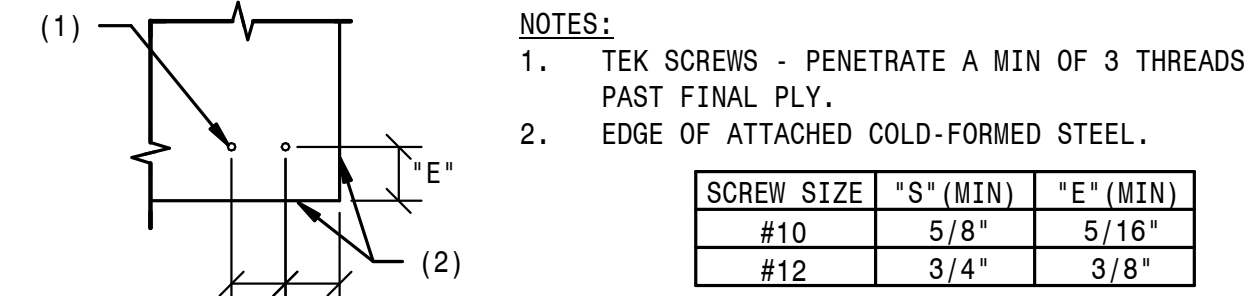
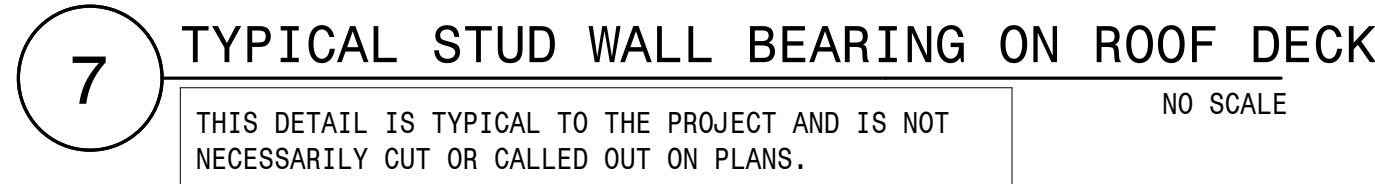
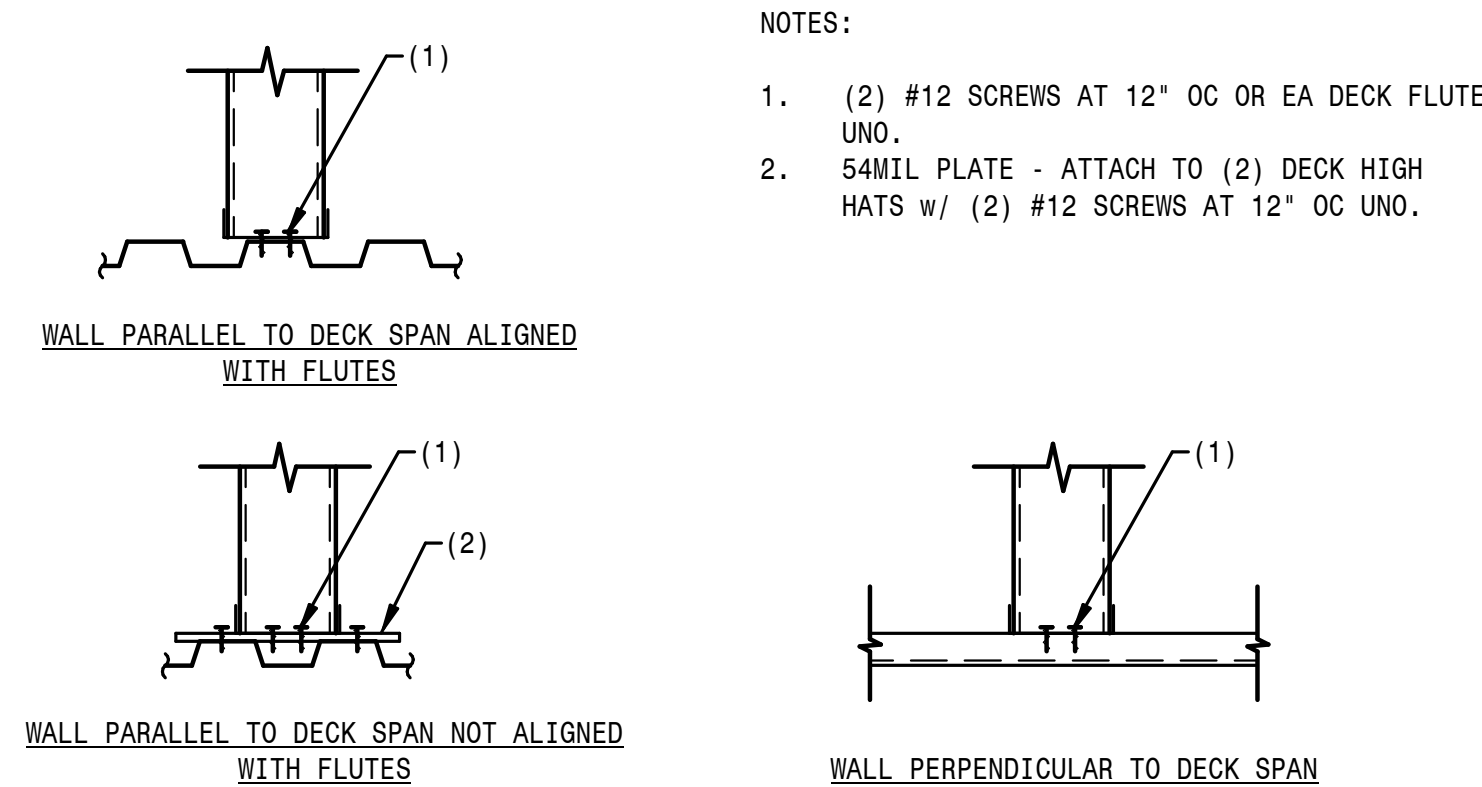
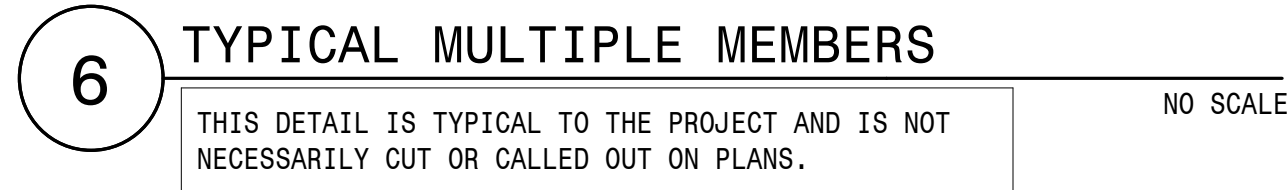
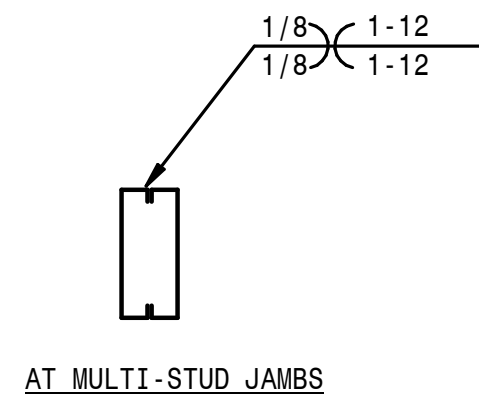
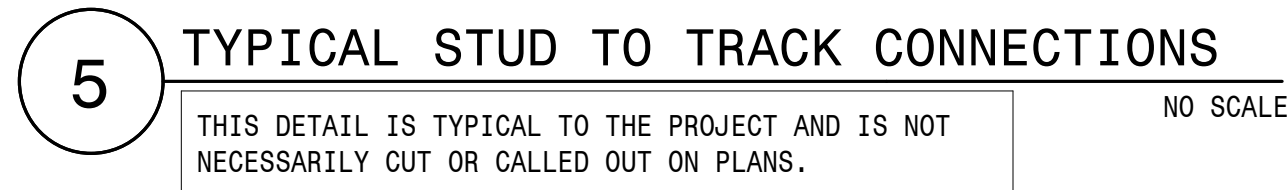
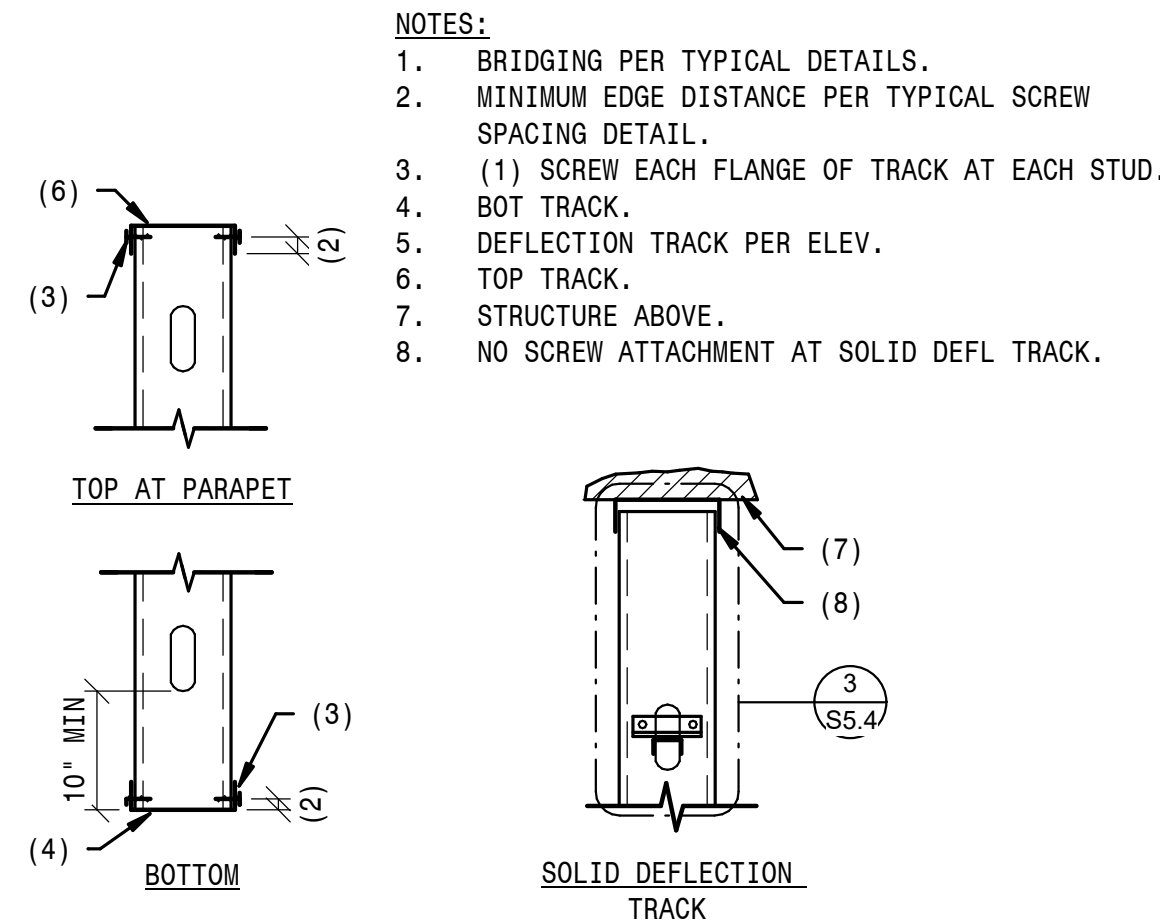
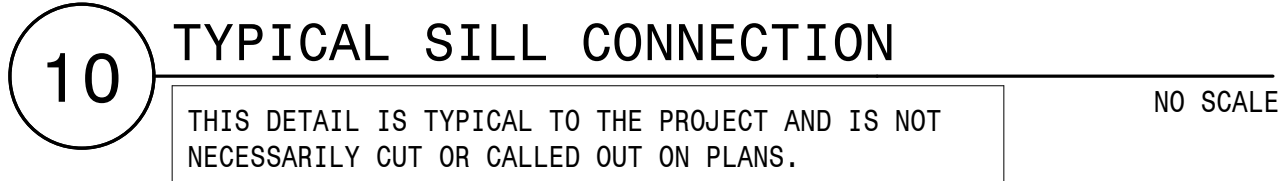
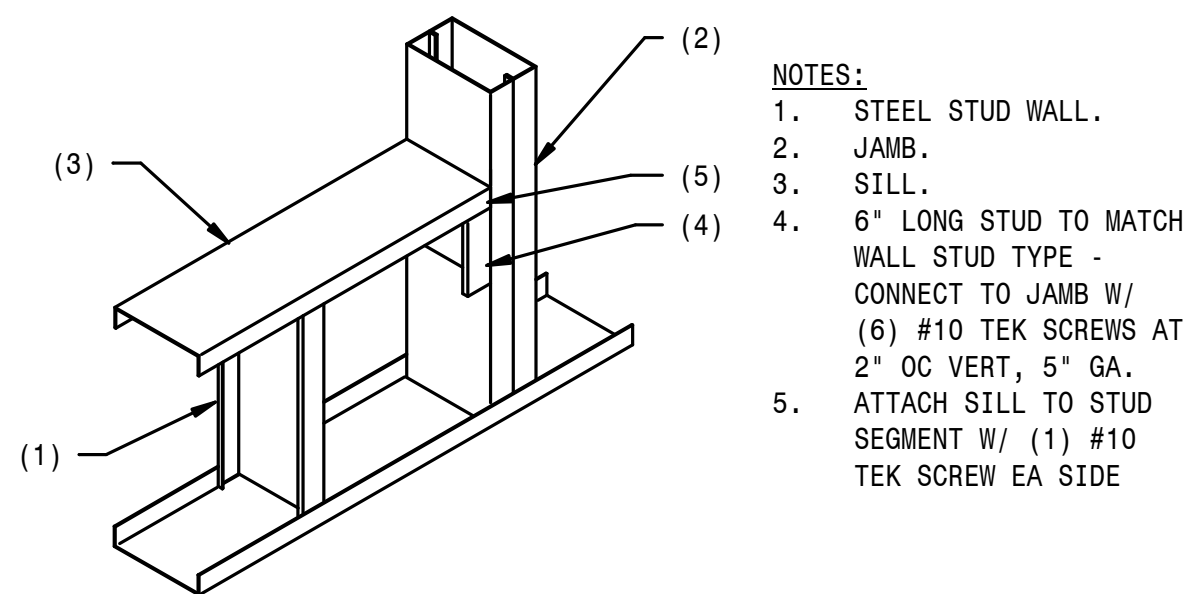
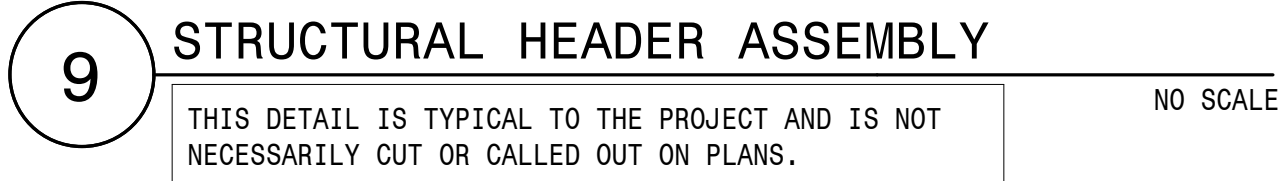
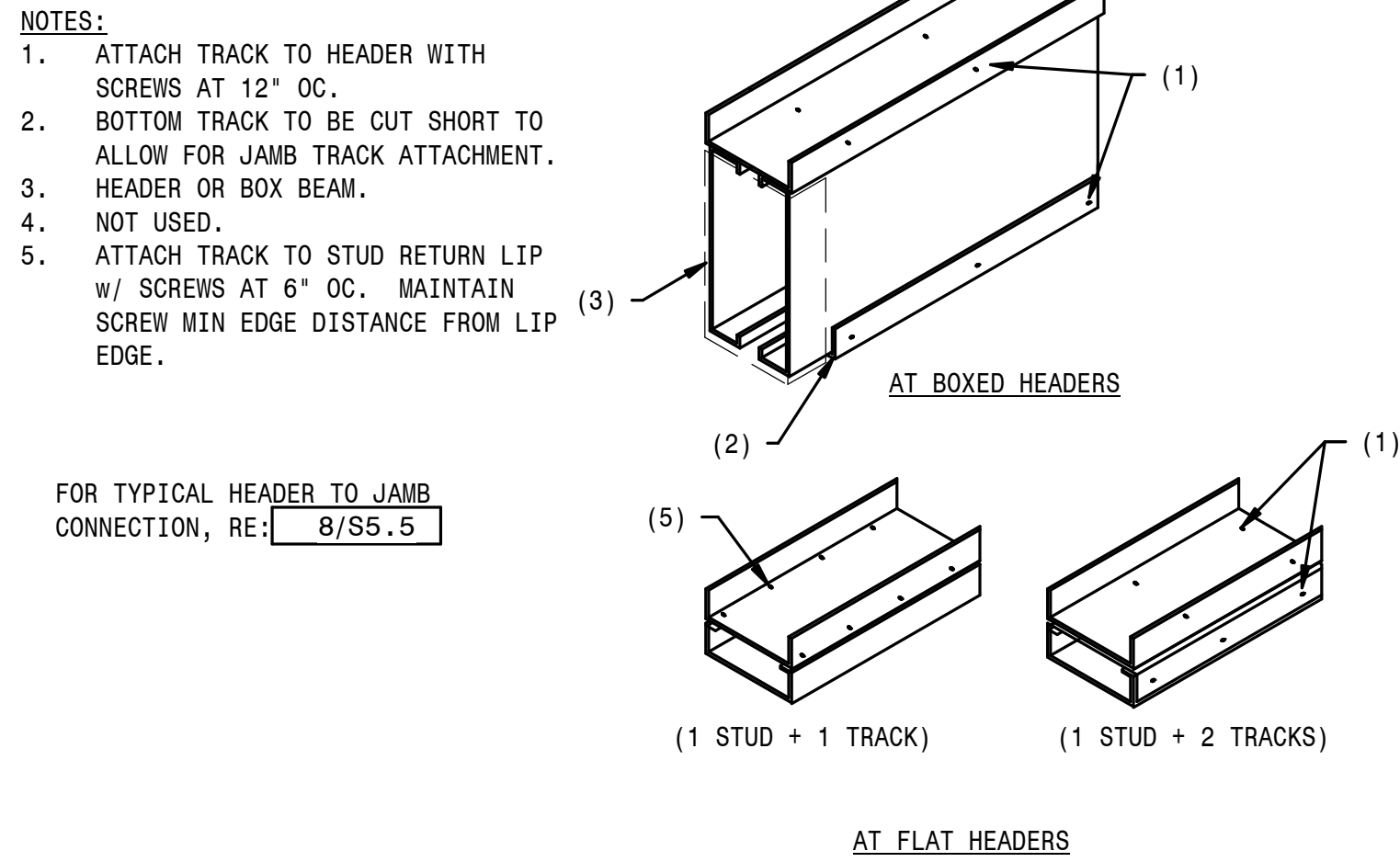
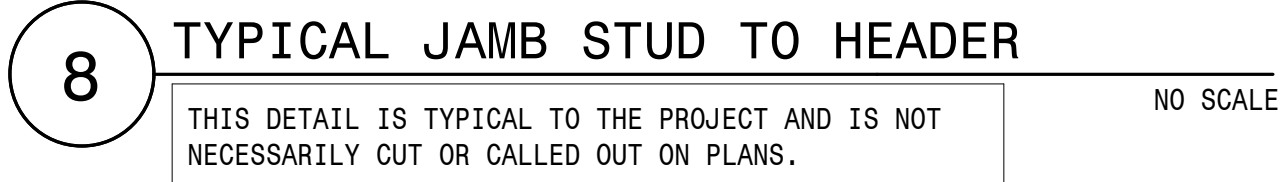
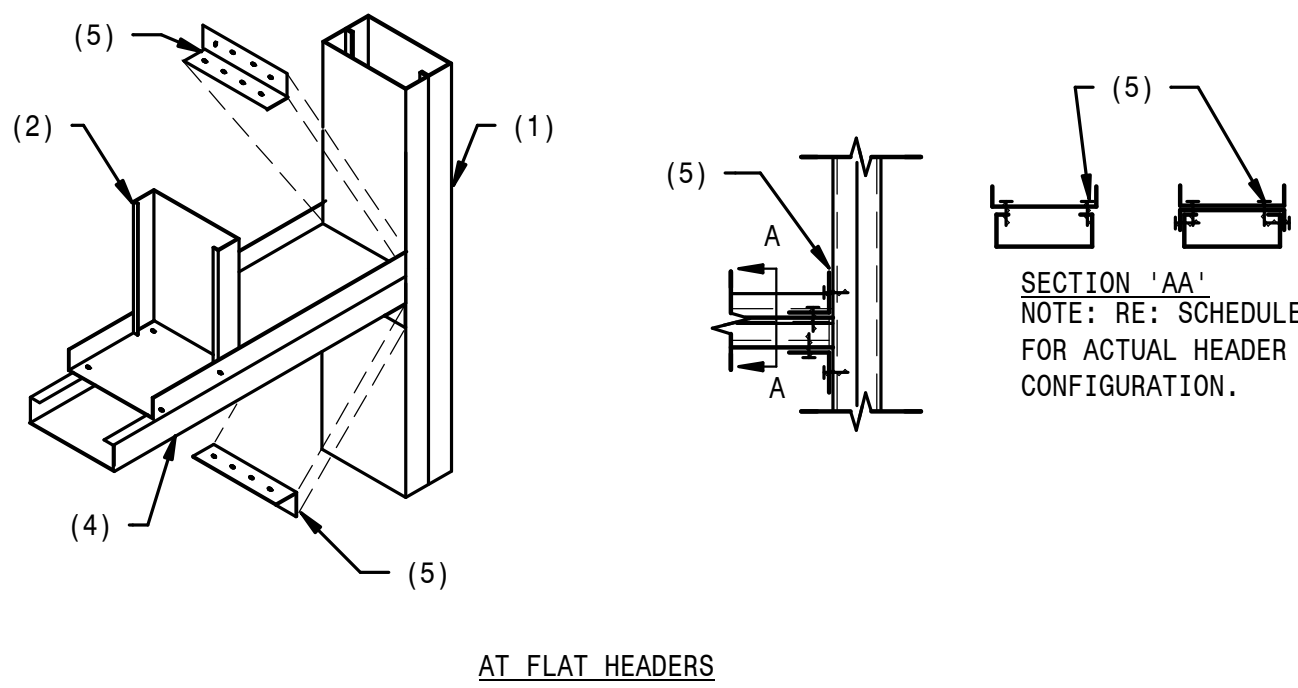
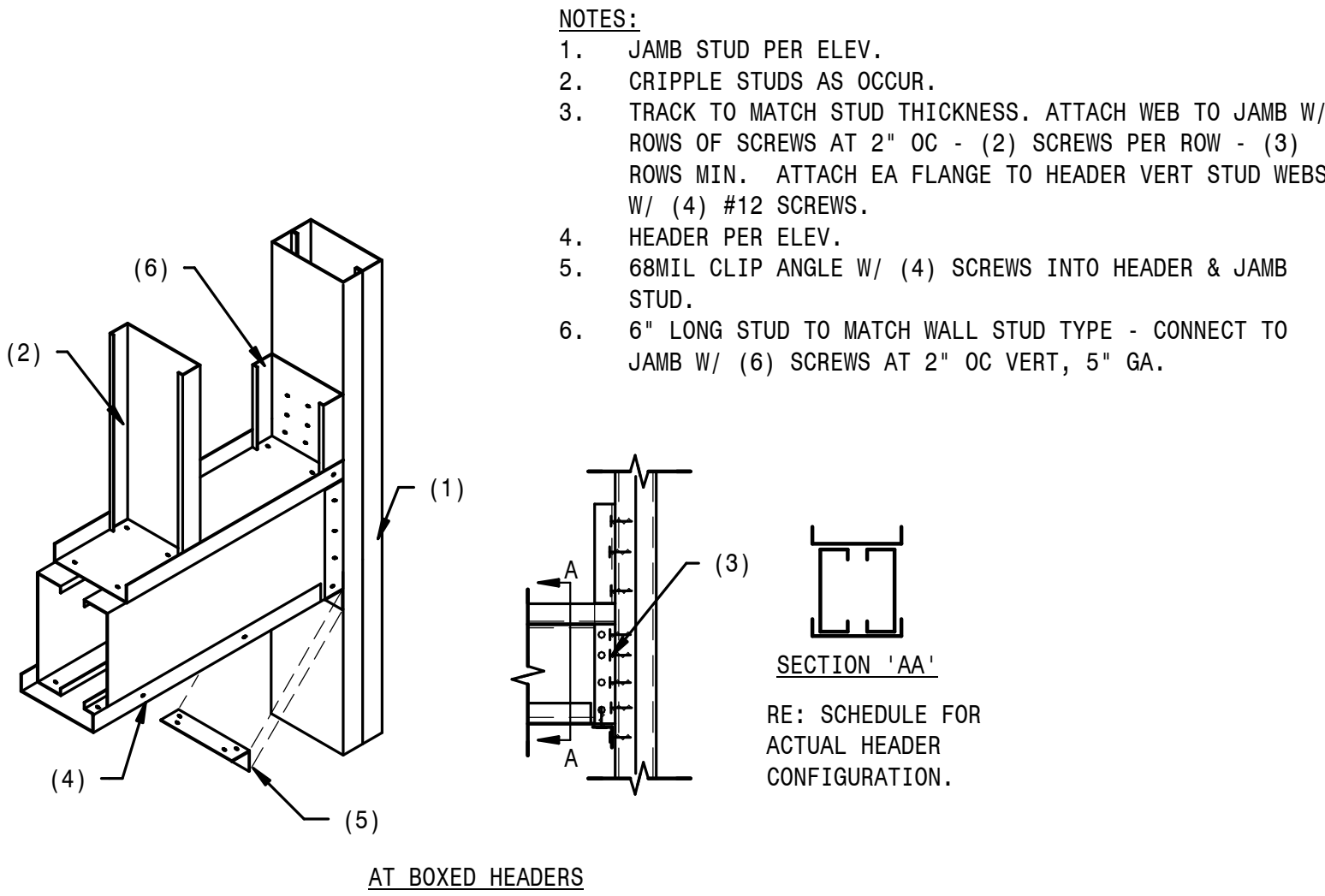
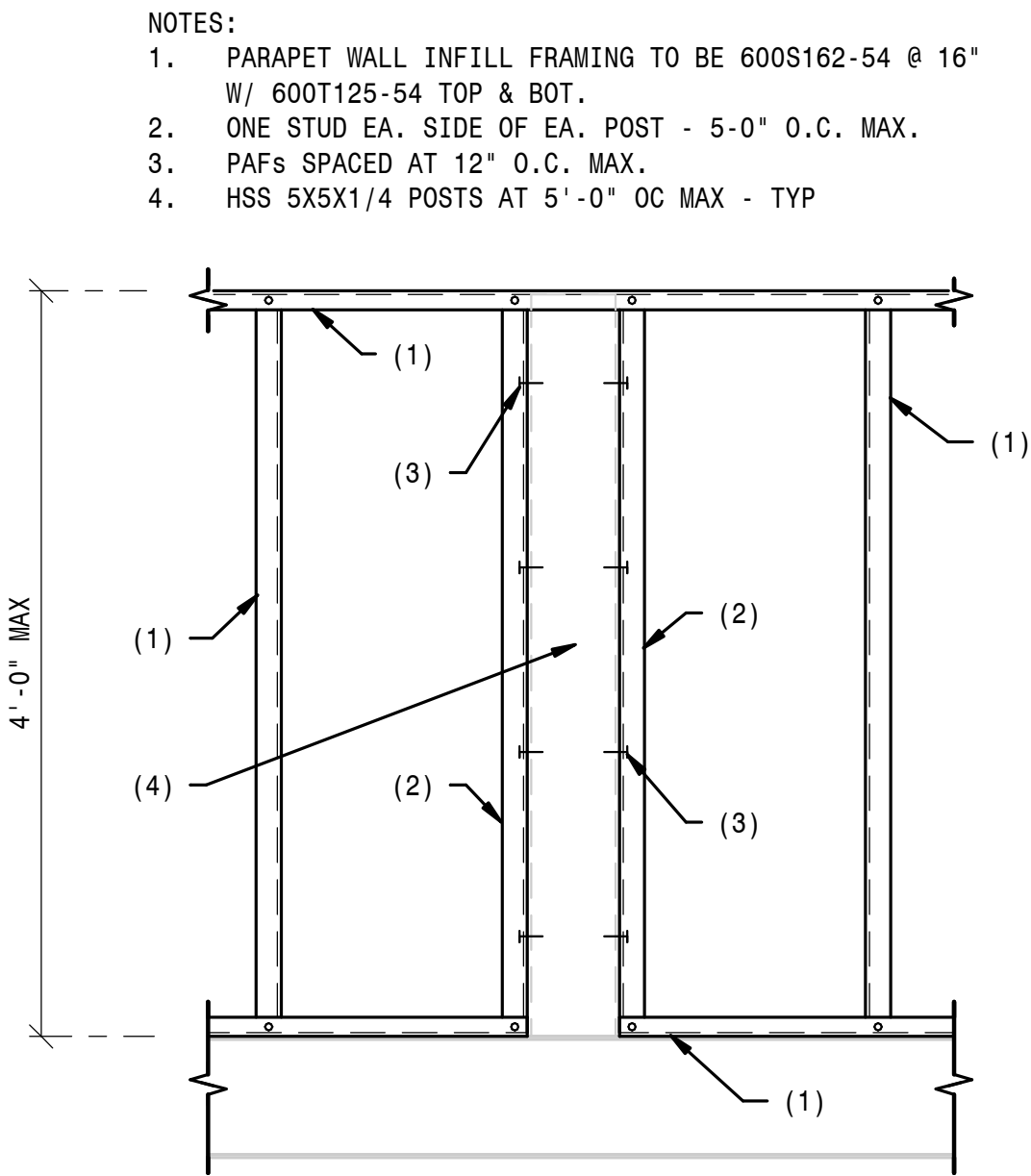
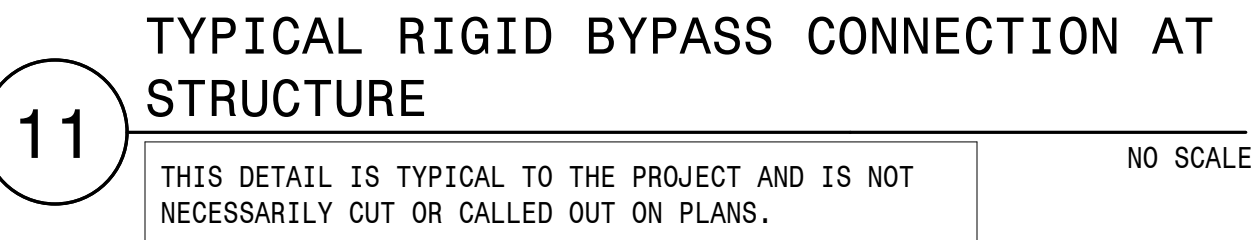
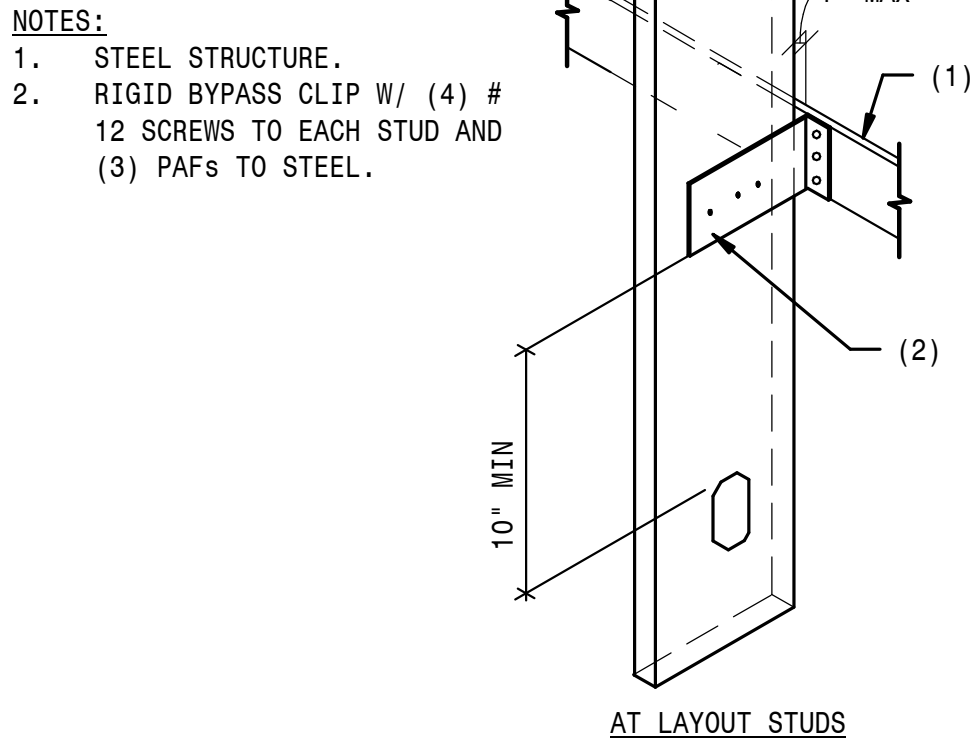


4 STRUCTURAL STUD BRIDGING
THIS DETAIL IS TYPICAL TO THE PROJECT AND IS NOT NECESSARILY CUT OR CALLED OUT ON PLANS. NO SCALE



RIGID BYPASS CLIP OPTIONS	
MFR	CLIP DESIGNATION
THE STEEL NETWORK (TSN)	STIFFCLIP LB

CLIP NOTES:
1. MATCH CLIP DEPTH WITH STUD DEPTH.
2. NOTIFY ENGINEER IF STUD STANDOFF EXCEEDS 1" FROM STRUCTURE



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JOB NO. 23-04900

BID SET

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FIRE STATION #8
15343 MONACO STREET
THORNTON, COLORADO, 80620

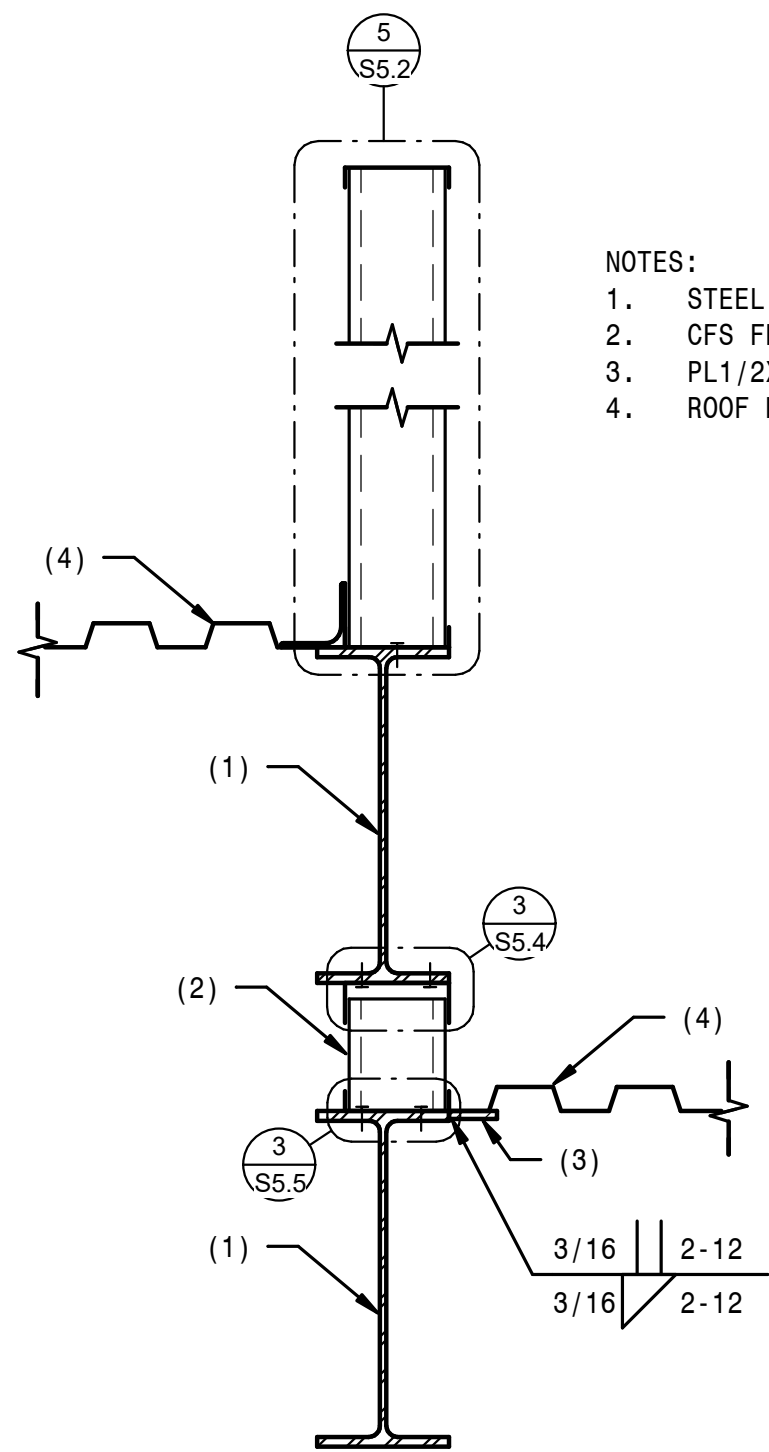
REVISION NAME	DATE

ISSUE DATE: 2024 0927

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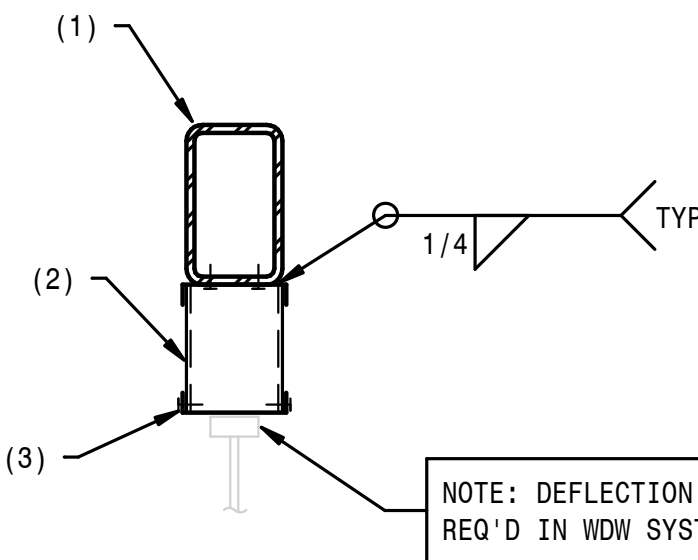
CFS DETAILS

S5.5



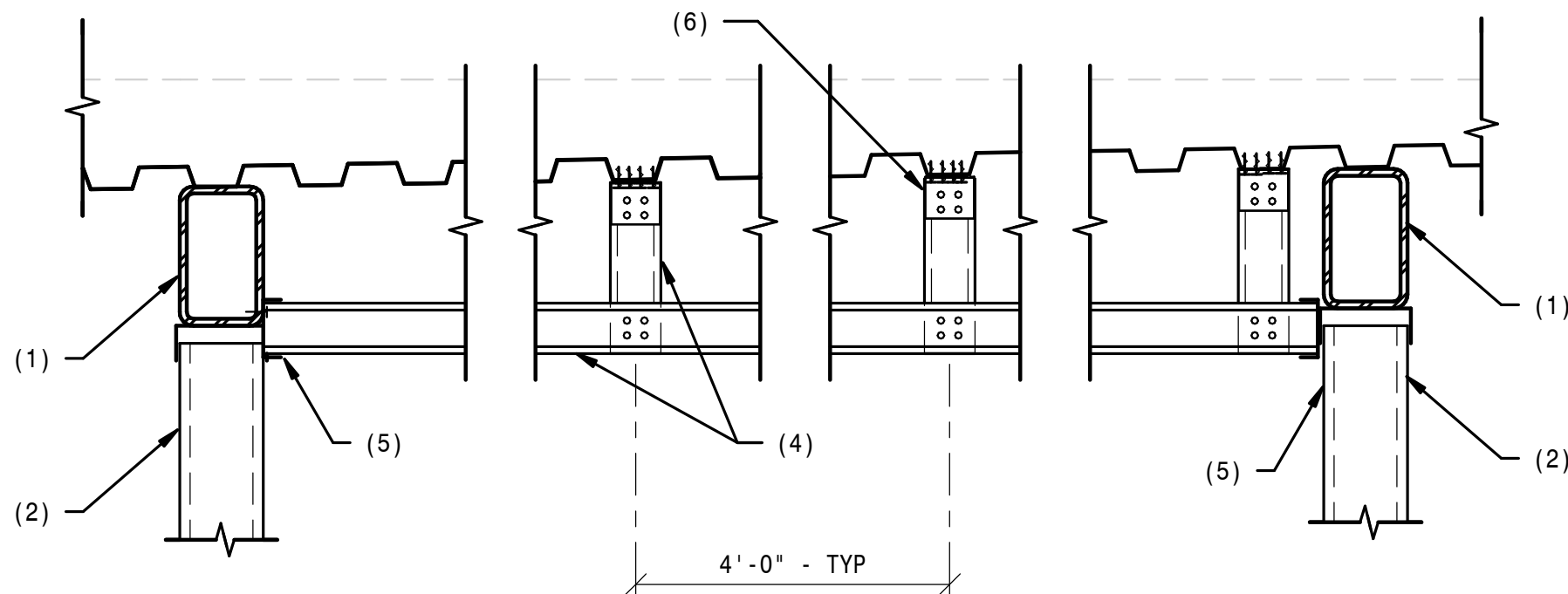
7 FRAMING AT HIGH/LOW BEAMS
NO SCALE

- NOTES:
1. STEEL BEAM PER PLAN.
 2. HSS6X6X1/4 AT EA CURTAIN WALL MULLION AND AT 5'-0" OC MAX AT STOREFRONT.
 3. CFS INFILL STUDS BTWN HSS.

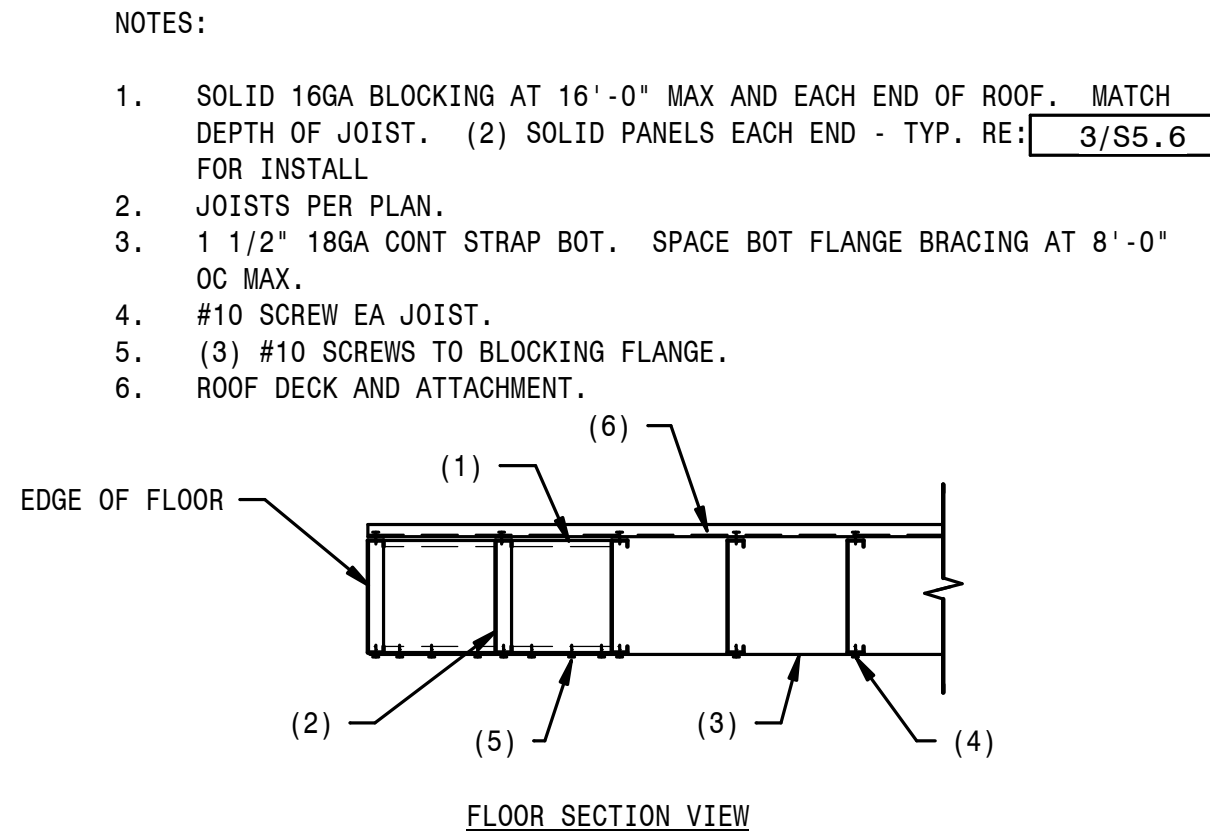


8 STOREFRONT/CURTAIN WALL SUPPORT FRAMING
NO SCALE

- NOTES:
1. STEEL BEAM PER PLAN.
 2. WALL FRAMING PER SCHEDULE.
 3. METAL DECK PER PLAN.
 4. 362S162-54 AT 16" OC - TYP.
 5. 362T125-54 W/ (2) PAFs TO HSS AT EA STUD
 6. TSN AL CLIP W/ (4) #12 SCREWS TO STUD AND (4) #12 SCREWS TO DECK - TYP.

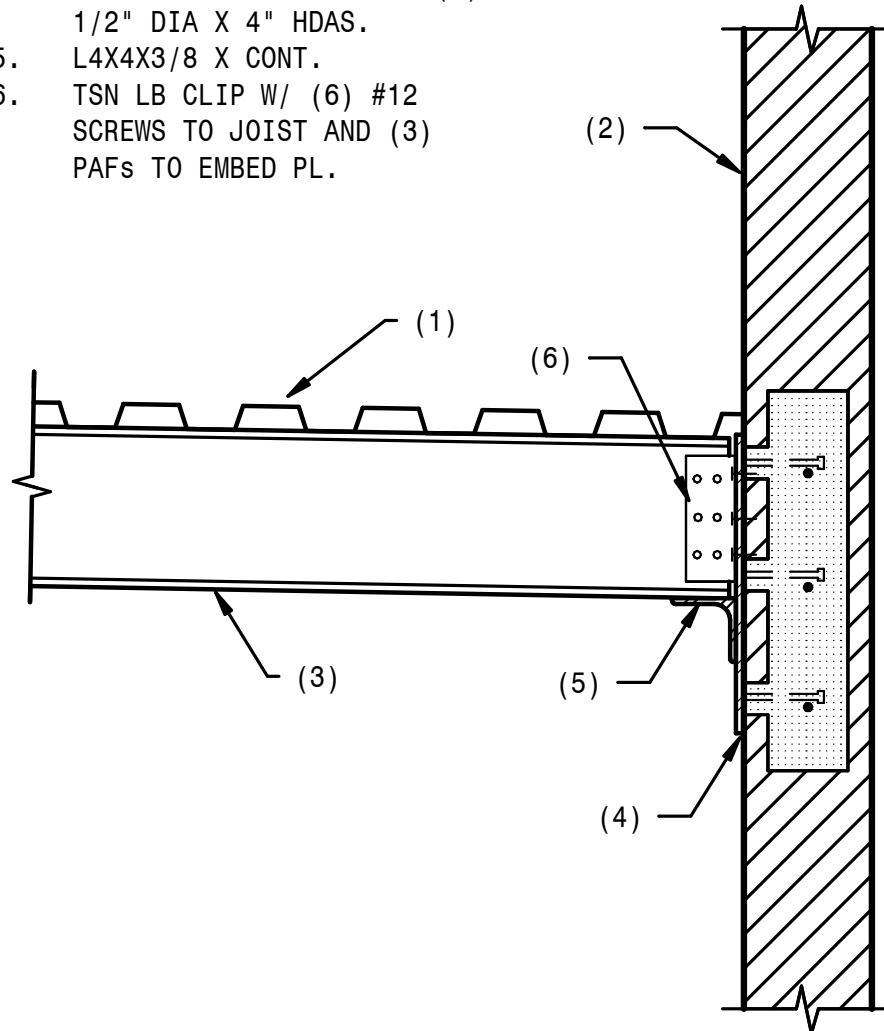


9 SOFFIT FRAMING AT LOW ROOF
NO SCALE

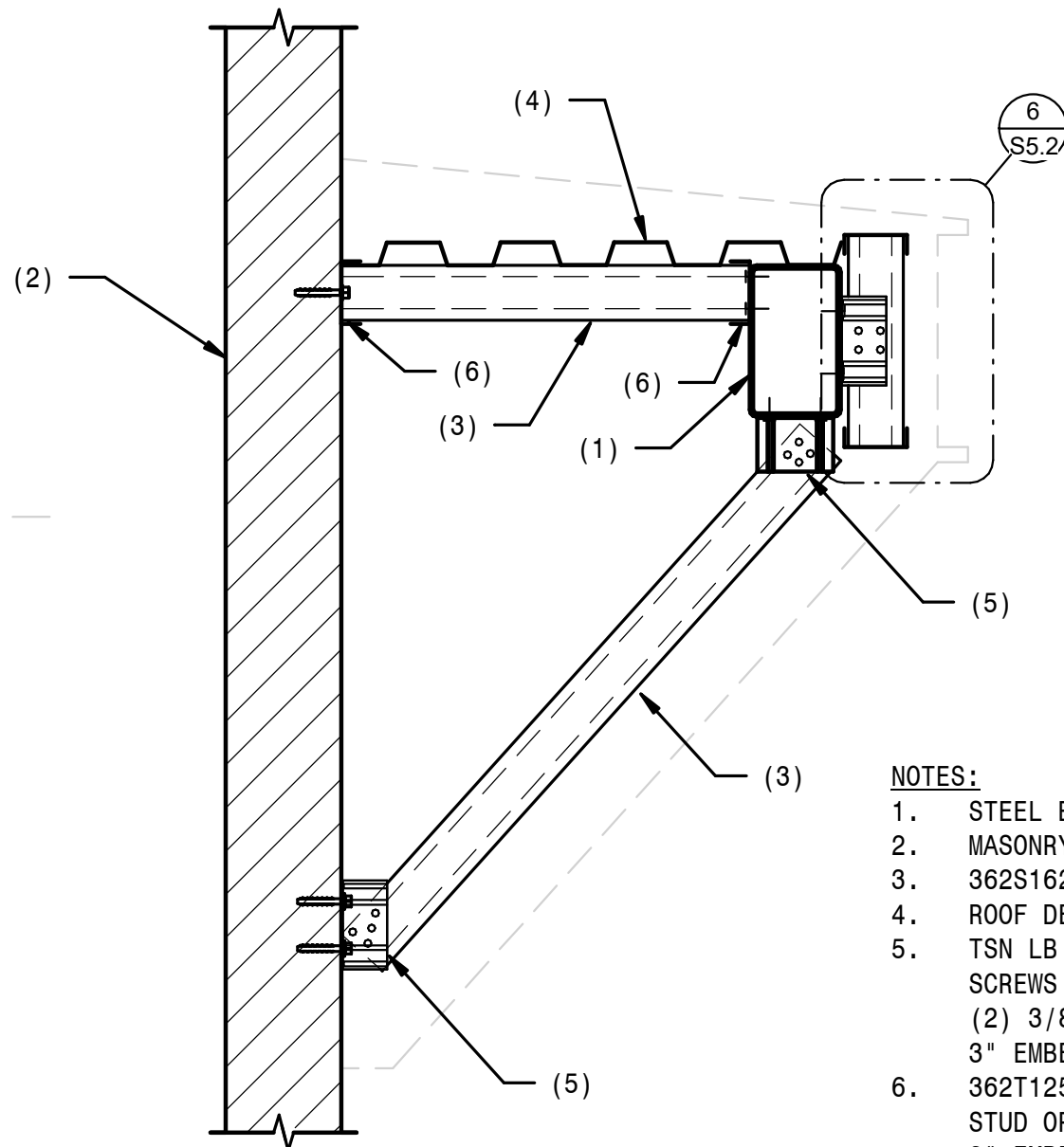


4 CFS JOIST AND RAFTER BRACING
NO SCALE

- NOTES:
1. ROOF DECK PER PLAN.
 2. CMU WALL PER PLAN.
 3. JOIST PER PLAN.
 4. EMBED PL 1/2x8X1'-6 W/ (6) 1/2" DIA X 4" HDAS.
 5. L4X4X3/8 X CONT.
 6. TSN LB CLIP W/ (6) #12 SCREWS TO JOIST AND (3) PAFs TO EMBED PL.

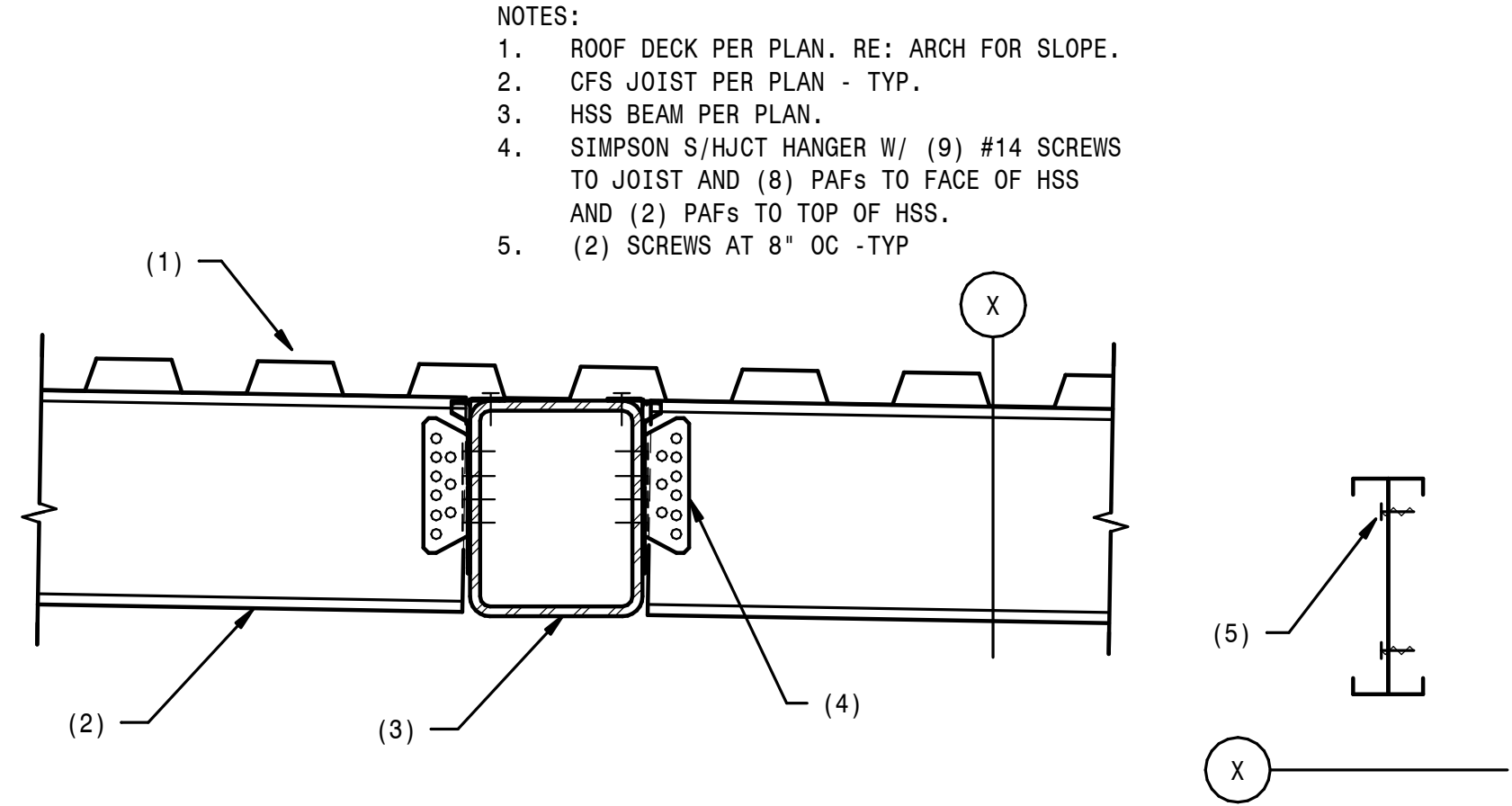


5 STEEL JOIST AT CMU WALL
NO SCALE



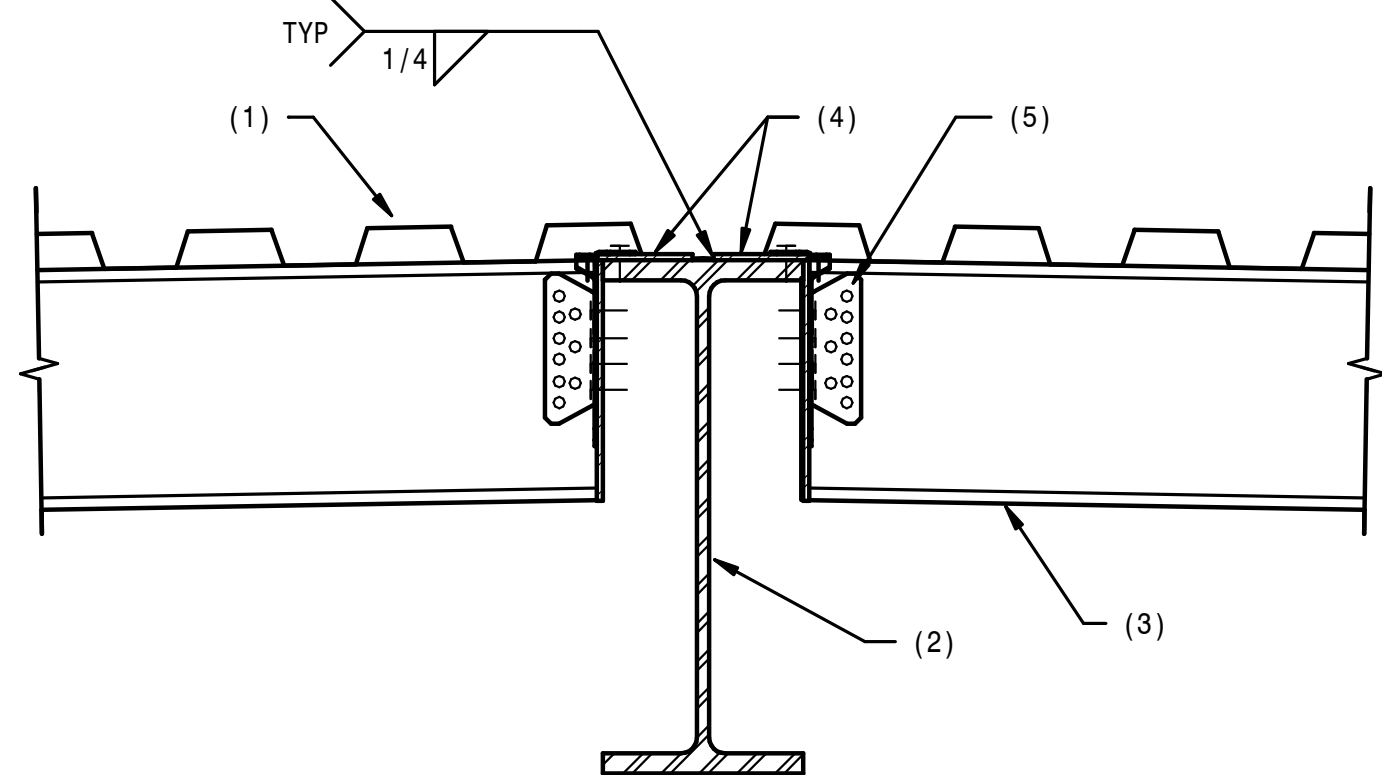
6 SOFFIT FRAMING
NO SCALE

- NOTES:
1. STEEL BEAM PER PLAN.
 2. MASONRY WALL.
 3. 362S162-54 AT 16" OC - TYP.
 4. ROOF DECK PER PLAN.
 5. TSN LB CLIP AT EA STUD W/ (4) #12 SCREWS TO STUD AND (2) PAFs TO HSS OR (2) 3/8" DIA TITEN HD SCREW ANCHORS W/ 3" EMBED TO MASONRY WALL.
 6. 362T125-54 W/ (2) PAFs TO HSS AT EA STUD OR (1) TITEN HD SCREW ANCHOR W/ 3" EMBED TO MASONRY WALL AT EA STUD.



1 STEEL JOIST AT HSS BEAM
NO SCALE

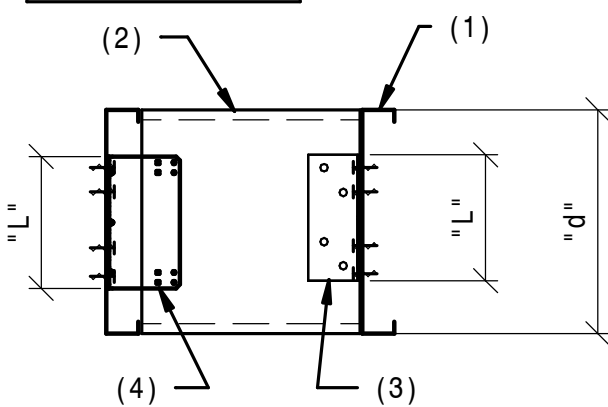
- NOTES:
1. ROOF DECK PER PLAN.
 2. STEEL BEAM PER PLAN.
 3. CFS JOIST PER PLAN.
 4. BENT PL1/4X3X1'-0 X 6" LONG AT EA JOIST.
 5. SIMPSON S/HJCT HANGER W/ (9) #14 SCREWS TO JOIST AND (8) PAFs TO FACE OF HSS AND (2) PAFs TO TOP OF HSS.



2 STEEL JOIST AT STEEL BEAM
NO SCALE

- NOTES:
1. JOIST PER PLAN.
 2. FULL DEPTH 54MIL BLOCKING.
 3. TSN AL CLIP OR BENT PL4x4x54MIL W/ (4) SCREWS TO JOIST AND (4) SCREWS TO BLOCKING.
 4. AT FLANGE SIDE, SIMPSON SJC CLIP W/ (4) SCREWS TO JOIST AND (8) SCREWS TO BLOCKING.

L = 0.5d MIN.



3 CFS SOLID JOIST BLOCKING
NO SCALE

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CFS DETAILS

S5.6